Mathematical Notation Catalogue for `Analysis/FabiusFunction`

Canonical symbols, exact definitions, local namespaces, and historical aliases

Encyclopedic companion to Vladimir Reshetnikov's `ProveIt` formalization

Notation contract 2026-09-01

Active corpus at the validation snapshot: 162 recursively inventoried `.tex` sources

Abstract

This catalogue is the normative dictionary for mathematical notation in the active `Analysis/FabiusFunction` documentation corpus. It distinguishes the bounded Fabius function, the signed global extension, and Rvachev's up-function; fixes number-system, probability, transform, binary, q-series, asymptotic, fractional-calculus, approximation, and operator conventions; and records every recurring collision discovered by a recursive audit of the current 162-source active `.tex` corpus. Each entry states its type, domain, codomain, indices, parameters, normalization, exceptional values, and formalization status. No phrase such as "with the usual convention" supplies missing data: when two usual conventions exist, both are defined and their conversion is printed. Historical papers and frontier reports remain mathematically interpretable through explicit alias and local-namespace crosswalks rather than through overloaded bare letters.

Source/PDF version notice. This TeX source contains the post-replay notation-catalogue union and its incoming presentation polish. The retained PDF predates that union, so source/PDF parity is not claimed; a fresh uninterrupted three-pass Libertinus rebuild is pending.

Normative scope. The command definitions in `docs/fabius-notation.tex`, version 2026-09-01, are normative for active project-authored LaTeX. This catalogue defines their mathematical meaning. The directory `docs/archive/` is not an active migration target and is used only as provenance. A historical source under `docs/papers/` may display its original notation only inside an explicit historical scope. A source-faithful whole-paper transcription opens that scope with the exact preamble marker `HISTORICAL-NOTATION-SCOPE`; a shorter quotation labels the historical convention at the quotation itself. The marker never exempts surrounding project prose, which uses the canonical symbols. Included tables and generated fragments inherit the namespace of their standalone parent document.

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1 Scope contract and how to use this catalogue

1.1 What counts as notation

The inventory includes a glyph or compound expression whenever its interpretation requires one or more mathematical choices: a domain, codomain, index set, endpoint rule, transform normalization, branch, sign, scaling, convergence mode, or proof-status boundary. Thus it includes not only named functions but also interval brackets, empty sums, 0^0, arrows of convergence, subscripts on transforms, and superscripts denoting derivatives or powers. Typography that changes no mathematical meaning—for example the insertion of braces in `\mathbb{R}`—is not a separate notation.

The catalogue has two levels.

1. A *global semantic symbol* has one meaning throughout the active corpus and, where available, one command in `fabius-notation.tex`.
2. A *local symbol* is scoped to a definition, theorem, table, or explicitly named document part. A local symbol must be declared where its scope begins and must not be cited outside that scope without qualification.

The local level is essential: a corpus this broad legitimately uses A, D, I, P, Q, and T for many unrelated temporary objects. Unification means making those scopes and meanings explicit, not forcing every matrix and polynomial to share a single name.

1.2 The no-implicit-parts checklist

Every new or revised notation declaration answers all applicable questions below.

1. What mathematical object is denoted: scalar, set, map, measure, sequence, matrix, operator, equivalence class, or proposition?
2. What are its domain and codomain? If it is indexed, what is the complete index set?
3. What is the exact defining formula, including all summation and product bounds?
4. Which parameters are fixed, which vary, and what inequalities or nonvanishing conditions do they satisfy?
5. Which normalization, sign, Fourier frequency, logarithm, complex-power branch, and measure convention is used?
6. What happens at endpoints, at an empty index set, at zero, and at a removable singularity?
7. Does equality mean pointwise equality, equality almost everywhere, equality of measures, equality in distribution, or asymptotic equivalence?
8. Is the assertion formally proved, a standard definition, derived only in a frontier report, conjectural, refuted, numerical, or historical?

1.3 Status vocabulary

Status	Exact meaning in this catalogue
Definition	A definition or typography contract; no theorem is asserted.
Lean-proved	The stated mathematical content has a named proved Lean declaration in the module identified by the crosswalk.
Classical	A standard mathematical convention used by the prose; its presence in the catalogue is not a claim that the project has formalized the general theorem.

Status	Exact meaning in this catalogue
Frontier-derived	Derived in a research-frontier document but not certified here as having an exact current Lean counterpart.
Conjectural	Proposed and explicitly unproved.
Refuted	A historical formula or normalization for which the project records a counterexample or negative theorem.
Historical alias	A symbol needed to read a source artifact; it is not canonical for new project prose.

2 Canonical quick reference

The first column is the literal shared command name. The second column is its rendering; the third fixes its semantics. Commands taking arguments show metavariables in braces.

Shared command	Rendering	Normative meaning
`\FabiusNotationVersion`	2026-09-01	Date-valued version of the shared notation contract; it is metadata, not a mathematical variable.
`\NaturalNumbers`	$\mathbb{N}_0$	Nonnegative integers $\{0, 1, 2, \ldots\}$.
`\PositiveIntegers`	$\mathbb{N}_{>0}$	Strictly positive integers $\{1, 2, 3, \ldots\}$.
`\IntegerNumbers`	$\mathbb{Z}$	Integers.
`\RationalNumbers`	$\mathbb{Q}$	Rational numbers.
`\RealNumbers`	$\mathbb{R}$	Real numbers.
`\ComplexNumbers`	$\mathbb{C}$	Complex numbers.
`\ExtendedRealNumbers`	$\overline{\mathbb{R}}$	Extended reals $\mathbb{R} \cup \{-\infty, +\infty\}$.
`\FabiusBounded`	F_{bd}	Bounded, CDF-normalized Fabius function.
`\FabiusGlobal`	F_{gl}	Signed global Fabius extension.
`\FabiusQuantile`	Q_F	Ordinary inverse of $F_{\mathrm{bd}}\|_{[0,1]}$.
`\FabiusClampedQuantile`	$Q_{F,\mathrm{cl}}$	Totalized clamped inverse on all real inputs.
`\RvachevUp`	up	Rvachev's even compactly supported up-function.
`\EulerE`	e	Base of the natural logarithm.
`\ImaginaryUnit`	i	The complex number satisfying $\mathrm{i}^2 = -1$.
`\Differential`	$\mathrm{d}x$	Upright differential in an integral or differential form.
`\AbsoluteValue{x}`	$\lvert x\rvert$	Absolute value or, when a field is explicitly complex, complex modulus.
`\Norm{x}`	$\lVert x\rVert$	The norm supplied by the explicitly named normed space.
`\Floor{x}`	$\lfloor x\rfloor$	Greatest integer not exceeding x.
`\Ceiling{x}`	$\lceil x\rceil$	Least integer not below x.
`\FractionalPart{x}`	$\{x\}$	$x - \lfloor x\rfloor$ for $x \in \mathbb{R}$.
`\PositivePart{x}`	$(x)_+$	$\max\{x, 0\}$ for an ordered scalar x.
`\IndicatorOf{A}`	$\mathbf{1}_A$	Indicator of a set A, equal to one on A and zero off A.
`\SetOf{x:P(x)}`	$\{x : P(x)\}$	Delimiters for a set specified by membership data or a predicate; the ambient type is stated locally.
`\InnerProduct{x,y}`	$\langle x, y\rangle$	Inner-product delimiters; the scalar field, space, and linear-slot convention are stated locally.
`\CardinalityOf{A}`	$\lvert A\rvert$	Finite cardinality or cardinal number according to the declared type; never measure or absolute value.

Shared command	Rendering	Normative meaning
`\DefinitionEquals`	$a \coloneqq b$	Definitional assignment: the left-hand symbol is introduced to mean exactly the right-hand object.
`\Sign`	sgn	Real sign function with $\operatorname{sgn}(0) = 0$.
`\IdentityOperator`	id	Identity map on the domain stated at the point of use.
`\SpanOperator`	span	Linear span over the stated scalar field.
`\TraceOperator`	tr	Matrix or trace-class operator trace.
`\RankOperator`	rank	Dimension of an image, over the stated field.
`\DiagonalOperator`	diag	Diagonal matrix/operator constructor.
`\ResidueOperator`	Res	Complex residue at a stated point.
`\LeastCommonMultiple`	lcm	Nonnegative least common multiple; finite-family and empty-family rules must be stated locally.
`\OrderOperator`	ord	An order invariant whose species (vanishing, pole, multiplicative, group-element, or recurrence order) is printed.
`\PrincipalLogarithm`	Log	Principal complex logarithm with argument in $(-\pi, \pi]$ and domain $\mathbb{C} \setminus (-\infty, 0]$.
`\FallingFactorial{x}{n}`	$(x)^{\underline{n}}$	Falling factorial $\prod_{j=0}^{n-1}(x-j)$, equal to one when $n = 0$.
`\RisingFactorial{x}{n}`	$(x)^{\overline{n}}$	Rising factorial $\prod_{j=0}^{n-1}(x+j)$, equal to one when $n = 0$.
`\SignedStirlingFirstKind{n}{k}`	$s(n, k)$	Signed coefficient of x^k in the falling factorial $(x)^{\underline{n}}$.
`\UnsignedStirlingFirstKind{n}{k}`	$\left[{n \atop k}\right]$	Number of permutations of n labels having k cycles.
`\StirlingSecondKind{n}{k}`	$\left\{{n \atop k}\right\}$	Number of partitions of n labels into k nonempty unordered blocks.
`\LahNumber{n}{k}`	$\mathsf{L}^{\mathrm{Lah}}_{n,k}$	Number of partitions of n labels into k nonempty linearly ordered lists.
`\TypeAEulerianNumber{n}{k}`	$\left\langle{n \atop k}\right\rangle$	Number of permutations of n labels with k ordinary descents.
`\TypeBEulerianNumber{n}{k}`	$\mathsf{E}^{\mathrm{Eul},B}_{n,k}$	Number of signed permutations of size n with k type-B descents, using the sentinel $\pi_0 = 0$.
`\SecondOrderEulerianNumber{n}{k}`	$\left\langle\!\left\langle{n \atop k}\right\rangle\!\right\rangle$	Number of order-n Stirling permutations with k ordinary descents.
`\TypeAEulerianPolynomial{n}`	$\mathsf{E}^{\mathrm{Eul},A}_{n}$	Type-A Eulerian row polynomial; its evaluation argument follows the command.
`\TypeBEulerianPolynomial{n}`	$\mathsf{E}^{\mathrm{Eul},B}_{n}$	Type-B Eulerian row polynomial; its evaluation argument follows the command.
`\SecondOrderEulerianPolynomial{n}`	$\mathsf{E}^{\mathrm{Eul},2}_{n}$	Second-order Eulerian row polynomial; its evaluation argument follows the command.
`\BellNumber{n}`	$\mathsf{B}^{\mathrm{Bell}}_{n}$	Number of all set partitions of n labelled elements.
`\ComplementaryBellNumber{n}`	$\mathsf{B}^{\mathrm{comp}}_{n}$	Signed block-parity sum $\sum_k (-1)^k \left\{{n \atop k}\right\}$.
`\TouchardPolynomial{n}`	$\mathsf{T}^{\mathrm{Tou}}_{n}$	Polynomial $\sum_k \left\{{n \atop k}\right\} x^k$; its evaluation argument follows.
`\ExponentialPartialBellPolynomial{n}{k}`	$\mathcal{B}^{\exp}_{n,k}$	Partial Bell polynomial in the exponential-series normalization.
`\ExponentialCompleteBellPolynomial{n}`	$\mathcal{B}^{\exp}_{n}$	Sum over all partial exponential Bell polynomials of total weight n.
`\GeometricBernoulliPolynomial{n}{q}`	$\mathrm{Ber}^{\mathrm{geom}}_{n,q}(x)$	Monic Appell polynomial that inverts averaging by the geometric uniform convolution; the evaluation argument follows the command.

Shared command	Rendering	Normative meaning
`\OrdinaryPartialBellPolynomial{n}{k}`	$\mathcal{B}^{\mathrm{ord}}_{n,k}$	Partial Bell polynomial in the ordinary-series normalization.
`\CoefficientExtraction{z}{n}`	$[z^n]$	Operator extracting the coefficient of z^n from the following series operand.
`\CycleCountOperator`	cyc	Number of cycles of the following permutation operand.
`\SetPartitionOperator`	Part	Set of set partitions of the following finite-set operand.
`\BitwiseAnd`	$a \mathbin{\&} b$	Bitwise AND on declared nonnegative integers.
`\ExponentialGeneratingFunctionOf{A}`	A^{egf}	Exponential generating function of the named coefficient family.
`\OrdinaryGeneratingFunctionOf{A}`	A^{ogf}	Ordinary generating function of the named coefficient family.
`\NormalizedReversionCoefficient{f}{j}`	$f^{\mathrm{rev,norm}}_j$	Normalized exponential-series input $f_{j+1}/((j+1)f_1)$ used in compositional-reversion formulas.
`\OrdinaryGeneratingCoefficient{x}{j}`	x^{ogf}_j	Explicit family-and-index ledger coefficient in an ordinary generating series.
`\PartitionLatticeMinimum`	$\bot_{\Pi}$	Discrete singleton partition, the minimum of a declared partition lattice.
`\PartitionLatticeMaximum`	$\top_{\Pi}$	One-block partition, the maximum of a declared partition lattice.
`\TraceBellArgument{k}`	s^{tr}_k	Scaled trace input $-(k-1)!\,\mathrm{tr}(A^k)$ for determinant/Bell identities.
`\FiniteField`	$\mathbb{F}_q$	Finite field of the declared prime-power order q.
`\CatalanNumber{n}`	$\mathsf{C}^{\mathrm{Cat}}_n$	Catalan number $\binom{2n}{n}/(n+1)$, visibly separated from complementary Bell numbers.
`\GregoryCoefficient{n}`	$\mathsf{G}^{\mathrm{Greg}}_n$	Coefficient of z^n in $z/\log(1+z)$.
`\GenocchiNumber{n}`	$\mathsf{G}^{\mathrm{Gen}}_n$	Genocchi number in the exponential generating convention specified below.
`\GenocchiPolynomial{n}`	$\mathsf{G}^{\mathrm{Gen}}_n$	Genocchi polynomial; its evaluation argument follows the command.
`\BarnesGFunction`	$\mathrm{G}_{\mathrm{Barnes}}$	Barnes G-function normalized by $\mathrm{G}_{\mathrm{Barnes}}(1)=1$ and $\mathrm{G}_{\mathrm{Barnes}}(z+1)=\Gamma(z)\mathrm{G}_{\mathrm{Barnes}}(z)$.
`\BellHessenbergMatrixH{n}`	$\mathsf{H}^{\mathrm{Bell}}_n$	First explicit Hessenberg determinant model for the complete exponential Bell polynomial.
`\BellHessenbergMatrixK{n}`	$\mathsf{K}^{\mathrm{Bell}}_n$	Second explicit Hessenberg determinant model for the complete exponential Bell polynomial.
`\AssociahedronByDimension{d}`	$\mathsf{K}^{\mathrm{Assoc}}_{\mathrm{dim}=d}$	Stasheff associahedron indexed by its actual dimension d, never by an inherited ambiguous source index.
`\LagrangeShiftCoordinate`	ζ_{Lag}	Local translated coordinate $z-b$ in a declared Lagrange-inversion scope.
`\SupportOperator`	supp	Pointwise-support operator token; its operand and ambient space must follow.
`\SupportOf{f}`	$\mathrm{supp}(f)$	Pointwise support $\{x : f(x) \neq 0\}$.
`\CoefficientSupportOf{F}`	$\mathrm{supp}_{\mathrm{coeff}}(F)$	Set of exponent indices or monomials carrying nonzero coefficients; the ambient index set is stated locally.
`\TopologicalSupportOf{f}`	$\mathrm{tsupp}(f)$	Closure of the pointwise support.
`\Convolution`	$f * g$	Convolution, with the underlying group and measure stated locally.
`\MetricDistance`	$\mathrm{dist}(x, y)$	Distance of the named metric space.
`\TotalVariationOf{a}`	$\mathrm{TV}(a)$	Total variation of one typed function, signed measure, or other locally declared object.
`\TotalVariationDistance`	$d_{\mathrm{TV}}(\mu, \nu)$	Total variation distance with the probability convention $\sup_A \lvert\mu(A) - \nu(A)\rvert$.

Shared command	Rendering	Normative meaning
`\Probability`	$\mathbb{P}$	Probability measure.
`\Expectation`	$\mathbb{E}$	Expectation with respect to the stated law.
`\Variance`	Var	Variance $\mathbb{E}[(X-\mathbb{E}X)^2]$.
`\Covariance`	Cov	Covariance of explicitly stated random variables.
`\LawOperator`	Law	Pushforward-law operator token; it is not a metric, density, or samplewise map.
`\LawOf{X}`	$\mathrm{Law}(X)$	Pushforward probability law of X.
`\EqualInLaw`	$X \overset{\text{law}}{=} Y$	Equality of probability laws, not pointwise equality.
`\ConvergesInLaw`	$X_n \xrightarrow{\text{law}} X$	Weak convergence of laws.
`\DyadicSigmaField`	$\mathcal{A}_n$	Level-n dyadic filtration σ-field; the generating random coordinates and base probability space are printed.
`\DecayOptimizationObjective{n}`	$\mathrm{Obj}_{\mathrm{decay},n}$	Exact level-n nonlinear objective in the Fourier-decay saddle problem, with variables and feasible set declared locally.
`\LinearizedDecayObjective{n}`	$\mathrm{Obj}^{\mathrm{lin}}_{\mathrm{decay},n}$	The explicitly linearized level-n decay objective; it is not identified with the nonlinear objective without an equality theorem.
`\FourierTwoPiOperator`	$\mathcal{F}_{2\pi}$	Forward Fourier operator with kernel $\mathrm{e}^{-2\pi \mathrm{i} x\xi}$.
`\FourierAngularOperator`	$\mathcal{F}_{\mathrm{ang}}$	Forward Fourier operator with kernel $\mathrm{e}^{-\mathrm{i} x\omega}$.
`\InverseFourierTwoPiOperator`	$\mathcal{F}_{2\pi}^{-1}$	Inverse of $\mathcal{F}_{2\pi}$ on a named inversion domain.
`\InverseFourierAngularOperator`	$\mathcal{F}_{\mathrm{ang}}^{-1}$	Inverse of $\mathcal{F}_{\mathrm{ang}}$, including the $1/(2\pi)$ measure factor.
`\FourierTwoPiTransformOf{f}`	$\mathcal{F}_{2\pi}[f]$	Operator-application form of the 2π-frequency Fourier transform.
`\FourierAngularTransformOf{f}`	$\mathcal{F}_{\mathrm{ang}}[f]$	Operator-application form of the angular-frequency Fourier transform.
`\FourierTwoPi{f}`	$\widehat{f}_{2\pi}$	Cycles-per-unit Fourier transform with kernel $\mathrm{e}^{-2\pi \mathrm{i} x\xi}$.
`\FourierAngular{f}`	$\widehat{f}_{\mathrm{ang}}$	Angular-frequency Fourier transform with kernel $\mathrm{e}^{-\mathrm{i} x\omega}$.
`\FourierSeriesCoefficient{h}{k}`	$\widehat{h}_{\mathrm{FS}}[k]$	The kth coefficient of a one-periodic function, with kernel $\mathrm{e}^{-2\pi \mathrm{i} kx}$ on $[0,1]$.
`\FourierSeriesCoefficientAtPeriod{h}{k}{T}`	$\widehat{h}_{\mathrm{FS};\,T}[k]$	The normalized kth coefficient of an explicitly T-periodic function; the period is part of the symbol.
`\LaplaceTransformSymbol`	$\mathcal{L}$	One-sided Laplace operator token unless a bilateral subscript is printed.
`\MellinTransformSymbol`	$\mathcal{M}$	Mellin-transform operator token on a stated strip of convergence.
`\LaplaceTransformOf{f}`	$\mathcal{L}[f]$	Laplace transform; one-sided or bilateral domain must be printed.
`\MellinTransformOf{f}`	$\mathcal{M}[f]$	Mellin transform $\int_0^\infty x^{s-1} f(x)\,\mathrm{d}x$ on its stated strip.
`\MomentGeneratingFunction{X}`	M_X	$z \mapsto \mathbb{E}[\mathrm{e}^{zX}]$ on its stated domain.
`\CharacteristicFunction{X}`	φ_X	$t \mapsto \mathbb{E}[\mathrm{e}^{\mathrm{i}tX}]$; a declared probability measure μ may replace X, meaning its positive-sign Fourier–Stieltjes transform.
`\SincRad`	$\mathrm{sinc}_{\mathrm{rad}}$	Radian sinc, $\sin t/t$, totalized to one at zero.
`\SincPi`	sinc_{π}	Normalized sinc, $\sin(\pi x)/(\pi x)$, totalized to one at zero.
`\BinaryDigitSum`	wt_2	Number of one-bits in a nonnegative integer.

Shared command	Rendering	Normative meaning
`\ThueMorseSignSymbol`	ε	The symbol reserved for the Thue–Morse sign sequence; use the argument-taking command in formulas.
`\ThueMorseSign{n}`	ε_n	$(-1)^{\mathrm{wt}_2(n)}$.
`\PadicValuation{p}`	ν_p	Finite p-adic valuation on nonzero integers or rationals; the operand domain determines the codomain $\mathbb{N}_0$ or $\mathbb{Z}$. Zero is undefined unless an explicitly named extended or software-totalized convention is in force.
`\TwoAdicValuation`	ν_2	The specialization ν_2, with the same operand-domain and zero-convention requirements.
`\QPochhammer{a}{q}{n}`	$(a;q)_n$	Finite or infinite q-shifted factorial with explicit base and length.
`\GeneralizedQPochhammer{a}{q}{\nu}`	$(a;q)_\nu$	Integer- or complex-order q-shifted factorial with an explicit logarithm branch and nonvanishing denominator when the order is not a natural number.
`\GaussianBinomial{n}{k}{q}`	$\begin{bmatrix} n \\ k \end{bmatrix}_q$	Gaussian binomial coefficient with explicit base.
`\GeneralizedGaussianBinomial{\alpha}{\beta}{q}`	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q$	Generalized Gaussian coefficient in the explicitly declared finite-product/rational, branch-dependent, or meromorphic regime.
`\QMultinomial{n}{n_1,\ldots,n_r}{q}`	$\begin{bmatrix} n \\ n_1,\ldots,n_r \end{bmatrix}_q$	Gaussian multinomial coefficient; the complete lower list and base are explicit.
`\QInteger{n}{q}`	$[n]_q$	q-integer with explicit base.
`\GeneralizedQNumber{z}{q}`	$[z]_q$	Branch-dependent generalized q-number for a declared scalar or complex exponent.
`\Polylogarithm`	Li	Polylogarithm with order supplied as a subscript.
`\LambertW`	W	Lambert W; a branch subscript is mandatory when more than one branch is possible.
`\WrightOmega`	ω	Wright omega, the inverse of $y \mapsto y + \log y$ on the declared real or complex branch.
`\LambertPolynomial{n}`	$\mathsf{L}_n^{\mathrm{Lam}}$	Canonical zero-based Lambert polynomial; $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$, and the evaluation variable follows outside the one-argument command.
`\ScaledLambertPolynomial{n}`	$\mathsf{L}_n^{\mathrm{Lam,sc}}$	Factorially scaled Lambert polynomial $n!\mathsf{L}_n^{\mathrm{Lam}}$ for $n \geq 1$; the evaluation variable follows the command.
`\MultiplicativeInverseOf{F}`	F^{-1}	Inverse of a unit F under multiplication.
`\CompositionalInverseOf{F}`	$F^{\langle -1 \rangle_\circ}$	Inverse of an admissible map F under composition; distinct from a reciprocal and a time-(-1) iterate.
`\PolynomialLogPowerSeriesRing{K}`	$\mathscr{P}\mathscr{L}_{\mathrm{pow},K}$	Polynomial-log power-series ring $K[L][[t]]$, with $t = X^{-1}$ and $L = \log X$.
`\PolynomialLogLaurentRing{K}`	$\mathscr{P}\mathscr{L}_{\mathrm{poly},K}$	Polynomial-log Laurent ring $K[L]((t))$, generally not a field.
`\RationalLogLaurentField{K}`	$\mathscr{P}\mathscr{L}_{\mathrm{rat},K}$	Laurent-series field $K(L)((t))$ with exact rational dependence on L.
`\IteratedLaurentLogField{K}`	$\mathscr{P}\mathscr{L}_{\mathrm{itLaur},K}$	Iterated Laurent field $K((L^{-1}))((t))$, with the completion order shown.
`\NearIdentityPolynomialLogGroup{K}`	$\mathcal{G}_{\mathrm{poly},K}$	Near-identity composition group $\{X + A : A \in K[L][[t]]\}$, subject to admissibility.
`\GammaFunction`	Γ	Euler Gamma function.

Shared command	Rendering	Normative meaning
`\RiemannZeta`	ζ	Riemann zeta function or its meromorphic continuation, as stated.
`\HurwitzZeta{s}{a}`	$\zeta(s,a)$	Hurwitz zeta function with order s and shift a both explicit.
`\BigO` (operator token; 0 TeX arguments)	$\mathcal{O}g$	Big-O applied to the following mathematical operand g, in a limit declared in the same sentence.
`\LittleO` (operator token; 0 TeX arguments)	og	Little-o applied to the following mathematical operand g, in a limit declared in the same sentence.
`\BigOAt{g}{a}`	$\mathcal{O}_a(g)$	Big-O with limiting regime a printed.
`\LittleOAt{g}{a}`	$o_a(g)$	Little-o with limiting regime a printed.
`\FormalBigO` (operator token; 0 TeX arguments)	$\mathcal{O}^{\mathrm{form}}$	Formal valuation-tail operator; it is not an analytic estimate and, in a Laurent ring, not an ideal-quotient congruence.
`\FormalBigOAt{t^N}{t}`	$\mathcal{O}_t^{\mathrm{form}}(t^N)$	Formal tail whose omitted difference has t-valuation at least N.

3 Foundational mathematical grammar

Shared notation contract version

Command. `\FabiusNotationVersion` expands to the ISO date 2026-09-01.
Meaning. It identifies the semantic API described by this catalogue.
Contract file. The corresponding source is `docs/fabius-notation.tex`. The version is not a publication date, source revision, theorem date, numeric parameter, or claim that every historical source was authored on that date.
Update rule. Any addition, removal, arity change, or mathematical change to a shared command updates this date and the quick reference, alphabetical index, rendered symbol index, retired-alias mapping, and relevant detailed entry in the same change.

The exponential base and imaginary unit

Exponential base. $\mathrm{e} \in \mathbb{R}$ is the unique positive number with $\log \mathrm{e} = 1$. The expression e^z is the complex exponential for $z \in \mathbb{C}$, not a local error variable raised to a power.
Imaginary unit. $\mathrm{i} \in \mathbb{C}$ is the root of $z^2 = -1$ with positive imaginary part. Thus $\mathrm{i}^2 = -1$ and $\bar{\mathrm{i}} = -\mathrm{i}$. The other root is $-\mathrm{i}$.
Typography. Both constants are upright. An italic e or i remains a local variable or index and acquires no constant meaning from context alone.

Number systems and inclusions

Canonical symbols. $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, $\mathbb{N}_{>0} = \{1, 2, 3, \ldots\}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$.
Inclusions. $\mathbb{N}_0 \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. These are the standard injective embeddings; a formula involving coercion into a larger system is interpreted only after the target has been named.
Endpoint convention. Neither $+\infty$ nor $-\infty$ is a real number. They belong only to $\overline{\mathbb{R}}$. Arithmetic involving them follows extended-order conventions only when explicitly stated.

Units of a ring or field. For a ring K with identity, $K^\times = \{a \in K : \text{there exists } b \in K \text{ with } ab = ba = 1\}$ is its multiplicative group of units. If K is a field, then $K^\times = K \setminus \{0\}$. The superscript $\times$ means neither Cartesian product nor a nonzero natural power in this context.
Retired ambiguity. A bare $\mathbb{N}$ is forbidden in normative prose because authors disagree about whether it contains zero.

Default metavariables and mandatory overrides

Discrete indices. Unless an entry says otherwise, $j, k, m, n, r, N \in \mathbb{N}_0$. A statement requiring positivity writes $n \in \mathbb{N}_{>0}$ or $n \geq 1$. The letter p denotes a prime only when the same sentence says "prime".
Real variables. The letters $x, y, t, u, s, \xi, \omega$ have no global type. Each definition declares whether they lie in $\mathbb{R}$, $\mathbb{C}$, an interval, or a measure space.
Parameters. The letters $a, b, q, \alpha, \beta, \lambda$ are never assigned a default domain. Their admissible set and all inequalities, such as $|q| < 1$ or $\alpha > 0$, are part of the declaration.
Functions and measures. Lower-case f, g, h denote local functions only after a domain and codomain are supplied. Greek μ, ν denote local measures only after a measurable space is supplied. No letter acquires mathematical meaning merely from its font.

Definition, equality, equivalence, and identification

Definition. The symbol $a := b$ introduces a as an abbreviation or new object whose value is exactly b. It is not an asymptotic statement.
Equality. The symbol $=$ means equality in the ambient type: equality of scalars, functions, sets, measures, matrices, or equivalence classes as determined by the declared type. Function equality is extensional; equality almost everywhere is written separately.
Almost-everywhere equality. For a measure μ, $f = g$ μ-a.e. means $\mu(\{x : f(x) \neq g(x)\}) = 0$. It does not license substitution into a pointwise endpoint formula without a continuous-representative theorem.
Law equality. For random variables X and Y, $X \overset{\text{law}}{=} Y$ abbreviates $\mathrm{Law}(X) = \mathrm{Law}(Y)$ and does not imply $X(\omega) = Y(\omega)$ on a shared sample space.
Isomorphism. Symbols $\cong$ and $\simeq$ require the category—sets, groups, rings, topological spaces, normed spaces, or probability spaces—to be stated locally.

Sets, set operations, and intervals

Set grammar. For a universe U, $x \in A$ means membership in $A \subseteq U$; $A \subseteq B$ permits equality and $A \subsetneq B$ does not. Union, intersection, complement, and set difference are $A \cup B$, $A \cap B$, $U \setminus A$, and $A \setminus B$.
Intervals. For $a, b \in \mathbb{R}$ with $a \leq b$, $[a, b] = \{x : a \leq x \leq b\}$, $(a, b) = \{x : a < x < b\}$, $[a, b) = \{x : a \leq x < b\}$, and $(a, b] = \{x : a < x \leq b\}$.
Empty reversed intervals. If $b < a$, each ordered interval above is empty. An oriented integral $\int_a^b$ is not an integral over that empty set; it changes sign under endpoint reversal.
Lean aliases. The Lean set constructors are `Set.Icc`, `Set.Ioo`, `Set.Ico`, and `Set.Ioc`, in the same order as the four printed intervals.

Maps, restrictions, composition, images, and preimages

Typed map. The declaration $f : A \to B$ includes domain A and codomain B. The restriction to $S \subseteq A$ is $f|_S : S \to B$.
Composition. For $f : A \to B$ and $g : B \to C$, $(g \circ f)(x) = g(f(x))$. Composition is evaluated right to left.
Identity map. For a declared set or type X, $\mathrm{id}_X : X \to X$ is the map $x \mapsto x$. The zero-argument command `\IdentityOperator` supplies only the operator token; the domain subscript X is source notation and must be printed whenever the surrounding typed equation does not determine it uniquely. Thus $f \circ \mathrm{id}_X = f$ and $\mathrm{id}_Y \circ f = f$ for every $f : X \to Y$.
Image and preimage. For $S \subseteq A$ and $T \subseteq B$, $f(S) = \{f(x) : x \in S\}$ and $f^{-1}(T) = \{x \in A : f(x) \in T\}$. The latter does not require f to be invertible.
Inverse map. The symbol $f^{-1} : B \to A$ denotes an inverse function only after f has been declared bijective. A superscript -1 on a set image is always the preimage, not reciprocal arithmetic.

Set, inner-product, cardinality, and definition delimiters

Set constructor. The command `\SetOf{body}` supplies only extensible braces. The body supplies either an explicit finite list or a typed predicate, for example $\{n \in \mathbb{N}_0 : n < N\}$. A colon and a vertical bar have the same set-builder meaning only after the ambient type has been printed. The command does not deduplicate a list, choose a universe, or turn a class into a set.
Inner product. The command `\InnerProduct{x,y}` supplies angle delimiters. In this catalogue $\langle x, y\rangle_H$ is linear in x and conjugate-linear in y on the complex Hilbert space H; a source adopting the opposite convention must say so at the first use and conjugate all adjoint formulas consistently. A duality pairing is instead labelled $\langle \phi, x\rangle_{X^*,X}$.
Cardinality. For a finite set A, $|A| \in \mathbb{N}_0$ is its number of elements and $|\varnothing| = 0$. For an infinite set it may denote a cardinal number only when the ambient set theory is explicit. Length, Lebesgue measure, probability, determinant, absolute value, and norm are never inferred from the same vertical bars.
Definitional equality. $a := b$ introduces the symbol a with type and value supplied by b. Reversing the direction uses $b =: a$; a theorem already requiring proof uses ordinary equality. Definitional equality in prose is not a claim of Lean kernel reduction unless an exact Lean definition is cited.
Empty and singleton cases. $\{\} = \varnothing$ when the body is an explicit empty list, whereas $\{x\}$ is a singleton. A displayed set-builder with an unsatisfiable predicate is empty by its predicate, not by typography.

Sequences, families, ranges, and finite indexing

Sequence. A sequence $(a_n)_{n\in\mathbb{N}_0}$ is a map $a : \mathbb{N}_0 \to A$ into a declared codomain A. A finite row $(a_{n,k})_{0\le k\le n}$ is defined only for the printed triangular range.
Half-open range. The project and Lean frequently use $j < n$ to mean $j \in \{0, 1, \dots, n-1\}$. Thus $\sum_{j<n}$ and $\prod_{j<n}$ are half-open at n.
Inclusive range. The notation $0 \le j \le n$ includes both endpoints and has $n + 1$ indices. It is never silently identified with $j < n$.
Indexed family. For a possibly infinite set I, $(a_i)_{i\in I}$ is a family indexed by I. Summability over I is unconditional unless an order is separately specified.

Sums, products, and neutral-element conventions

Finite operations. For a finite set S, $\sum_{i\in S} a_i$ and $\prod_{i\in S} a_i$ are taken in the stated additive and multiplicative structures.
Empty cases. The empty sum is 0 and the empty product is 1. Consequently $\sum_{j<n}$ is zero and $\prod_{j<n}$ is one when $n = 0$.
Infinite series. The notation $\sum_{n=0}^{\infty} a_n$ includes an ordering by $\mathbb{N}_0$. For a family over an arbitrary type, the project uses unconditional summability; rearrangement requires absolute or unconditional convergence.
Infinite products. The product $\prod_{n=0}^{\infty} a_n$ is the limit of the ordered partial products $\prod_{n<N} a_n$ as $N \to \infty$. Local uniform convergence is stated when the factors depend on a complex variable.

Powers, factorials, and binomial coefficients

Natural powers. For an element x of a monoid and $n \in \mathbb{N}_0$, $x^0 = 1$ and $x^{n+1} = x^n x$. In particular, the natural-power convention is $0^0 = 1$.
Integer powers. For $n \in \mathbb{Z}$, x^n requires an invertible nonzero x when $n < 0$; the ambient field may totalize 0^{-1} algebraically, but an analytic formula never relies on that totalization without saying so.
Complex powers. For z^α with noninteger α, the chosen branch of $\log z$ and its cut are mandatory. The principal convention is $z^\alpha = \exp(\alpha \operatorname{Log} z)$ with $\arg z \in (-\pi, \pi]$.
Factorial. For $n \in \mathbb{N}_0$, $0! = 1$ and $n! = \prod_{1\le j\le n} j$. The odd double factorial satisfies $(-1)!! = 1$ and $(2n+1)!! = \prod_{j=0}^{n}(2j+1)$.
Binomial coefficient. For $n, k \in \mathbb{N}_0$, $\binom{n}{k} = n!/(k!(n-k)!)$ when $k \le n$ and $\binom{n}{k} = 0$ when $k > n$.
Falling factorial. For a scalar x and $n \in \mathbb{N}_0$, $(x)^{\underline{n}} := \prod_{j=0}^{n-1}(x-j)$; the empty product gives $(x)^{\underline{0}} = 1$. Thus $(x)^{\underline{n}} = \Gamma(x+1)/\Gamma(x-n+1)$ only where the quotient is defined or after a separately justified continuation.
Rising factorial. For a scalar x and $n \in \mathbb{N}_0$, $(x)^{\overline{n}} := \prod_{j=0}^{n-1}(x+j)$ and $(x)^{\overline{0}} = 1$. It equals the ordinary Pochhammer symbol $(x)_n$ in this catalogue; the base-q analogue is always `\QPochhammer` and never shares this two-argument syntax.

The principal logarithm and order operators

Principal logarithm. The command `\PrincipalLogarithm` denotes the analytic branch

$$\operatorname{Log} z = \log|z| + \mathrm{i} \operatorname{Arg} z, \qquad -\pi < \operatorname{Arg} z \le \pi,$$

on $\mathbb{C} \setminus (-\infty, 0]$. In particular it is undefined at zero and has distinct boundary values on the two sides of the negative real axis. The real symbol $\log x$ for $x > 0$ is its restriction and carries no branch ambiguity.
Complex powers. The principal power is $z^\alpha = \exp(\alpha \operatorname{Log} z)$ on the same cut domain. Algebraic laws such as $(zw)^\alpha = z^\alpha w^\alpha$ are not used across a branch crossing without an explicit argument calculation.
Vanishing order. For a nonzero holomorphic f near z_0, $\operatorname{ord}_{z_0}(f) = m \in \mathbb{N}_0$ when $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$. For a pole, a signed divisor order is written $\operatorname{ord}_{z_0}^{\mathrm{div}}(f) < 0$ so that it cannot be mistaken for the nonnegative zero order.
Initial q-order. For a nonzero formal Laurent series $f(q) = \sum_{n\ge n_0} c_n q^n \in R((q))$, where R is an integral domain and $c_{n_0} \neq 0$,

$$\operatorname{ord}_{q=0}(f) := n_0.$$

For $f = 0$ the order is $+\infty$ when the extended convention is explicitly in force; otherwise it is left undefined. This subscripted operator records the expansion variable and base point, and is not the 2-adic valuation $\nu_2(\,\cdot\,)$ or a group-element order.

Multiplicative order. For a group element a, put $S_G(a) = \{n \in \mathbb{N}_{>0} : a^n = e_G\}$. When this set is nonempty,

$$\operatorname{ord}_G(a) = \min S_G(a).$$

When it is empty, $\operatorname{ord}_G(a) = \infty$. The group subscript is mandatory whenever an analytic order also occurs.
Other orders. Differential-equation order, recurrence order, matrix order, and asymptotic truncation order are prose-qualified quantities. The shared operator provides typography, not permission to leave the species or base point implicit.

Absolute value, norms, and distance

Absolute value. For $x \in \mathbb{R}$, $|x| = \max\{x, -x\}$. For $z \in \mathbb{C}$, $|z| = \sqrt{z\bar{z}}$.
Norm. The expression $\|x\|$ is meaningful only after a normed space has been named. For a function, $\|f\|_\infty$ means an essential supremum or a pointwise supremum exactly as the surrounding declaration states.
Distance. In a metric space (X, d), $\operatorname{dist}(x, y) = d(x, y)$. Probability metrics such as total variation and Wasserstein distance are given their own decorated symbols and never use a bare dist without the metric name.

Floor, ceiling, fractional part, sign, and positive part

Definitions. For $x \in \mathbb{R}$, $\lfloor x \rfloor = \max\{n \in \mathbb{Z} : n \le x\}$, $\lceil x \rceil = \min\{n \in \mathbb{Z} : x \le n\}$, $\{x\} = x - \lfloor x \rfloor \in [0, 1)$, $(x)_+ = \max\{x, 0\}$, and $\operatorname{sgn}(x) = -1, 0, 1$ according as $x < 0$, $x = 0$, or $x > 0$.
Nearest integer. A nearest-integer or nearest-dyadic operation must separately state its tie-breaking rule; neither floor nor ceiling implies one.
Use in powers. For $\alpha > 0$, $(x)_+^\alpha$ is zero when $x \le 0$. For $\alpha = 0$, the natural-power value at $x \le 0$ is one only if a formula explicitly uses natural exponentiation; real powers require a separate endpoint rule.

Indicator and half-weighted interval indicator

Ordinary indicator. For a set A in a universe U, $\mathbf{1}_A : U \to \{0, 1\}$ is one on A and zero on $U \setminus A$.
Half-weighted indicator. When adjacent closed intervals meet, the project uses a separately named half-endpoint indicator taking value $1/2$ at each boundary and 1 in the interior. It is not denoted by the ordinary $\mathbf{1}_A$.
Measure-theoretic caveat. Changing endpoint values does not change a Lebesgue integral, but it does change pointwise finite-spline identities and exact values at grid points.

Pointwise support, topological support, and essential support

Pointwise support. For a function $f : X \to A$ with zero in A, $\operatorname{supp}(f) = \{x \in X : f(x) \ne 0\}$. The bare command `\SupportOperator` is the same operator token before an explicitly supplied operand; `\SupportOf` supplies the parentheses and is preferred in prose.
Topological support. For topological X, $\operatorname{tsupp}(f) = \overline{\operatorname{supp}(f)}$. Thus a boundary point can belong to the topological support even when f and all of its derivatives vanish there.
Measure support. For a Borel measure μ, $\operatorname{supp} \mu$ is the complement of the largest open μ-null set. This is not obtained by evaluating a measure as a pointwise function.

Essential support. Essential support depends on a background measure and ignores null changes; it must be labeled "essential".

Derivatives, iterated derivatives, and integrals

Derivative. For $f : \mathbb{R} \to E$ into a real normed space, $f'(x)$ denotes the Fréchet derivative evaluated on 1 when it exists. Higher derivatives are $f^{(n)}$, with $f^{(0)} = f$.
Partial derivatives. For a multivariable function, $\partial_j f$ differentiates in the jth coordinate. A gradient, Jacobian, Hessian, or directional derivative is named explicitly and includes its inner-product or coordinate convention.
Differential. In $\int_a^b f(x)\,\mathrm{d}x$, the upright $\mathrm{d}x$ identifies the variable of integration and is not a free multiplicative variable.
Oriented interval integral. For integrable f, $\int_b^a f(x)\,\mathrm{d}x = -\int_a^b f(x)\,\mathrm{d}x$ and $\int_a^a f(x)\,\mathrm{d}x = 0$.
Whole-space integral. An integral $\int_{\mathbb{R}}$ is with respect to Lebesgue measure unless another measure is printed. An expectation is written $\mathbb{E}$, not as an unlabeled integral.

Convolution of functions, measures, and sequences

Functions. For integrable functions $f, g : \mathbb{R} \to \mathbb{C}$, $(f * g)(x) = \int_{\mathbb{R}} f(x-y)g(y)\,\mathrm{d}y$.
Measures. For finite Borel measures μ, ν on $\mathbb{R}$, $(\mu * \nu)(A) = \int\int \mathbf{1}_A(x+y)\,\mu(\mathrm{d}x)\nu(\mathrm{d}y)$. It is the law of $X + Y$ for independent X, Y with laws μ, ν.
Document-local dyadic uniform laws. In the New Frontiers report, `\DyadicUniformLaw{n}` has TeX arity one and prints $\mathsf{U}_n^{\mathrm{dyad}}$. Its mathematical argument is $n \in \mathbb{N}_0$, and it denotes the probability measure

$$\mathsf{U}_n^{\mathrm{dyad}} := \mathrm{Unif}\left[-2^{-n-1}, 2^{-n-1}\right], \qquad \frac{\mathrm{d}\mathsf{U}_n^{\mathrm{dyad}}}{\mathrm{d}x} = 2^n \mathbf{1}_{[-2^{-n-1}, 2^{-n-1}]}(x).$$

The interval has length 2^{-n}, so the displayed density has total mass one; endpoint values of the density are immaterial. If the coordinate variables are independent with these laws, the finite convolution $\mathsf{U}_0^{\mathrm{dyad}} * \cdots * \mathsf{U}_N^{\mathrm{dyad}}$ is their sum law and converges weakly as $N \to \infty$ to the Rvachev probability law, supported on $[-1, 1]$. This local family is a sequence of measures, not a valuation or an unnamed U-variable.
Document-local reciprocal-integer law. In the reciprocal-integer-divisor report, `\ReciprocalIntegerLaw{M}` has TeX arity one and prints $\mathsf{R}_M^{\mathrm{law}}$. Fix an integer $M \geq 2$, let D_M be uniform on the explicitly listed M-point set

$$\{(M-1-2j)/M : j \in \mathbb{N}_0,\ 0 \leq j < M\},$$

let $(D_{M,n})_{n \in \mathbb{N}_0}$ be independent copies of D_M, and define

$$Y_M := \sum_{n=0}^{\infty} \frac{D_{M,n}}{2^{n+1}}, \qquad \mathsf{R}_M^{\mathrm{law}} := \mathrm{Law}(Y_M).$$

The series converges absolutely, the law is symmetric and atomless, and

$$\mathrm{supp}\Big(\mathsf{R}_M^{\mathrm{law}}\Big) = \left[-\frac{M-1}{M}, \frac{M-1}{M}\right].$$

Under the report's positive-sign 2π characteristic-function convention, put $\Phi(z) = \int \mathrm{e}^{2\pi \mathrm{i} z x}\,\mu_{\mathrm{up}}(\mathrm{d}x)$. The characteristic function of $\mathsf{R}_M^{\mathrm{law}}$ is the unique entire continuation of

$$Q_M(z) = \frac{\Phi(z)}{\Phi(z/M)} = \prod_{n=0}^{\infty} \frac{\mathrm{sinc}_{\mathrm{rad}}(\pi z/2^n)}{\mathrm{sinc}_{\mathrm{rad}}(\pi z/(M 2^n))};$$

the quotient notation alone is not used at a common zero before this removable continuation is specified. Let X and X' be random variables satisfying $\mathrm{Law}(X) = \mathrm{Law}(X') = \mu_{\mathrm{up}}$, on a coupling for which the pair (X, X') is independent of Y_M. Then $X \overset{\mathrm{law}}{=} M^{-1}X' + Y_M$; only the independence of X' and Y_M is needed for this distributional decomposition. With the boundary definitions $Y_1 = 0$, $\mathsf{R}_1^{\mathrm{law}} = \delta_0$, and $Q_1 = 1$, the multiplicative cocycle is, for $M, N \in \mathbb{N}_{>0}$,

$$\mathsf{R}_{MN}^{\mathrm{law}} = \mathsf{R}_M^{\mathrm{law}} * (x \mapsto x/M)_\# \mathsf{R}_N^{\mathrm{law}}, \qquad Q_{MN}(z) = Q_M(z)Q_N(z/M).$$

These are manuscript-level human-proved claims in that report, not a claimed Lean formalization.
Conjectural automatic-scale factor pair. The automatic-scale report uses exactly the local pair ρ_1, ρ_2 for centered, symmetric, compactly supported probability measures: centered means $\int x\, \rho_j(\mathrm{d}x) = 0$, and symmetric means $(x \mapsto -x)_\# \rho_j = \rho_j$, for $j \in \{1, 2\}$. Its factorization hypothesis is $\rho_1 * \rho_2 = \mu_{\mathrm{up}}$. Each angular-frequency characteristic function $\phi_j(z) = \int \mathrm{e}^{\mathrm{i}zx}\, \rho_j(\mathrm{d}x)$ is assumed entire and in the Cartwright class—that is, of finite exponential type with $\int_{\mathbb{R}} \log^+ |\phi_j(x)|/(1 + x^2)\, \mathrm{d}x < \infty$—and its zeros are assumed to belong to the zero multiset of the Rvachev characteristic function. The additional completely-regular-growth hypothesis is part of the stated conjecture, not a consequence of compact support. Under those hypotheses the conjecture predicts a factorization by whole dyadic sinc factors, up to a finite B-spline factor and interchange of ρ_1 and ρ_2. The subscripts enumerate two law factors; they are neither 2-adic valuations nor the spectral radii denoted by other local ρ-symbols.
Sequences. For sequences indexed by $\mathbb{N}_0$, the Cauchy convolution is $(a * b)_n = \sum_{k=0}^{n} a_k b_{n-k}$. It is distinct from bilateral convolution over $\mathbb{Z}$.
Power. The notation f^{*n} means n-fold convolution, with $f^{*0} = \delta_0$ for measures and the appropriate convolution unit for the ambient algebra.

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4 The semantic Fabius–Rvachev objects

The bounded Fabius function F_{bd}

Command. `\FabiusBounded`.
Type. A smooth function $F_{\mathrm{bd}} : \mathbb{R} \to [0, 1] \subset \mathbb{R}$.
Normalization. $F_{\mathrm{bd}}(x) = 0$ for $x \le 0$, $F_{\mathrm{bd}}(x) = 1$ for $x \ge 1$, $F_{\mathrm{bd}}(1 - x) = 1 - F_{\mathrm{bd}}(x)$ for every real x, and $F'_{\mathrm{bd}}(x) = 2F_{\mathrm{bd}}(2x)$ on the left half of the unit interval.
Shape. Its restriction $F_{\mathrm{bd}}|_{[0,1]}$ is continuous and strictly increasing from zero to one. Its constant tails are part of the object, not an informal extension.
Lean crosswalk. The real-valued object is `fabiusReal`; the abstract specification is `IsFabius`; construction and uniqueness are exposed through `fabius` and `fabius_spec`. Status: **Lean-proved**.
Forbidden aliases. The bare capital letter, the former calligraphic glyph, and the former decorated-capital glyph are not canonical because the active historical corpus also uses them for transforms, families, and the signed global extension. This catalogue names those aliases textually rather than re-emitting their retired control sequences.

Rvachev's up-function up

Command. `\RvachevUp`.
Type and definition. The function $\mathrm{up} : \mathbb{R} \to \mathbb{R}$ is defined by

$$\mathrm{up}(x) = F_{\mathrm{bd}}(1 - |x|).$$

This single formula uses the constant tails of F_{bd} and therefore gives $\mathrm{up}(x) = 0$ whenever $|x| \ge 1$.

Normalization. It is even, nonnegative, smooth, has integral one, satisfies $\mathrm{up}(0) = 1$, and has topological support $[-1, 1]$. Every derivative vanishes at $x = \pm 1$.
Differential law. For every $x \in \mathbb{R}$, $\mathrm{up}'(x) = 2(\mathrm{up}(2x + 1) - \mathrm{up}(2x - 1))$.
Lean crosswalk. `rvachevUp` in `Basic.lean`, with differential, support, Fourier, and shape theorems in their dedicated modules. Status: **Lean-proved**.
Historical aliases. The source papers print an upright word "up", use legacy commands named `\up` or `\Up`, or use the local glyph φ. These spellings are historical evidence, not commands to be re-emitted in project-authored formulas; normative prose uses only up.

The signed global extension F_{gl}

Command. `\FabiusGlobal`.
Type. A smooth function $F_{\mathrm{gl}} : \mathbb{R} \to \mathbb{R}$. It is not a cumulative distribution function and is not constrained to $[0, 1]$.
Canonical translate formula. For $x \in \mathbb{R}$,

$$F_{\mathrm{gl}}(x) = \sum_{n=0}^{\infty} \varepsilon_n \ \mathrm{up}(x - 2n - 1).$$

At each fixed x, compact support leaves only finitely many nonzero summands; hence this formula is pointwise locally finite even when written as an infinite sum.
Identifications. For $x \in [0, 1]$, $F_{\mathrm{gl}}(x) = F_{\mathrm{bd}}(x)$. Globally, $F_{\mathrm{gl}}'(x) = 2F_{\mathrm{gl}}(2x)$. Repeated differentiation gives $F_{\mathrm{gl}}^{(m)}(x) = 2^{\binom{m+1}{2}} F_{\mathrm{gl}}(2^m x)$ for $m \in \mathbb{N}_0$.
Lean declaration. `extendedFabius`.
Lean module and status. Translate-cell and calculus results are in `GlobalExtension.lean`. Status: **Lean-proved**.
Retired aliases. The former calligraphic glyph, the former decorated-capital glyph, and the plain capital glyph of the 2017 arithmetic paper are historical aliases. They must be translated semantically, never by a blind global replacement.

The ordinary and clamped Fabius inverses

Commands. `\FabiusQuantile` and `\FabiusClampedQuantile`.
Ordinary inverse. Because $F_{\mathrm{bd}}|_{[0,1]} : [0, 1] \to [0, 1]$ is a strictly increasing bijection,

$$Q_F : [0, 1] \to [0, 1], \qquad Q_F(y) = \left(F_{\mathrm{bd}}|_{[0,1]}\right)^{-1}(y).$$

Thus $F_{\mathrm{bd}}(Q_F(y)) = y$ for $y \in [0, 1]$ and $Q_F(F_{\mathrm{bd}}(x)) = x$ for $x \in [0, 1]$.
Clamped inverse. For $y \in \mathbb{R}$, define $\mathrm{clamp}(y) = \min\{1, \max\{0, y\}\}$ and

$$Q_{F,\mathrm{cl}}(y) = Q_F(\mathrm{clamp}(y)).$$

Consequently it is zero on $(-\infty, 0]$ and one on $[1, \infty)$.
Lean crosswalk. `fabiusInv` and `fabiusOrderIso` in `FabiusInverse.lean`. Status: **Lean-proved**.
Retired aliases. Bare I, G, J, the formal inverse expression, and a legacy inverse macro occur in older prose. They are local or historical only.

The abstract predicate `IsFabius`

Kind. A proposition about a candidate bounded function, not a fourth Fabius function.
Role. It packages the exact tails, reflection, smoothness/differentiability, and left-half differential law required by the formal development. A theorem quantified over an arbitrary f with `IsFabius` f is more general than a theorem stated only for F_{bd}.
Authority. The Lean declaration's type is authoritative for its precise fields, coercions, and domains. A prose summary is not permission to omit a hypothesis.
Status. **Lean-proved**: existence supplies the canonical witness and uniqueness identifies every witness with it.

Canonical relations among the three functions

Fold. For every $x \in \mathbb{R}$, $\text{up}(x) = F_{\text{bd}}(1 - |x|)$.
Density relation. For $x \in \mathbb{R}$, $F'_{\text{bd}}(x) = 2\,\text{up}(2x - 1)$. Both sides vanish outside the unit interval.
Unit-interval translate. For $x \in [0, 1]$, $F_{\text{bd}}(x) = \text{up}(x - 1)$. This is not a global identity.
Global agreement. For $x \in [0, 1]$, $F_{\text{gl}}(x) = F_{\text{bd}}(x)$. Outside that interval the left side is signed and the right side has constant tails.
Audit rule. Every use of a differential, translation, or scaling identity states whether its subject is F_{bd}, up, or F_{gl} and prints the domain on which it holds.

Dyadic points, reduced levels, and derivative cells

Dyadic point. A dyadic rational is $x = m/2^n$ with $m \in \mathbb{Z}$ and $n \in \mathbb{N}_0$. A unit-interval dyadic additionally has $0 \le m \le 2^n$.
Reduced positive level. For an interior dyadic, "reduced at level n" means $0 < m < 2^n$ and m is odd. The oddness prevents the same rational from being represented at a smaller positive denominator level.
Cell. A level-n dyadic cell is an interval between consecutive multiples of 2^{-n}; open, closed, or half-open endpoints are printed because derivative and spline formulas can distinguish them.
Derivative scaling. At a reduced interior dyadic $m/2^n$, derivatives above a finite order vanish according to the exact global-scaling theorem. The catalogue does not infer local analyticity from the terminating Taylor jet.
Representation independence. An exact evaluator accepting (m, n) must prove invariance under $(m, n) \mapsto (2m, n + 1)$ rather than assuming a reduced input.

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5 Probability, measures, and distributional notation

Probability spaces and random variables

Probability space. $(\Omega, \mathscr{A}, \mathbb{P})$ denotes a probability space: Ω is the sample set, $\mathscr{A}$ is a σ-algebra on Ω, and $\mathbb{P} : \mathscr{A} \to [0, 1]$ is countably additive with $\mathbb{P}(\Omega) = 1$.
Random variable. A real random variable is an $\mathscr{A}/\mathcal{B}(\mathbb{R})$-measurable map $X : \Omega \to \mathbb{R}$. A random vector $X : \Omega \to \mathbb{R}^d$ is measurable for the Borel σ-algebra of $\mathbb{R}^d$.
Events. Braces inside probability are inverse images: $\mathbb{P}\{X \le x\}$ abbreviates $\mathbb{P}(X^{-1}((-\infty, x]))$. They do not imply that X has a density.

Almost surely. $X = Y$ almost surely, written $X = Y$ a.s., means $\mathbb{P}\{\omega : X(\omega) = Y(\omega)\} = 1$. This is weaker than pointwise equality and different from equality in law.
Independence. Random variables $(X_j)_{j\in J}$ are independent when

$$\mathbb{P}\left(\bigcap_{j\in K}\{X_j \in B_j\}\right) = \prod_{j\in K} \mathbb{P}\{X_j \in B_j\}$$

for every finite $K \subseteq J$ and every family of Borel sets B_j. *i.i.d.* means independent and identically distributed.

Law, pushforward, equality in law, and weak convergence

Law. For a measurable $X : \Omega \to S$,

$$\mathrm{Law}(X) := \mathrm{Law}(X) := X_{\#}\mathbb{P}, \qquad \mathrm{Law}(X)\,(B) = \mathbb{P}(X^{-1}(B)),$$

where S is a measurable space, B is measurable, and $X_{\#}$ denotes pushforward. Thus `\LawOperator` is the operator token and `\LawOf` is its argument-delimited form; both take a random variable and return a probability measure.
Equality in law. $X \overset{\text{law}}{=} Y$ means $\mathrm{Law}(X) = \mathrm{Law}(Y)$. It neither asserts that X and Y share a sample space nor that they are equal almost surely.
Convergence in law. $X_n \xrightarrow{\text{law}} X$ means

$$\int_S h \,\mathrm{d}\,\mathrm{Law}(X_n) \longrightarrow \int_S h \,\mathrm{d}\,\mathrm{Law}(X)$$

for every bounded continuous $h : S \to \mathbb{R}$, with S a metric space. For real-valued variables this is equivalent to convergence of distribution functions at every continuity point of the limiting distribution function.
Measure notation. $\mu_n \Rightarrow \mu$ is allowed for weak convergence of measures; it must not be reused for implication in the same displayed formula.
Lean crosswalk. Probability measures and pushforwards use Mathlib's `ProbabilityMeasure`, `Measure.map`, and distribution APIs. A prose probabilistic representation is labelled **prose-derived** unless an exact Lean theorem is cited.

The canonical dyadic filtration

Command. `\DyadicSigmaField`. The printed base glyph is $\mathcal{A}$; it is never a generic event, matrix, Appell family, or algebra when the command is used.
Probability space. Fix a probability space $(\Omega, \mathscr{A}, \mathbb{P})$ and an explicitly ordered coordinate family $(U_j)_{j\in\mathbb{N}_{>0}}$. For $n \in \mathbb{N}_0$ define

$$\mathcal{A}_n := \sigma(U_1, \dots, U_n), \qquad \mathcal{A}_0 := \{\varnothing, \Omega\}.$$

If a document instead generates from binary digits, Rademacher variables, or dyadic partition cells, it replaces the displayed generators while retaining the level-zero and monotonicity clauses.
Filtration law. $\mathcal{A}_n \subseteq \mathcal{A}_{n+1} \subseteq \mathscr{A}$ for every n. Conditional expectation $\mathbb{E}[X \mid \mathcal{A}_n]$ is defined only for an integrable X (or the appropriate nonnegative extended variant) and is an equivalence class modulo almost sure equality.

Limit field. $\mathcal{A}_\infty := \sigma(\bigcup_{n\in\mathbb{N}_0} \mathcal{A}_n)$. Equality with the ambient $\mathscr{A}$ is an additional generation hypothesis, not part of the notation.
Collision policy. An unqualified calligraphic A remains local. The shared command is reserved for this filtration, and every subscript is a level index in $\mathbb{N}_0 \cup \{\infty\}$ rather than a matrix dimension or event label.

Distribution functions, densities, and atoms

Distribution function. For a real random variable X,

$$F_X(x) := \mathbb{P}\{X \le x\}, \qquad x \in \mathbb{R}.$$

It is right-continuous, nondecreasing, and has limits 0 and 1 at negative and positive infinity. The subscript is compulsory unless the distribution is uniquely fixed in the surrounding entry.
Density. A Borel function $f_X : \mathbb{R} \to [0,\infty)$ is a density when $\mathrm{Law}(X)\,(B) = \int_B f_X(x)\,\mathrm{d}x$ for every Borel set B. Then $F_X(x) = \int_{-\infty}^x f_X(t)\,\mathrm{d}t$. A distribution need not possess a density.
Atom. The mass at a is $\mathbb{P}\{X = a\} = F_X(a) - F_X(a^-)$, where $F_X(a^-) = \lim_{x\uparrow a} F_X(x)$. A Dirac measure at a is $\delta_a(B) = \mathbf{1}_B(a)$.
Signed measures. A signed measure ν is not a probability law. Its total variation measure is $|\nu|$; the scalar $\nu(\mathbb{R})$ is its total mass and may be negative.
Endpoints. A uniform law on $[a,b]$, $a < b$, has density $(b-a)^{-1}\mathbf{1}_{[a,b]}(x)$. Changing the values at the two endpoints changes neither the associated absolutely continuous measure nor any Lebesgue integral.

Expectation, moments, variance, and covariance

Expectation. For an integrable real random variable,

$$\mathbb{E}[X] := \int_\Omega X(\omega)\,\mathrm{d}\mathbb{P}(\omega) = \int_{\mathbb{R}} x\,\mathrm{d}\,\mathrm{Law}(X)\,(x).$$

For a nonnegative X, the extended integral is permitted and may equal $+\infty$.
Complex expectation. For an integrable $X : \Omega \to \mathbb{C}$, meaning $\mathbb{E}[|X|] < \infty$,

$$\mathbb{E}[X] := \mathbb{E}[\Re X] + \mathrm{i}\mathbb{E}[\Im X] = \int_\Omega X\,\mathrm{d}\mathbb{P},$$

where the last integral is the complex Bochner integral. This is the interpretation used when a characteristic-function integrand is complex-valued.
Raw moments. $m_n(X) := \mathbb{E}[X^n]$ for $n \in \mathbb{N}_0$ whenever the expectation exists; $m_0(X) = 1$, including on the event $X = 0$, by the polynomial convention $x^0 = 1$ for every x.
Central moments. $\mu_n(X) := \mathbb{E}[(X - \mathbb{E}X)^n]$, assuming the needed integrability. Raw m_n and central μ_n must not be interchanged.
Variance and covariance. For square-integrable real variables,

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2], \qquad \mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)].$$

Thus $\mathrm{Var}(X) = \mathrm{Cov}(X,X) \ge 0$.
Cumulants. When $\log M_X(t)$ exists near 0, $\kappa_n(X)$ is the coefficient defined by $\log M_X(t) = \sum_{n\ge1} \kappa_n(X)t^n/n!$. The logarithm is the real logarithm on the positive MGF, not an unstated complex branch.

Canonical probabilistic model of the Fabius distribution

Input variables. Let $(U_k)_{k\in\mathbb{N}_{>0}}$ be independent with $U_k \sim \mathrm{Uniform}[0,1]$. The index begins at 1.
Random series. Define

$$X := \sum_{k=1}^{\infty} 2^{-k} U_k.$$

The series converges absolutely for every sample point because $0 \le U_k \le 1$ and $\sum_{k\ge 1} 2^{-k} = 1$; hence $X \in [0,1]$ almost surely.
Distribution. $F_{\mathrm{bd}}(x) = \mathbb{P}\{X \le x\}$ for $x \in \mathbb{R}$. The density is $f_X(x) = 2\,\mathrm{up}(2x-1)$, so the distribution identity reproduces $F'_{\mathrm{bd}}(x) = 2\,\mathrm{up}(2x-1)$.
Centered variable. For $Y = 2X - 1$ one has $Y \in [-1,1]$ almost surely and Y has density up. Equivalently, $\mathrm{up}(t) = \frac{1}{2} f_X((t+1)/2)$.
Finite approximants. $X_N = \sum_{k=1}^{N} 2^{-k} U_k$ has support $[0, 1-2^{-N}]$ and $X_N \to X$ almost surely, in every L^p for $1 \le p < \infty$, and in law. These distinct modes are printed when a bound depends on one of them.
Status. The random-series representation is a semantic bridge used in the corpus. Each downstream theorem retains its own proof-status label; the representation alone does not formalize a probabilistic argument in Lean.

Geometrically weighted uniform series

Parameters. Unless a complex analytic continuation is explicitly declared, $q \in (0,1)$. Let $U_0, U_1, \dots$ be independent and uniformly distributed on $[0,1]$.
Normalized series. The probability-normalized variable is

$$X_q := (1-q) \sum_{n=0}^{\infty} q^n U_n.$$

The weights sum to 1, so $X_q \in [0,1]$ almost surely. At $q = \frac{1}{2}$ its law agrees with the canonical Fabius law after the index shift $U_n \leftrightarrow U_{n+1}$.
Unnormalized series. Some frontier documents instead study $S_q = \sum_{n\ge 0} q^n U_n = X_q/(1-q)$ with support $[0, (1-q)^{-1}]$. The letter S_q and its support must be printed whenever this normalization is used.
Characteristic function. Independence gives

$$\varphi_{X_q}(t) = \prod_{n=0}^{\infty} \frac{\mathrm{e}^{\mathrm{i}t(1-q)q^n} - 1}{\mathrm{i}t(1-q)q^n},$$

with every factor assigned its continuous value 1 at $t = 0$.
Complex q. For $q \in \mathbb{C}$, the same product may define an analytic function for $|q| < 1$, but it is no longer the characteristic function of a real probability variable unless additional reality hypotheses are given.

Total variation, Wasserstein, and Kolmogorov distances

One-object total variation. The command `\TotalVariationOf{a}` denotes the total variation of one object a. Its type must be stated locally. For a real function on $[u,v]$,

$$\mathrm{TV}(f) := \sup_{u=x_0<\cdots<x_n=v} \sum_{j=1}^{n} \left| f(x_j) - f(x_{j-1}) \right|.$$

For a finite signed measure σ, it denotes $|\sigma|(X)$, equivalently the total-variation norm under the convention stated in the source. These are one-object functionals, not distances between two probability measures.
Total variation. For probability measures μ, ν on the same measurable space,

$$d_{\mathrm{TV}}(\mu,\nu) := \sup_A |\mu(A) - \nu(A)| = \frac{1}{2}\int |f-g| \,\mathrm{d}\lambda$$

when both have densities f, g with respect to λ. Some literature omits the factor $1/2$; that alternative is never silently substituted.
Kolmogorov distance. For real laws with CDFs F_X, F_Y, $d_{\mathrm{K}}(F_X, F_Y) = \sup_{x\in\mathbb{R}} |F_X(x) - F_Y(x)|$.
Wasserstein distance. For $p \in [1,\infty)$ and laws with finite pth moments,

$$W_p(\mu,\nu) = \left(\inf_{\pi\in\Pi(\mu,\nu)} \int |x-y|^p \,\mathrm{d}\pi(x,y)\right)^{1/p},$$

where $\Pi(\mu,\nu)$ is the set of couplings. The product coupling makes $\Pi(\mu,\nu)$ nonempty for probability measures on standard Borel spaces.
Type rule. A metric symbol takes two laws or measures, not two random variables, unless the law operation is explicitly stated in the local scope.

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6 Fourier, Laplace, Mellin, and generating transforms

6.1 The two Fourier conventions

The 2π-frequency Fourier transform

Commands. `\FourierTwoPiOperator` is the operator token, `\FourierTwoPiTransformOf{f}` is its bracketed application, and `\FourierTwoPi{f}` is its normalization-labelled hat. These are three presentations of one forward transform, not three transforms.
Forward transform. For $f \in L^1(\mathbb{R})$,

$$\widehat{f}_{2\pi}(\xi) := \mathcal{F}_{2\pi}[f]\,(\xi) := \mathcal{F}_{2\pi}[f](\xi) := \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,\mathrm{d}x, \qquad \xi \in \mathbb{R}.$$

The transform variable is ξ and carries cycles per unit.
Inverse. Under a hypothesis guaranteeing inversion,

$$f(x) = \int_{\mathbb{R}} \widehat{f}_{2\pi}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\,\mathrm{d}\xi.$$

There is no scalar prefactor in either direction. The operator implementing this inverse is `\InverseFourierTwoPiOperator`. It is written only on a named inversion domain; it does not assert pointwise inversion for arbitrary L^1 data. For L^2, both formulas are interpreted through Plancherel extension.
Rules. With justified derivatives and integrals,

$$\widehat{f'}_{2\pi}(\xi) = 2\pi\mathrm{i}\xi\,\widehat{f}_{2\pi}(\xi), \qquad \widehat{f*g}_{2\pi} = \widehat{f}_{2\pi}\,\widehat{g}_{2\pi}.$$

Translation by a contributes $\mathrm{e}^{-2\pi\mathrm{i}a\xi}$; dilation obeys $\widehat{f(a\,\cdot)}_{2\pi}(\xi) = |a|^{-1}\widehat{f}_{2\pi}(\xi/a)$ for real $a \neq 0$.

Measures. For a finite Borel measure μ, $\widehat{\mu}_{2\pi}(\xi) = \int e^{-2\pi i x\xi}\, d\mu(x)$. This has no density assumption.

The angular-frequency Fourier transform

Operator token. `\FourierAngularOperator` denotes the transform operator without supplying its operand.
Bracketed application. `\FourierAngularTransformOf{f}` denotes the transform applied to f and prints the operand in brackets.
Normalization-labelled hat. `\FourierAngular{f}` denotes the same transform of f with the angular-frequency normalization printed on the hat.
Forward transform. For $f \in L^1(\mathbb{R})$,

$$\widehat{f}_{\mathrm{ang}}(\omega) := \mathcal{F}_{\mathrm{ang}}[f]\,(\omega) := \mathcal{F}_{\mathrm{ang}}[f](\omega) := \int_{\mathbb{R}} f(x)e^{-i\omega x}\, dx, \qquad \omega \in \mathbb{R}.$$

Measures. For a finite Borel measure μ,

$$\widehat{\mu}_{\mathrm{ang}}(\omega) := \int_{\mathbb{R}} e^{-i\omega x}\, d\mu(x), \qquad \omega \in \mathbb{R}.$$

This is a Fourier–Stieltjes transform and requires no density.
Inverse. Under an inversion hypothesis,

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}_{\mathrm{ang}}(\omega)e^{i\omega x}\, d\omega.$$

The forward direction has no prefactor and the inverse has $1/(2\pi)$.
Inverse command. `\InverseFourierAngularOperator` denotes precisely this inverse on a stated inversion domain.
Rules. $\widehat{f'}_{\mathrm{ang}}(\omega) = i\omega\widehat{f}_{\mathrm{ang}}(\omega)$ and $\widehat{f * g}_{\mathrm{ang}} = \widehat{f}_{\mathrm{ang}}\,\widehat{g}_{\mathrm{ang}}$.
Conversion. The two catalogued transforms satisfy

$$\widehat{f}_{2\pi}(\xi) = \widehat{f}_{\mathrm{ang}}(2\pi\xi), \qquad \widehat{f}_{\mathrm{ang}}(\omega) = \widehat{f}_{2\pi}(\omega/(2\pi)).$$

The same identities hold, with f replaced throughout by a finite Borel measure μ, for the two Fourier–Stieltjes transforms just defined. A formula may change variables between the conventions only by displaying this rescaling.

Characteristic functions and Fourier signs

Definition. For a real random variable X with probability law μ,

$$\varphi_X(t) := \mathbb{E}[e^{itX}] = \varphi_\mu(t) := \int_{\mathbb{R}} e^{itx}\, d\mu(x), \qquad t \in \mathbb{R}.$$

Thus the command accepts either a declared random variable or a declared probability measure. The argument's type fixes the interpretation, and $\varphi_X = \varphi_{\mathrm{Law}(X)}$ pointwise. It is not used for an arbitrary signed or complex measure without a local declaration. The probabilistic sign is positive.
Fourier conversion. Because both catalogued Fourier transforms use a negative sign,

$$\varphi_X(t) = \widehat{\mathrm{Law}(X)}_{\mathrm{ang}}(-t) = \widehat{\mathrm{Law}(X)}_{2\pi}\left(-\frac{t}{2\pi}\right).$$

Moments. If $\mathbb{E}|X|^n < \infty$, then $\varphi_X^{(n)}(0) = \mathrm{i}^n \mathbb{E}[X^n]$. Analyticity at zero requires stronger hypotheses than the existence of all moments.
Independent sums. For independent X, Y, $\varphi_{X+Y} = \varphi_X\, \varphi_Y$. The equality is pointwise in t.

The two sinc normalizations

Radian sinc. For $z \in \mathbb{C}$,

$$\operatorname{sinc}_{\mathrm{rad}}(z) := \begin{cases} \sin z / z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The value at zero fills the removable singularity, and the result is entire.
Normalized sinc. For $z \in \mathbb{C}$,

$$\operatorname{sinc}_{\pi}(z) := \begin{cases} \sin(\pi z)/(\pi z), & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Conversion. $\operatorname{sinc}_\pi(z) = \operatorname{sinc}_{\mathrm{rad}}(\pi z)$ and $\operatorname{sinc}_{\mathrm{rad}}(z) = \operatorname{sinc}_\pi(z/\pi)$.
Zeros. $\operatorname{sinc}_{\mathrm{rad}}$ vanishes at πk, while sinc_π vanishes at k, for $k \in \mathbb{Z} \setminus \{0\}$. This distinction is essential in zero-set and sampling arguments.
Retired alias. A bare sinc operator is forbidden in normative prose. Historical occurrences must be annotated with one of the two definitions.

Fourier products for Rvachev's up-function

2π-frequency form. With $\int \mathrm{up} = 1$ and the 2π convention,

$$\widehat{\mathrm{up}}_{2\pi}(\xi) = \prod_{n=0}^{\infty} \operatorname{sinc}_{\mathrm{rad}}\left(\frac{\pi\xi}{2^n}\right) = \prod_{n=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{\xi}{2^n}\right).$$

Angular form. Equivalently,

$$\widehat{\mathrm{up}}_{\mathrm{ang}}(\omega) = \prod_{j=1}^{\infty} \operatorname{sinc}_{\mathrm{rad}}\left(\frac{\omega}{2^j}\right).$$

The angular product begins at $j = 1$; the 2π product begins at $n = 0$.
At zero. Every factor is assigned value 1 at zero, so both products equal 1 at the origin.
Convergence. The products converge locally uniformly on $\mathbb{C}$, hence define entire functions. A logarithm of the product is used only on a zero-free domain with an explicit branch.
Zeros. Multiplicities are computed by counting all pairs (n, k) for which $\xi/2^n = k \in \mathbb{Z} \setminus \{0\}$, not by treating the displayed factors as having disjoint zero sets.

Fourier series on the unit circle

Command. `\FourierSeriesCoefficient{h}{k}`. The printed subscript "FS" distinguishes a discrete Fourier-series coefficient from both continuous line-transform normalizations. For a period other than one, use `\FourierSeriesCoefficientAtPeriod{h}{k}{T}`; its third argument is mandatory and appears in the printed symbol.
Coefficient. For a 1-periodic integrable function h,

$$\widehat{h}_{\mathrm{FS}}[k] := \int_0^1 h(x) \mathrm{e}^{-2\pi \mathrm{i} k x}\, \mathrm{d}x, \qquad k \in \mathbb{Z}.$$

Series. The formal Fourier series is $\sum_{k\in\mathbb{Z}} \widehat{h}_{\mathrm{FS}}[k]\, \mathrm{e}^{2\pi \mathrm{i} kx}$. Its mode of convergence—pointwise, L^2, uniform, distributional, or a named summability method—is part of every theorem.
Separation from line transform. Square brackets and an integer index mark a Fourier-series coefficient. Parentheses and a real frequency mark $\widehat{h}_{2\pi}(\xi)$ or $\widehat{h}_{\mathrm{ang}}(\omega)$ on the line.
Period T. For a T-periodic function, $T > 0$, $\widehat{h}_{\mathrm{FS};T}[k] = T^{-1} \int_0^T h(x)\mathrm{e}^{-2\pi \mathrm{i} kx/T}\, \mathrm{d}x$. The period, the factor T^{-1}, and the kernel scale are never implicit. A finite measure or an unnormalized Fourier–Stieltjes coefficient is a different local object and does not use either function-coefficient command.

One-sided and bilateral Laplace transforms

Commands. `\LaplaceTransformSymbol` is the operator token and `\LaplaceTransformOf{f}` supplies a bracketed operand. An unsubscripted token is one-sided by this catalogue's convention.
One-sided. For $f : [0, \infty) \to \mathbb{C}$,

$$\mathcal{L}[f]\,(s) := \int_0^\infty \mathrm{e}^{-sx} f(x)\, \mathrm{d}x, \qquad s \in \mathbb{C}$$

where the integral converges.
Bilateral. For $f : \mathbb{R} \to \mathbb{C}$,

$$\mathcal{L}_{\mathrm{bi}}[f](s) := \int_{-\infty}^\infty \mathrm{e}^{-sx} f(x)\, \mathrm{d}x.$$

The subscript bi is compulsory; it prevents a support assumption from being smuggled into an identity.
Abscissa. The abscissa of absolute convergence is $\sigma_a = \inf\{\sigma \in \mathbb{R} : \int_0^\infty \mathrm{e}^{-\sigma x}|f(x)|\, \mathrm{d}x < \infty\}$, with infimum in $\overline{\mathbb{R}}$.
Probability relation. For a nonnegative random variable X, $\mathbb{E}[\mathrm{e}^{-sX}]$ is the Laplace–Stieltjes transform of its law. $M_X(t) = \mathbb{E}[\mathrm{e}^{tX}]$ is obtained at $s = -t$ only where both sides are finite.

Moment-generating and exponential-generating functions

Moment-generating function. For a real random variable X,

$$M_X(t) := \mathbb{E}[\mathrm{e}^{tX}]$$

for real or complex t at which the expectation is absolutely finite. It is not declared on all of $\mathbb{C}$ merely because every formal moment exists.
Moment expansion. If the MGF is analytic at zero, $M_X(t) = \sum_{n=0}^\infty \mathbb{E}[X^n]t^n/n!$ locally.
Exponential-generating function. For an abstract sequence (a_n), $A_{\mathrm{egf}}(z) = \sum_{n\geq 0} a_n z^n/n!$. It is an MGF only after a random variable with $a_n = \mathbb{E}[X^n]$ has been specified.
Ordinary-generating function. $A_{\mathrm{ogf}}(z) = \sum_{n\geq 0} a_n z^n$. The subscripts egf and ogf are retained wherever both occur.

Mellin transform

Commands. `\MellinTransformSymbol` is the operator token and `\MellinTransformOf{f}` supplies the operand. Neither command chooses a convergence strip or analytic continuation.
Definition. For $f : (0, \infty) \to \mathbb{C}$,

$$\mathcal{M}[f]\,(s) := \int_0^\infty f(x)x^{s-1}\, \mathrm{d}x, \qquad s \in \mathbb{C},$$

on the maximal strip of absolute convergence.

Complex power. For $x > 0$, x^{s-1} means $\exp((s-1)\log x)$ with the real logarithm $\log x$; no complex branch is involved.
Scaling. For real $a > 0$, $\mathcal{M}[f(a\,\cdot)]\,(s) = a^{-s}\mathcal{M}[f]\,(s)$.
Continuation. A meromorphic continuation of the integral-defined transform is identified by the same symbol only after its initial strip and uniqueness of continuation have been stated. Poles, residues, and contour shifts refer to that continuation.

Cauchy and Stieltjes transforms

Cauchy transform. For a finite Borel measure μ on $\mathbb{R}$,

$$G_\mu(z) := \int_{\mathbb{R}} \frac{1}{z-x}\,\mathrm{d}\mu(x), \qquad z \in \mathbb{C} \setminus \operatorname{supp}(\mu)\,.$$

Alternative sign. Some sources define $\widetilde{G}_\mu(z) = \int (x-z)^{-1}\,\mathrm{d}\mu(x) = -G_\mu(z)$. The catalogue uses the $z - x$ denominator; a quoted alternative carries a tilde and the minus-sign conversion.
Stieltjes transform. For a measure on $[0,\infty)$, $S_\mu(s) = \int_0^\infty (s+x)^{-1}\,\mathrm{d}\mu(x)$ for $s \in \mathbb{C}\setminus(-\infty,0]$ when finite. Denominator signs must be accounted for before relating it to G_μ.
Boundary values. Symbols $G_\mu(x \pm \mathrm{i}0)$ mean non-tangential limits from the upper or lower half-plane when those limits exist; they are not evaluation at a complex number with an infinitesimal component.

Transforms of distributions and generalized functions

Tempered distributions. For $T \in \mathscr{S}'(\mathbb{R})$, its 2π-Fourier transform is defined by duality $\langle \widehat{T}_{2\pi}, \phi\rangle = \langle T, \widehat{\phi}_{2\pi}\rangle$ after fixing the test-function transform above. The pairing is bilinear, not a Hermitian inner product.
Dirac identities. $\widehat{\delta_a}_{2\pi}(\xi) = \mathrm{e}^{-2\pi\mathrm{i}a\xi}$ and $\widehat{1}_{2\pi} = \delta_0$ in $\mathscr{S}'$.
Derivatives. $\widehat{D^m T}_{2\pi} = (2\pi\mathrm{i}\xi)^m \widehat{T}_{2\pi}$, where D is distributional derivative.
Audit rule. A distributional identity is never presented as an absolutely convergent integral identity. The ambient test-function space and duality convention appear at the first use.

Nonlinear and linearized Fourier-decay objectives

Commands. For $n \in \mathbb{N}_0$ the exact optimization functional is $\mathrm{Obj}_{\mathrm{decay},n}$ and its first-order or otherwise explicitly specified linearization is $\mathrm{Obj}^{\mathrm{lin}}_{\mathrm{decay},n}$. Their semantic names prevent either from being mistaken for a Fabius function, a Fourier operator, or a filtration.
Required declaration. At the first use a document supplies a variable space V_n, a feasible set $\mathcal{C}_n \subseteq V_n$, a codomain (normally $\mathbb{R} \cup \{+\infty\}$), and formulas

$$\mathrm{Obj}_{\mathrm{decay},n} : \mathcal{C}_n \to \overline{\mathbb{R}}, \qquad \mathrm{Obj}^{\mathrm{lin}}_{\mathrm{decay},n} : T_{v_0}\mathcal{C}_n \to \overline{\mathbb{R}}.$$

The base point v_0, constraints, frequency normalization, logarithm base, weights, and all n-dependence are part of those formulas. The shared commands intentionally do not smuggle in one draft's choice.
Linearization contract. A label "linearized" identifies a new objective. An expansion such as $\mathrm{Obj}_{\mathrm{decay},n}(v_0 + h) = \mathrm{Obj}_{\mathrm{decay},n}(v_0) + \mathrm{Obj}^{\mathrm{lin}}_{\mathrm{decay},n}(h) + r_n(h)$ states the norm and limit in which $r_n(h) = o_{h\to 0}(\|h\|)$. Dropping r_n produces an approximation, not an identity.
Optimization values. $\inf_{v\in\mathcal{C}_n} \mathrm{Obj}_{\mathrm{decay},n}(v)$ is an extended-real value. A symbol for a minimizer is introduced only after existence; uniqueness is an additional theorem. The optimum of the linearized problem is not substituted for the nonlinear optimum without a bound.

Status. The commands are **definitions of notation**. Particular decay bounds, saddle locations, convexity assertions, and asymptotic optima inherit the proof-status of the document that states them.

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7 Binary arithmetic, Thue–Morse data, and exact combinatorics

Binary digit sum and bits

Binary expansion. Every $n \in \mathbb{N}_0$ has the unique finite expansion $n = \sum_{j=0}^{\infty} b_j(n) 2^j$ with $b_j(n) \in \{0,1\}$ and all but finitely many $b_j(n)$ zero. Leading zeros are allowed but contribute nothing.
Digit sum. The canonical binary weight is

$$\mathrm{wt}_2(n) := \sum_{j=0}^{\infty} b_j(n), \qquad n \in \mathbb{N}_0.$$

Thus $\mathrm{wt}_2(0) = 0$.
Bit extraction. $b_j(n) = \lfloor n/2^j \rfloor - 2\lfloor n/2^{j+1} \rfloor$ for $j \in \mathbb{N}_0$. Negative integers require an explicitly selected finite-word or 2-adic representation and are outside this global definition.
Retired aliases. The former digit-sum, population-count, and bare-weight spellings may occur in quotations. Normative prose uses `\BinaryDigitSum`; an unsubscripted weight is especially unsafe because it can denote an integration weight.

Base-q digits, digit sums, digit characters, and digit power sums

Digits. For an integer base $q \geq 2$ and $n \in \mathbb{N}_0$, the j-th base-q digit is

$$d_j(n) := \lfloor n/q^j \rfloor \bmod q \in \{0, \dots, q-1\}, \qquad j \in \mathbb{N}_0,$$

so that $n = \sum_{j \geq 0} d_j(n)\, q^j$. At $q = 2$ this is the bit $b_j(n)$ of the binary-weight entry. The base is always written explicitly; it is never inferred from context.
Digit sum. $s_q(n) := \sum_{j \geq 0} d_j(n)$. At $q = 2$ normative prose writes $\mathrm{wt}_2(n)$, not $s_2(n)$; a document that abbreviates $s_2 := \mathrm{wt}_2$ must say so once. These are plain-math conventions, not shared commands.
Digit window. A statement over the block $0 \leq n < q^m$ may use the window product $\prod_{j<m} w\big(d_j(n)\big)$ for a digit weight w; leading zeros inside the window are weighted by $w(0)$, which is why identities stated with the window need no normalisation $w(0) = 1$, whereas s_q ignores leading zeros.
Digit character. For ζ in a commutative ring, $u_n := \zeta^{s_q(n)}$. The recursion $u_{qn+r} = \zeta^r u_n$ $(r < q)$ and the digit product $\sum_{n<q^m} u_n z^n = \prod_{j<m} \sum_{r<q} \zeta^r z^{rq^j}$ hold for every ζ; the hypothesis that ζ be a nontrivial q-th root of unity, $\sum_{r<q} \zeta^r = 0$, enters only through the vanishing of each factor at $z = 1$. At $q = 2$, $\zeta = -1$ gives $u_n = \varepsilon_n$.
Digit power sums. $P_d(m) := \sum_{n<q^m} u_n\, n^d$. Under $\sum_{r<q} \zeta^r = 0$, $P_d(m) = 0$ for $d < m$ (Prouhet cancellation), and with $\zeta^q = 1$ the sharp value is $(\zeta - 1)^m P_m(m) = m!\, q^{m(m+1)/2}$, stated division-free so that it holds in any commutative ring.
Lean crosswalk. `digitAt` is $d_j(n)$, `windowWeight` the window product, `digitPowerSum` is $P_d(m)$; the digit product is `sum_zeta_pow_digits_sum_mul_pow` and its arbitrary-weight form `sum_windowWeight_mul_pow` (`BaseDigitProduct.lean`); cancellation and the sharp moment are `digitPowerSum_eq_zero_of_lt` and `sub_one_pow_mul_digitPowerSum_self` (`BaseDigitProuhet.lean`). The digit sum itself is Mathlib's `(Nat.digitsqn).sum`.

Thue–Morse bit and sign sequences

Bit sequence. $t_n \coloneqq \mathrm{wt}_2(n) \bmod 2 \in \{0,1\}$ for $n \in \mathbb{N}_0$.
Sign sequence. The canonical sign is

$$\varepsilon_n \coloneqq (-1)^{\mathrm{wt}_2(n)} = (-1)^{t_n} \in \{-1,1\}.$$

Consequently $\varepsilon_0 = 1$.
Command layer. `\ThueMorseSignSymbol` is a zero-argument reserved glyph token rendering ε; it names the sign sequence inside operator labels, DFT displays, and prose where no index is being evaluated. `\ThueMorseSign{n}` takes exactly one argument $n \in \mathbb{N}_0$ and renders the indexed value ε_n. The former is not a constant value and the latter must not receive a second raw subscript.
Recurrences. $t_{2n} = t_n$, $t_{2n+1} = 1 - t_n$ and $\varepsilon_{2n} = \varepsilon_n$, $\varepsilon_{2n+1} = -\varepsilon_n$.
Generating product. For every $N \in \mathbb{N}_0$,

$$\sum_{n=0}^{2^N-1} \varepsilon_n\, z^n = \prod_{j=0}^{N-1} (1 - z^{2^j}).$$

At $N = 0$ both sides are 1: the sum contains the sole term $n = 0$ and the product is empty.
Rademacher–sine representation. For every $n \in \mathbb{N}_0$,

$$\varepsilon_n = \prod_{j=0}^{\infty} \operatorname{sgn}\Bigl(\sin \frac{\pi(n+\frac{1}{2})}{2^j}\Bigr),$$

and the j-th factor equals $(-1)^{b_j(n)}$ exactly. Every factor with $2^j > n$ is 1, so the product has finite multiplicative support and is asserted without a convergence hypothesis. The half-unit shift is essential: the unshifted product $\prod_j \operatorname{sgn}(\sin(\pi n/2^j))$ contains the factor $\operatorname{sgn}(\sin \pi n) = 0$ at $j = 0$ and is identically zero.
Square wave. For every $x \in \mathbb{R}$, $\sin(\pi x) = (-1)^{\lfloor x \rfloor} \sin(\pi\,\{x\})$; the identity holds at the integers as well, where both sides vanish. Off the integers $\sin(\pi\,\{x\}) > 0$, hence $\operatorname{sgn}(\sin \pi x) = (-1)^{\lfloor x \rfloor}$ there. At the Rademacher points $x = (n+\frac{1}{2})/2^j$ one has $\lfloor x \rfloor = \lfloor n/2^j \rfloor$ and $\{x\} \neq 0$.
Lean crosswalk. The formal sign sequence uses declarations in `ThueMorse.lean` and related prefix modules. Exact theorem names are recorded in the crosswalk appendix; prose identities are not upgraded beyond their verified status. The Rademacher–sine representation is `thueMorseSign_eq_tprod_sign_sin` with factor law `sign_sin_rademacherPoint`, and the factored square wave is `sin_pi_mul_eq_neg_one_zpow_floor`, all in `RademacherSine.lean`.

Thue–Morse autocorrelations, normalized and limiting

Window and dyadic autocorrelation. For $N, k \in \mathbb{N}_0$, $R_N(k) \coloneqq \sum_{n<N} \varepsilon_n\, \varepsilon_{n+k}$, and at the dyadic window $A_m(k) \coloneqq R_{2^m}(k)$. These are integers; $R_N(0) = N$ and $|R_N(k)| \le N$.
Exact recursions. $R_{2K}(2r) = 2R_K(r)$ and $R_{2K}(2r+1) = -(R_K(r) + R_K(r+1))$ for every even window; nothing uses that the window is a power of two.
Normalized and limiting values. $\eta_m(k) \coloneqq A_m(k)/2^m \in [-1,1]$, so that $\eta_{m+1}(2r) = \eta_m(r)$ and $\eta_{m+1}(2r+1) = -(\eta_m(r) + \eta_m(r+1))/2$ exactly. The limit $\eta(k) \coloneqq \lim_{m\to\infty} \eta_m(k)$ exists for every k and is the unique solution of $\eta(0) = 1$, $\eta(2r) = \eta(r)$, $\eta(2r+1) = -(\eta(r) + \eta(r+1))/2$; in particular $\eta(1) = \eta(2) = -\frac{1}{3}$, $\eta(3) = \frac{1}{3}$, $\eta(5) = 0$. The symbol η is document-local to the Thue–Morse spectral discussion; it is not a shared command and must not be reused for a different sequence in the same document.

Lean crosswalk. `thueMorseWindowAutocorrelation` is $R_N(k)$, `thueMorseAutocorrelation` is $A_m(k)$ (`ThueMorseAutocorrelation.lean`); `normalizedAutocorrelation` is $\eta_m(k)$, `limitingAutocorrelation` is $\eta(k)$ defined by the recursion, and `tendsto_normalizedAutocorrelation` is the convergence (`ThueMorseAutocorrelationLimit.lean`).

Thue–Morse block sums and the nested dyadic integral

Block sum. For $f : \mathbb{R} \to \mathbb{R}$, $x, h \in \mathbb{R}$ and $m \in \mathbb{N}_0$, the block sum is $\sum_{n<2^m} \varepsilon_n\, f(x + nh)$. It is the m-fold dyadic difference $\prod_{j<m}(I - T_{2^j h})f$ evaluated at x, with T_a the translation by a.
Nested dyadic integral. The iterated integral $\int_0^h \int_0^{2h} \cdots \int_0^{2^{m-1}h} g(x + t_0 + \cdots + t_{m-1})\, dt_{m-1} \cdots dt_0$ is written with the outermost variable on $[0, h]$ and the step doubling inward; its box volume is $h^m 2^{0+1+\cdots+(m-1)} = h^m 2^{\binom{m}{2}}$. Because the integrand depends only on the sum of the variables, any other nesting order gives the same value, but the displayed order is the one the recursion produces.
Cubature identity. For f of class C^m and every real x, h, the block sum equals $(-1)^m$ times the nested dyadic integral of $f^{(m)}$; no sign condition on h is needed for the identity. Consequences stated with $h \geq 0$: the block sum is nonnegative when $(-1)^m f^{(m)} \geq 0$ on $[x, x + (2^m - 1)h]$, and is bounded by $C\, h^m 2^{\binom{m}{2}}$ when $|f^{(m)}| \leq C$ there.
Lean crosswalk. `thueMorseBlockSum`, `nestedDyadicIntegral`, `thueMorseBlockSum_eq_nestedDyadicInte` `thueMorseBlockSum_nonneg`, `abs_thueMorseBlockSum_le`, all in `ThueMorseCubature.lean`.

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p-adic and 2-adic valuations

Nonzero integers. Fix a prime p. Every nonzero $n \in \mathbb{Z}$ has a unique factorization $n = p^k u$ with $k \in \mathbb{N}_0$, $u \in \mathbb{Z}$, and $p \nmid u$. The finite integer valuation is

$$\nu_p(n) \coloneqq k = \max\{r \in \mathbb{N}_0 : p^r \mid n\} \in \mathbb{N}_0.$$

The value depends only on $|n|$, so signs contribute no power of p.
Nonzero rationals. For $x \in \mathbb{Q}^\times$ choose any representation $x = a/b$ with nonzero $a, b \in \mathbb{Z}$ and define

$$\nu_p(x) \coloneqq \nu_p(a) - \nu_p(b) \in \mathbb{Z}.$$

This integer is independent of the chosen fraction: multiplying numerator and denominator by the same nonzero integer adds and subtracts the same valuation. Equivalently, $x = p^k u/v$ for a unique $k \in \mathbb{Z}$ after requiring $u, v \in \mathbb{Z} \setminus \{0\}$ and $p \nmid uv$. Negative values therefore record powers of p in the denominator.
Algebraic laws. For nonzero rationals x, y,

$$\nu_p(xy) = \nu_p(x) + \nu_p(y), \qquad \nu_p(x/y) = \nu_p(x) - \nu_p(y).$$

If also $x + y \neq 0$, then $\nu_p(x + y) \geq \min\{\nu_p(x), \nu_p(y)\}$, with equality when the two valuations differ.
Zero conventions. The classical finite maps above have domains $\mathbb{Z} \setminus \{0\}$ and $\mathbb{Q}^\times$; their value at zero is undefined. An explicitly extended valuation instead has codomain $\mathbb{Z} \cup \{+\infty\}$ and is defined by $\overline{\nu_p}(0) = +\infty$, with $k < +\infty$ and $k + (+\infty) = +\infty$ for every $k \in \mathbb{Z}$. A finite value at zero is never inferred from the printed glyph: it belongs only to a named software API or another locally declared totalization.
Two-adic specialization. $\nu_2(x) = \nu_2(x)$ on the declared nonzero integer or rational domain. For a nonzero integer, $n = 2^{\nu_2(n)} n_{\text{odd}}$ with n_{odd} odd and $\nu_2(n) \in \mathbb{N}_0$. On rationals its codomain is $\mathbb{Z}$; for example, $\nu_2(3/8) = -3$ and $\nu_2(-12) = 2$.

Prime-power Pascal row. For a prime p, $m \in \mathbb{N}_0$, and $j \in \mathbb{N}_{>0}$ with $j \le p^m$, the finite-natural valuation satisfies

$$\nu_p\binom{p^m}{j} + \nu_p(j) = m, \qquad \nu_p\binom{p^m}{j} = m - \nu_p(j).$$

The right endpoint $j = p^m$ and the boundary $m = 0$ are included. The strict dyadic-comb specialization has $0 < j < 2^m$. These formulas do not cover the distinct endpoint-flat weights $\binom{2^m-2}{j-1}$.
Lean crosswalk. In `PrimePowerBinomialValuation.lean`, the arbitrary-prime identities are the declarations

```
primePowerChoose_padicValNat_add
```

```
primePowerChoose_padicValNat
```

The strict dyadic specialization is

```
twoPowChoose_padicValNat
```

Mathlib's three relevant functions are distinct totalized APIs: `padicValNat` : $\mathbb{N}_0 \to \mathbb{N}_0$, `padicValInt` : $\mathbb{Z} \to \mathbb{N}_0$, and `padicValRat` : $\mathbb{Q} \to \mathbb{Z}$ after the prime parameter has been supplied. Each assigns zero the finite value 0; on nonzero inputs its codomain agrees with the corresponding finite valuation above. Thus a theorem using one of these Lean names does not use the $+\infty$ convention, and the three codomains are not interchangeable. Status of the displayed project identities: **Lean-proved**.
Scoped historical spelling. The source-faithful transcription `docs/papers/arXiv-1702.06487v3/157-Arithmeti` carries the explicit `HISTORICAL-NOTATION-SCOPE:` marker and retains the raw source form `\nu_2(r)`. In that paper the operand is a nonzero rational number and the value is the exponent of 2 in its prime factorization, hence an element of $\mathbb{Z}$ that may be negative. That spelling is interpretive historical evidence only; active normative prose uses `\TwoAdicValuation` and states its domain and zero convention.
Order notation. For a zero or pole of a meromorphic function at z_0, $\operatorname{ord}_{z_0}(f)$ is a different valuation-like object; positive values denote zeros and negative values poles in this catalogue.

Dyadic blocks and prefix sums

Block. The level-N dyadic block of indices is $B_N = \{0, 1, \ldots, 2^N - 1\}$ for $N \in \mathbb{N}_0$.
Signed prefix sum. For a function $P : \mathbb{N}_0 \to \mathbb{C}$,

$$\Sigma_N(P) \coloneqq \sum_{n=0}^{2^N-1} \varepsilon_n \, P(n).$$

Prouhet cancellation. If P is a polynomial of degree $< N$, then $\Sigma_N(P) = 0$. At degree N the surviving value depends on the leading coefficient and the exact sign convention; it must not be quoted without that coefficient.
Arbitrary prefix. $S_M(P) = \sum_{n=0}^{M-1} \varepsilon_n P(n)$ for $M \in \mathbb{N}_0$. The half-open upper bound ensures $S_0(P) = 0$.
Shifted block. A sum over $a2^N \le n < (a+1)2^N$ prints both $a \in \mathbb{N}_0$ and $N \in \mathbb{N}_0$; its sign is not inferred from the unshifted identity without applying the binary carry law.

Finite discrete Fourier transforms

Forward DFT. For a vector $a = (a_0, \ldots, a_{N-1}) \in \mathbb{C}^N$, $N \in \mathbb{N}_{>0}$,

$$\widehat{a}_k \coloneqq \sum_{n=0}^{N-1} a_n \mathrm{e}^{-2\pi \mathrm{i} kn/N}, \qquad k = 0, \ldots, N-1.$$

Inverse. $a_n = N^{-1}\sum_{k=0}^{N-1}\widehat{a}_k \mathrm{e}^{2\pi \mathrm{i}kn/N}$. The normalization is forward-unscaled, inverse-N^{-1}.
Root-of-unity form. With $\zeta_N = \mathrm{e}^{2\pi\mathrm{i}/N}$, $\widehat{a}_k = \sum_n a_n \zeta_N^{-kn}$. The symbol ζ_N is a primitive root chosen with positive angle.
Thue–Morse specialization. For $N = 2^m$ and $a_n = \varepsilon_n$, the product formula evaluates the DFT at $z = \zeta_N^{-k}$. Multiplicities of zeros require counting vanishing factors, not only identifying a vanishing product.
Document-local command. In the Thue–Morse atlas, `\ThueMorseDFTCoefficient{m}{k}` has the two arguments $m \in \mathbb{N}_0$ and $k \in \mathbb{Z}$ in that order and means

$$\mathrm{DFT}_{2^m}[\varepsilon][k] := \sum_{n=0}^{2^m-1} \varepsilon_n \, \mathrm{e}^{-2\pi \mathrm{i}kn/2^m}.$$

The value depends only on k modulo 2^m; the canonical representative range is $0 \le k < 2^m$. The forward transform is unscaled, the exponential sign is negative, and inversion therefore carries the factor 2^{-m} and the positive exponential. This operator is a finite DFT, not either line-Fourier transform and not a non-Fourier accent.

Zero sets, multiplicities, and runs

Order of a zero. For analytic f not identically zero near z_0, $\mathrm{ord}_{z_0}(f) = m \in \mathbb{N}_0$ means $f(z) = (z - z_0)^m g(z)$ with $g(z_0) \ne 0$.
Run in a sequence. For a sequence (a_n), a zero run of length ℓ from r means $a_r = \cdots = a_{r+\ell-1} = 0$ and, when the adjacent indices exist, $a_{r-1} \ne 0$ and $a_{r+\ell} \ne 0$.
Boundary runs. At $r = 0$, only the right maximality condition is required. For a finite sequence ending at M, a terminal run has no right-neighbor condition.
Separation. A multiplicity of a zero of a generating polynomial is not a run of zero coefficients. The corpus contains both phenomena and keeps ord_{z_0} separate from ℓ_{run}.

Compositions, partitions, and multi-indices

Weak composition. A weak composition of n into k parts is $\alpha = (\alpha_1, \ldots, \alpha_k) \in \mathbb{N}_0^k$ with $|\alpha| = \sum_{j=1}^k \alpha_j = n$.
Composition. A composition has $\alpha_j \in \mathbb{N}_{>0}$. For $n = 0$, the unique composition with zero parts is the empty tuple; there is no positive-part composition into $k > 0$ parts.
Partition. A partition $\lambda \vdash n$ is a finite nonincreasing sequence of positive integers summing to n. Its length is $\ell(\lambda)$ and its multiplicity of part j is $m_j(\lambda)$.
Multi-index. For $\alpha \in \mathbb{N}_0^d$, $|\alpha| = \sum_j \alpha_j$, $\alpha! = \prod_j \alpha_j!$, $x^\alpha = \prod_j x_j^{\alpha_j}$, and $\partial^\alpha = \prod_j \partial_{x_j}^{\alpha_j}$. Products over $d = 0$ are empty and equal 1.
Factorials. The shared $(x)^{\underline{n}} = x(x-1)\cdots(x-n+1)$ and $(x)^{\overline{n}} = x(x+1)\cdots(x+n-1)$ both equal 1 at $n = 0$.

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8 Combinatorial coefficient calculus and inversion

Signed and unsigned Stirling numbers, second-kind numbers, and Lah numbers

Index domain and zero extension. Unless a formula says otherwise, $n, k \in \mathbb{N}_0$. Each triangular family is extended by zero when $k < 0$, $n < 0$, or $k > n$. Every empty degree-zero basis expansion has the single value at $(n, k) = (0, 0)$ equal to 1. Thus the boundary conventions are part of the notation, not omitted side conditions.

Signed first kind. The signed Stirling number of the first kind is defined by

$$(x)^{\underline{n}} = x(x-1)\cdots(x-n+1) = \sum_{k=0}^{n} s(n,k)\, x^k.$$

Consequently

$$s(0,0) = 1, \qquad s(n,0) = 0 \quad (n > 0),$$

and

$$s(n+1,k) = s(n,k-1) - ns(n,k)\,.$$

The sign is $(-1)^{n-k}$ whenever the corresponding unsigned count is nonzero.

Unsigned first kind. The unsigned cycle number is

$$\begin{bmatrix} n \\ k \end{bmatrix} := (-1)^{n-k} s(n,k)\,.$$

It counts permutations of an n-element set having exactly k disjoint cycles. Its recurrence is

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = \begin{bmatrix} n \\ k-1 \end{bmatrix} + n\begin{bmatrix} n \\ k \end{bmatrix}.$$

The bracket is not a coefficient-extraction bracket and is not a Gaussian binomial coefficient.

Second kind. The Stirling number $\left\{ {n \atop k} \right\}$ counts partitions of an n-element labelled set into exactly k nonempty, unordered blocks. It is characterized by

$$x^n = \sum_{k=0}^{n} \left\{ {n \atop k} \right\} (x)^{\underline{k}}$$

and by

$$\left\{ {n+1 \atop k} \right\} = k\left\{ {n \atop k} \right\} + \left\{ {n \atop k-1} \right\}.$$

The boundary values are $\left\{ {0 \atop 0} \right\} = 1$, $\left\{ {n \atop 0} \right\} = 0$ for $n > 0$, and $\left\{ {n \atop n} \right\} = 1$.

Inverse basis changes. The lower-triangular matrices of signed first-kind and second-kind numbers are inverses:

$$\sum_{j=k}^{n} s(n,j) \left\{ {j \atop k} \right\} = \delta_{n,k} = \sum_{j=k}^{n} \left\{ {n \atop j} \right\} s(j,k)\,.$$

This is matrix inversion over the integer-indexed triangular arrays; it is not functional or multiplicative inversion.

Unsigned Lah number. For $1 \le k \le n$,

$$\mathsf{L}^{\mathrm{Lah}}_{n,k} := \binom{n-1}{k-1} \frac{n!}{k!}.$$

It counts partitions of an n-element labelled set into k nonempty linearly ordered lists, with the collection of lists itself unordered. Put $\mathsf{L}^{\mathrm{Lah}}_{0,0} = 1$ and set it to zero outside $0 \le k \le n$.

Lah basis changes. With the preceding boundary convention,

$$(x)^{\overline{n}} = \sum_{k=0}^{n} \mathsf{L}^{\mathrm{Lah}}_{n,k}\,(x)^{\underline{k}},$$

$$(x)^{\underline{n}} = \sum_{k=0}^{n} (-1)^{n-k} \mathsf{L}^{\mathrm{Lah}}_{n,k}\,(x)^{\overline{k}}.$$

The label Lah prevents the family from colliding with a Lambert polynomial, logarithm, Laplace transform, or other L-family.
Shared commands. The normative source commands are `\SignedStirlingFirstKind`, `\UnsignedStirlingFirst` `\StirlingSecondKind`, and `\LahNumber`.

Bell numbers, Touchard polynomials, and the two Bell-polynomial normalizations

Bell number. The Bell number is

$$\mathsf{B}^{\mathrm{Bell}}_{n} := \sum_{k=0}^{n} \left\{ {n \atop k} \right\}, \qquad \mathsf{B}^{\mathrm{Bell}}_{0} = 1.$$

It counts all set partitions of an n-element labelled set. The superscript Bell is printed because an unlabelled B_n is also widely used for Bernoulli numbers and polynomials.
Document-local Bell block multiplicity. In the q-Pochhammer and q-binomial monograph, `\BellBlockMultiplicity{r}` has TeX arity one and prints m^{Bell}_{r}. For a fixed $n \in \mathbb{N}_0$, the complete-Bell summation variable is the entire vector

$$(m^{\mathrm{Bell}}_{r})_{1\le r\le n} \in \mathbb{N}_0^{n}, \qquad \sum_{r=1}^{n} r\, m^{\mathrm{Bell}}_{r} = n.$$

Here m^{Bell}_{r} is the number of size-r blocks, or equivalently the multiplicity with which the rth cumulant input is selected. In the monograph's Fabius-value specialization its contribution is

$$\prod_{r=1}^{n} \frac{1}{m^{\mathrm{Bell}}_{r}!} \left(\frac{\kappa_r}{r!}\right)^{m^{\mathrm{Bell}}_{r}}.$$

For $n = 0$ the index set has the unique empty vector in $\mathbb{N}_0^{0}$, the weighted sum is zero, and the empty product is one. The semantic superscript prevents this nonnegative multiplicity from being mistaken for a valuation or an unrestricted integer index; the command is local to that monograph and is not part of the shared API.
Touchard polynomial. For $n \in \mathbb{N}_0$,

$$\mathsf{T}^{\mathrm{Tou}}_{n}(x) := \sum_{k=0}^{n} \left\{ {n \atop k} \right\} x^k.$$

Thus $\mathsf{T}^{\mathrm{Tou}}_{n}(1) = \mathsf{B}^{\mathrm{Bell}}_{n}$ and $\mathsf{T}^{\mathrm{Tou}}_{0}(x) = 1$. Its exponential generating function is

$$\sum_{n\ge 0} \mathsf{T}^{\mathrm{Tou}}_{n}(x) \frac{z^n}{n!} = \exp\bigl(x(\mathrm{e}^{z} - 1)\bigr).$$

The tag Tou distinguishes it from every other T_n-family.

Complementary Bell number. The signed block-parity sum is

$$\mathsf{B}_n^{\mathrm{comp}} := \sum_{k=0}^{n} (-1)^k \begin{Bmatrix} n \\ k \end{Bmatrix} = \mathsf{T}_n^{\mathrm{Tou}}(-1).$$

Its EGF and recurrence are

$$\sum_{n\geq 0} \mathsf{B}_n^{\mathrm{comp}} \frac{z^n}{n!} = \exp(1 - \mathrm{e}^z),$$

$$\mathsf{B}_{n+1}^{\mathrm{comp}} = -\sum_{j=0}^{n} \binom{n}{j} \mathsf{B}_j^{\mathrm{comp}}.$$

The source aliases C_n, U_n, and a hatted or tilded Bell number are retired because C_n also denoted a Catalan number and accents have unrelated meanings elsewhere.
Exponential partial Bell polynomial. For $0 \leq k \leq n$,

$$\mathcal{B}_{n,k}^{\mathrm{exp}}(x_1, x_2, \ldots) := \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum_{j\geq 1} m_j = k \\ \sum_{j\geq 1} j m_j = n}} \frac{n!}{\prod_{j\geq 1} m_j!\,(j!)^{m_j}} \prod_{j\geq 1} x_j^{m_j}.$$

Only $x_1, \ldots, x_{n-k+1}$ can occur. The empty-index boundary is $\mathcal{B}_{0,0}^{\mathrm{exp}} = 1$; it is zero for impossible indices. Equivalently,

$$\frac{1}{k!} \left(\sum_{j\geq 1} x_j \frac{t^j}{j!} \right)^k = \sum_{n\geq k} \mathcal{B}_{n,k}^{\mathrm{exp}}(x_1, x_2, \ldots) \frac{t^n}{n!}.$$

Exponential complete Bell polynomial. The complete polynomial is

$$\mathcal{B}_n^{\mathrm{exp}}(x_1, \ldots, x_n) := \sum_{k=0}^{n} \mathcal{B}_{n,k}^{\mathrm{exp}}(x_1, \ldots, x_{n-k+1}),$$

with $\mathcal{B}_0^{\mathrm{exp}} = 1$. Its generating identity is

$$\exp\left(\sum_{j\geq 1} x_j \frac{t^j}{j!} \right) = \sum_{n\geq 0} \mathcal{B}_n^{\mathrm{exp}}(x_1, \ldots, x_n) \frac{t^n}{n!}.$$

Ordinary partial Bell polynomial. For an ordinary series $X(t) = \sum_{j\geq 1} x_j t^j$, define

$$\mathcal{B}_{n,k}^{\mathrm{ord}}(x_1, x_2, \ldots) := \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum m_j = k \\ \sum j m_j = n}} \frac{k!}{\prod_{j\geq 1} m_j!} \prod_{j\geq 1} x_j^{m_j}.$$

Then

$$X(t)^k = \sum_{n\geq k} \mathcal{B}_{n,k}^{\mathrm{ord}}(x_1, x_2, \ldots) t^n.$$

The printed ord is essential: this polynomial differs from the exponential normalization by factorials.
Conversion of normalizations. For all valid n, k,

$$\mathcal{B}_{n,k}^{\mathrm{ord}}(x_1, x_2, \ldots) = \frac{k!}{n!} \mathcal{B}_{n,k}^{\mathrm{exp}}(1!x_1, 2!x_2, 3!x_3, \ldots).$$

This equality explains the factorial conversion; it does not identify the two polynomial families.

Faà di Bruno specialization. For sufficiently differentiable functions f, g and $n \geq 1$,

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n} f(g(x)) = \sum_{k=1}^{n} f^{(k)}(g(x)) \mathcal{B}^{\exp}_{n,k}\big(g'(x), g''(x), \ldots, g^{(n-k+1)}(x)\big).$$

The $n = 0$ case is simply $f(g(x))$. Derivatives therefore use the exponential Bell normalization, whereas ordinary series composition uses the ordinary normalization.

Geometric Bernoulli polynomials for geometric uniform convolution

Probability data. Fix $q \in (0, 1)$, put $a = 1 - q$, and let $Z_0, Z_1, \ldots$ be independent random variables, each uniformly distributed on $[0, 1]$. Define

$$V_q := a \sum_{j=0}^{\infty} q^j Z_j.$$

The series converges absolutely almost surely, satisfies $0 \leq V_q \leq 1$, and has total deterministic weight $a \sum_{j \geq 0} q^j = 1$. Its moment-generating function is the entire function

$$M_q(t) := \mathbb{E}\mathrm{e}^{tV_q} = \prod_{j=0}^{\infty} \frac{\mathrm{e}^{aq^j t} - 1}{aq^j t},$$

where every factor has its removable value 1 at $t = 0$.
Defining exponential generating function. For $n \in \mathbb{N}_0$ and $x \in \mathbb{C}$, the canonical symbol $\mathrm{Ber}^{\mathrm{geom}}_{n,q}(x)$ is defined by

$$\frac{\mathrm{e}^{xt}}{M_q(t)} = \sum_{n=0}^{\infty} \mathrm{Ber}^{\mathrm{geom}}_{n,q}(x) \frac{t^n}{n!}.$$

The identity may be read analytically near $t = 0$, where $M_q(0) = 1$, or formally in $\mathbb{C}[x][[t]]$. It defines a monic polynomial of degree n, with $\mathrm{Ber}^{\mathrm{geom}}_{0,q}(x) = 1$. The command arguments are the degree n and deformation parameter q; the polynomial evaluation argument x follows the command.
Appell and averaging-inverse laws. For $n \geq 1$,

$$\frac{\mathrm{d}}{\mathrm{d}x} \mathrm{Ber}^{\mathrm{geom}}_{n,q}(x) = n \mathrm{Ber}^{\mathrm{geom}}_{n-1,q}(x).$$

For every $n \in \mathbb{N}_0$,

$$\mathbb{E}\, \mathrm{Ber}^{\mathrm{geom}}_{n,q}(x + V_q) = x^n.$$

Thus this family is the polynomial inverse of the averaging operator $T_q p(x) = \mathbb{E}p(x + V_q)$; it is not a Bell-polynomial family.
Difference and reflection laws. The product decomposition $M_q(t) = ((\mathrm{e}^{at} - 1)/(at)) M_q(qt)$ gives, for $n \geq 1$,

$$\mathrm{Ber}^{\mathrm{geom}}_{n,q}(x + a) - \mathrm{Ber}^{\mathrm{geom}}_{n,q}(x) = anq^{n-1} \mathrm{Ber}^{\mathrm{geom}}_{n-1,q}(x/q).$$

Since V_q and $1 - V_q$ have the same distribution,

$$\mathrm{Ber}^{\mathrm{geom}}_{n,q}(1 - x) = (-1)^n \mathrm{Ber}^{\mathrm{geom}}_{n,q}(x).$$

Cumulant representation. Write the cumulant expansion as

$$\log M_q(t) = \sum_{j \geq 1} \kappa_j(V_q) \frac{t^j}{j!}.$$

Then

$$\mathrm{Ber}^{\mathrm{geom}}_{n,q}(x) = \mathcal{B}^{\exp}_n\big(x - \kappa_1(V_q), -\kappa_2(V_q), \ldots, -\kappa_n(V_q)\big), \qquad \kappa_1(V_q) = \tfrac{1}{2}.$$

This formula uses a complete exponential Bell polynomial to represent the Appell family; it does not identify the two families.

Classical limit. As $q \downarrow 0$, $a \to 1$, V_q converges to one uniform random variable, and

$$\frac{\mathrm{e}^{xt}}{M_q(t)} \longrightarrow \frac{t\mathrm{e}^{xt}}{\mathrm{e}^t - 1}.$$

Consequently $\mathrm{Ber}^{\mathrm{geom}}_{n,q}(x)$ converges coefficientwise to the classical Bernoulli polynomial $B_n(x)$ in this generating-function convention. The endpoint $q = 0$ is a limit statement, not part of the declared parameter domain $(0, 1)$.

Type-A, type-B, and second-order Eulerian numbers and row polynomials

Type-A number. For $n \geq 1$, $\left\langle {n \atop k} \right\rangle$ counts permutations of $\{1, \dots, n\}$ with exactly k ordinary descents, where a descent is an index $i \in \{1, \dots, n-1\}$ satisfying $\pi_i > \pi_{i+1}$. It is zero outside $0 \leq k \leq n-1$, with $\left\langle {0 \atop 0} \right\rangle = 1$, and obeys

$$\left\langle {n \atop k} \right\rangle = (k+1)\left\langle {n-1 \atop k} \right\rangle + (n-k)\left\langle {n-1 \atop k-1} \right\rangle.$$

Type-A row polynomial. Set

$$\mathsf{E}^{\mathrm{Eul},A}_0(t) = 1, \qquad \mathsf{E}^{\mathrm{Eul},A}_n(t) = \sum_{k=0}^{n-1} \left\langle {n \atop k} \right\rangle t^k \quad (n \geq 1).$$

The type tag prevents this row polynomial from colliding with an arbitrary $A_n(t)$.
Type-B descent convention. A signed permutation of size n is a word $\pi_1 \cdots \pi_n$ whose absolute values form a permutation of $\{1, \dots, n\}$. Put $\pi_0 = 0$. A type-B descent is an $i \in \{0, \dots, n-1\}$ for which $\pi_i > \pi_{i+1}$. The number with k such descents is $\mathsf{E}^{\mathrm{Eul},B}_{n,k}$, not the type-A angle-bracket number.
Type-B recurrence and polynomial. With $\mathsf{E}^{\mathrm{Eul},B}_{0,0} = 1$ and zero extension,

$$\mathsf{E}^{\mathrm{Eul},B}_{n,k} = (2k+1)\mathsf{E}^{\mathrm{Eul},B}_{n-1,k} + (2n-2k+1)\mathsf{E}^{\mathrm{Eul},B}_{n-1,k-1}.$$

The row polynomial is

$$\mathsf{E}^{\mathrm{Eul},B}_n(t) := \sum_{k=0}^{n} \mathsf{E}^{\mathrm{Eul},B}_{n,k} t^k,$$

and satisfies

$$\sum_{m \geq 0} (2m+1)^n t^m = \frac{\mathsf{E}^{\mathrm{Eul},B}_n(t)}{(1-t)^{n+1}}.$$

Second-order number. A Stirling permutation of order n is a permutation of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ in which every entry between the two copies of i is larger than i. The number having k ordinary descents is $\left\langle\!\left\langle {n \atop k} \right\rangle\!\right\rangle$. For $n \geq 1$,

$$\left\langle\!\!\left\langle {n \atop k} \right\rangle\!\!\right\rangle = (2n-k-1)\left\langle\!\!\left\langle {n-1 \atop k-1} \right\rangle\!\!\right\rangle + (k+1)\left\langle\!\!\left\langle {n-1 \atop k} \right\rangle\!\!\right\rangle.$$

The degree-zero convention is $\left\langle\!\left\langle {0 \atop 0} \right\rangle\!\right\rangle = 1$.
Second-order row polynomial. Define

$$\mathsf{E}^{\mathrm{Eul},2}_0(t) = 1, \qquad \mathsf{E}^{\mathrm{Eul},2}_n(t) = \sum_{k=0}^{n-1} \left\langle\!\!\left\langle {n \atop k} \right\rangle\!\!\right\rangle t^k \quad (n \geq 1).$$

Then

$$\mathsf{E}^{\mathrm{Eul},2}_{n+1}(t) = (2nt+1)\mathsf{E}^{\mathrm{Eul},2}_n(t) - t(t-1)\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{E}^{\mathrm{Eul},2}_n(t),$$

and $\mathsf{E}^{\mathrm{Eul},2}_n(1) = (2n-1)!!$ for $n \geq 1$.

Collision boundary. Single angle brackets mean type A; double angle brackets mean second order; the explicitly tagged $\mathsf{E}^{\mathrm{Eul},B}$ means type B. None of these symbols is an inner product.

Coefficient extraction, EGF/OGF labels, reversion coefficients, and trace inputs

Coefficient extraction. If $F(z) = \sum_{m\in I} a_m z^m$, where $I = \mathbb{N}_0$ for an ordinary formal series or $I \subseteq \mathbb{Z}$ for a Laurent series, then

$$[z^n]\,F(z) := a_n.$$

The first argument is the series variable and the second is its exponent; the series operand follows the bracket. For a multivariate series, $[z_1^{n_1}\cdots z_d^{n_d}]F$ extracts the indicated multi-index coefficient.
Exponential generating function. For a sequence $A = (a_n)_{n\geq 0}$,

$$A^{\mathrm{egf}}(z) := \sum_{n\geq 0} a_n \frac{z^n}{n!}.$$

The superscript egf is part of the printed symbol and is not a Fourier hat.
Ordinary generating function. For the same sequence,

$$A^{\mathrm{ogf}}(z) := \sum_{n\geq 0} a_n z^n.$$

The EGF and OGF are different transforms even when they are evaluated at the same variable.
Ordinary coefficient ledger. If

$$X(z) = \sum_{j\geq 1} x_j \frac{z^j}{j!} \quad \text{and} \quad X^{\mathrm{ogf}}(z) = \sum_{j\geq 1} x_j^{\mathrm{ogf}} z^j,$$

then

$$x_j^{\mathrm{ogf}} = \frac{x_j}{j!}.$$

The family argument x and the index j are both explicit; this is not a reversion coefficient.
Normalized reversion coefficient. Suppose

$$f(w) = \sum_{j\geq 1} f_j \frac{w^j}{j!}, \qquad f_1 \neq 0, \qquad g(z) = f^{\langle -1\rangle}(z) = \sum_{n\geq 1} g_n \frac{z^n}{n!},$$

where $f(g(z)) = z = g(f(z))$. For $j \geq 1$, the reports use

$$f_j^{\mathrm{rev,norm}} := \frac{f_{j+1}}{(j+1)f_1}$$

as an input to Bell-polynomial inverse formulas. Its superscript rev, norm records both reversion and normalization. It is distinct from an ordinary-series coefficient and from a factorially rescaled transfer coefficient.
Scaled trace input for Bell polynomials. For a square matrix A and $k \geq 1$,

$$s_k^{\mathrm{tr}} := -(k-1)!\,\mathrm{tr}(A^k).$$

This is the k-th scaled trace argument used in determinant/Bell identities. In particular s_2^{tr} is not the binary digit sum $\mathrm{wt}_2(2)$ and not a signed Stirling number.

Finite field. The shared symbol $\mathbb{F}_q$ denotes the finite field with q elements, where $q = p^r$ is a prime power. A bare $\mathbb{F}$ denotes an unspecified finite base field only when its order is immaterial or declared in the same scope.
Small operators. $\operatorname{cyc}(\sigma)$ is the number of cycles of a permutation σ; $\operatorname{Part}(S)$ is the set of set partitions of S; and $a \mathbin{\&} b$ is bitwise AND of the nonnegative binary expansions of integers a, b. These operators are not multiplication, intersection, or a generic trace.

Partition-lattice endpoints and Möbius inversion

Ambient poset. For a finite set S, let $\Pi(S)$ be the set of partitions of S, ordered by refinement: $\pi \le \sigma$ means every block of π is contained in a block of σ. When $S = \{1, \dots, n\}$, write Π_n.
Minimum and maximum. The minimum $\bot_\Pi$ is the discrete partition into singleton blocks; the maximum $\top_\Pi$ is the one-block partition. Their Π subscripts are semantic and distinguish them from numeric $0, 1$, Boolean values, or endpoints of an unrelated poset.
Möbius function. For any locally finite poset, the Möbius function is determined by

$$\mu(x, x) = 1, \qquad \sum_{x \le z \le y} \mu(x, z) = 0 \quad (x < y).$$

For the partition lattice,

$$\mu_{\Pi_n}(\pi, \top_\Pi) = (-1)^{|\pi|-1}(|\pi| - 1)!,$$

where $|\pi|$ is the number of blocks, and

$$\mu_{\Pi_n}(\bot_\Pi, \top_\Pi) = (-1)^{n-1}(n-1)! \quad (n \ge 1).$$

Moment–cumulant use. For random variables indexed by S, a moment factors over blocks of π, and Möbius inversion on $\Pi(S)$ recovers the corresponding joint cumulant. The lattice endpoints are combinatorial objects; no numerical substitution for their glyphs is valid.

Roman Stirling extensions, restricted partitions, Fubini numbers, Narayana numbers, and generalized factorial coefficients

Why the qualifiers are structural. Each family in this entry changes either the index domain, the admissible blocks, the order carried by blocks, or the coefficient basis. Its decoration is therefore mathematical data. None is an unqualified Stirling number, Bell number, Catalan number, or Gaussian binomial coefficient.
Roman Stirling extension. The ordinary combinatorial triangles in the Stirling-and-Lah entry have nonnegative indices. A report that explicitly announces the *Roman extension* instead continues the triangular recurrences to integer indices and imposes the dualities

$$\begin{bmatrix} n \\ k \end{bmatrix} = \begin{Bmatrix} -k \\ -n \end{Bmatrix}, \qquad \begin{Bmatrix} n \\ k \end{Bmatrix} = \begin{bmatrix} -k \\ -n \end{bmatrix}.$$

These are defining equations for a reciprocal integer-indexed array. When an index is negative, neither side retains its permutation-cycle or set-partition counting interpretation. In the mixed quadrant $n \ge 1$, $k \ge 0$, the reports use

$$\begin{bmatrix} -n \\ k \end{bmatrix} = \frac{1}{n!} \sum_{j=1}^{n} (-1)^{n-j} \binom{n}{j} \frac{1}{j^k}.$$

The phrase *Roman extension* must occur in the same scope; negative arguments do not silently change the default domain of the shared Stirling commands.

Negative-index Bell values. Only after declaring that Roman extension, the reports define, for $k \in \mathbb{N}_{>0}$,

$$\mathsf{B}^{\mathrm{Bell}}_{-k} := \sum_{n \geq 1} \begin{bmatrix} -n \\ k \end{bmatrix} = \mathrm{e}^{-1} \sum_{j \geq 1} \frac{1}{j^k j!}.$$

Both series converge absolutely. This analytically continued sequence is not the cardinality of partitions of a negative-size set, and it is not obtained by substituting a negative integer into the Dobinski formula without the displayed definition.

r-Stirling numbers of the second kind. For fixed $r \in \mathbb{N}_0$,

$$\begin{Bmatrix} n \\ k \end{Bmatrix}_r$$

counts partitions of $\{1, \dots, n\}$ into k nonempty blocks such that the distinguished labels $1, \dots, r$ lie in pairwise distinct blocks. For $n, k \in \mathbb{N}_0$, its nonzero counting domain is $n \geq r$ and $r \leq k \leq n$; its boundary value is $\left\{ {r \atop r} \right\}_r = 1$; and it is zero outside that domain, including when $n < r$, $k < r$, or $k > n$. For $n > r$,

$$\begin{Bmatrix} n \\ k \end{Bmatrix}_r = k \begin{Bmatrix} n-1 \\ k \end{Bmatrix}_r + \begin{Bmatrix} n-1 \\ k-1 \end{Bmatrix}_r.$$

The mixed bivariate EGF, including its shift by r, is

$$\sum_{n \geq r} \sum_{k \geq r} \begin{Bmatrix} n \\ k \end{Bmatrix}_r y^k \frac{z^{n-r}}{(n-r)!} = y^r \mathrm{e}^{rz} \exp\bigl(y(\mathrm{e}^z - 1)\bigr).$$

Associated Stirling numbers. For $r \in \mathbb{N}_{>0}$, $\mathsf{S}^{\geq r}(n, k)$ counts partitions of an n-element labelled set into k blocks, every block having size at least r. Set $\mathsf{S}^{\geq r}(0, 0) = 1$; it is zero if $n < rk$, if $k < 0$, if $n < 0$, or if $k = 0 < n$. Thus the empty set has one zero-block partition, whereas a nonempty set has none. For $n \geq 0$,

$$\mathsf{S}^{\geq r}(n+1, k) = k\mathsf{S}^{\geq r}(n, k) + \binom{n}{r-1} \mathsf{S}^{\geq r}(n-r+1, k-1),$$

where the zero extension interprets every out-of-domain term. Equivalently,

$$\sum_{n \geq 0} \mathsf{S}^{\geq r}(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(\mathrm{e}^z - \sum_{j=0}^{r-1} \frac{z^j}{j!} \right)^k.$$

The historical source glyph $\mathsf{S}_r(n, k)$ is read with this minimum-block-size meaning only inside its declared section.

Reduced or gap Stirling numbers. For $d \in \mathbb{N}_{>0}$, $\mathsf{S}^{[\mathrm{gap}\, d]}(n, k)$ counts partitions of $\{1, \dots, n\}$ into k blocks in which two labels in the same block differ by at least d. In the source range $n \geq k \geq d$,

$$\mathsf{S}^{[\mathrm{gap}\, d]}(n, k) = \begin{Bmatrix} n-d+1 \\ k-d+1 \end{Bmatrix}.$$

The $d = 1$ specialization is the ordinary second-kind number. Outside the stated range the combinatorial definition, not the displayed shifted formula, governs; no additional boundary values are inferred.

Fubini or ordered Bell numbers. The visibly qualified number $\mathsf{F}^{\mathrm{Fub}}_n$ counts linearly ordered lists of the blocks of a set partition. For $n \in \mathbb{N}_0$,

$$\mathsf{F}^{\mathrm{Fub}}_n = \sum_{k=0}^{n} k! \begin{Bmatrix} n \\ k \end{Bmatrix}, \qquad \sum_{n \geq 0} \mathsf{F}^{\mathrm{Fub}}_n \frac{z^n}{n!} = \frac{1}{2 - \mathrm{e}^z},$$

and

$$\mathsf{F}_0^{\mathrm{Fub}} = 1, \qquad \mathsf{F}_n^{\mathrm{Fub}} = \sum_{j=1}^{n} \binom{n}{j} \mathsf{F}_{n-j}^{\mathrm{Fub}} \quad (n \geq 1).$$

It is not the unlabelled Bell number $\mathsf{B}_n^{\mathrm{Bell}}$: ordering the blocks is the extra structure.
Narayana numbers. For $n \in \mathbb{N}_{>0}$ and $1 \leq k \leq n$, the Narayana number is

$$\mathsf{N}_{n,k}^{\mathrm{Nar}} := \frac{1}{n} \binom{n}{k} \binom{n}{k-1}.$$

It counts Dyck paths of semilength n with exactly k peaks and satisfies

$$\sum_{k=1}^{n} \mathsf{N}_{n,k}^{\mathrm{Nar}} = \mathsf{C}_n^{\mathrm{Cat}}, \qquad \mathsf{N}_{n,k}^{\mathrm{Nar}} = \mathsf{N}_{n,n+1-k}^{\mathrm{Nar}}.$$

No value $\mathsf{N}_{0,0}^{\mathrm{Nar}}$ is introduced by this convention; an empty Dyck path, when needed, is represented separately by the constant term of the relevant generating function.
Generalized factorial coefficients. Let $f : \mathbb{N}_0 \to \mathbb{C}$, let $x, t \in \mathbb{C}$ with $t \neq 0$, and let $n \geq 1$ be an integer. The report-local product

$$(x)_{n;f,t} := \prod_{j=1}^{n-1} \left(x + \frac{f(j)}{t^j} \right)$$

has degree $n-1$, with the $n=1$ empty product equal to 1. Its coefficient triangle is

$$\begin{bmatrix} n \\ k \end{bmatrix}_{f,t} := \left[x^{k-1} \right] (x)_{n;f,t}, \qquad 1 \leq k \leq n.$$

The four data n, k, f, t are all part of the object. These brackets are not unsigned Stirling brackets, Gaussian binomials, or a q-Pochhammer symbol, and no $n = 0$ value is inferred without a separate declaration.

Catalan, complementary Bell, Gregory, Genocchi, Barnes, Hessenberg, associahedral, and shifted-coordinate symbols

Catalan number. For $n \in \mathbb{N}_0$,

$$\mathsf{C}_n^{\mathrm{Cat}} := \frac{1}{n+1} \binom{2n}{n}.$$

It counts, among many equinumerous families, full plane binary trees with n internal vertices and triangulations of a convex $(n+2)$-gon. It is not the complementary Bell number, even though one source used C_n for both.
Gregory coefficient. The Gregory coefficients are defined by the ordinary Taylor expansion

$$\frac{z}{\log(1+z)} = 1 + \sum_{n \geq 1} \mathsf{G}_n^{\mathrm{Greg}} z^n \qquad (|z| < 1 \text{ analytically}).$$

The equality also defines them formally over a characteristic-zero coefficient field.
Genocchi polynomial and number. The Genocchi polynomials satisfy

$$\frac{2t\mathrm{e}^{xt}}{\mathrm{e}^t + 1} = \sum_{n \geq 0} \mathsf{G}_n^{\mathrm{Gen}}(x) \frac{t^n}{n!},$$

and

$$\mathsf{G}_n^{\mathrm{Gen}} := \mathsf{G}_n^{\mathrm{Gen}}(0).$$

They are unrelated to the Gregory coefficients despite the legacy shared letter G_n.

Barnes G-function. The Barnes function is denoted $\mathrm{G}_{\mathrm{Barnes}}(z)$ and normalized by

$$\mathrm{G}_{\mathrm{Barnes}}(1) = 1, \qquad \mathrm{G}_{\mathrm{Barnes}}(z+1) = \Gamma(z)\mathrm{G}_{\mathrm{Barnes}}(z).$$

For $N \in \mathbb{N}_{>0}$, $\mathrm{G}_{\mathrm{Barnes}}(N+1) = \prod_{k=1}^{N-1} k!$. It is a function, not either G_n coefficient family.
Bell Hessenberg matrices. For $1 \le i, j \le n$, define

$$(\mathsf{H}_n^{\mathrm{Bell}})_{ij} = \begin{cases} \binom{n-i}{j-i} x_{j-i+1}, & j \ge i, \\ -1, & j = i-1, \\ 0, & j < i-1, \end{cases}$$

and

$$(\mathsf{K}_n^{\mathrm{Bell}})_{ij} = \begin{cases} x_{j-i+1}/(j-i)!, & j \ge i, \\ -(i-1), & j = i-1, \\ 0, & j < i-1. \end{cases}$$

Then

$$\det \mathsf{H}_n^{\mathrm{Bell}} = \det \mathsf{K}_n^{\mathrm{Bell}} = \mathcal{B}_n^{\exp}(x_1, \dots, x_n).$$

The labels H and K describe two determinant models for the same Bell polynomial; neither matrix is an associahedron.
Associahedron indexed by dimension. The symbol $\mathsf{K}^{\mathrm{Assoc}}_{\dim=d}$ denotes the d-dimensional Stasheff associahedron. Its vertices correspond to triangulations of a convex $(d+3)$-gon, equivalently to full parenthesizations of $d+2$ factors. The lossless legacy-to-canonical crosswalk is: K_n of declared dimension $n-1$ becomes $\mathsf{K}^{\mathrm{Assoc}}_{\dim=n-1}$; K_n of declared dimension $n-2$ becomes $\mathsf{K}^{\mathrm{Assoc}}_{\dim=n-2}$; the corresponding shifted K_{n+1} becomes $\mathsf{K}^{\mathrm{Assoc}}_{\dim=n-1}$; and $\mathcal{K}_N$ for triangulations of an N-gon becomes $\mathsf{K}^{\mathrm{Assoc}}_{\dim=N-3}$. The unified formulae-and-inversion report is internally mixed: one section declares K_n to have dimension $n-1$, while a later section declares K_{n+1} to have dimension $n-1$, which is the incompatible rule $\dim K_m = m-2$. No source index may be copied without first applying its declared dimension formula.
Lagrange shift coordinate. In a shifted Lagrange-inversion calculation, the local coordinate

$$\zeta_{\mathrm{Lag}} := z - b$$

is a translated variable. It is not the Riemann zeta function $\zeta(s)$, a Hurwitz zeta value $\zeta(s,a)$, a root of unity, an incidence-algebra kernel, or a generic complex coordinate outside that declared scope.
Zeta-family boundary. Five roles must remain separate: $\zeta(s)$ is the one-argument Riemann zeta function; $\zeta(s,a)$ is the two-argument Hurwitz zeta function; $\zeta_{\mathrm{Lag}} = z - b$ is a local translated coordinate; $\zeta_P(x,y) = 1$ for $x \le y$ is the incidence-algebra zeta function of a locally finite poset P, whose convolution inverse is the Möbius function; and a root-of-unity variable satisfies $\zeta^N = 1$ for a declared positive integer N. Boolean-subset zeta transforms are the specialization of the poset construction to the subset lattice. Their shared historical glyph does not identify the objects.
Governance rule. A raw C_n, G_n, K_n, $L(n,k)$, $T_n(x)$, or local ζ is insufficient wherever two listed meanings coexist. The semantic commands in this entry are normative and their printed tags are intentionally visible.

9 q-series, products, and arithmetic functions

The q parameter, formal series, and analytic regimes

Polynomial and formal regimes. The letter q may be an indeterminate. Then $R[q]$ is the polynomial ring and

$$R[[q]] = \left\{ \sum_{n=0}^{\infty} a_n q^n : a_n \in R \right\}$$

is the ring of formal power series over a commutative ring R. The infinity sign in a formal series records an unbounded coefficient sequence, not analytic convergence.
Equality and Cauchy product. Formal equality is coefficientwise: $\sum_{n\geq 0} a_n q^n = \sum_{n\geq 0} b_n q^n$ exactly when $a_n = b_n$ for every n. Addition is coefficientwise and multiplication is

$$\left(\sum_{n\geq 0} a_n q^n\right)\left(\sum_{n\geq 0} b_n q^n\right) = \sum_{n\geq 0}\left(\sum_{j=0}^{n} a_j b_{n-j}\right) q^n.$$

The inner sum is finite, so no convergence hypothesis is hidden in this definition.
Formal Laurent series. The ring $R((q))$ consists of $\sum_{n\geq n_0} a_n q^n$ with $n_0 \in \mathbb{Z}$ and only finitely many negative powers. It is distinct from a bilateral analytic Laurent series with infinitely many terms in both directions.
Terminating rational regime. A terminating identity may be interpreted in the rational-function field $K(q)$, or after clearing explicitly named denominators in $K[q]$. Termination removes infinite-series convergence questions but does not remove poles caused by a zero denominator.
Analytic regime. For infinite products and unilateral or bilateral series, an explicit common regime is $q \in \mathbb{C}$ with $0 < |q| < 1$; products permitting $q = 0$ say so separately. Probability models normally strengthen this to $q \in (0, 1)$. Every convergence disk or annulus also includes the remaining parameters.
Modular parameter and nome. For theta and eta functions, $\tau \in \mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ and the nome is

$$q_\tau = \mathrm{e}^{\pi \mathrm{i} \tau}.$$

Thus $|q_\tau| < 1$. Some sources use $\mathrm{e}^{2\pi\mathrm{i}\tau} = q_\tau^2$; a formula must identify which nome is in force. Under $\tau \mapsto -1/\tau$, the transformed nome is $\widetilde{q}_\tau = \exp(-\pi\mathrm{i}/\tau)$.
Roots of unity. At $q = \zeta$ with $\zeta^N = 1$, denominators in rational q-expressions can vanish. Evaluation means polynomial evaluation after cancellation only when a polynomial representative has been proved; otherwise the singular value is excluded.
Base-role rule. If a document also uses q for a denominator, quantile, frequency, conjugate exponent, or finite-field cardinality, the series base is q_{base} and the field cardinality is Q. Semantic subscripts take priority over traditional brevity.

Finite, infinite, integer-index, and complex-order q-Pochhammer symbols

Finite definition. For a, q in a commutative ring and $n \in \mathbb{N}_0$,

$$(a;q)_n := \prod_{j=0}^{n-1} (1 - aq^j).$$

The product is empty at $n = 0$, so $(a;q)_0 = 1$ for every a, q, including values that make later factors vanish.

Several parameters. For a finite ordered list $\boldsymbol{a} = (a_1, \dots, a_r)$,

$$(a_1, \dots, a_r; q)_n := \prod_{i=1}^{r} (a_i; q)_n .$$

For $r = 0$ this is the empty product 1. The punctuation separates a parameter list from the base; it does not make the a_i into additional bases.
Infinite definition. For $a, q \in \mathbb{C}$ and $|q| < 1$,

$$(a; q)_\infty := \prod_{j=0}^{\infty} (1 - aq^j),$$

with analytic meaning as a locally uniformly convergent limit of finite products. The literal index ∞ denotes an infinite product, not an extended-natural argument to the finite definition.
Positive and negative integer indices. For $m \in \mathbb{N}_{>0}$,

$$(a; q)_{-m} := \frac{1}{(aq^{-m}; q)_m} = \frac{(-q/a)^m q^{\binom{m}{2}}}{(q/a; q)_m},$$

where all displayed denominators are nonzero. Together with the finite definition this gives $(a; q)_{m+n} = (a; q)_m \, (aq^m; q)_n$ for integer indices whenever all three factors are defined. For $n \in \mathbb{N}_0$, $(a; q)_n = (a; q)_n$; the two command names record whether the governing domain is finite-natural or generalized.
Complex order. Fix $q \in \mathbb{C} \setminus \{0\}$, $|q| < 1$, and a logarithm value L_q satisfying $\exp(L_q) = q$. Put $q^\nu = \exp(\nu L_q)$. For $\nu \in \mathbb{C}$,

$$(a; q)_\nu := \frac{(a; q)_\infty}{(aq^\nu; q)_\infty}$$

where the denominator is nonzero. Changing the branch can change q^ν, hence the value. For $\nu \in \mathbb{Z}$ the definition agrees with the integer-index convention on their common domain.
Complex concatenation and unit shift. With the logarithm branch fixed as above, and whenever every displayed factor is defined,

$$(a; q)_{\alpha+\beta} = (a; q)_\alpha \, (aq^\alpha; q)_\beta .$$

Taking $\beta = 1$ gives

$$(a; q)_{\alpha+1} = (1 - aq^\alpha) \, (a; q)_\alpha .$$

Both identities use the same chosen value of $\log q$; changing branches in only one factor would change the statement.
The base $q = 0$. For finite indices, $(a; 0)_0 = 1$ and $(a; 0)_n = 1 - a$ for $n \geq 1$, using $0^0 = 1$ in the first factor and $0^j = 0$ for $j > 0$. Negative and complex orders at $q = 0$ are not inferred from formulas containing q^{-m} or a logarithm of q.
Factorization. For $m, n \in \mathbb{N}_0$,

$$(a; q)_{m+n} = (a; q)_m \, (aq^m; q)_n .$$

This is a finite product identity over a commutative ring and needs no division.
Retired signatures. Former two-argument macros hid the base and are retired. The shared `\QPochhammer` always receives (a, q, n) in that order and is used for a finite-natural length or the literal infinite-product index. The shared `\GeneralizedQPochhammer` has the same argument order and visible glyph, but its third argument is a declared integer or complex order governed by the preceding branch and denominator conditions.

q-integers and q-factorials

Definition. For $n \in \mathbb{N}_0$,

$$[n]_q := \begin{cases} (1-q^n)/(1-q), & q \neq 1, \\ n, & q = 1. \end{cases}$$

Equivalently $[n]_q = 1 + q + \cdots + q^{n-1}$ for $n > 0$, and $[0]_q = 0$.
q-factorial. $[n]_q! := \prod_{j=1}^{n} [j]_q$, with $[0]_q! = 1$. The exclamation mark belongs to the notation and does not apply an ordinary factorial to the polynomial $[n]_q$.
Limit. $\lim_{q\to 1} [n]_q = n$ and $\lim_{q\to 1} [n]_q! = n!$ for fixed n.
Alternative symmetric integer. The symmetric quantity $(q^n - q^{-n})/(q - q^{-1})$ is written $[n]_{q,\mathrm{sym}}$ if needed; it is not the canonical `\QInteger`.

Generalized q-numbers and their inverse

Branch data. Fix $q \in \mathbb{C} \setminus \{0, 1\}$ and a named logarithm value L_q satisfying $\exp(L_q) = q$. For $z \in \mathbb{C}$, define $q^z = \exp(zL_q)$. If $q \in (0, 1)$, the default is the real logarithm $L_q = \log q < 0$.
Definition. The shared command `\GeneralizedQNumber{z}{q}` prints

$$[z]_q := \frac{1-q^z}{1-q}.$$

At $q = 1$, the separately assigned continuous value is $[z]_1 = z$. For $z = n \in \mathbb{N}_0$, this agrees with $[n]_q$.
The excluded base $q = 0$. For an arbitrary complex exponent z, the branch formula requires $q \neq 0$, and this catalogue assigns no canonical value to 0^z or to $[z]_0$. If $z = n \in \mathbb{N}_0$, the finite polynomial convention belongs instead to `\QInteger`: $[0]_0 = 0$ and $[n]_0 = 1$ for $n \geq 1$.
Zeros and branch dependence. For $q \neq 1$, $[z]_q = 0$ exactly when $\exp(zL_q) = 1$. In the real regime $0 < q < 1$ and $z \in \mathbb{R}$, this means $z = 0$. For complex z, changing L_q can change both the function and its zero lattice.
Inverse on a chosen branch. If $w = [z]_q$, then

$$q^z = 1 - (1-q)w, \qquad z = \frac{\log_*(1-(1-q)w)}{L_q},$$

where $\log_*$ is a specifically chosen local branch and its domain excludes zero and the branch cut. The formula produces one inverse branch, not a global single-valued inverse on $\mathbb{C}$.
Classical limit. For fixed z and a compatible logarithm branch near $q = 1$, $[z]_q \to z$. A uniform limit over unbounded z-sets is not implicit.

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q-falling powers and q-Newton bases

Definition. For scalars x, a, q in a commutative ring and $m \in \mathbb{N}_0$,

$$(x-a)_m^{\langle q \rangle} := \prod_{j=0}^{m-1} (x - q^j a), \qquad (x-a)_0^{\langle q \rangle} = 1.$$

The monograph-local command is `\QFallingPower{x}{a}{q}{m}`. All four inputs are semantic; in particular the base is never inferred from a surrounding chapter.
Relation to q-Pochhammer. When x is invertible,

$$(x-a)_m^{\langle q \rangle} = x^m (a/x; q)_m.$$

The product definition remains valid without invertibility and is therefore primary.

Jackson derivative. With the derivative D_q from the `q-difference` entry,

$$D_q(x-a)_m^{\langle q\rangle} = [m]_q(x-a)_{m-1}^{\langle q\rangle}$$

for $m \ge 1$. Iteration supplies $D_q^j(x-a)_m^{\langle q\rangle} = [m]_q!/[m-j]_q!\,(x-a)_{m-j}^{\langle q\rangle}$ only for $0 \le j \le m$ and where the quotient notation has a valid interpretation.
Basis role. For fixed a, q, these are monic polynomials of distinct degrees. Thus the first $N+1$ form a basis for polynomials of degree at most N. The phrase q-Newton basis refers to this declared family, not to an unspecified ordinary falling-factorial basis.

Gaussian binomial coefficients

Definition. For $n, k \in \mathbb{N}_0$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \begin{cases} \dfrac{(q;q)_n}{(q;q)_k\ (q;q)_{n-k}}, & 0 \le k \le n, \\ 0, & k > n, \end{cases}$$

interpreted as its polynomial in q, so removable singularities are filled.
Boundary values. $\begin{bmatrix} n \\ 0 \end{bmatrix}_q = \begin{bmatrix} n \\ n \end{bmatrix}_q = 1$, including $n = 0$.
Recurrence. For $1 \le k \le n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k}\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

Retired signatures. The corpus formerly used incompatible argument orders for its abbreviated Gaussian-binomial macro. The canonical `\GaussianBinomial` receives upper index, lower index, then base.
Limit. $\begin{bmatrix} n \\ k \end{bmatrix}_1 = \binom{n}{k}$, understood by polynomial evaluation, not substitution into the displayed rational quotient.

MacMahon's q-Catalan polynomial

Document-local command and type. The q-Pochhammer monograph's local `\MacMahonQCatalanPolynomial{n}{q}` has TeX arity two and prints $C_n^{\mathrm{Mac}}(q)$. The first argument is $n \in \mathbb{N}_0$. The second argument names the displayed polynomial indeterminate—normally q, or X when matching the Lean polynomial variable—rather than supplying a denominator at which pointwise division is to be performed. Intrinsically, the family is $C_n^{\mathrm{Mac}}(X) \in \mathbb{Z}[X]$; scalar values are obtained afterward by polynomial evaluation.
Exact polynomial quotient. It is the unique polynomial satisfying

$$[n+1]_q\ C_n^{\mathrm{Mac}}(q) = \begin{bmatrix} 2n \\ n \end{bmatrix}_q, \qquad C_n^{\mathrm{Mac}}(q) = \frac{\begin{bmatrix} 2n \\ n \end{bmatrix}_q}{[n+1]_q} \quad \text{in } \mathbb{Z}[q].$$

The second display is exact polynomial division: cyclotomic divisibility first proves that $[n+1]_q$ divides the central Gaussian polynomial, and the monicity of the divisor gives the quotient in $\mathbb{Z}[q]$. It is not the pointwise quotient of two scalar-valued functions. In particular, at a zero of $[n+1]_q$, its value is the evaluation of the already constructed quotient polynomial—the removable value after cancellation—and not an undefined $0/0$.

Degree and boundary value. For every $n \in \mathbb{N}_0$,

$$\deg C_n^{\mathrm{Mac}} = n(n-1), \qquad C_0^{\mathrm{Mac}}(q) = 1.$$

The degree formula uses natural-number subtraction at $n = 0$. The explicit boundary value confirms that the polynomial is the nonzero constant 1, so its degree is 0 rather than an exceptional zero-polynomial degree.
Classical specialization. Evaluation of the quotient polynomial at $q = 1$ gives

$$C_n^{\mathrm{Mac}}(1) = \mathsf{C}_n^{\mathrm{Cat}} = \frac{1}{n+1}\binom{2n}{n}.$$

This is polynomial evaluation after cancellation, since the defining exact identity specializes to $(n+1)C_n^{\mathrm{Mac}}(1) = \binom{2n}{n}$.
Coefficient signs. The coefficients are nonnegative integers, by the separate classical combinatorial interpretation using an appropriate major-index statistic on Dyck paths of semilength n. Integrality from cyclotomic divisibility alone does not prove nonnegativity. The current Lean module proves the exact integral quotient, degree, and specialization at one, but does not prove coefficientwise nonnegativity; that last assertion therefore has classical/cited, not Lean-proved, status in this catalogue.
Collision policy. The ordinary Catalan number remains $\mathsf{C}_n^{\mathrm{Cat}}$. The bare notation $C_n(q)$ is not canonical: the literature contains inequivalent q-Catalan families, and the corpus also assigns C_n to unrelated coefficient sequences. The semantic superscript and the full local command identify specifically MacMahon's quotient convention. This command is local to the monograph and is intentionally absent from the shared quick-reference and alphabetical command indexes.
Lean crosswalk. In `QCatalan.lean`, `qCatalan` n is the polynomial in $\mathbb{Z}[X]$; `qInt_X_dvd_gaussianBinomial_int` proves the required divisibility, `qInt_X_mul_qCatalan` is the exact product identity, `qCatalan_natDegree` proves degree $n(n-1)$, and `qCatalan_eval_one` identifies evaluation at one with Mathlib's `catalan` n. These named results have status **Lean-proved**; none of them states coefficientwise nonnegativity.

Generalized Gaussian binomial coefficients

Finite-product/rational definition. Fix a nonzero complex base q, a logarithm value L_q, and $q^\alpha = \exp(\alpha L_q)$. For $\alpha \in \mathbb{C}$ and $k \in \mathbb{N}_0$, define

$$\begin{bmatrix}\alpha\\k\end{bmatrix}_q := \frac{\left(q^{\alpha-k+1};q\right)_k}{(q;q)_k},$$

where the denominator is nonzero. The shared `\GeneralizedGaussianBinomial` receives the upper parameter, lower parameter, and base, in that order. For an integral upper parameter this is a rational identity in q; for a nonintegral upper parameter it also depends on the chosen logarithm branch.
Integer specialization. When $\alpha = n \in \mathbb{N}_0$ and $k \in \mathbb{N}_0$, the generalized expression equals $\left[\begin{smallmatrix}n\\k\end{smallmatrix}\right]_q$ for every k, including $k > n$, where both sides are the zero polynomial. At a base where the displayed rational quotient is formally $0/0$, this is polynomial evaluation after cancellation, not literal division. The out-of-range zero convention is not silently extended to an arbitrary complex upper parameter.
Two complex arguments. For the two-complex-argument definition, fix $0 < q < 1$ and use the real logarithm, so $q^z = \exp(z \log q)$. For $\alpha, \beta \in \mathbb{C}$, define

$$\begin{bmatrix}\alpha\\\beta\end{bmatrix}_q := \frac{\left(q^{\beta+1};q\right)_\infty\left(q^{\alpha-\beta+1};q\right)_\infty}{(q;q)_\infty\left(q^{\alpha+1};q\right)_\infty},$$

whenever the denominator is nonzero. Equivalently, using the catalogue's q-gamma convention,

$$\begin{bmatrix}\alpha\\ \beta\end{bmatrix}_q = \frac{\Gamma_q(\alpha+1)}{\Gamma_q(\beta+1)\Gamma_q(\alpha-\beta+1)}$$

wherever that meromorphic quotient is defined. This is not a polynomial in q, and it has no automatic out-of-range zero rule. When $\beta = k \in \mathbb{N}_0$, tail cancellation recovers the preceding finite-product definition.

No implicit regime change. Polynomial identities, rational identities, branch-dependent complex formulas, and meromorphic continuations are four distinct claims. A source moving between them states the domain and continuation theorem rather than reusing the bracket alone.

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q-multinomials, inversions, and lattice path area

Definition and domain. For $r \in \mathbb{N}_0$, $n, n_1, \ldots, n_r \in \mathbb{N}_0$, and $n_1 + \cdots + n_r = n$,

$$\begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_q := \frac{(q;q)_n}{(q;q)_{n_1} \cdots (q;q)_{n_r}},$$

interpreted as its polynomial in q. For $r = 0$, the constraint forces $n = 0$ and the value is 1; for $r = 1$, the constraint forces $n_1 = n$ and the value is again 1. The quotient represents an element of $\mathbb{Z}[q]$, so evaluation at a root of unity or any other base means polynomial evaluation after cancellation, not literal division by possibly vanishing factors. If an input is negative or the sum constraint fails, the expression is undefined unless that source separately declares a zero extension.

Factorization. For a valid composition,

$$\begin{bmatrix} n \\ n_1, \ldots, n_r \end{bmatrix}_q = \begin{bmatrix} n \\ n_1 \end{bmatrix}_q \begin{bmatrix} n - n_1 \\ n_2, \ldots, n_r \end{bmatrix}_q.$$

The shared `\QMultinomial` receives n, the complete lower list, and q; the list length and sum are not macro-inferred data.

Word inversion statistic. Let $W(n_1, \ldots, n_r)$ be the words on the ordered alphabet $1 < \cdots < r$ in which i occurs n_i times. For $w = w_1 \cdots w_n$,

$$\mathrm{inv}(w) = \left|\{(i,j) : 1 \le i < j \le n,\ w_i > w_j\}\right|.$$

Then the q-multinomial is $\sum_{w \in W(n_1,\ldots,n_r)} q^{\mathrm{inv}(w)}$.

Binary lattice-path convention. For a binary word, 0 is an east step and 1 is a north step. Its area is any of the equal quantities

$$\sum_{i:w_i=0} |\{j < i : w_j = 1\}| = \sum_{j:w_j=1} |\{i > j : w_i = 0\}| = \mathrm{inv}(w),$$

the number of unit cells below the path in its m-by-k rectangle. Hence $\mathrm{area}(0^m 1^k) = 0$ and $\mathrm{area}(1^k 0^m) = mk$. A complementary-area convention must say so and replaces q by q^{-1} with the corresponding normalizing power.

Finite fields, general linear groups, subspaces, and flags

Prime-power parameter. Here $Q = p^e$, where p is prime and $e \in \mathbb{N}_{>0}$. The symbol $\mathbb{F}_Q$ is a field with exactly Q elements, unique up to noncanonical field isomorphism. Upper-case Q prevents a collision with the q-series base.

Vector spaces and zero dimension. For $n \in \mathbb{N}_0$, $\mathbb{F}_Q^n$ is the n-dimensional coordinate vector space. $\mathbb{F}_Q^0 = \{0\}$ is the one-element zero vector space, not the empty set.

General linear group. For $r \in \mathbb{N}_0$,

$$\mathrm{GL}_r(\mathbb{F}_Q) = \{A \in \mathrm{Mat}_{r\times r}(\mathbb{F}_Q) : \det A \neq 0\}.$$

Its cardinality is $\prod_{j=0}^{r-1}(Q^r - Q^j)$; at $r = 0$, the empty product is 1 and $\mathrm{GL}_0(\mathbb{F}_Q)$ contains the unique empty identity matrix.

Subspace count. For $0 \le k \le n$,

$$\left|\{U \le \mathbb{F}_Q^n : \dim_{\mathbb{F}_Q} U = k\}\right| = \begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

The bracket is evaluated at the prime power Q; it is not a second definition of the field.

Flags. For $n_1 + \cdots + n_r = n$, the Q-multinomial counts flags

$$0 = V_0 \subseteq V_1 \subseteq \cdots \subseteq V_r = \mathbb{F}_Q^n, \qquad \dim(V_i/V_{i-1}) = n_i.$$

All inclusions and quotient dimensions are part of the object being counted.

Weight character. If a finite-dimensional representation V has an integral-weight decomposition $V = \bigoplus_{m\in\mathbb{Z}} V_m$, its monograph-local character is

$$\mathrm{ch}_x(V) := \sum_{m\in\mathbb{Z}} (\dim V_m) x^m.$$

The sum is finite, repeated weights contribute multiplicity, x is a formal variable, and $\mathrm{ch}_x(0) = 0$. The local command `\WeightCharacter` receives the representation and the formal variable.

Möbius function of a finite subspace lattice

Poset definition. For a finite-dimensional $\mathbb{F}_Q$-space V, let $\mathcal{L}(V)$ be its poset of linear subspaces ordered by inclusion. Its Möbius function is defined on comparable pairs by

$$\mu_{\mathcal{L}(V)}(U,U) = 1, \qquad \sum_{U\subseteq W\subseteq Z} \mu_{\mathcal{L}(V)}(U,W) = 0 \quad (U \subsetneq Z).$$

The ambient space and both interval endpoints are mandatory inputs.

Closed form. If $U \subseteq Z$ and $d = \dim_{\mathbb{F}_Q}(Z/U)$, then

$$\mu_{\mathcal{L}(V)}(U,Z) = (-1)^d Q^{\binom{d}{2}}.$$

For incomparable U, Z, the incidence-algebra extension is 0; alternatively the two-argument poset Möbius function is simply left undefined off comparable pairs. The chosen convention must be stated when off-interval inputs can occur.

Semantic command. The monograph-local `\SubspaceMobius{V}{U}{Z}` prints the ambient lattice and both endpoints. A bare μ is reserved neither for this function nor for the arithmetic Möbius function because the corpus also uses μ for measures and means.

Unilateral and bilateral basic hypergeometric series

Unilateral definition. The unilateral basic hypergeometric series is

$$ {}_r\phi_s\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z\right) := \sum_{n=0}^{\infty} \frac{\prod_{j=1}^{r} (a_j; q)_n}{(q;q)_n \prod_{j=1}^{s} (b_j; q)_n} \left((-1)^n q^{\binom{n}{2}}\right)^{1+s-r} z^n, $$

provided no denominator factor vanishes. A terminating instance is a finite rational identity. A nonterminating analytic instance additionally states a convergence domain, normally with $0 < |q| < 1$.

Normalization warning. References differ on the $((-1)^n q^{\binom{n}{2}})^{1+s-r}$ factor. Every occurrence in this corpus either expands the definition above or cites this entry.

Termination. If an upper parameter is q^{-N} with $N \in \mathbb{N}_0$, the series terminates at $n = N$ unless an earlier denominator singularity makes the expression undefined.

Bilateral definition. With integer-index q-Pochhammers already defined,

$$ {}_r\psi_s\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z\right) := \sum_{n\in\mathbb{Z}} \frac{\prod_{j=1}^{r} (a_j; q)_n}{\prod_{j=1}^{s} (b_j; q)_n} \left((-1)^n q^{\binom{n}{2}}\right)^{s-r} z^n. $$

This is never identified with the unilateral symbol. Every analytic use supplies an annulus of convergence and excludes denominator zeros.

Balanced series. A terminating ${}_{r+1}\phi_r(a_1, \dots, a_{r+1}; b_1, \dots, b_r; q, q)$ is called balanced when

$$ b_1 \cdots b_r = q\, a_1 \cdots a_{r+1}. $$

If $a_{r+1} = q^{-N}$, the condition is equivalently $b_1 \cdots b_r = q^{1-N} a_1 \cdots a_r$. Both the argument $z = q$ and termination index N are part of the label.

Well-poised and very-well-poised. The series ${}_r\phi_{r-1}(a_1, \dots, a_r; b_1, \dots, b_{r-1}; q, z)$ is well-poised when

$$ qa_1 = a_2 b_1 = a_3 b_2 = \cdots = a_r b_{r-1}. $$

It is very-well-poised when, in addition, a chosen square root satisfies $a_2 = q\sqrt{a_1}$ and $a_3 = -q\sqrt{a_1}$. The paired signs make the resulting parameter pair independent of replacing the square root by its negative, but a derivation that separates the two parameters still names the choice.

Ramanujan's ${}_1\psi_1$ summation. For $0 < |q| < 1$ and $|b/a| < |z| < 1$, with all denominator products nonzero,

$$ {}_1\psi_1\left(\begin{matrix} a \\ b \end{matrix}; q, z\right) = \frac{(q, b/a, az, q/(az); q)_\infty}{(b, q/a, z, b/(az); q)_\infty}. $$

The annulus is essential: it controls the two tails of the bilateral series and cannot be replaced by the unilateral condition $|z| < 1$ alone.

---

q-difference operators

Jackson derivative. For $q \neq 1$ and $x \neq 0$,

$$ D_q f(x) := \frac{f(x) - f(qx)}{(1-q)x}. $$

At $x = 0$, $D_q f(0)$ is assigned $f'(0)$ only when that derivative exists and the local definition explicitly makes the continuous extension.

Shift operator. $T_q f(x) = f(qx)$. Thus $D_q = (1 - T_q)/((1 - q)x)$ as an identity on functions where division by x is valid.
Base direction. The expression $(f(qx) - f(x))/((q - 1)x)$ is algebraically identical. The genuinely different operator $(f(qx) - f(x))/((q - 1)qx)$ is labelled separately.
Classical limit. If f is differentiable at x, then $D_q f(x) \to f'(x)$ as $q \to 1$ through real positive values. Uniformity in x is never inferred without a theorem.

Jackson q-integrals and differentials

Integral from zero. For $0 < q < 1$, a scalar endpoint a, and a function for which the series converges,

$$\int_0^a f(t)\,\mathrm{d}_q t := a(1 - q) \sum_{n=0}^{\infty} q^n f(aq^n).$$

For $a = 0$ the value is 0. The monograph-local `\JacksonDifferential{q}{t}` prints $\mathrm{d}_q t$, so neither the base nor integration variable is implicit.
General oriented interval. Define

$$\int_a^b f(t)\,\mathrm{d}_q t := \int_0^b f(t)\,\mathrm{d}_q t - \int_0^a f(t)\,\mathrm{d}_q t.$$

This is an oriented difference, not a measure integral over an empty set when $b < a$. For complex endpoints it samples two geometric rays and requires convergence on both.
Fundamental theorem. If $G(aq^n) \to G(0)$ and the telescoping series is legitimate, then

$$\int_0^a D_q G(t)\,\mathrm{d}_q t = G(a) - G(0).$$

The limit hypothesis is part of the statement; the isolated formula for D_q does not create continuity at zero.
Integration by parts. Under absolute convergence sufficient to rearrange the sampled series,

$$\int_0^a f(t) D_q g(t)\,\mathrm{d}_q t = f(a)g(a) - f(0)g(0) - \int_0^a g(qt) D_q f(t)\,\mathrm{d}_q t.$$

The shifted factor $g(qt)$ is essential and distinguishes Jackson integration from the ordinary Leibniz rule.
Power integral. For $a > 0$, $s \in \mathbb{C}$, and a fixed branch for a^{s+1}, the geometric series gives

$$\int_0^a t^s\,\mathrm{d}_q t = \frac{a^{s+1}}{[s+1]_q}$$

when $|q^{s+1}| < 1$. In the real regime $0 < q < 1$, this includes $\operatorname{Re} s > -1$.

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q-gamma, q-digamma, and q-beta functions

$0 < q < 1$ gamma convention. For $0 < q < 1$, define the meromorphic function

$$\Gamma_q(z) := (1 - q)^{1-z} \frac{(q; q)_\infty}{(q^z; q)_\infty}, \qquad q^z = \exp(z \log q).$$

The power of $1 - q > 0$ uses the real logarithm. For $z > 0$ it is finite and positive, $\Gamma_q(1) = 1$, and $\Gamma_q(z + 1) = [z]_q \Gamma_q(z)$.

Poles and zeros. In the real-base convention, the denominator vanishes when $q^{z+j} = 1$ for some $j \in \mathbb{N}_0$. Thus the meromorphic pole lattice is

$$z = -j + \frac{2\pi \mathrm{i} k}{\log q}, \qquad j \in \mathbb{N}_0,\ k \in \mathbb{Z},$$

subject to cancellation only if proved. The function has no zeros because its numerator is nonzero.
$q > 1$ *reciprocal-base convention.* For real $q > 1$, the separately declared convention is

$$\Gamma_q(z) = (q-1)^{1-z} q^{z(z-1)/2} \frac{(q^{-1}; q^{-1})_\infty}{(q^{-z}; q^{-1})_\infty},$$

with real logarithms for positive bases. It is not obtained by inserting $q > 1$ into the divergent $0 < q < 1$ product.
q*-digamma.* At points where Γ_q is finite and nonzero,

$$\psi_q(z) := \frac{\mathrm{d}}{\mathrm{d}z} \log \Gamma_q(z) = \frac{\Gamma_q'(z)}{\Gamma_q(z)}.$$

The logarithmic derivative is single-valued even though a local logarithm of Γ_q was used to motivate it. This ψ_q is not a characteristic function or a wave function.
q*-beta.* For $x, y > 0$ and $0 < q < 1$,

$$B_q(x,y) := \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)} = \int_0^1 t^{x-1} \frac{(qt;q)_\infty}{(q^y t; q)_\infty} \, \mathrm{d}_q t.$$

The integral formula includes the Jackson differential and real powers determined by $t \in (0,1]$; a complex continuation uses the gamma quotient, not an undeclared pointwise integral branch.
Lean crosswalk. In `QBetaIntegral.lean`, the total real-valued definition is `qBeta` $q\,x\,y$. Under the explicit hypotheses $0 < q < 1$ and $x, y > 0$, `qBeta_eq_prod` proves the infinite-product evaluation, `qBeta_eq_qGamma` proves the displayed gamma quotient, `qBeta_comm` and `qBeta_pos` prove symmetry and positivity, and `qBeta_add_one_left` and `qBeta_add_one_right` prove the two positive-domain recurrences. The definition is total because Lean's real operations and `jacksonIntegral` are total; the named evaluation theorems do not silently extend beyond their positivity hypotheses. Status: **Lean-proved with the displayed hypotheses**. This module does not formalize the meromorphic continuation or the classical $q \uparrow 1$ limits.
Classical limits. For fixed z away from poles, $\Gamma_q(z) \to \Gamma(z)$, $\psi_q(z) \to \psi(z)$, and $B_q(x,y) \to B(x,y)$ as $q \uparrow 1$, on compact subsets where the corresponding uniform convergence theorem applies.

Euler q-exponentials and factorial-scaled variants

Pochhammer-scaled pair. For $0 < |q| < 1$,

$$e_q(z) := \sum_{n=0}^{\infty} \frac{z^n}{(q;q)_n} = \frac{1}{(z;q)_\infty} \quad (|z| < 1),$$

and

$$E_q(z) := \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} z^n}{(q;q)_n} = (-z;q)_\infty \quad (z \in \mathbb{C}).$$

The lower-case series has poles after meromorphic continuation; the upper-case product is entire in z.

q-factorial-scaled pair. Since $[n]_q! = (q;q)_n/(1-q)^n$, define

$$\mathrm{e}_q^J(z) = \sum_{n=0}^{\infty} \frac{z^n}{[n]_q!} = \frac{1}{((1-q)z;q)_\infty},$$

$$\mathrm{E}_q^J(z) = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} z^n}{[n]_q!} = (-(1-q)z;q)_\infty.$$

The superscript J labels the Jackson-factorial scaling; it is not a power.
Reciprocity. On the common domain,

$$e_q(z)E_q(-z) = 1, \qquad \mathrm{e}_q^J(z)\mathrm{E}_q^J(-z) = 1.$$

Changing between the two pairs requires the explicit substitution $z \mapsto (1-q)z$; their identical spoken name does not make their arguments interchangeable.
Classical normalization. For fixed z, $\mathrm{e}_q^J(z) \to \mathrm{e}^z$ and $\mathrm{E}_q^J(z) \to \mathrm{e}^z$ as $q \uparrow 1$. The unscaled e_q and E_q require an argument rescaling to have that limit.

Multiplicative and bilateral Jacobi theta normalizations

Multiplicative product. For $0 < |q| < 1$ and $z \in \mathbb{C}^\times$,

$$\vartheta_{\mathrm{mult}}(z;q) := (z;q)_\infty (q/z;q)_\infty (q;q)_\infty.$$

The monograph-local `\MultiplicativeTheta` receives z and q; its semantic subscript distinguishes this function from the additive Jacobi functions $\vartheta_2, \vartheta_3, \vartheta_4$.
Plus-sign bilateral series. On the same domain,

$$\Theta_{\mathrm{bil}}(z;q) := \sum_{n\in\mathbb{Z}} q^{\binom{n}{2}} z^n.$$

The series converges normally on compact subsets of $\mathbb{C}^\times$. Its local command `\BilateralTheta` receives argument and base.
Exact normalization crosswalk. Jacobi's triple product gives

$$\vartheta_{\mathrm{mult}}(z;q) = \Theta_{\mathrm{bil}}(-z;q).$$

The argument negation is mandatory: suppressing it changes every odd Laurent coefficient.
Quasi-periodicity and zeros. One has

$$\Theta_{\mathrm{bil}}(qz;q) = z^{-1}\Theta_{\mathrm{bil}}(z;q), \qquad \vartheta_{\mathrm{mult}}(qz;q) = -z^{-1}\vartheta_{\mathrm{mult}}(z;q).$$

The bilateral zeros are the simple points $z = -q^m$, while the multiplicative-product zeros are the simple points $z = q^m$, $m \in \mathbb{Z}$.
Inverse near a zero. At $z = q^m$, $\partial_z \vartheta_{\mathrm{mult}}(q^m;q) \neq 0$, so the local inverse at target $y = 0$ begins

$$z = q^m + \frac{y}{\partial_z \vartheta_{\mathrm{mult}}(q^m;q)} + \mathcal{O}(y^2).$$

This is one local analytic branch around a specified zero, not a global inverse of a quasi-periodic function.

Jacobi theta functions, products, heat equation, and modular inversion

Parameters and prescribed lifts. Let $\tau \in \mathbb{H}$, $z \in \mathbb{C}$,

$$q = \mathrm{e}^{\pi \mathrm{i}\tau}, \qquad \zeta = \mathrm{e}^{2\pi \mathrm{i}z}.$$

Every rational power is the prescribed lift $q^{\alpha} = \mathrm{e}^{\pi \mathrm{i}\alpha\tau}$, and $\zeta^{1/2}$ means $\mathrm{e}^{\pi \mathrm{i}z}$. Neither is obtained from an unspecified logarithm of its displayed complex value.
Series definitions. The three normalizations used in the monograph are

$$\begin{aligned}
\vartheta_3(z \mid \tau) &= \sum_{n\in\mathbb{Z}} \mathrm{e}^{\pi \mathrm{i}n^2\tau + 2\pi \mathrm{i}nz} = \sum_{n\in\mathbb{Z}} q^{n^2}\zeta^{n},\\
\vartheta_4(z \mid \tau) &= \sum_{n\in\mathbb{Z}} (-1)^n \mathrm{e}^{\pi \mathrm{i}n^2\tau + 2\pi \mathrm{i}nz},\\
\vartheta_2(z \mid \tau) &= \sum_{n\in\mathbb{Z}} \mathrm{e}^{\pi \mathrm{i}(n+1/2)^2\tau + 2\pi \mathrm{i}(n+1/2)z}.
\end{aligned}$$

The series and all termwise derivatives converge normally on compact subsets of $\mathbb{C} \times \mathbb{H}$.
Product forms. With the prescribed q, ζ,

$$\begin{aligned}
\vartheta_3(z \mid \tau) &= (q^2, -q\zeta, -q/\zeta; q^2)_\infty,\\
\vartheta_4(z \mid \tau) &= (q^2, q\zeta, q/\zeta; q^2)_\infty,\\
\vartheta_2(z \mid \tau) &= q^{1/4}\zeta^{1/2}(q^2, -q^2\zeta, -\zeta^{-1}; q^2)_\infty.
\end{aligned}$$

These formulas fix the nome, the argument exponential, and the quarter- and half-power lifts simultaneously.
Heat equation. For $j \in \{2, 3, 4\}$,

$$\frac{\partial \vartheta_j}{\partial \tau} = \frac{1}{4\pi \mathrm{i}} \frac{\partial^2 \vartheta_j}{\partial z^2}.$$

The factor $4\pi\mathrm{i}$ belongs to the convention $\exp(2\pi \mathrm{i}nz)$; changing the argument scale changes the equation.
Square-root branch. Because $-\mathrm{i}\tau$ lies in the right half-plane, $(-\mathrm{i}\tau)^{1/2}$ is its principal holomorphic square root, normalized by $(-\mathrm{i}\cdot \mathrm{i}t)^{1/2} = t^{1/2}$ for $t > 0$.
Modular inversion. With that square root,

$$\vartheta_3\left(\frac{z}{\tau}\,\middle|\,-\frac{1}{\tau}\right) = (-\mathrm{i}\tau)^{1/2}\mathrm{e}^{\pi \mathrm{i}z^2/\tau}\vartheta_3(z \mid \tau).$$

At $z = 0$, modular inversion sends $(\vartheta_2, \vartheta_3, \vartheta_4)$ to $(\vartheta_4, \vartheta_3, \vartheta_2)$, multiplied componentwise by the same $(-\mathrm{i}\tau)^{1/2}$.
Quadratic-series crosswalk. With $\Theta_{\mathrm{quad}}(x; Q)$ defined below,

$$\begin{aligned}
\vartheta_3(z \mid \tau) &= \Theta_{\mathrm{quad}}(\zeta; q) = \Theta_{\mathrm{bil}}(q\zeta; q^2),\\
\vartheta_4(z \mid \tau) &= \Theta_{\mathrm{quad}}(-\zeta; q) = \vartheta_{\mathrm{mult}}(q\zeta; q^2),\\
\vartheta_2(z \mid \tau) &= q^{1/4}\zeta^{1/2}\Theta_{\mathrm{bil}}(q^2\zeta; q^2).
\end{aligned}$$

Dedekind eta function

Definition. For $\tau \in \mathbb{H}$ and $q = \mathrm{e}^{\pi \mathrm{i} \tau}$,

$$\eta(\tau) := \mathrm{e}^{\pi \mathrm{i} \tau/12} \prod_{n=1}^{\infty} (1 - \mathrm{e}^{2\pi \mathrm{i} n \tau}) = q^{1/12}(q^2; q^2)_\infty.$$

The twelfth power is the prescribed exponential $\mathrm{e}^{\pi \mathrm{i} \tau/12}$, not a free root choice.
Nonvanishing. The product converges and none of its factors vanishes on $\mathbb{H}$; hence η is holomorphic and nonzero there.
Theta product. At zero argument,

$$\vartheta_2(0 \mid \tau)\vartheta_3(0 \mid \tau)\vartheta_4(0 \mid \tau) = 2\eta(\tau)^3.$$

Modular transformations. With the same principal square root as for the Jacobi functions,

$$\eta(\tau + 1) = \mathrm{e}^{\pi \mathrm{i}/12}\eta(\tau), \qquad \eta(-1/\tau) = (-\mathrm{i}\tau)^{1/2}\eta(\tau).$$

The multiplier in the translation formula is part of the normalization.

Ramanujan's general theta function and bilateral quadratic theta

Ramanujan theta. For $a, b \in \mathbb{C}$ with $|ab| < 1$,

$$\Theta_{\mathrm{Ram}}(a, b) := \sum_{n \in \mathbb{Z}} a^{n(n+1)/2} b^{n(n-1)/2}.$$

The exponents are integers. Zeroth powers equal 1, including at a zero base. The paired, division-free form is

$$\Theta_{\mathrm{Ram}}(a, b) = 1 + \sum_{n=1}^{\infty} (a^n + b^n)(ab)^{n(n-1)/2}.$$

Product and degenerate base. Ramanujan's product is

$$\Theta_{\mathrm{Ram}}(a, b) = (-a, -b, ab; ab)_\infty.$$

At base zero, $(c; 0)_\infty := 1 - c$, so if $ab = 0$, both sides equal $1 + a + b$. The monograph-local `\RamanujanTheta` receives a, b; it never hides the product base ab.
Bilateral crosswalk. When $a, b \neq 0$,

$$\Theta_{\mathrm{Ram}}(a, b) = \Theta_{\mathrm{bil}}(a; ab).$$

The paired definition supplies the continuous algebraic interpretation when one of the two parameters is zero.
Quadratic theta. For $0 < |Q| < 1$ and $x \in \mathbb{C}^\times$,

$$\Theta_{\mathrm{quad}}(x; Q) := \sum_{n \in \mathbb{Z}} Q^{n^2} x^n = \Theta_{\mathrm{bil}}(Qx; Q^2).$$

The local `\BilateralQuadraticTheta` receives x, Q. Upper-case Q here is a locally typed theta base, not automatically a finite-field cardinality.
Schröter lattice dissection. For $a, b \in \mathbb{N}_{>0}$, $0 < |q| < 1$, and $x, y \in \mathbb{C}^\times$,

$$\begin{aligned}\Theta_{\mathrm{quad}}(x; q^a)\Theta_{\mathrm{quad}}(y; q^b) = \sum_{k=0}^{a+b-1} y^k q^{bk^2} \Theta_{\mathrm{quad}}(xyq^{2bk}; q^{a+b}) \\ \times \Theta_{\mathrm{quad}}(y^a x^{-b} q^{2abk}; q^{ab(a+b)}).\end{aligned}$$

The finite range has exactly $a + b$ residue classes. The identity comes from the bijection $n = r - bs$, $m = k + r + as$ on the class $m - n \equiv k \pmod{a + b}$; absolute convergence justifies rearrangement.

Theta coefficients, sums of squares, and Gaussian integers

Representation count. For $d, n \in \mathbb{N}_0$,

$$r_d(n) \coloneqq \left|\{(x_1, \dots, x_d) \in \mathbb{Z}^d : x_1^2 + \dots + x_d^2 = n\}\right|.$$

For $d = 0$, the unique empty tuple gives $r_0(0) = 1$ and $r_0(n) = 0$ for $n > 0$.
Universal generating function. Coefficientwise in $\mathbb{Z}[[q]]$, and analytically for $|q| < 1$,

$$\left(\sum_{m\in\mathbb{Z}} q^{m^2}\right)^d = \sum_{n=0}^{\infty} r_d(n)q^n.$$

A divisor-sum formula for a particular d requires additional arithmetic or modular input; it does not follow merely from this Cauchy product.
Character modulo four. For $m \in \mathbb{Z}$,

$$\chi_4(m) = \begin{cases} 0, & 2 \mid m, \\ 1, & m \equiv 1 \pmod 4, \\ -1, & m \equiv 3 \pmod 4. \end{cases}$$

It is completely multiplicative and periodic modulo 4. The subscript distinguishes it from a generic characteristic function.
Two- and four-square formulas. For $n \in \mathbb{N}_{>0}$,

$$r_2(n) = 4\sum_{d|n} \chi_4(d), \qquad r_4(n) = 8\sum_{\substack{d|n \\ 4\nmid d}} d.$$

Separately, $r_2(0) = r_4(0) = 1$; the positive-divisor sums above are not evaluated at zero.
Gaussian integers. The ring $\mathbb{Z}[\mathrm{i}] = \{x + \mathrm{i}y : x, y \in \mathbb{Z}\}$ has norm

$$N(x + \mathrm{i}y) = x^2 + y^2, \qquad N(\alpha\beta) = N(\alpha)N(\beta).$$

It is Euclidean for N, hence a unique-factorization domain, and its units are $\{\pm 1, \pm\mathrm{i}\}$.
Rational-prime splitting. One has $2 = -\mathrm{i}(1+\mathrm{i})^2$. A rational prime $p \equiv 3 \pmod 4$ remains irreducible, while a prime $p \equiv 1 \pmod 4$ factors as $p = \varpi\overline{\varpi}$, where the Gaussian prime ϖ is unique up to a unit and conjugation. The letter ϖ, not π, prevents collision with the circle constant in theta and Fourier formulas.

Cyclotomic polynomials and roots of unity

Primitive root. $\zeta_N = \mathrm{e}^{2\pi\mathrm{i}/N}$ for $N \in \mathbb{N}_{>0}$ is the chosen positively oriented primitive Nth root.
Cyclotomic polynomial. $\Phi_N(x) \in \mathbb{Z}[x]$ is the monic polynomial whose roots are ζ_N^k for $1 \le k \le N$ and $\gcd(k, N) = 1$.
Factorization. $x^N - 1 = \prod_{d|N} \Phi_d(x)$, where d ranges over the positive divisors of N.
Order of a root. For $\zeta^N = 1$, $\operatorname{ord}(\zeta) = \min\{d \in \mathbb{N}_{>0} : \zeta^d = 1\}$. This group-theoretic order is distinct from the analytic $\operatorname{ord}_{z_0}(f)$ of a zero.
Evaluation policy. A rational q-series expression is not evaluated at a root of unity until all denominator zeros and possible cancellations have been audited.

Cyclotomic valuation. For $d \in \mathbb{N}_{>0}$ and a nonzero polynomial $F \in \mathbb{Z}[q]$,

$$\nu_{d,q}^{\mathrm{cyc}}(F) := \max\{e \in \mathbb{N}_0 : \Phi_d(q)^e \text{ divides } F\}.$$

It is the exponent of the polynomial $\Phi_d(q)$, not a d-adic valuation of an integer. It is left undefined at $F = 0$, rather than being silently assigned $+\infty$.
Semantic command. The monograph-local `\CyclotomicValuation{d}{q}{F}` prints the cyclotomic index, polynomial variable, and polynomial. The variable input is necessary when several polynomial rings are present.
Shifted-factorial factorization. For $n \in \mathbb{N}_0$,

$$(q;q)_n = (-1)^n \prod_{d=1}^{n} \Phi_d(q)^{\lfloor n/d \rfloor}.$$

Consequently cyclotomic divisibility of ratios of shifted factorials is checked by subtracting these explicit nonnegative exponents after polynomiality has been established.
Lean crosswalk: carry and divisibility. In `CyclotomicDivisibility.lean`, `cyclotomic_exponent_eq_one_iff` proves, for $k \le n$ and $0 < d$, that the Gaussian cyclotomic exponent is one exactly when $n \bmod d < k \bmod d$. The theorem `cyclotomic_dvd_gaussianBinomial_iff` proves the corresponding divisibility criterion for Φ_d and the universal Gaussian polynomial over $\mathbb{Q}[X]$. These theorems formalize the displayed exponent and carry criterion, but they do not define the document-local three-argument `\CyclotomicValuation` as a general Lean function. Status: **Lean-proved with the stated natural-number hypotheses**.
Lean crosswalk: primitive-root blocks and q-Lucas. For a primitive dth root ζ in an integral domain, `PrimitiveRootBlock.lean` proves the interior vanishing theorem `gaussianBinomial_isPrimitiveRoot_eq_` and the complete block identity `finiteQPochhammerIn_isPrimitiveRoot`, including the hypothesis $0 < d$:

$$\begin{bmatrix} d \\ k \end{bmatrix}_\zeta = 0 \quad (0 < k < d), \qquad \prod_{j=0}^{d-1}(1 - y\zeta^j) = 1 - y^d \quad (y \text{ in the coefficient ring}).$$

In `QLucas.lean`, `gaussianBinomial_q_lucas` proves, for $a, r, b, s \in \mathbb{N}_0$ with $b, s < d$,

$$\begin{bmatrix} ad + b \\ rd + s \end{bmatrix}_\zeta = \binom{a}{r} \begin{bmatrix} b \\ s \end{bmatrix}_\zeta.$$

Finally, `gaussianBinomial_mul_isPrimitiveRoot` in `CyclotomicDivisibility.lean` specializes this to $\left[\begin{smallmatrix} an \\ bn \end{smallmatrix}\right]_\zeta = \binom{a}{b}$ for a primitive nth root and $0 < n$. All three modules work over the algebraic hypotheses written in their declarations; they do not by themselves authorize evaluation of an uncancelled rational q-series quotient at a root of unity. Status: **Lean-proved with the displayed hypotheses**.

Coefficientwise convergence, q-adic order, and formal congruence

Coefficientwise convergence. For formal series $F_N(q) = \sum_{j \ge j_0} a_{N,j} q^j$ and $F(q) = \sum_{j \ge j_0} a_j q^j$, the notation $F_N \to F$ coefficientwise means: for every fixed j, there exists N_j such that $a_{N,j} = a_j$ for all $N \ge N_j$. No norm, pointwise evaluation, or uniform analytic convergence follows from this definition.
q-adic order. For a nonzero formal Laurent series, the q-adic order is the already defined $\mathrm{ord}_{q=0}(F)$, its least exponent with nonzero coefficient. A sequence tends q-adically to zero when these orders tend to $+\infty$.

Formal congruence. For $M \in \mathbb{Z}$ and $N \in \mathbb{N}_0$,

$$F(q) \equiv G(q) \pmod{q^N}$$

means $F - G \in q^N R[[q]]$, equivalently equality of the coefficients of $q^0, \ldots, q^{N-1}$. A coefficientwise congruence modulo M instead means $a_j \equiv b_j \pmod{M}$ for every j; the two moduli must never be conflated.
I-adic generalization. If a commutative ring A is complete and separated for an ideal I and $q \in I$, infinite products may converge coefficientwise I-adically in $A[[z]]$. This is an algebraic topology. It does not assert convergence after embedding A into $\mathbb{C}$.

Bailey pairs, transforms, lowering, and chains

Bailey pair. Fix an explicit base q. Two sequences $(\alpha_n)_{n\geq 0}$ and $(\beta_n)_{n\geq 0}$ form a Bailey pair relative to a when

$$\beta_n = \sum_{r=0}^{n} \frac{\alpha_r}{(q;q)_{n-r}(aq;q)_{n+r}} \qquad (n \in \mathbb{N}_0),$$

with all denominators nonzero. The base is part of the governing declaration even though traditional terminology names only the relative parameter a.
Full transform. Let $\rho_1\rho_2 \neq 0$, put $A = aq/(\rho_1\rho_2)$, and assume $(aq;q)_m$, $(aq/\rho_1;q)_m$, and $(aq/\rho_2;q)_m$ are nonzero for $m \geq 1$. Define

$$\begin{aligned}\alpha_n^{\mathrm{BT}} &= \frac{(\rho_1,\rho_2;q)_n A^n}{(aq/\rho_1, aq/\rho_2;q)_n}\alpha_n, \\ \beta_n^{\mathrm{BT}} &= \sum_{j=0}^{n} \frac{(\rho_1,\rho_2;q)_j (A;q)_{n-j} A^j}{(q;q)_{n-j}(aq/\rho_1, aq/\rho_2;q)_n}\beta_j.\end{aligned}$$

Then the transformed pair is again relative to a. The superscript BT is a semantic label, not exponentiation.
Parameter lowering. Assume $(a;q)_m \neq 0$ for $m \geq 1$. Put

$$\alpha_0^{\mathrm{BL}} = \alpha_0, \qquad \alpha_n^{\mathrm{BL}} = (1-a)\left(\frac{\alpha_n}{1-aq^{2n}} - \frac{aq^{2n-2}\alpha_{n-1}}{1-aq^{2n-2}}\right) \quad (n \geq 1),$$

and $\beta_n^{\mathrm{BL}} = \beta_n$. With the displayed denominators nonzero, this is a Bailey pair relative to a/q. The label BL means Bailey lowering.
Iterated chain. For $t \in \mathbb{N}_0$, the depth-zero pair is the original pair. For $t \geq 1$,

$$\alpha_n^{\mathrm{BC}(t)} = a^{tn} q^{tn^2}\alpha_n,$$

$$\beta_n^{\mathrm{BC}(t)} = \sum_{n \geq r_1 \geq \cdots \geq r_t \geq 0} \frac{a^{r_1+\cdots+r_t} q^{r_1^2+\cdots+r_t^2}\beta_{r_t}}{(q;q)_{n-r_1}\prod_{\ell=1}^{t-1}(q;q)_{r_\ell - r_{\ell+1}}}.$$

The finite weakly decreasing index simplex, the chain depth t, and the relative parameter are never inferred from α or β alone.
Local semantic commands. The monograph uses superscripts BT, BL, and BC(t) through local commands so a transformed sequence cannot be confused with the input Bailey pair or with unrelated sequences called α, β.

Rogers–Ramanujan series, continuants, and the Andrews–Gordon family

Single Rogers family. For $m \in \mathbb{N}_0$ and $|q| < 1$,

$$F_m^{\mathrm{RR}}(q) := \sum_{n=0}^{\infty} \frac{q^{n^2+mn}}{(q;q)_n}.$$

Thus $F_0^{\mathrm{RR}} = G$ and $F_1^{\mathrm{RR}} = H$ in the traditional notation. The semantic superscript prevents a collision with Fabius functions and generic generating functions.
Recurrence and products. Absolute convergence permits

$$F_m^{\mathrm{RR}}(q) = F_{m+1}^{\mathrm{RR}}(q) + q^{m+1} F_{m+2}^{\mathrm{RR}}(q).$$

The two Rogers–Ramanujan identities are

$$F_0^{\mathrm{RR}}(q) = \frac{1}{(q, q^4; q^5)_\infty}, \qquad F_1^{\mathrm{RR}}(q) = \frac{1}{(q^2, q^3; q^5)_\infty}.$$

Finite continuant. For $N \in \mathbb{N}_0$,

$$C_N^{\mathrm{cont}} = 1 + \cfrac{q}{1 + \cfrac{q^2}{\ddots + \cfrac{q^N}{1}}} = \frac{P_N^{\mathrm{cont}}}{Q_N^{\mathrm{cont}}},$$

where

$$P_{-1}^{\mathrm{cont}} = P_0^{\mathrm{cont}} = 1, \qquad Q_{-1}^{\mathrm{cont}} = 0, \quad Q_0^{\mathrm{cont}} = 1,$$
$$P_N^{\mathrm{cont}} = P_{N-1}^{\mathrm{cont}} + q^N P_{N-2}^{\mathrm{cont}}, \qquad Q_N^{\mathrm{cont}} = Q_{N-1}^{\mathrm{cont}} + q^N Q_{N-2}^{\mathrm{cont}}.$$

The rational continuant defines the truncation even where a nested intermediate denominator vanishes but the final rational function has a removable singularity.
Gaussian finitizations. The exact polynomials are

$$P_N^{\mathrm{cont}} = \sum_{j=0}^{\lfloor (N+1)/2 \rfloor} q^{j^2} \begin{bmatrix} N+1-j \\ j \end{bmatrix}_q,$$

$$Q_N^{\mathrm{cont}} = \sum_{j=0}^{\lfloor N/2 \rfloor} q^{j^2+j} \begin{bmatrix} N-j \\ j \end{bmatrix}_q.$$

Locally uniformly for $|q| < 1$, they tend respectively to $F_0^{\mathrm{RR}}(q)$ and $F_1^{\mathrm{RR}}(q)$.
Continued fraction and fifth-root branch. The single-valued ratio satisfies

$$\frac{F_1^{\mathrm{RR}}(q)}{F_0^{\mathrm{RR}}(q)} = \cfrac{1}{1 + \cfrac{q}{1 + \cfrac{q^2}{1 + \cfrac{q^3}{1 + \ddots}}}}.$$

On a simply connected subdomain of $0 < |q| < 1$, choose one branch of $q^{1/5}$ and define

$$R_{\mathrm{RR}}(q) := q^{1/5} \frac{F_1^{\mathrm{RR}}(q)}{F_0^{\mathrm{RR}}(q)} = q^{1/5} \frac{(q, q^4; q^5)_\infty}{(q^2, q^3; q^5)_\infty}.$$

Only the prefactor is branch-dependent; the ratio and continued fraction are single-valued.

Andrews–Gordon indexing. Let $k \geq 2$, $1 \leq i \leq k$, and $n_1, \ldots, n_{k-1} \in \mathbb{N}_0$. Put $N_j = n_j + \cdots + n_{k-1}$. Then

$$\sum_{n_1,\ldots,n_{k-1}\geq 0} \frac{q^{N_1^2+\cdots+N_{k-1}^2+N_i+\cdots+N_{k-1}}}{(q;q)_{n_1}\cdots(q;q)_{n_{k-1}}} = \frac{(q^i, q^{2k+1-i}, q^{2k+1}; q^{2k+1})_\infty}{(q;q)_\infty}.$$

When $i = k$, the linear tail $N_i + \cdots + N_{k-1}$ is the empty sum 0. The modulus is $2k+1$; an even-modulus identity is not obtained by a nonintegral substitution for k.
Recorded-equality marker. The monograph-local relation $\overset{\text{record}}{=}$ marks an identity recorded with provenance but not proved at that location. It has the same asserted mathematical equality only when the accompanying status and hypotheses support it; the marker itself is never a proof.

Borwein product dissections, reciprocity, and two-ended stabilization

Product. For $n \in \mathbb{N}_{>0}$,

$$P_n^{\mathrm{Bor}}(q) := \frac{(q;q)_{3n}}{(q^3;q^3)_n} = \prod_{j=1}^{n}(1-q^{3j-2})(1-q^{3j-1}).$$

Its degree is $3n^2$. The superscript Bor distinguishes this product from generic polynomials and continuant numerators.
Residue-three dissection. There are unique $A_n^{\mathrm{Bor}}, B_n^{\mathrm{Bor}}, C_n^{\mathrm{Bor}} \in \mathbb{Z}[x]$ such that

$$P_n^{\mathrm{Bor}}(q) = A_n^{\mathrm{Bor}}(q^3) - qB_n^{\mathrm{Bor}}(q^3) - q^2C_n^{\mathrm{Bor}}(q^3).$$

The two minus signs are part of the normalization and do not assert a sign for any coefficient.
Reciprocity. For $n \geq 1$,

$$\begin{aligned} A_n^{\mathrm{Bor}}(x) &= x^{n^2}A_n^{\mathrm{Bor}}(x^{-1}), \\ B_n^{\mathrm{Bor}}(x) &= x^{n^2-1}C_n^{\mathrm{Bor}}(x^{-1}), \\ C_n^{\mathrm{Bor}}(x) &= x^{n^2-1}B_n^{\mathrm{Bor}}(x^{-1}). \end{aligned}$$

These are Laurent-polynomial identities. In particular, $B_n^{\mathrm{Bor}}(1) = C_n^{\mathrm{Bor}}(1)$ and $A_n^{\mathrm{Bor}}(1) = 2B_n^{\mathrm{Bor}}(1)$.
Coefficient names. For $r \in \mathbb{N}_0$,

$$a_{n,r}^{\mathrm{Bor}} = [x^r]A_n^{\mathrm{Bor}}(x), \quad b_{n,r}^{\mathrm{Bor}} = [x^r]B_n^{\mathrm{Bor}}(x), \quad c_{n,r}^{\mathrm{Bor}} = [x^r]C_n^{\mathrm{Bor}}(x).$$

Outside the polynomial degree, coefficient extraction gives 0.
Exact stabilization. For fixed $d \in \mathbb{N}_0$ and $n \geq d+1$,

$$a_{n,d}^{\mathrm{Bor}} = a_{d+1,d}^{\mathrm{Bor}}, \quad b_{n,d}^{\mathrm{Bor}} = b_{d+1,d}^{\mathrm{Bor}}, \quad c_{n,d}^{\mathrm{Bor}} = c_{d+1,d}^{\mathrm{Bor}},$$

and reciprocity gives

$$a_{n,n^2-d}^{\mathrm{Bor}} = a_{d+1,d}^{\mathrm{Bor}}, \quad b_{n,n^2-1-d}^{\mathrm{Bor}} = c_{d+1,d}^{\mathrm{Bor}}, \quad c_{n,n^2-1-d}^{\mathrm{Bor}} = b_{d+1,d}^{\mathrm{Bor}}.$$

These identities state eventual coefficient constancy at both ends; they make no coefficientwise positivity assertion.
Semantic commands. The local `\BorweinProduct` names the product. `\BorweinComponentA`, `\BorweinComponentB`, and `\BorweinComponentC` name the three residue components. Their corresponding coefficient commands display the family, level, and coefficient index.

Divisibility, gcd, lcm, and standard arithmetic functions

Divisibility. $a \mid b$ for $a, b \in \mathbb{Z}$ means $b = ac$ for some $c \in \mathbb{Z}$. In a sum $d \mid n$, the divisor d is positive unless otherwise stated.
GCD and LCM. $\gcd(a, b) \in \mathbb{N}_0$ is the nonnegative greatest common divisor. The zero-argument command `\LeastCommonMultiple` supplies the operator token in $\operatorname{lcm}(a_1, \ldots, a_r)$; its parenthesized integer inputs are source arguments, not TeX macro arguments. Its value is the unique nonnegative generator of the ideal $a_1\mathbb{Z} \cap \cdots \cap a_r\mathbb{Z}$ and is therefore insensitive to signs.
Zero and family conventions. For $r = 0$ the empty intersection is $\mathbb{Z}$. Explicitly, $\operatorname{lcm}(a, b) = \operatorname{lcm}(|a|, |b|)$, $\gcd(0, 0) = 0$, $\operatorname{lcm}(0, b) = 0$ for every b, a singleton family has LCM $|a|$, and the empty family has LCM 1. The empty-family value is the unit for adjoining further inputs and is required in the dyadic-denominator formulas.
Möbius function. $\mu_{\mathrm{M}}(n) = 0$ if a prime square divides n, and $(-1)^r$ if n is a product of r distinct primes; its domain is $\mathbb{N}_{>0}$ and $\mu_{\mathrm{M}}(1) = 1$. The subscript prevents a collision with measures and means.
Euler totient. $\varphi_{\mathrm{E}}(n) = |\{1 \le k \le n : \gcd(k, n) = 1\}|$, $n \in \mathbb{N}_{>0}$. The subscript separates it from characteristic functions.
Divisor sums. $\sigma_\alpha(n) = \sum_{d|n} d^\alpha$ for $n \in \mathbb{N}_{>0}$ and $\alpha \in \mathbb{C}$, with positive divisors and $d^\alpha = \exp(\alpha \log d)$.

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10 Asymptotics, logarithms, Lambert W, and saddle analysis

Limits and named asymptotic regimes

Limit annotation. A limit variable and destination are part of every asymptotic assertion: $n \to \infty$, $x \to 0^+$, $z \to z_0$ in a sector, or $q \uparrow 1$ are different regimes. A subscript may abbreviate a regime only after it has been declared.
One-sided limits. $x \to a^+$ means $x > a$ and $x \to a$; $x \to a^-$ means $x < a$. For $q \uparrow 1$, q is real and approaches 1 from below.
Uniformity. An estimate uniform for $y \in Y$ means its threshold and implied constant may be chosen independently of y. Pointwise asymptotics never imply this.
Complex sectors. A sector is printed as $S(\theta_0, \alpha, r) = \{z : 0 < |z| < r,\ |\arg z - \theta_0| < \alpha\}$, with a selected continuous argument. Statements at a branch point include the slit and sector.
Parameter order. For iterated limits, notation such as $n \to \infty$ followed by $q \uparrow 1$ records the order. A joint limit instead states a relation such as $(1 - q)n \to \tau$.

Big O, little o, and comparison relations

Big O. $f = \mathcal{O}_{x\to a}(g)$ means that there are $C > 0$ and a punctured neighborhood in the declared domain such that $|f(x)| \le C|g(x)|$. If g vanishes arbitrarily near a, the inequality definition remains authoritative.
Little o. $f = o_{x\to a}(g)$ means $f(x)/g(x) \to 0$ where the quotient is defined, or equivalently the epsilon inequality when that avoids isolated zeros.
Asymptotic equivalence. $f(x) \sim g(x)$ as $x \to a$ means $f(x)/g(x) \to 1$. It requires g to be nonzero in a punctured neighborhood.
Two-sided comparability. $f \asymp g$ in a regime means $c|g| \le |f| \le C|g|$ for constants $0 < c \le C < \infty$. This is not asymptotic equivalence.

Vinogradov notation. $f \ll g$ means $f = \mathcal{O}(g)$ and $f \gg g$ means $g \ll f$. A dependency subscript, for example $\ll_\alpha$, lists every parameter on which the implied constant may depend.
Shared commands. Normative prose prefers `\BigOAt` and `\LittleOAt`; bare `\BigO` and `\LittleO` are reserved for formulas whose regime is printed immediately outside the expression.

Asymptotic sequences and expansions

Asymptotic sequence. Functions (ϕ_n) form an asymptotic sequence in a regime when $\phi_{n+1} = o(\phi_n)$ for every n.
Expansion. $f \sim \sum_{n\geq 0} a_n\phi_n$ means that for every $N \in \mathbb{N}_{>0}$,

$$f - \sum_{n=0}^{N-1} a_n\phi_n = o(\phi_{N-1})$$

in the stated regime. It does not assert convergence of the infinite series.
Poincaré scale. For powers at infinity, $f(x) \sim \sum_{n\geq 0} a_n x^{-\lambda_n}$ requires $\lambda_0 < \lambda_1 < \cdots$ and x in a domain where the powers have been defined.
Exponentially small terms. A transseries term such as $\mathrm{e}^{-A/\epsilon}\epsilon^\beta(\log\epsilon)^m$ lists the sign of $\operatorname{Re}(A/\epsilon)$, the logarithm branch, and the ordering used to compare exponential scales.
Truncation. An optimally truncated expansion names its truncation index $N(\epsilon)$ and the remainder norm; the infinite formal series is not assigned its optimal-truncation error.

Real logarithm, principal logarithm, and arguments

Real logarithm. $\log x$ for $x > 0$ is the real natural logarithm with base e.
Principal branch. The shared command denotes

$$\operatorname{Log} z = \log|z| + \mathrm{i}\operatorname{Arg} z, \qquad \operatorname{Arg} z \in (-\pi, \pi],$$

for $z \in \mathbb{C}\setminus(-\infty, 0]$, with the boundary convention $\operatorname{Arg}(-r) = \pi$ when a boundary value is explicitly taken from above. The analytic domain itself excludes the nonpositive real axis.
Other branches. On a simply connected zero-free domain D, a branch $\log_D z$ is an analytic function satisfying $\exp(\log_D z) = z$. Its normalization at one base point is stated.
Complex powers. $z^\alpha := \exp(\alpha\operatorname{Log} z)$ when the principal branch is intended. If $\alpha \in \mathbb{Z}$, the algebraic power has its global meaning and no branch is needed.
Boundary notation. $\operatorname{Log}(x \pm \mathrm{i}0)$ denotes a specified limiting value. It is not silently substituted into a formula proved only inside the slit plane.

Lambert W and its branches

Defining equation. A branch of W satisfies

$$W_k(z)\mathrm{e}^{W_k(z)} = z,$$

where $k \in \mathbb{Z}$ labels the branch. The inverse is multivalued before a branch is selected.
Principal branch. W_0 is real on $[-\mathrm{e}^{-1}, \infty)$, analytic at 0, and satisfies $W_0(0) = 0$. Its standard complex cut is $(-\infty, -\mathrm{e}^{-1}]$.

Lower real branch. W_{-1} is real on $[-\mathrm{e}^{-1}, 0)$, takes values in $(-\infty, -1]$, and approaches $-\infty$ as $z \uparrow 0$.
Branch point. At $z = -\mathrm{e}^{-1}$ the two real branches meet at -1 and have a square-root branch point. A Taylor series in $z + \mathrm{e}^{-1}$ is therefore not asserted there.
Asymptotic branch. For large z away from cuts,

$$W_k(z) = L_1 - L_2 + \frac{L_2}{L_1} + \cdots, \quad L_1 = \log z + 2\pi \mathrm{i} k, \quad L_2 = \log L_1,$$

with both logarithm branches and the sector specified.
Bare symbol policy. A branchless $W(z)$ is permitted only when the domain makes the real principal branch unambiguous; all complex and negative-real uses carry a subscript.

Polylogarithm, gamma, and zeta functions

Polylogarithm. For $|z| < 1$,

$$\mathrm{Li}_s(z) := \sum_{n=1}^{\infty} \frac{z^n}{n^s}, \qquad s \in \mathbb{C}.$$

Elsewhere the symbol denotes its analytically continued branch, with the z-plane cut and logarithm convention stated.
Gamma function. For $\operatorname{Re} s > 0$, $\Gamma(s) = \int_0^\infty x^{s-1} \mathrm{e}^{-x}\, \mathrm{d}x$. Its meromorphic continuation has simple poles at $0, -1, -2, \ldots$ and no zeros. For $n \in \mathbb{N}_0$, $\Gamma(n+1) = n!$.
Riemann zeta. For $\operatorname{Re} s > 1$, $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, extended meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$.
Hurwitz zeta. $\zeta(s, a) = \sum_{n=0}^{\infty} (n + a)^{-s}$ initially for $\operatorname{Re} s > 1$ and $a \notin \{0, -1, -2, \ldots\}$, with the branch of each power specified if a is complex. The two arguments of `\HurwitzZeta` are the order and shift, respectively.
Collision policy. The same Greek glyph may label a root of unity or a zeta function only with a semantic subscript or arguments that make its type immediate.

Coefficient extraction and formal series

Coefficient operator. For $A(z) = \sum_{n \in \mathbb{Z}} a_n z^n$,

$$[z^n] A(z) := a_n.$$

The bracket is an operator, not multiplication by z^n.
Cauchy formula. If A is analytic on and inside a positively oriented circle $|z| = r$,

$$[z^n] A(z) = \frac{1}{2\pi \mathrm{i}} \int_{|z|=r} \frac{A(z)}{z^{n+1}}\, \mathrm{d}z, \qquad n \in \mathbb{N}_0.$$

Laurent coefficients. For an annular Laurent series the same operator permits $n \in \mathbb{Z}$; the annulus and contour winding number are stated.
Formal interpretation. In $R[[z]]$, coefficient extraction is algebraic and does not assert analytic convergence. Cauchy's formula belongs only to the analytic interpretation.

Residues and contour orientation

Residue. For meromorphic f near z_0, $\operatorname{Res}_{z=z_0} f(z)$ is the coefficient of $(z-z_0)^{-1}$ in the Laurent expansion.
Simple pole. If $f = g/h$, g, h analytic, $h(z_0) = 0$, and $h'(z_0) \neq 0$, then $\operatorname{Res}_{z=z_0} f = g(z_0)/h'(z_0)$.
Contour. Unless an arrow says otherwise, a simple closed contour is positively oriented. Then $\int_\gamma f(z)\,\mathrm{d}z = 2\pi\mathrm{i}\sum \operatorname{Ind}(\gamma, z_j)\operatorname{Res}_{z=z_j} f$.
Indentations and cuts. A keyhole contour states the upper and lower boundary values of every multivalued factor and the direction of each segment. An omitted arc is discarded only with a displayed bound.

Saddles, phase functions, and steepest descent

Phase-amplitude form. A saddle integral is written

$$I(\lambda) = \int_\Gamma A(z,\lambda)\mathrm{e}^{\lambda\Phi(z)}\,\mathrm{d}z,$$

with λ's limiting regime, the oriented contour Γ, and analytic domains of A and Φ stated.
Saddle. A simple saddle z_* satisfies $\Phi'(z_*) = 0$ and $\Phi''(z_*) \neq 0$. Degenerate saddles state the first nonzero derivative order.
Descent direction. A steepest-descent path has $\operatorname{Im}\Phi$ constant and $\operatorname{Re}\Phi$ decreasing away from the saddle. The square root in the Gaussian factor is fixed by the contour orientation.
Leading term. For a single nondegenerate contributing saddle, the schematic factor

$$A(z_*,\lambda)\mathrm{e}^{\lambda\Phi(z_*)}\sqrt{\frac{2\pi}{-\lambda\Phi''(z_*)}}$$

is meaningful only after selecting the square-root branch and verifying the contour deformation. It is not a universal sign-free formula.
Multiple saddles. A set $\mathcal{S}(\lambda)$ lists all saddles; a contributing subset additionally depends on the original contour and Stokes region. Dominance is decided by $\operatorname{Re}(\lambda\Phi(z_*))$, not by modulus of z_*.

Lambert-W parametrization of saddle families

Typical reduction. When a saddle equation is rearranged to $u\mathrm{e}^u = z(\lambda)$, its solutions are $u_k = W_k(z(\lambda))$, $k \in \mathbb{Z}$.
Back-substitution. If $u = \alpha z + \beta$, then $z_k = (W_k(z(\lambda)) - \beta)/\alpha$ with $\alpha \neq 0$. Every scaling and sign is retained; a Lambert solution is not the original saddle until this substitution is made.
Branch enumeration. The integer k labels Lambert branches, not necessarily the order in which saddle contributions dominate. A separate ordering permutation is used if needed.
Coalescence. At a branch point, two branch-labelled saddles may coalesce. Uniform asymptotics then use an Airy or higher canonical model rather than summing two singular Gaussian approximations.
Status. Lambert-W saddle catalogues in frontier documents are **frontier/conjectural** unless an exact contour theorem and error bound are cited.

Singularity analysis and transfer notation

Dominant singularity. For a power series with radius $\rho > 0$, a dominant singularity lies on $|z| = \rho$. There may be more than one; the symbol ρ alone does not choose one.
Delta domain. A dented domain at positive ρ is

$$\Delta(\rho,\eta,\theta) = \{z : |z| < \rho + \eta,\ z \neq \rho,\ |\arg(z-\rho)| > \theta\},$$

with $\eta > 0$ and $0 < \theta < \pi/2$. Alternate angular parametrizations must be expanded.
Basic transfer. For $\alpha \notin \{0, -1, -2, \ldots\}$,

$$[z^n](1 - z/\rho)^{-\alpha} \sim \frac{n^{\alpha-1}}{\Gamma(\alpha)}\rho^{-n}.$$

The power uses the branch analytic at $z = 0$ in the dented domain.
Periodic singularities. Contributions from every dominant singularity are summed with their phases. Replacing them by the positive-real contribution can erase oscillation or exact zeros.
Remainder. A transfer theorem states the regularity and domain hypotheses that turn a local expansion into a coefficient error estimate; a formal local expansion alone does not supply it.

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11 Fractional calculus, Volterra operators, and regularity classes

Riemann–Liouville fractional integrals

Left-sided operator. For $\alpha > 0$, lower terminal $a \in \mathbb{R}$, and suitable f,

$$(I^{\alpha}_{a+}f)(x) \coloneqq \frac{1}{\Gamma(\alpha)}\int_a^x (x-t)^{\alpha-1} f(t)\ \mathrm{d}t, \qquad x > a.$$

Right-sided operator. For upper terminal b,

$$(I^{\alpha}_{b-}f)(x) \coloneqq \frac{1}{\Gamma(\alpha)}\int_x^b (t-x)^{\alpha-1} f(t)\ \mathrm{d}t, \qquad x < b.$$

Order zero. $I^0_{a+} = I^0_{b-} = \mathrm{id}$ by extension, not by substitution into the singular gamma-integral formula.
Semigroup. Under integrability hypotheses, $I^{\alpha}_{a+}I^{\beta}_{a+} = I^{\alpha+\beta}_{a+}$ for $\alpha, \beta > 0$. Terminals must match.
Complex order. For complex α with $\operatorname{Re}\alpha > 0$, the kernel power uses the real logarithm because $x - t > 0$.

Riemann–Liouville and Caputo derivatives

Riemann–Liouville. Let $\alpha > 0$ and $m = \lceil \alpha \rceil$. The left derivative is

$$D^{\alpha}_{a+}f \coloneqq \frac{\mathrm{d}^m}{\mathrm{d}x^m} I^{m-\alpha}_{a+} f.$$

Caputo. Under sufficient regularity,

$${}^{\mathrm{C}}D^{\alpha}_{a+}f \coloneqq I^{m-\alpha}_{a+} f^{(m)}.$$

These operators differ by explicit endpoint terms and are not interchangeable.

Integer order. If $\alpha = m \in \mathbb{N}_0$, both reduce to the ordinary mth derivative only under the appropriate endpoint/regularity interpretation.
Endpoint data. A fractional differential equation declares whether initial conditions are ordinary derivatives $f^{(j)}(a)$, fractional traces, or integral conditions. The operator symbol alone does not determine them.
Fourier derivative. On the whole line, a multiplier definition such as $\widehat{D^\alpha f}_{\mathrm{ang}}(\omega) = (\mathrm{i}\omega)^\alpha \widehat{f}_{\mathrm{ang}}(\omega)$ requires a branch for the complex power and is a third, separately named construction.

Volterra integral operators and resolvents

Operator. For a kernel K on $a \le t \le x \le b$,

$$(V_K f)(x) := \int_a^x K(x,t) f(t)\ \mathrm{d}t.$$

Convolution kernel. If $K(x,t) = k(x-t)$ and $a = 0$, this is the one-sided convolution $(k *_+ f)(x) = \int_0^x k(x-t) f(t)\ \mathrm{d}t$. It is not the whole-line `\Convolution`.
Iterates. $V_K^0 = \mathrm{id}$ and $V_K^{n+1} = V_K \circ V_K^n$. A superscript here is operator composition, not a pointwise power.
Resolvent. When the Neumann series converges, $(\mathrm{id} - \lambda V_K)^{-1} = \sum_{n=0}^\infty \lambda^n V_K^n$. The topology and allowed λ are stated.
Kernel resolvent. A resolvent kernel $R_\lambda(x,t)$ represents the operator resolvent after its identity term is separated; conventions that include a Dirac identity kernel are labelled.

Lebesgue, Sobolev, and Hölder spaces

Lebesgue space. $L^p(E)$, $1 \le p < \infty$, consists of almost-everywhere classes with norm $\|f\|_{L^p(E)} = (\int_E |f|^p)^{1/p}$; L^∞ uses essential supremum. The measure is Lebesgue measure unless printed otherwise.
Smoothness. $C^k(E)$ has continuous derivatives through order $k \in \mathbb{N}_0$; $C^\infty(E) = \bigcap_k C^k(E)$. On a closed interval, derivatives at endpoints are one-sided restrictions of derivatives on a neighborhood unless another convention is stated.
Hölder space. For $0 < \alpha \le 1$,

$$[f]_{C^{0,\alpha}(E)} = \sup_{x \ne y \in E} \frac{|f(x) - f(y)|}{|x-y|^\alpha}.$$

$C^{k,\alpha}$ applies this seminorm to the kth derivative and includes lower sup norms in its norm.
Sobolev space. $W^{k,p}(E)$ uses weak derivatives through order k in L^p. H^s means the L^2-based Sobolev space, with Fourier normalization specified when a Bessel-potential norm is displayed.
Local spaces. The suffix loc means membership after restriction to every compact subset of the open domain; it imposes no boundary control.

Gevrey classes, flatness, and nonanalyticity

Gevrey class. A smooth f is locally Gevrey of order $s \ge 1$ when every compact K admits $C, A > 0$ such that

$$\sup_{x \in K} |f^{(n)}(x)| \le C A^n (n!)^s \quad (n \in \mathbb{N}_0).$$

For several variables, $n!$ is replaced by the multi-index factorial.

Analytic class. Order $s = 1$ corresponds locally to real analyticity under the standard derivative criterion. Membership for $s > 1$ does not imply analyticity.
Flatness. f is flat at a if $f^{(n)}(a) = 0$ for every $n \in \mathbb{N}_0$. A nonzero smooth function may be flat, so a zero Taylor series does not identify the function near a.
Fabius endpoints. up is flat at ± 1 and F_{bd} is flat at the transition points 0 and 1 relative to its constant tails. This is compatible with smoothness and with failure of analyticity there.
Quasianalyticity. A Denjoy–Carleman class is declared with its derivative weight sequence M_n; whether it is quasianalytic is a theorem about that sequence, not synonymous with being Gevrey.

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12 Exact Fabius moments and arithmetic normalizations

Even full moments of the up-function

Definition. For $n \in \mathbb{N}_0$,

$$c_n := \int_{\mathbb{R}} t^{2n} \operatorname{up}(t)\, \mathrm{d}t = \int_{-1}^{1} t^{2n} \operatorname{up}(t)\, \mathrm{d}t.$$

The equality of the two integrals uses the support of up.
Initial value. $c_0 = 1$ because up is normalized to integral one. The full odd moments vanish by evenness.
Fourier expansion. With the 2π convention,

$$\widehat{\operatorname{up}}_{2\pi}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n}.$$

The series is entire; z may be complex.
Canonical integer numerator. Define

$$N_n^{(c)} := c_n (2n+1)!! \prod_{j=1}^{n} (2^{2j} - 1).$$

The empty product gives $N_0^{(c)} = 1$. Historical prose used a capital-letter sequence that collided with Fabius functions; new prose uses the semantic superscript.
Lean declarations. `moment` n and `momentIntegral` n.
Lean modules and status. Exact arithmetic results are in `Arithmetic.lean`; analytic results are in `AnalyticMoments.lean`. Status: **Lean-proved**.

Normalized half moments

Definition. Set $d_0 = 1$ and, for $n \in \mathbb{N}_{>0}$,

$$d_n := n \int_0^1 t^{n-1} \operatorname{up}(t)\, \mathrm{d}t.$$

The separate definition at zero avoids the meaningless factor $0 \int_0^1 t^{-1} \operatorname{up}(t)\, \mathrm{d}t$.
Binomial bridge. For $n \in \mathbb{N}_0$,

$$d_n = 2^{-n} \sum_{0 \le k \le n/2} \binom{n}{2k} c_k,$$

where the bound means $k = 0, \ldots, \lfloor n/2 \rfloor$.

Dyadic-value bridge. For $n \in \mathbb{N}_0$,

$$F_{\mathrm{bd}}(2^{-n}) = \frac{d_n}{n!\,2^{\binom{n}{2}}}.$$

An inverse-dyadic value means evaluation at an inverse power of two, not a value of the inverse function.
Canonical integer numerator. Define

$$N_n^{(d)} := d_n(n+1)! \prod_{j=1}^{n}(2^j - 1).$$

Again the empty product gives $N_0^{(d)} = 1$. A historical capital-letter numerator sequence is recorded as a retired alias rather than reused.
Two-adic value. For $n \geq 1$, the reduced numerator and denominator of $2d_n$ are odd, equivalently $\nu_2(d_n) = -1$ for the rational valuation extended additively to quotients.
Lean crosswalk. `halfMoment` n and `halfMomentIntegral` n. Status: **Lean-proved**.

Reshetnikov's integer normalization

Definition. For $n \in \mathbb{N}_{>0}$,

$$R_n^{\mathrm{Res}} := 2^{\binom{n-1}{2}}(2n)!\,F_{\mathrm{bd}}(2^{-n}) \prod_{j=1}^{\lfloor n/2 \rfloor}(2^{2j} - 1).$$

Type and theorem. $R_n^{\mathrm{Res}} \in \mathbb{N}_{>0}$ and is odd. Its superscript is mandatory in cross-topic documents because R_n also denotes polynomial remainders, resolvents, and recurrence polynomials.
Factors. The power of two is triangular, the factorial is $(2n)!$, and the Mersenne-type product stops at $\lfloor n/2 \rfloor$. Removing or shifting any of these produces a different normalization.
Lean crosswalk. `reshetnikov` n and the corresponding integrality, positivity, and oddness theorems in `Arithmetic.lean` and `PaperStatements.lean`. Status: **Lean-proved**.

Dyadic common denominators

Grid. For $n \in \mathbb{N}_{>0}$, let $A_n = \{a \in \mathbb{N}_{>0} : a < 2^n,\ a \text{ odd}\}$.
Reduced denominator. For $r \in \mathbb{Q}$, write $r = p/q$ with $q \in \mathbb{N}_{>0}$ and $\gcd(p,q) = 1$; then $\mathrm{den}(r) = q$, including $\mathrm{den}(0) = 1$.
Definition. The semantic dyadic denominator is

$$D_n^{\mathrm{dyad}} := \mathrm{lcm}\{\mathrm{den}(\mathrm{up}(a/2^n)) : a \in A_n\}.$$

At $n = 0$ the index set is empty; the catalogue assigns $D_0^{\mathrm{dyad}} = 1$, the LCM identity, when a zero-level value is needed.
Reflection. The same LCM is obtained from the corresponding bounded-Fabius values after applying the exact reflection/translation relation.
Collision policy. A bare D_n is never globally canonical: it also denotes determinants and asymptotic main terms. The semantic superscript is required outside a single-purpose local scope.
Lean declaration. `dyadicDenominator` n.
Lean modules and status. The definition is in `Arithmetic.lean`.
Bounds are in `DenominatorBound.lean`. Status: **Lean-proved**.

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13 Approximation, orthogonal polynomials, splines, and quadrature

Weights and polynomial inner products

Measure form. For a positive Borel measure μ with finite needed moments,

$$\langle f, g\rangle_\mu \coloneqq \int f(x)\overline{g(x)}\ \mathrm{d}\mu(x).$$

The inner product is linear in the first argument and conjugate-linear in the second in this catalogue.
Weight form. If $\mathrm{d}\mu(x) = w(x)\,\mathrm{d}x$ on an interval I, write $\langle f, g\rangle_w = \int_I f(x)\overline{g(x)}w(x)\ \mathrm{d}x$, with $w \geq 0$ almost everywhere.
Norm. $\|f\|_\mu = \sqrt{\langle f, f\rangle_\mu}$. It is a norm only after quotienting functions equal μ-almost everywhere or restricting to a nondegenerate function space.
Fabius weights. When the measure has density up on $[-1, 1]$, write $\langle f, g\rangle_{\mathrm{up}}$ after declaring that measure. An unlabeled upright-word subscript is retired because it bypasses the semantic command. A bounded-Fabius Stieltjes measure is a different but related object.

Orthogonal, monic, and orthonormal polynomials

Orthogonal family. Polynomials $(p_n)_{n\in\mathbb{N}_0}$ satisfy $\deg p_n = n$ and $\langle p_n, p_m\rangle_\mu = 0$ for $n \neq m$.
Monic normalization. P_n is monic when its coefficient of x^n is 1. This fixes an orthogonal polynomial uniquely when the moment functional is positive definite.
Orthonormal normalization. π_n is orthonormal when $\langle \pi_n, \pi_m\rangle_\mu = \delta_{nm}$ and its leading coefficient is positive. Thus $\pi_n = P_n/\|P_n\|_\mu$.
Three-term recurrence. For monic polynomials,

$$P_{n+1}(x) = (x - \alpha_n)P_n(x) - \beta_n P_{n-1}(x),$$

with $P_{-1} = 0$, $P_0 = 1$, $\alpha_n \in \mathbb{R}$, and $\beta_n > 0$ for $n \geq 1$. Orthonormal recurrence coefficients use a different normalization.
Classical names. Legendre, Chebyshev, Jacobi, and Gegenbauer polynomials carry their standard family parameters only after their normalization is displayed; several incompatible leading-value conventions exist.

Legendre and Chebyshev normalizations

Legendre. P_n^{Leg} is defined by $P_n^{\mathrm{Leg}}(1) = 1$ and orthogonality for weight 1 on $[-1, 1]$. Its leading coefficient is $2^{-n}\binom{2n}{n}$, so it is not monic for $n > 1$.
First Chebyshev family. $T_n(\cos\theta) = \cos(n\theta)$, $T_n(1) = 1$, orthogonal for $(1 - x^2)^{-1/2}$ on $[-1, 1]$.
Second Chebyshev family. $U_n(\cos\theta) = \sin((n + 1)\theta)/\sin\theta$, with the continuous endpoint values, orthogonal for $(1 - x^2)^{1/2}$.
Coefficient convention. A Legendre coefficient may be the raw inner product $\langle f, P_n^{\mathrm{Leg}}\rangle$ or the expansion coefficient $(2n + 1)\langle f, P_n^{\mathrm{Leg}}\rangle/2$. The corpus prints which one it means.
Collision. The letter P_n is local: it also denotes probability, prefix polynomials, and Padé denominators. A family superscript is required in a cross-topic statement.

Legendre Gaunt coefficients and the squared zero-row Wigner $3j$ datum

Indices and normalization. Throughout this entry $i, j, k \in \mathbb{N}_0$, and $P_n^{\mathrm{Leg}}(1) = 1$. The Gaunt coefficient is

$$G_{ijk} := \int_{-1}^{1} P_i^{\mathrm{Leg}}(x) P_j^{\mathrm{Leg}}(x) P_k^{\mathrm{Leg}}(x)\ \mathrm{d}x.$$

It is not a coefficient for monic or orthonormal Legendre polynomials.
Admissible support. Write $\mathsf{Adm}(i, j, k)$ for the simultaneous conditions

$$i + j + k \equiv 0 \pmod 2, \qquad i \le j + k, \quad j \le i + k, \quad k \le i + j.$$

Thus admissibility means even total degree and all three weak triangle inequalities; none is implicit. On this support, $s := (i + j + k)/2$ is a natural number and $s - i, s - j, s - k \in \mathbb{N}_0$.
Total rational square datum. Define

$$W_{ijk}^{2,0} := \begin{cases} \dfrac{\binom{2(s-i)}{s-i}\binom{2(s-j)}{s-j}\binom{2(s-k)}{s-k}}{(2s+1)\binom{2s}{s}}, & \mathsf{Adm}(i, j, k), \\ 0, & \text{otherwise.} \end{cases}$$

This is a total element of $\mathbb{Q}$ and equals the square of the integer-index zero-row Wigner $3j$ datum. The superscript $2, 0$ records "square" and "zero magnetic row"; it does not select a signed Wigner symbol.
Gaunt and product-linearization identities. The exact relations are

$$G_{ijk} = 2W_{ijk}^{2,0}, \qquad [P_k^{\mathrm{Leg}}]\big(P_i^{\mathrm{Leg}} P_j^{\mathrm{Leg}}\big) = (2k + 1)W_{ijk}^{2,0},$$

where the bracket denotes the coefficient of P_k^{Leg} in the finite Legendre-basis expansion. Off the admissible support both sides of the first identity vanish.
Triangle coordinates and factorial form. Equivalently, there are $a, b, c \in \mathbb{N}_0$ with $i = b + c$, $j = a + c$, and $k = a + b$. Then

$$W_{ijk}^{2,0} = \frac{\binom{2a}{a}\binom{2b}{b}\binom{2c}{c}}{(2(a + b + c) + 1)\binom{2(a+b+c)}{a+b+c}},$$

and expansion of each central binomial coefficient gives the source's equivalent factorial quotient.
Rvachev Gram use family. Substitution into the finite Rvachev–Legendre Gaunt expansion gives rational Gram entries, rational Gram-matrix entries, and—under a bounded Fabius object plus its Fabius-specification proof—real up-law Gram entries as finite sums of canonical Rvachev–Legendre coefficients times $W_{ij,2r}^{2,0}$. This finite use does not authorize an infinite Legendre-series interchange.
Formal boundary. Only the total rational square of the integer-index, zero-magnetic-row datum is present. There is no sign or Condon–Shortley phase choice, no half-integer spin, no nonzero magnetic index, no general $3j$, $6j$, or $9j$ symbol, and no Wigner orthogonality or recoupling theory.
Lean crosswalk. `legendreGauntAdmissible` and `legendreWignerThreeJZeroSqRat` in `LegendreGauntClosedForm.lean` implement the support predicate and total square datum. The central identities are

```
legendreGauntRat_eq_two_mul_wignerThreeJZeroSqRat
legendreGaunt_eq_two_mul_wignerThreeJZeroSqRat
legendreProductLinearizationCoeffRat_eq_mul_wignerThreeJZeroSqRat
```

`FabiusLegendreGauntClosedForm.lean` supplies the finite rational and real Rvachev Gram-entry sums, with `IsFabius` required for the analytic real formula. Status: **Lean-proved with the displayed hypotheses**.

Moment matrices and Hankel determinants

Moment sequence. For a finite measure μ, $m_n = \int x^n \, \mathrm{d}\mu(x)$ for $n \in \mathbb{N}_0$ whenever finite.
Hankel matrix. The order-N moment matrix is

$$H_N(\mu) \coloneqq (m_{i+j})_{0 \le i,j < N} \in \mathbb{C}^{N \times N}, \qquad N \in \mathbb{N}_0.$$

Thus H_0 is the unique 0×0 matrix.
Determinant. $\Delta_N(\mu) = \det H_N(\mu)$ and $\Delta_0(\mu) = 1$. A positive measure with infinite support has $\Delta_N > 0$ for every $N \ge 1$.
Shifted matrix. $H_N^{(r)} = (m_{i+j+r})_{0 \le i,j < N}$ for $r \in \mathbb{N}_0$. The shift belongs in the superscript, not in an unstated change of the moment sequence.
Fabius parity. For the centered up measure, odd moments vanish and even moments are c_n. Reindexing the even block must show whether its entry is c_{i+j} or c_{i+j+r}.

Jacobi matrices and spectral measures

Matrix. For orthonormal polynomials with $x\pi_n = a_{n+1}\pi_{n+1} + b_n\pi_n + a_n\pi_{n-1}$, $a_n > 0$, the Jacobi operator is the tridiagonal matrix

$$J = \begin{pmatrix} b_0 & a_1 & 0 & \cdots \\ a_1 & b_1 & a_2 & \ddots \\ 0 & a_2 & b_2 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}.$$

Indexing. Rows and columns begin at 0; a_0 does not appear. A finite truncation J_N uses indices $0, \ldots, N-1$.
Spectral measure. The measure μ satisfies $\langle e_0, p(J)e_0 \rangle = \int p(x) \, \mathrm{d}\mu(x)$ for polynomials p, under the spectral theorem.
Resolvent. $G_\mu(z) = \langle e_0, (z \,\mathrm{id} - J)^{-1} e_0 \rangle$ matches the Cauchy-transform denominator $z - x$.
Frontier status. Any claimed closed recurrence coefficients for the Fabius measure remain **frontier/conjectural** unless linked to a proved Lean declaration.

Knots, B-splines, degree, and order

Knot vector. A knot vector is a nondecreasing finite sequence $\tau = (\tau_0, \ldots, \tau_m)$. Repeated knots are allowed and their multiplicity affects smoothness.
Degree versus order. A B-spline of degree $p \in \mathbb{N}_0$ has order $p + 1$. The corpus always says degree or order; a bare index is not assumed to be one or the other.
Cox–de Boor base case. $B_{i,0}(x) = \mathbf{1}_{[\tau_i, \tau_{i+1})}(x)$, except that a designated final endpoint may be closed to make a partition of unity on the full knot span.
Recursion. For $p \ge 1$,

$$B_{i,p}(x) = \frac{x - \tau_i}{\tau_{i+p} - \tau_i} B_{i,p-1}(x) + \frac{\tau_{i+p+1} - x}{\tau_{i+p+1} - \tau_{i+1}} B_{i+1,p-1}(x),$$

where a term with zero denominator is defined to be zero.
Support. $\operatorname{supp}(B_{i,p}) \subseteq [\tau_i, \tau_{i+p+1}]$. Equality needs nondegenerate knot spans.

Cardinal splines. Uniform integer knots and a translated normalization are declared before using the shorter cardinal B-spline symbol.

Divided differences

Distinct nodes. For pairwise distinct $x_0, \ldots, x_n$,

$$[x_0, \ldots, x_n]f = \sum_{j=0}^{n} \frac{f(x_j)}{\prod_{0 \le k \le n,\ k \ne j}(x_j - x_k)}.$$

Recursive form. $[x_j]f = f(x_j)$ and $[x_0, \ldots, x_n]f = ([x_1, \ldots, x_n]f - [x_0, \ldots, x_{n-1}]f)/(x_n - x_0)$.
Repeated nodes. If nodes coalesce and f is sufficiently differentiable, $[\underbrace{x, \ldots, x}_{n+1}]f = f^{(n)}(x)/n!$. This is a continuous extension, not substitution into a zero denominator.
Spline relation. B-splines expressed as divided differences state the truncated-power normalization and knot multiplicities.

Quadrature nodes, weights, and exactness

Rule. An N-node quadrature rule for μ is $Q_N(f) = \sum_{j=1}^{N} w_{j,N} f(x_{j,N})$, with nodes and weights indexed from 1.
Degree of exactness. It has degree of exactness d when $Q_N(p) = \int p \, \mathrm{d}\mu$ for all polynomials of degree at most d, and this fails for at least one polynomial of degree $d+1$.
Gaussian rule. For a positive measure, Gaussian nodes are the zeros of the degree-N orthogonal polynomial; the rule has positive weights and degree $2N-1$.
Endpoint variants. Gauss–Radau fixes one endpoint and Gauss–Lobatto fixes both. Their exactness degrees and endpoint weights differ from the Gaussian rule and are printed separately.
Error. An expression $E_N(f) = \int f \, \mathrm{d}\mu - Q_N(f)$ fixes the sign. The derivative form of an error requires the exact smoothness interval and normalization constant.
Shifted dyadic Rvachev self-sampling. Let F be a bounded Fabius object satisfying the Fabius specification, $N \in \mathbb{N}_0$, $P \in \mathbb{R}[X]$ with $\deg P \le N$, and $\theta \in \mathbb{R}$. Then

$$\int_{\mathbb{R}} P(x)\,\mathrm{up}_F(x)\;\mathrm{d}x = 2^{-N} \sum_{k \in \mathbb{Z}} P\big(2^{-N}(\theta + k)\big)\,\mathrm{up}_F\big(2^{-N}(\theta + k)\big)\,.$$

The right side is formally a bilateral infinite sum, but compact support makes it finite. The theorem proves exactness for every polynomial of degree at most N; it does not assert failure in degree $N+1$, so it does not by itself establish that the degree of exactness is exactly N.
Lean crosswalk. In `PolynomialCombExactness.lean`, the displayed rule is the declaration

`integral_polynomial_mul_rvachevUp_eq_dyadic_tsum`

Its normalized source identity is

`tsum_shifted_polynomial_eq_integral`

Both retain the bounded-Fabius and Fabius-specification hypotheses. Status: **Lean-proved**.

Padé approximation

Definition. For a formal series $f(z) = \sum_{n\geq 0} a_n z^n$, a type $[L/M]$ Padé approximant is a rational function $P_L(z)/Q_M(z)$ with $\deg P_L \leq L$, $\deg Q_M \leq M$, $Q_M(0) = 1$, and

$$Q_M(z)f(z) - P_L(z) = \mathcal{O}_{z\to 0}\big(z^{L+M+1}\big) .$$

Existence and uniqueness. A normalized pair may fail to exist with exact degrees or may have a common factor. The rational function is unique under the standard normality hypothesis, which is stated when used.
Poles. Zeros of Q_M after cancellation are poles of the approximant. Uncancelled numerator-denominator zero pairs are not counted as poles.
Collision policy. P, Q are local polynomial names; they do not denote probability or the Fabius quantile inside this entry. Outside it, the type-$[L/M]$ subscript or semantic label is retained.

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14 Linear algebra, operators, kernels, and spectral notation

Vectors, matrices, and indices

Vectors. A column vector is $x = (x_1, \dots, x_n)^{\mathsf{T}} \in \mathbb{K}^n$, where $\mathbb{K}$ is explicitly $\mathbb{R}$ or $\mathbb{C}$. Algorithmic zero-based coordinates are marked separately.
Matrices. $A = (a_{ij})_{1\leq i\leq m,\, 1\leq j\leq n} \in \mathbb{K}^{m\times n}$. The first dimension is the number of rows.
Transpose and adjoint. A^{T} is transpose, $\overline{A}$ entrywise conjugation, and $A^* = \overline{A}^{\mathsf{T}}$ conjugate transpose. For real matrices, $A^* = A^{\mathsf{T}}$.
Identity and zeros. I_n is the $n \times n$ identity and $0_{m\times n}$ the rectangular zero matrix. A bare 0 inherits type only inside a displayed typed equation.
Empty size. The unique 0×0 matrix has determinant 1, trace 0, and rank 0. Matrix products with a zero inner dimension are zero matrices by empty sums.

Trace, rank, determinant, and diagonal

Trace. For $A \in \mathbb{K}^{n\times n}$, $\operatorname{tr} A = \sum_{j=1}^{n} a_{jj}$, with $\operatorname{tr} A = 0$ at $n = 0$.
Trace-class extension. For a trace-class operator T on a real or complex Hilbert space and any orthonormal basis (e_j), $\operatorname{tr} T \coloneqq \sum_j \langle Te_j, e_j\rangle$. Trace class makes this series absolutely convergent and its value independent of the orthonormal basis; in finite dimension it agrees with the matrix trace above. No trace is inferred for an arbitrary bounded infinite-dimensional operator.
Rank. $\operatorname{rank} A$ is the dimension of the image of the associated linear map. Row rank and column rank agree over a field.
Determinant. $\det A$ is defined only for a square matrix or an endomorphism of a finite free module with a chosen basis-free determinant construction.
Diagonal. $\operatorname{diag}(x_1, \dots, x_n)$ is the square diagonal matrix with those diagonal entries. Applying `\DiagonalOperator` to a matrix in order to extract a vector is forbidden; extraction is written $\operatorname{diagvec}(A)$ to prevent an input-type overload.
Span. For vectors v_j, $\operatorname{span}_{\mathbb{K}}\{v_j : j \in J\}$ is their linear span over $\mathbb{K}$. The scalar field is printed when not obvious.

Tensor, Kronecker, and Hadamard products

Tensor product. $V \otimes_{\mathbb{K}} W$ is the algebraic tensor product of vector spaces; completed topological tensor products carry a norm or topology subscript.
Kronecker product. For matrices $A = (a_{ij})$ and B, $A \otimes B$ is the block matrix $(a_{ij}B)$. Its dimensions are the pairwise products of the input dimensions.
Hadamard product. $A \circ B = (a_{ij}b_{ij})$ for matrices of identical shape. It is not composition.
Composition. $S \circ T$ means first T, then S. When Hadamard products occur in the same section, they use $\odot$ instead of $\circ$.
Powers. $A^0 = I_n$ for square $A \in \mathbb{K}^{n\times n}$, including the zero matrix. Entrywise powers are written $A^{\circ k}$.

Vector, matrix, and operator norms

Euclidean norm. For $x \in \mathbb{C}^n$, $\|x\|_2 = (\sum_j |x_j|^2)^{1/2}$.
Induced norm. For $A : \mathbb{K}^n \to \mathbb{K}^m$, $\|A\|_{p\to q} = \sup_{x\neq 0} \|Ax\|_q/\|x\|_p$. The spectral norm is $\|A\|_{2\to 2}$.
Frobenius norm. $\|A\|_{\mathrm{F}} = (\sum_{i,j} |a_{ij}|^2)^{1/2} = (\operatorname{tr} A^*A)^{1/2}$.
Unsubscripted norm. The shared `\Norm` supplies delimiters only. Each local scope declares the norm behind them; it is not globally the Euclidean or supremum norm.
Condition number. For invertible A, $\kappa_p(A) = \|A\|_{p\to p}\|A^{-1}\|_{p\to p}$; for singular A it is $+\infty$.

Spectrum, eigenvalues, and resolvents

Finite spectrum. For $A \in \mathbb{C}^{n\times n}$, $\sigma(A) = \{\lambda \in \mathbb{C} : \det(\lambda I_n - A) = 0\}$.
Multiplicity. Algebraic multiplicity is root multiplicity in the characteristic polynomial; geometric multiplicity is $\dim\ker(A - \lambda I_n)$. They are not conflated.
Spectral radius. $\rho(A) = \max_{\lambda\in\sigma(A)} |\lambda|$ for $n > 0$; for the 0×0 matrix the spectrum is empty and the catalogue sets $\rho(A) = 0$ when a total convention is required.
Operator spectrum. For a bounded operator T on a complex Banach space, $\sigma(T) = \{\lambda : \lambda\,\mathrm{id} - T \text{ is not boundedly invertible}\}$.
Resolvent. $R_T(z) = (z\,\mathrm{id} - T)^{-1}$ for $z \notin \sigma(T)$. This operator-valued R_T is distinct from R_n^{Res}.

Hankel and Toeplitz matrices

Hankel. A finite Hankel matrix has entries $H_{ij} = a_{i+j}$, with declared index origins. In this catalogue $0 \le i, j < N$ is the default.
Toeplitz. A Toeplitz matrix has entries $T_{ij} = a_{i-j}$ and therefore requires a two-sided sequence for zero-based finite indices.
Moment interpretation. A Hankel matrix (m_{i+j}) is positive semidefinite when (m_n) is a moment sequence of a positive real measure.
Operator interpretation. Infinite Hankel or Toeplitz matrices define operators only after a sequence space and boundedness/domain conditions are supplied.
Collision. A transform, matrix, and determinant may all be called H in source literature. Cross-topic prose uses $H_N(\mu)$, $T_N(a)$, or a semantic operator subscript.

Minors and total positivity

Minor. For row index set I and column index set J of equal finite size, $A[I, J]$ is the corresponding submatrix and $\Delta_{I,J}(A) = \det A[I, J]$. Both index sets are listed in increasing order.
Totally nonnegative. A real matrix is totally nonnegative if every minor is nonnegative.
Totally positive. It is totally positive if every minor that is not forced to vanish by the matrix shape is positive. For a rectangular matrix, all square orders up to $\min(m, n)$ are included.
Strict versus order r. The suffix TP_r restricts to minors of order at most r; TN_r is the nonnegative analogue. Neither implies unqualified total positivity when r is finite below matrix size.
Kernel version. A kernel $K(x, y)$ is totally positive when determinants $\det(K(x_i, y_j))$ have the specified sign for strictly ordered point tuples. Weak orders and repeated nodes require a confluent formulation.
Status. Total-positivity claims for Fabius moment or collocation matrices are **frontier/conjectural** unless their exact minor family is linked to a proof.

Positive-definite kernels and RKHS notation

Positive definite. A Hermitian kernel $K : X \times X \to \mathbb{C}$ is positive semidefinite if $\sum_{i,j} \overline{c_i} c_j K(x_i, x_j) \geq 0$ for all finite choices.
Strictness. Strict positive definiteness requires positivity for nonzero coefficient vectors when the points are pairwise distinct.
RKHS. The reproducing-kernel Hilbert space $\mathcal{H}_K$ satisfies $f(x) = \langle f, K(\,\cdot\,, x)\rangle_{\mathcal{H}_K}$ under the catalogue's first-slot-linear convention. References using second-slot linearity conjugate this identity.
Gram matrix. $G_X = (K(x_i, x_j))_{i,j=1}^{N}$. Its rank may be smaller than N even for a positive-semidefinite kernel.
Collision. Kernel K, compact set K, and field $\mathbb{K}$ may coexist only with type-bearing subscripts or a local declaration.

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15 Research-frontier namespaces

The entries in this section define the symbols needed to read the semi-formalized frontier documents. They do not assert that the proposed Fabius specializations are theorems. Every such specialization carries one of the proof-status labels in section 1.3.

Appell and Sheffer polynomial families

Appell family. A polynomial sequence $(A_n)_{n\in\mathbb{N}_0}$ is Appell when A_0 is a nonzero constant and $A_n'(x) = nA_{n-1}(x)$ for $n \geq 1$.
Generating function. With the normalization $A_0 = 1$,

$$\sum_{n=0}^{\infty} A_n(x)\frac{z^n}{n!} = g(z)\mathrm{e}^{xz}, \qquad g(0) = 1,$$

as a formal identity or analytic identity on a printed disk.
Associated law. If $g(z) = 1/M_X(z)$, the family is associated with a law only where the reciprocal series exists; this does not imply that all A_n are orthogonal.
Sheffer family. A Sheffer sequence has EGF $g(z)\exp(xf(z))$ with $g(0) \neq 0$, $f(0) = 0$, and $f'(0) \neq 0$. Appell is the case $f(z) = z$.

Collision. The local A_n is neither a matrix nor an event. A generic Appell family retains a declared generator g; the Rvachev reciprocal-moment family is A_n^{Rv} outside its defining section and is specified in the next entry.

Rvachev moments, Appell polynomials, and polynomial deconvolution

Probability input. Let F be a bounded Fabius object satisfying the project's Fabius specification, and put $u_F = \mathrm{up}_F$. The associated probability measure has density $u_F(y)\ \mathrm{d}y$, total mass 1, even symmetry, and support contained in $[-1,1]$. Every analytic identity below retains this hypothesis on F; the notation alone does not manufacture a canonical solution.
Raw moments. For $n \in \mathbb{N}_0$,

$$m_n^{\mathrm{Rv}} := \int_{\mathbb{R}} y^n u_F(y)\ \mathrm{d}y.$$

Even symmetry gives $m_{2j+1}^{\mathrm{Rv}} = 0$. The project stores exact rational values in `rvachevRawMomentRat`; casting that sequence to $\mathbb{R}$ is a separate typed operation.
Binomial-convolution reciprocal. The rational sequence $(\gamma_n^{\mathrm{Rv}})_{n\geq 0}$ is uniquely defined by

$$\sum_{k=0}^{n} \binom{n}{k} m_k^{\mathrm{Rv}} \gamma_{n-k}^{\mathrm{Rv}} = \delta_{n,0} \qquad (n \in \mathbb{N}_0).$$

This is reciprocal under exponential-generating-function convolution. It is not pointwise reciprocal $1/m_n^{\mathrm{Rv}}$, and it remains meaningful algebraically without assuming convergence of an analytic reciprocal moment-generating function.
Rvachev–Appell family. Define

$$A_n^{\mathrm{Rv}}(x) := \sum_{k=0}^{n} \binom{n}{k} \gamma_{n-k}^{\mathrm{Rv}} x^k.$$

Then A_n^{Rv} is monic of exact degree n, and $(A_n^{\mathrm{Rv}})' = nA_{n-1}^{\mathrm{Rv}}$ for $n \geq 1$. The Lean names are `rvachevAppellPolynomialRat` over $\mathbb{Q}$ and `rvachevAppellPolynomial` over $\mathbb{R}$.
Polynomial deconvolution operator. For $P(x) = \sum_{n=0}^{d} p_n x^n \in \mathbb{R}[x]$, define

$$(\mathscr{D}_{\mathrm{up}} P)(x) := \sum_{n=0}^{d} p_n A_n^{\mathrm{Rv}}(x).$$

Thus $\mathscr{D}_{\mathrm{up}}$ is a real-linear map on polynomials and sends x^n to A_n^{Rv}. It preserves the natural degree and leading coefficient of every nonzero polynomial and is injective. The semantic subscript names the smoothing law being inverted; $\mathscr{D}$, $\mathcal{D}$, and $\mathscr{D}_u$ without that subscript are not corpus-wide aliases.
Smoothing identities. For $x \in \mathbb{R}$,

$$\int_{\mathbb{R}} A_n^{\mathrm{Rv}}(x+y) u_F(y)\ \mathrm{d}y = x^n, \qquad \int_{\mathbb{R}} (\mathscr{D}_{\mathrm{up}} P)(x+y) u_F(y)\ \mathrm{d}y = P(x).$$

Evenness of u_F also permits $x - y$ in place of $x + y$. For $n \in \mathbb{N}_{>0}$,

$$\int_{\mathbb{R}} A_n^{\mathrm{Rv}}(y) u_F(y)\ \mathrm{d}y = 0.$$

These statements are pointwise polynomial identities, not merely almost-everywhere relations.

Mesh synthesis. Let $M \in \mathbb{N}_{>0}$ and $\deg P \le \nu_2(M)$. Then

$$\sum_{k\in\mathbb{Z}} (\mathscr{D}_{\mathrm{up}}P)(k/M)u_F(x-k/M) = MP(x).$$

Compact support makes the bilateral sum pointwise finite. For $x \in [-1,1]$, the exact uniform truncation is

$$P(x) = \frac{1}{M} \sum_{-2M<k<2M} (\mathscr{D}_{\mathrm{up}}P)(k/M)u_F(x-k/M).$$

The strict inequalities omit both endpoints; their atoms vanish by support.
Arbitrary phase. For every $\theta, x \in \mathbb{R}$, under the same M and degree hypotheses,

$$P(x) = \frac{1}{M} \sum_{k\in\mathbb{Z}} (\mathscr{D}_{\mathrm{up}}P)\left(x - \frac{\theta+k}{M}\right) u_F\left(\frac{\theta+k}{M}\right).$$

A summand can be nonzero only if $|\theta + k| \le M$, so the displayed bilateral sum is finite. The shifts contain explicit division by M; juxtaposition does not mean $x - \theta - k/M$.
Dyadic specialization and frontier boundary. At $M = 2^N$, every polynomial of degree at most N is covered. The proposed extension to degree $N+1$ at phases $\theta \equiv 0, 1/2 \pmod 1$ for even N, or $\theta \equiv 1/4, 3/4 \pmod 1$ for odd N, remains a frontier assertion and is not included in the Lean-proved theorem above.
Lean crosswalk. The exact names are listed one per semantic role:

Raw moment: `rvachevRawMomentRat`.
Reciprocal moment: `rvachevReciprocalMomentRat`.
Rational Appell family: `rvachevAppellPolynomialRat`.
Real Appell family: `rvachevAppellPolynomial`.
Polynomial deconvolution: `rvachevDeconvolvedPolynomial`.
Bundled linear map: `rvachevDeconvolutionLinearMap`.

These definitions are in `RvachevMomentAppell.lean`.
Mesh and phase reproduction are in `RvachevPolynomialSynthesis.lean`. Status: **Lean-proved with the displayed hypotheses**; no analytic reciprocal-MGF convergence or infinite differential-operator expansion is claimed.

Polynomial–logarithmic variables, coefficient classes, and completions

Commands. The shared semantic commands are `\PolynomialLogPowerSeriesRing`, `\PolynomialLogLaurentR`
`\RationalLogLaurentField`, `\IteratedLaurentLogField`, and `\NearIdentityPolynomialLogGroup`.
Their arguments name the coefficient field. A document-local calligraphic letter is not a synonym for any of these commands unless that document explicitly defines it.
Base data. Fix a field K of characteristic zero, a large variable X, and the generators

$$t := X^{-1}, \qquad L := \log X.$$

For real asymptotics the default regime is $X \to +\infty$. For complex asymptotics a sector and a branch of $\log$ must be supplied. In the formal algebra, t and L are independent generators; their analytic relation is recorded by the distinguished derivation $D_X t = -t^2$ and $D_X L = t$. The letters x, ℓ, λ used in individual reports are local renamings of X, L, not additional global generators.
Polynomial-log power-series ring. Define

$$\mathscr{P}\mathscr{L}_{\mathrm{pow},K} := K[L][[t]].$$

An element is a unique series

$$F = \sum_{n \ge 0} p_n(L) t^n, \qquad p_n \in K[L].$$

Each coefficient block is a finite polynomial in L; no uniform bound on $\deg_L p_n$ is implicit. The ideal of strictly positive outer order is $tK[L][[t]]$.
Polynomial-log Laurent ring. Define

$$\mathscr{PL}_{\mathrm{poly},K} \coloneqq K[L]((t)) = \bigcup_{m \ge 0} t^{-m} K[L][[t]].$$

Every element has a unique representation

$$F = \sum_{n \ge n_0} p_n(L) t^n, \qquad n_0 \in \mathbb{Z}, \quad p_n \in K[L].$$

Only finitely many negative powers of t may occur. Equivalently, only finitely many positive powers of X occur. This ring is an integral domain but, in general, not a field.
Unit criterion. For nonzero

$$F = t^{n_0}\big(p_{n_0}(L) + p_{n_0+1}(L)t + \cdots\big),$$

F is a unit of $\mathscr{PL}_{\mathrm{poly},K}$ exactly when $p_{n_0}(L) \in K^\times$, that is, when its leading coefficient block is a nonzero scalar. A nonzero polynomial such as $L+1$ is not a unit in $K[L]$. Special divisibility may keep a particular quotient in the polynomial-log ring, but there is no unconditional division theorem by an arbitrary nonzero leading polynomial.
Rational-log Laurent field. Unrestricted exact division of nonzero coefficient blocks is performed in

$$\mathscr{PL}_{\mathrm{rat},K} \coloneqq K(L)((t)).$$

This field retains rational dependence on L exactly. A scalar such as a chosen value of $\log c$ must belong to K; otherwise K must first be extended.
Iterated Laurent-log field. Expanding rational functions at $L = \infty$ gives

$$\mathscr{PL}_{\mathrm{itLaur},K} \coloneqq K((L^{-1}))((t)).$$

An element has the form

$$\sum_{n \ge n_0} t^n \sum_{j \ge j_n} a_{n,j} L^{-j}.$$

For each fixed outer level n, the inner Laurent series contains only finitely many positive powers of L, but it may contain infinitely many negative powers. The iteration order is part of the type:

$$K((L^{-1}))((t)) \ne K((t))((L^{-1}))$$

in general, because the two fields impose different support conditions. The embedding $K(L)((t)) \hookrightarrow K((L^{-1}))((t))$ means Laurent expansion of each rational coefficient at $L = \infty$; it changes an exact coefficient such as $1/(L+1)$ into $L^{-1} - L^{-2} + L^{-3} - \cdots$.
Completion order and finite data. A finite representation in the iterated field needs an outer t-cutoff and an inner L^{-1}-cutoff, or one explicitly defined staircase in the lexicographic monomial order. An outer cutoff alone does not make an infinite Laurent-log coefficient block finite.

Puiseux-log extension. For a fixed $s \in \mathbb{N}_{>0}$,

$$K[L]\big((t^{1/s})\big)$$

allows exponents in $s^{-1}\mathbb{Z}$. Formally,

$$D_X(t^\alpha p(L)) = t^{\alpha+1}\big(p'(L) - \alpha p(L)\big), \qquad \alpha \in s^{-1}\mathbb{Z}.$$

An analytic realization additionally requires a branch of $X^{1/s}$. The union over all s allows rational exponents but has no single denominator bound.
Hahn power-log extension. Let Γ be an ordered additive subgroup of $\mathbb{R}$. A Hahn power-log series has schematic form

$$\sum_{\alpha \in S} p_\alpha(L) t^\alpha, \qquad p_\alpha \in K[L],$$

where $S \subseteq \Gamma$ is well ordered in the direction of increasing smallness. The support condition is what makes multiplication locally finite. If complex exponents are admitted, ordering merely by real part is insufficient: a tie-breaking order and a support convention must be stated.
Near-identity group. Define

$$\mathcal{G}_{\mathrm{poly},K} := \{X + A : A \in \mathscr{P}\mathscr{L}_{\mathrm{pow},K}\}.$$

Under the admissibility hypotheses in the composition entry, this is a group under composition. It is not a multiplicative group of coefficient series.

Coefficient support, dominance, valuation, and leading data

Coefficient support command. The shared command is `\CoefficientSupportOf`. It is distinct from `\SupportOf`, which is reserved for pointwise function support. For

$$F = \sum_{n \ge n_0} p_n(L) t^n = \sum_{\substack{n \in \mathbb{Z} \\ j \in \mathbb{N}_0}} a_{n,j} t^n L^j,$$

define

$$\mathrm{supp}_{\mathrm{coeff}}(F) := \{(n,j) \in \mathbb{Z} \times \mathbb{N}_0 : a_{n,j} \neq 0\}.$$

Every fiber over a fixed n is finite, and the projection to the n-coordinate is bounded below. The zero series has empty coefficient support. This is support in an exponent-index set, not the pointwise support of a function on a topological space.
Dominance order. On the positive ray as $X \to +\infty$,

$$t^n L^j \succ t^m L^k \quad \Longleftrightarrow \quad n < m \quad \text{or} \quad (n = m \text{ and } j > k).$$

Thus outer inverse-power order is compared first; logarithmic degree refines only a fixed outer block. Negative powers of L belong to an enlarged Laurent-log coefficient field, not to $K[L]((t))$.
Outer valuation. For nonzero $F = \sum_{n \ge n_0} p_n(L) t^n$, define

$$\nu_t(F) := \min\{n \in \mathbb{Z} : p_n \neq 0\}, \qquad \nu_t(0) := +\infty.$$

For nonzero inputs,

$$\nu_t(FG) = \nu_t(F) + \nu_t(G), \qquad \nu_t(F+G) \ge \min\{\nu_t(F), \nu_t(G)\}.$$

The second inequality may be strict when the leading blocks cancel.

Leading block, monomial, and scalar. Let

$$p_{\nu_t(F)}(L) = aL^d + \text{lower powers}, \qquad a \in K^\times.$$

Then the leading coefficient block is the whole polynomial $p_{\nu_t(F)}$; the leading monomial is $t^{\nu_t(F)}L^d$; and the scalar leading coefficient is a. These are three different objects. For $F = 0$, the degree, leading monomial, and leading coefficient are undefined unless a separate extended convention is declared.
Rank-two valuation. The refined monomial valuation is

$$v(t^n L^j) = (n, -j)$$

with lexicographic order. Hence for nonzero F,

$$v(F) = \big(\nu_t(F), -\deg_L p_{\nu_t(F)}\big).$$

The negative sign makes larger logarithmic degree more dominant within a fixed outer block.
Log-degree profile. Extend absent coefficient blocks by zero and define

$$d_F(n) = \begin{cases} \deg_L p_n, & p_n \neq 0, \\ -\infty, & p_n = 0. \end{cases}$$

The profile d_F is additional data; it is not recoverable from the coarse number $\nu_t(F)$. Linear bounds such as $d_F(n) \leq an + b$ are hypotheses or proved invariants, never part of the ring definition.
Hahn support. For a Hahn extension the support is a subset of the chosen exponent group and must be well ordered. Multiplication uses Minkowski addition of supports. Well ordering or the relevant local-finiteness condition ensures that every output coefficient receives only finitely many contributions.

Exclusive truncation, formal tails, analytic remainders, and summability

Exclusive outer truncation. For $N \in \mathbb{Z}$, define

$$\operatorname{Trunc}_{<N} F := \sum_{n<N} p_n(L)t^n.$$

It keeps every logarithmic monomial in every retained coefficient block. If $N \leq n_0$, the sum is empty and equals 0. The zero series also truncates to 0. "Through outer order t^{N-1}" is the exclusive cutoff $< N$.
Inclusive outer truncation. The report-6 local notation

$$[F]_{\leq N} := \sum_{n \leq N} p_n(L)t^n$$

is inclusive. Therefore

$$[F]_{\leq N} = \operatorname{Trunc}_{<N+1} F.$$

Any transfer between reports must account for this one-block difference.
Formal tail command. The command `\FormalBigOAt` is defined by

$$F = G + \mathcal{O}_t^{\text{form}}\big(t^N\big) \quad\Longleftrightarrow\quad \nu_t(F - G) \geq N \quad\Longleftrightarrow\quad F - G \in t^N K[L][[t]].$$

It asserts coefficient agreement below outer order N. It asserts no numerical bound, convergence, uniformity in L, or analytic realization. The token `\FormalBigO` is the same operator without supplied arguments.

Why the Laurent tail is not an ideal. In $\mathscr{PL}_{\mathrm{poly},K} = K[L]((t))$, t is a unit. Therefore the principal ideal generated by t^N is the whole Laurent ring. The subset $t^N K[L][[t]]$ is an additive valuation-filtration tail, not an ideal under arbitrary Laurent multiplication. Consequently:

- in $K[L][[t]]$, reduction modulo the ideal (t^N) is a genuine ring quotient;
- in $K[L]((t))$, use $\nu_t(F - G) \geq N$, $\mathcal{O}_t^{\mathrm{form}}(t^N)$, or equality of exclusive truncations;
- quotient-algebra language becomes legitimate after removing the finite Laurent principal part and restricting to a power-series tail ring.

Coefficient extraction. For the double expansion above,

$$[t^n]F = p_n(L), \qquad [t^n L^j]F = a_{n,j}.$$

An absent coefficient is 0. Bracket argument order is not uniform in the source reports: one local macro prints $[M]F$, another prints $[F]_M$. Shared prose must state the series and monomial roles explicitly.

Separate logarithmic cutoff. Bounding $\deg_L p_n$ is an additional approximation, not part of the outer t-adic topology. Discarding a high power of L in a retained t-block requires an explicit degree bound. An iterated Laurent-log coefficient requires a second inner cutoff or a defined two-dimensional staircase.

Analytic big-O command. For actual functions on a domain S,

$$R(X) = \mathcal{O}_{X\to\infty,\, X\in S}(A(X))$$

means that there exist $C > 0$ and a sufficiently late part of the stated domain such that $|R(X)| \leq C\,|A(X)|$ there. In complex work the sector, closed subsector if uniformity is intended, and every logarithmic or fractional-power branch are part of the statement. The limit must not be left to context.

Three different O-roles. The corpus distinguishes:

$$\mathcal{O}_t^{\mathrm{form}}(t^N) \quad \text{(formal valuation tail)}, \qquad \mathcal{O}_{X\to\infty}(A(X)) \quad \text{(analytic estimate)}, \qquad \mathcal{O}(N^2) \quad \text{(algorithmic complexit}$$

No one of these meanings may be inferred from the other two.

Completeness and summability. A sequence is t-adically Cauchy when every coefficient block below every fixed cutoff eventually stabilizes. A family $(S_r)_{r\geq 0}$ is formally summable when $\nu_t(S_r) \to +\infty$; below any cutoff only finitely many members then contribute. This coefficientwise property is the formal substitute for numerical convergence. On a Laurent ring, a Cauchy sequence must also have a common eventual lower support bound; otherwise it need not define a Laurent series with a finite principal part.

Large-variable differentiation and shifted logarithmic operators

Distinguished derivation. With $t = X^{-1}$ and $L = \log X$,

$$D_X = t\,\partial_L - t^2\,\partial_t.$$

Here ∂_t and ∂_L are formal partial derivatives holding the other generator fixed. The formula is the unique derivation satisfying $D_X t = -t^2$ and $D_X L = t$.

One coefficient block. For $n \in \mathbb{Z}$ and $p \in K[L]$,

$$D_X\big(t^n p(L)\big) = t^{n+1}\big(\partial_L - n\big)p(L).$$

The Euler derivation preserves outer order:

$$X D_X\big(t^n p(L)\big) = t^n\big(\partial_L - n\big)p(L).$$

Repeated derivative operator. For $k \in \mathbb{N}_0$, define

$$\Delta_{n,k} \coloneqq \prod_{j=0}^{k-1} (\partial_L - n - j), \qquad \Delta_{n,0} \coloneqq 1.$$

The $k = 0$ product is empty and acts as the identity. The factors commute because each is a polynomial in ∂_L. Then

$$D_X^k\big(t^n p(L)\big) = t^{n+k} \Delta_{n,k} p(L).$$

Valuation behavior. For nonzero F,

$$\nu_t(D_X F) \geq \nu_t(F) + 1.$$

Equality can fail through annihilation or cancellation. If the leading term is $at^n L^d$, then:

- if $n \neq 0$, the derivative begins with $-na\, t^{n+1} L^d$;
- if $n = 0$ and $d > 0$, it begins with $ad\, tL^{d-1}$;
- if $n = d = 0$, the leading constant is annihilated and the next nonconstant term decides the order.

In particular, $D_X K = 0$ and $D_X(K[L][[t]]) \subseteq tK[L][[t]]$.
Formal partial versus analytic derivative. The formal ∂_t used for coefficient extraction holds L fixed. It is not the analytic derivative with respect to t after substituting $L = -\log t$. Likewise, a bare D may denote D_X, ∂_L, or differentiation in a Lambert-polynomial dummy variable; the differentiated variable must be printed.
Antiderivative resonance. Solving

$$(\partial_L - n)p(L) = q(L)$$

for a polynomial p is triangular and unique when $n \neq 0$. When $n = 0$ it reduces to polynomial integration and introduces an arbitrary scalar constant. This is the formal source of the constant of integration and must be handled separately at the resonant block.

Admissible composition and near-identity substitution

Monomial-leading inner argument. Let

$$F(Y) = \sum_{n \geq a} p_n(\log Y) Y^{-n}$$

and

$$G(X) = cX^\rho(1 + U(X)), \qquad c \in K^\times, \quad \rho \in \mathbb{N}_{>0}, \quad U \in tK[L][[t]].$$

Choose a value of $\log c$ in the coefficient field and put $V = \log(1 + U)$. Then

$$(F \circ G)(X) = \sum_{n \geq a} p_n(\rho L + \log c + V)\, c^{-n} t^{\rho n} (1 + U)^{-n}.$$

Every coefficient below a fixed outer cutoff is a finite sum. If $\rho = r/s > 0$ is rational, the result belongs to a Puiseux-log extension after adjoining $t^{1/s}$; a general ordered real exponent requires a Hahn exponent group.

Generator substitution. For

$$G = X + H = X(1 + tH), \qquad H \in K[L][[t]],$$

composition changes both generators:

$$t \circ G = \frac{t}{1 + tH}, \qquad L \circ G = L + \log(1 + tH).$$

The substitution $t \mapsto t + H$ is incorrect because t denotes X^{-1}, not X. Updating L while leaving t fixed is also incorrect.
Exact formal Taylor formula. For $F \in K[L]((t))$ and $H \in K[L][[t]]$,

$$F \circ (X + H) = \sum_{k \geq 0} \frac{H^k}{k!} D_X^k F.$$

This is an exact identity of formal series, not an analytic approximation. If $a = \nu_t(F)$, then only finitely many k, in particular $k \leq N - a$, can affect the coefficient of t^N.
Finite coefficient formula. Write

$$F = \sum_{n \geq a} p_n(L) t^n, \qquad H^k = \sum_{m \geq 0} h_{k,m}(L) t^m,$$

with $h_{0,0} = 1$ and $h_{0,m} = 0$ for $m > 0$. Then

$$[t^r]\big(F \circ (X + H)\big) = \sum_{\substack{n \geq a,\ k,m \geq 0 \\ n+k+m=r}} \frac{h_{k,m}(L)}{k!} \Delta_{n,k} p_n(L).$$

The displayed affine constraint makes the sum finite.
Convolution-polynomial boundary cases. If $H = \sum_{j \geq 0} h_j t^j$, define

$$C_{m,r} := \sum_{\substack{j_1,\dots,j_r \geq 0 \\ j_1+\cdots+j_r=m}} h_{j_1} \cdots h_{j_r}.$$

The empty-product conventions are

$$C_{0,0} = 1, \qquad C_{m,0} = 0 \quad (m > 0).$$

Equivalently, $H^r = \sum_{m \geq 0} C_{m,r} t^m$, and

$$C_{m,r+1} = \sum_{j=0}^{m} h_j C_{m-j,r}.$$

If $h_0 = 0$, then $C_{m,r} = 0$ whenever $m < r$.
Associativity domain. The identity $(F \circ G) \circ H = F \circ (G \circ H)$ is asserted only when every displayed composition exists in the selected support class. For substitution maps, right substitution reverses the apparent order of operator composition. Formal closure is a theorem about support and local finiteness, not a license to compose arbitrary transseries.
Analytic admissibility. For analytic realizations, the image of the inner map must eventually lie in the outer expansion's ray or sector, must avoid every relevant logarithmic branch cut, and must carry the omitted outer remainder below the requested output order. These conditions are additional to the formal coefficient calculation.

Compositional inverse, multiplicative reciprocal, and residual certificates

Commands. The shared commands are `\MultiplicativeInverseOf` and `\CompositionalInverseOf`. They render

$$F^{-1} = F^{-1}, \qquad F^{\langle -1\rangle_\circ} = F^{\langle -1\rangle_\circ}.$$

The first is inverse under multiplication and satisfies $FF^{-1} = 1$. The second is inverse under composition and satisfies

$$F \circ F^{\langle -1\rangle_\circ} = F^{\langle -1\rangle_\circ} \circ F = X.$$

Negative monomial powers, the branch label W_{-1}, inverse transforms, and time-(-1) iterates are not instances of either replacement merely because they contain the glyph -1.

Near-identity inverse. For $F = X + A \in \mathcal{G}_{\mathrm{poly},K}$, there is a unique $F^{\langle -1\rangle_\circ} = X + B$ in the same group. The correction B is the unique fixed point

$$B = -A \circ (X + B).$$

Starting with $B_0 = 0$, the error gains at least one outer block per iteration because $D_X A \in tK[L][[t]]$.

Lagrange–Bürmann formula. The formal inverse is

$$(X + A)^{\langle -1\rangle_\circ}(X) = X + \sum_{r\geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(A(X)^r)\,.$$

The sum starts at $r = 1$, where the derivative order is zero. The r-th summand has outer order at least $r - 1$, which makes the family formally summable.

Coefficient formula. If

$$A^r = \sum_{m\geq 0} A_{r,m}(L)t^m, \qquad F^{\langle -1\rangle_\circ} = X + \sum_{N\geq 0} b_N(L)t^N,$$

then

$$b_N(L) = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_L - s) \right] A_{r,N-r+1}(L).$$

When the lower product bound exceeds the upper bound, in particular at $r = 1$, the product is empty and acts as the identity.

Residual with all arguments explicit. For an ambient map F and a candidate inverse H, define

$$\mathrm{Err}_{\mathrm{rev}}(F, H) := F \circ H - X.$$

A report-local one-argument macro may hide F in a subscript; shared prose must supply both F and H because the residual is not determined by H alone.

Finite certificate. For

$$G_N = X + \sum_{j=0}^{N} b_j(L)t^j,$$

G_N is a right inverse through outer block N exactly when

$$F \circ G_N - X = \mathcal{O}_t^{\mathrm{form}}\left(t^{N+1}\right).$$

The left certificate is

$$G_N \circ F - X = \mathcal{O}_t^{\mathrm{form}}\left(t^{N+1}\right).$$

An exact one-sided inverse in the group is two-sided by uniqueness. At finite precision both residuals are useful because they detect different substitution-order mistakes.

Triangular correction. If

$$F \circ G_N - X = r_N(L)t^N + \mathcal{O}_t^{\mathrm{form}}\left(t^{N+1}\right),$$

then

$$G_{N+1} = G_N - r_N(L)t^N$$

removes the residual at level N when the leading slope is 1. For a leading slope $a \in K^\times$, divide the correction by a.

Newton reversion. Newton's update is

$$G_{\mathrm{new}} = G - \frac{F \circ G - X}{(D_X F) \circ G}.$$

For $F = X + A$, the denominator is $1 + \mathcal{O}_t^{\mathrm{form}}(t)$ and is therefore a unit. The residual order approximately doubles when all intermediate arithmetic carries sufficient guard precision.

Analytic residual transfer. If f is continuously differentiable and one-to-one on an interval, $x_* = f^{\langle -1\rangle\circ}(y)$, and $|f'(x)| \geq m > 0$ between x_N and x_*, then

$$|x_N - x_*| \leq \frac{|f(x_N) - y|}{m}.$$

This analytic theorem requires injectivity and a derivative lower bound; a formal residual alone does not prove existence of an analytic inverse branch.

Wright omega, the canonical Lambert polynomials, and Stirling coefficients

Commands. The shared commands are `\WrightOmega`, `\LambertPolynomial`, and `\ScaledLambertPolynomial`. The polynomial command takes only an index; its evaluation argument is written after the command.

Real Wright omega. The map $y \mapsto y + \log y$ is a bijection from $(0, \infty)$ to $\mathbb{R}$. Its real inverse is

$$\omega(x) + \log \omega(x) = x, \qquad \omega(x) = W_0(\mathrm{e}^x).$$

The real function is defined for every $x \in \mathbb{R}$; the polynomial-log expansion below concerns $x \to +\infty$. Complex Wright omega requires an unwinding-number or analytic-continuation convention.

Canonical zero-based family. For $n \in \mathbb{N}_0$, define

$$\mathsf{L}_n^{\mathrm{Lam}}(u) := \frac{(-1)^{n+1}}{(n+1)!}\left[\prod_{j=0}^{n-1}(\partial_u - j)\right]u^{n+1}.$$

At $n = 0$ the product is empty, so

$$\mathsf{L}_0^{\mathrm{Lam}}(u) = -u.$$

For $n \geq 1$, the first values are

$$\mathsf{L}_1^{\mathrm{Lam}}(u) = u, \qquad \mathsf{L}_2^{\mathrm{Lam}}(u) = \frac{u^2}{2} - u, \qquad \mathsf{L}_3^{\mathrm{Lam}}(u) = \frac{u^3}{3} - \frac{3}{2}u^2 + u.$$

Omega expansion. With $L = \log x$,

$$\omega(x) \sim x + \sum_{n\geq 0}\frac{\mathsf{L}_n^{\mathrm{Lam}}(L)}{x^n} = x - L + \sum_{n\geq 1}\frac{\mathsf{L}_n^{\mathrm{Lam}}(L)}{x^n}.$$

The two displays are identical. The former is the canonical zero-based normalization; the latter merely separates the $n = 0$ logarithmic block.

Recurrence and boundary values. For $n \in \mathbb{N}_0$,

$$\mathsf{L}_{n+1}^{\mathrm{Lam}}(u) = -\mathsf{L}_n^{\mathrm{Lam}}(u) + n \int_0^u \mathsf{L}_n^{\mathrm{Lam}}(v)\ \mathrm{d}v.$$

Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathsf{L}_{n+1}^{\mathrm{Lam}}(u) = n\mathsf{L}_n^{\mathrm{Lam}}(u) - \frac{\mathrm{d}}{\mathrm{d}u}\mathsf{L}_n^{\mathrm{Lam}}(u), \qquad \mathsf{L}_n^{\mathrm{Lam}}(0) = 0.$$

For $n \geq 1$,

$$\deg \mathsf{L}_n^{\mathrm{Lam}} = n, \quad [u^n]\mathsf{L}_n^{\mathrm{Lam}} = \frac{1}{n}, \quad \frac{\mathrm{d}}{\mathrm{d}u}\mathsf{L}_n^{\mathrm{Lam}}(u)\bigg|_{u=0} = (-1)^{n-1}.$$

The statement $\deg \mathsf{L}_0^{\mathrm{Lam}} = 0$ would be false because $\mathsf{L}_0^{\mathrm{Lam}} = -u$ has degree 1; therefore the degree law is intentionally restricted to $n \geq 1$.

Signed and unsigned Stirling numbers. Define signed Stirling numbers of the first kind by

$$z^{\underline{n}} = z(z-1)\cdots(z-n+1) = \sum_{k=0}^{n} s(n,k)z^k.$$

The $n = 0$ product is empty, so $s(0,0) = 1$. Set $s(n,k) = 0$ for $k < 0$ or $k > n$. Then

$$s(n+1,k) = s(n,k-1) - ns(n,k), \qquad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k}s(n,k).$$

The bracket glyph here denotes an unsigned cycle number only when its two arguments are integer indices; it is neither coefficient extraction nor a Gaussian binomial coefficient.

Explicit coefficients. For $n \geq 1$,

$$\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n} (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} \frac{u^m}{m!}.$$

If $\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n} \lambda_{n,m}u^m$, then

$$\lambda_{n,m} = (-1)^{n-m}\frac{1}{m!}\begin{bmatrix} n \\ n-m+1 \end{bmatrix},$$

with $\lambda_{n,0} = \lambda_{n,n+1} = 0$ and $\lambda_{1,1} = 1$.

Scaled family. For $n \geq 1$,

$$\mathsf{L}_n^{\mathrm{Lam,sc}}(u) := n!\mathsf{L}_n^{\mathrm{Lam}}(u) \in \mathbb{Z}[u].$$

Its recurrence is

$$\mathsf{L}_{n+1}^{\mathrm{Lam,sc}}(u) = -(n+1)\mathsf{L}_n^{\mathrm{Lam,sc}}(u) + n(n+1)\int_0^u \mathsf{L}_n^{\mathrm{Lam,sc}}(v)\ \mathrm{d}v.$$

The scaled family is a storage normalization; it does not define a different asymptotic coefficient sequence.

Generating equation. If

$$Q(s,u) = \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(u)s^n,$$

then $Q(0,u) = 0$ and

$$Q = -\log(1 - su + sQ).$$

This equation determines the positive-index coefficient family uniquely in the formal power-series ring $K[u][[s]]$.

Lambert W branches, lower-branch scaling, and analytic status

Large complex branch coordinates. Fix an integer branch label k, a branch of the logarithm, and a sector on which

$$\xi_k := \operatorname{Log} z + 2\pi \mathrm{i} k, \qquad \ell_k := \log_{\mathcal{B}_k}(\xi_k)$$

are defined and $|\xi_k| \to \infty$. Then the formal large-branch expansion is

$$W_k(z) \sim \xi_k - \ell_k + \sum_{n\ge 1} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n} = \xi_k + \sum_{n\ge 0} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n}.$$

The polynomial family is branch-independent; the coordinates ξ_k, ℓ_k, their branches, and the analytic domain are not.

Formal versus analytic assertion. A finite polynomial sum in ξ_k^{-1} is a formal truncation until an analytic theorem supplies an estimate such as

$$W_k(z) - \left(\xi_k - \ell_k + \sum_{n=1}^{N} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n}\right) = \mathcal{O}_{|\xi_k|\to\infty,\ \xi_k\in S_k}\left(\ell_k^{N+1}/\xi_k^{N+1}\right).$$

The sector S_k, distance from branch cuts, and compatibility of the two logarithms are hypotheses. A formal residual does not by itself establish a uniform analytic constant.

Residual correction. For

$$w_N = \xi_k - \ell_k + \sum_{n=1}^{N} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n}, \qquad R_N = w_N + \log_{\mathcal{B}_k}(w_N) - \xi_k,$$

one Newton correction is

$$\delta_N = -\frac{R_N}{1 + 1/w_N}.$$

Taylor's theorem transfers a sufficiently small residual to an inverse error only in a region avoiding $w = 0$ and the chosen logarithmic cut.

Principal real branch. For $z \to +\infty$, take $\xi_0 = \log z > 0$ and $\ell_0 = \log\log z$ with the real logarithm. This specialization yields the familiar principal-branch expansion of $W_0(z)$.

Lower real branch near zero. The report-6 local symbol

$$\varepsilon_{\mathrm{br}} > 0, \qquad \varepsilon_{\mathrm{br}} \to 0$$

is a branch-point-independent small parameter for $W_{-1}(-\varepsilon_{\mathrm{br}})$. Put

$$S = \log(1/\varepsilon_{\mathrm{br}}), \qquad L = \log S.$$

Then the large variable is S, and the expansion of the negative branch is obtained from the inverse of $u \mapsto u - \log u$. Its alternating signs do not define a second Lambert-polynomial family.

Square-root branch point. At $z = -\mathrm{e}^{-1}$, the derivative of $w \mapsto w\mathrm{e}^{w}$ vanishes at $w = -1$; the inverse is therefore not an ordinary power-log series in $z + \mathrm{e}^{-1}$. A local coordinate such as

$$p = \sqrt{2(1 + \mathrm{e}z)}$$

produces a Puiseux series with the sign distinguishing the two local branches. The square-root branch and continuation path must be stated.

Convergence boundary. The polynomial recurrence and every finite residual identity are formal algebra. Convergence or asymptotic validity of the infinite series is a separate analytic claim. A statement of analytic validity must name the branch, limiting sector or ray, and remainder estimate; the notation catalogue does not promote an unverified source claim to a theorem.

Precision records, epsilon roles, and collision policy in the six reports

Approximation marker. The report-6 local command `\ApproximationOf{`F`}` prints

$$F_{\mathrm{approx}}.$$

It denotes a declared approximate or truncated representative of F; it does not specify the cutoff, error norm, or formal valuation. Those data must be supplied separately. It is not a hat accent and is not synonymous with an estimator unless a statistical model is declared.

Inclusive truncation marker. The same report's local `\trunc{`F`}{`N`}` prints $[F]_{\le N}$ and is inclusive. Other reports use exclusive $\mathrm{Trunc}_{<N} F$. The relation is $[F]_{\le N} = \mathrm{Trunc}_{<N+1} F$.

Nested inverse-log scale. For

$$L_0 = X, \qquad L_{j+1} = \log L_j,$$

the report-6 local symbol

$$\varepsilon_{\log,j} := L_j^{-1}$$

is the reciprocal of the j-th iterated logarithm. Its subscript $\log, j$ is semantic and must not be dropped when other epsilon roles coexist.

Lower-branch epsilon. The symbol $\varepsilon_{\mathrm{br}}$ is the positive small argument in $W_{-1}(-\varepsilon_{\mathrm{br}})$, with $\varepsilon_{\mathrm{br}} \to 0^+$. It is not an error tolerance.

Newton residual scale. The symbol $\varepsilon_{\mathrm{Newt}}$ denotes a local residual scale in an analytic Newton estimate. Under a nonvanishing derivative, one normalized Newton step changes a first-order residual scale to a quadratic scale $\varepsilon_{\mathrm{Newt}}^2$. This is neither the branch parameter nor an inverse-log generator.

Coefficient bracket collision. The glyph $[\,\cdot\,]$ has four unrelated roles in the broader corpus:

$$[t^n]F, \qquad [F]_{\le N}, \qquad \begin{bmatrix} n \\ k \end{bmatrix}, \qquad \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

They mean coefficient extraction, inclusive truncation, an unsigned Stirling cycle number, and a Gaussian binomial coefficient, respectively. Argument types and semantic macros disambiguate them.

Letter L collision. In this namespace $L = \log X$ is a logarithmic coordinate. The Lambert family is always written $\mathsf{L}_n^{\mathrm{Lam}}$, never as a bare L_n. A local pair $L_1 = \log z$, $L_2 = \log L_1$ denotes iterated logarithms, not the first two Lambert polynomials.

Generic versus Lambert coefficients. A generic coefficient block is $p_n(L)$. The distinguished Lambert coefficient is $\mathsf{L}_n^{\mathrm{Lam}}(L)$. The bare family P_n is not globally reserved because several source reports used it for both roles.

Outer t versus q-series base. The polynomial-logarithmic outer coordinate is $t = X^{-1}$. A free q is reserved for a genuine q-series base when q-series notation is present. A bound summation index, rational exponent, or locally declared scalar may still be named q, but it acquires no q-series meaning from the letter alone.

Inverse collision. A reciprocal is F^{-1}; a compositional inverse is $F^{\langle -1 \rangle_\circ}$. The glyph $F^{[-1]}$ may denote a time-(-1) iterate only in an explicitly declared iteration context and must not be silently identified with reversion.

Derivative collision. Write D_X, ∂_L, ∂_t, or ∂_u according to the differentiated variable. The formal partial ∂_t holds L fixed; it is not the analytic derivative after $L = -\log t$.

Class-name collision. The six reports formerly assigned the same bare calligraphic letters to different rings. Shared text uses the exact semantic commands:

command	defined class
$\mathscr{PL}_{\mathrm{pow},K}$	$K[L][[t]]$
$\mathscr{PL}_{\mathrm{poly},K}$	$K[L]((t))$
$\mathscr{PL}_{\mathrm{rat},K}$	$K(L)((t))$
$\mathscr{PL}_{\mathrm{itLaur},K}$	$K((L^{-1}))((t))$
$\mathcal{G}_{\mathrm{poly},K}$	$\{X + A : A \in K[L][[t]]\}$.

A document-local shorthand may expand to one of these commands, but the bare glyph has no corpus-wide type.
Sparse implementation record. A finite one-log representative is a sparse map $n \mapsto p_n(L)$ together with:

- the smallest and largest retained outer exponents;
- the coefficient field and logarithmic-degree policy;
- any inner cutoff for Laurent-log coefficients;
- branch and exponent-group metadata;
- the declared formal tail or analytic error certificate.

No finite record stores the infinite object itself.

Lagrange and cardinal interpolation

Nodes. Let $x_0, \dots, x_N$ be pairwise distinct scalars. The cardinal polynomial at node j is

$$\ell_j(x) = \prod_{\substack{0 \le k \le N \\ k \ne j}} \frac{x - x_k}{x_j - x_k}, \qquad \ell_j(x_k) = \delta_{jk}.$$

Interpolant. $I_N f(x) = \sum_{j=0}^{N} f(x_j)\ell_j(x)$ is the unique polynomial of degree at most N matching f at all nodes.
Empty case. For $N = 0$, the product is empty and $\ell_0 = 1$. There is no $N = -1$ interpolant unless a separate empty-data convention is introduced.
Document-local Fabius geometric-Newton coefficient. In the geometric-comb synthesis volume, `\FabiusGeometricNewtonCoefficient {k}{q}` has TeX arity two and prints $\mathsf{c}_k^{\mathrm{F,Newt}}(q)$. Its mathematical arguments satisfy $k \in \mathbb{N}_0$ and $q \in (0, 1)$. With the bounded Fabius function F_{bd} and the ordered, pairwise distinct nodes $1, q, \dots, q^k$, it is the divided difference

$$\mathsf{c}_k^{\mathrm{F,Newt}}(q) \coloneqq [F_{\mathrm{bd}}; 1, q, \dots, q^k]$$

and hence the coefficient of

$$N_k(x; 1, q) \coloneqq \prod_{r=0}^{k-1} (x - q^r)$$

in the Newton interpolant. At $k = 0$ the product is empty, $N_0 = 1$, and $\mathsf{c}_0^{\mathrm{F,Newt}}(q) = F_{\mathrm{bd}}(1) = 1$. The exact coefficient table in that volume specializes the second argument to $q = 1/2$; retaining q in the command distinguishes the general geometric comb from this half-base table. The subscript is a Newton-basis degree, not a valuation, and the command is document-local rather than a shared Fabius API.
Generic Newton Lean crosswalk. In `NewtonInterpolation.lean`, `newtonCoeff` and `newtonPoly` define the triangular Newton coefficients and their degree-n polynomial over an arbitrary field. The theorem `eval_newtonPoly` proves interpolation at node i when the product of its differences from the earlier nodes is nonzero; `newtonPoly_eq_interpolate` and `newtonCoeff_eq_sum` use injectivity on the finite node range to identify the Lagrange interpolant and the divided-difference sum. For geometric nodes, `nodal_range_pow` proves the q-Pochhammer form of the Newton basis under $q \neq 0$, `prod_erase_pow_sub_pow` evaluates the nodal denominator for $j \le k$, and `newtonCoeff_pow_eq_sum` proves the explicit coefficient sum under injectivity of $j \mapsto q^j$ on the required range. Instantiating the generic data by $v(j) = q^j$ and $y(j) = F_{\mathrm{bd}}(q^j)$ gives the mathematical object printed by `\FabiusGeometricNewtonCoefficient`; there is no separate Lean declaration

with that document-local command name. Lean's definitions remain total when a denominator vanishes, while the interpolation and divided-difference theorems retain their nonvanishing or injectivity hypotheses. Status: **Lean-proved at the generic level with the stated hypotheses**.
Barycentric weights. $w_j = (\prod_{k \neq j}(x_j - x_k))^{-1}$ and $I_N f(x) = (\sum_j w_j f(x_j)/(x - x_j))/(\sum_j w_j/(x - x_j))$ away from nodes, with the value at a node assigned by continuity. Scaling every w_j by the same nonzero constant changes neither interpolant.
Cardinal infinite grids. An infinite cardinal function L satisfying $L(k) = \delta_{0k}$ for $k \in \mathbb{Z}$ is a different object and requires a function space and convergence mode for $\sum_k f(k)L(x-k)$.
Finite Rvachev decoder. Let s be a finite index set, $v : s \to \mathbb{R}$ a node map, and ℓ_i the Lagrange basis polynomial for $i \in s$. With $\mathscr{D}_{\text{up}}$ the Rvachev deconvolution operator, $M \in \mathbb{N}_{>0}$, and $k \in \mathbb{Z}$, define

$$D_{ki}^{s,v,M} := (\mathscr{D}_{\text{up}}\ell_i)(k/M).$$

For nodal data $y : s \to \mathbb{R}$, the corresponding atom coefficient is

$$a_k^{s,v,y,M} := \frac{1}{M}\sum_{i \in s} D_{ki}^{s,v,M} y_i.$$

The superscripts display the finite node family, node locations, data, and mesh; a source that suppresses them must declare those objects in the immediately governing scope.
Exact finite synthesis. For a bounded Fabius object F satisfying the Fabius specification, $i \in s$, $M \neq 0$, $|s| - 1 \leq \nu_2(M)$, and $x \in [-1, 1]$,

$$\frac{1}{M}\sum_{-2M<k<2M} D_{ki}^{s,v,M}\ \mathrm{up}_F(x - k/M) = \ell_i(x).$$

The index set is the open integer interval $-2M < k < 2M$, including neither endpoint. Summing against y_i reconstructs the complete Lagrange interpolant from the coefficients $a_k^{s,v,y,M}$. Node injectivity is unnecessary for the polynomial reconstruction identity but is required for the Kronecker-delta cardinal interpretation; the unit decoder-row-sum theorem additionally assumes $s \neq \varnothing$.
Lean crosswalk. `lagrangeRvachevDecoder` and `lagrangeRvachevAtomCoefficient` in `LagrangeRvachevSynthesis.lean` implement the two displayed definitions. The exact reconstruction, node-evaluation, coefficient/deconvolution, full-interpolation, and row-sum results are the named theorems

```
normalized_sum_Ioo_lagrangeRvachevDecoder_mul_shifted_rvachevUp
normalized_sum_Ioo_lagrangeRvachevDecoder_eval_node
lagrangeRvachevAtomCoefficient_eq_deconvolved_interpolate
sum_Ioo_lagrangeRvachevAtomCoefficient_mul_shifted_rvachevUp
sum_lagrangeRvachevDecoder_eq_one
```

Lean's definitions remain total at $M = 0$ through field division; the reconstruction theorem requires $M \neq 0$. No optimized decoder, bundled matrix inverse, Gaussian q-binomial closed form, or minimum-variation solution is supplied. Status: **Lean-proved with the displayed hypotheses**.

Sampling, reconstruction, and band limitation

Band limitation. Under the 2π Fourier convention, f is bandlimited to $[-B, B]$ when $\mathrm{supp}\left(\widehat{f}_{2\pi}\right) \subseteq [-B, B]$.

Critical lattice. The Shannon lattice spacing is $(2B)^{-1}$ for $B > 0$. With suitable L^2 or regularity hypotheses,

$$f(x) = \sum_{k\in\mathbb{Z}} f\left(\frac{k}{2B}\right) \operatorname{sinc}_\pi(2Bx - k).$$

The theorem states whether convergence is in L^2, uniform, or pointwise.
Oversampling. A denser lattice adds frame redundancy; it does not change the definition of `\SincPi`.
Fabius caveat. Compact support in physical space makes the Fourier transform entire, not bandlimited. Sampling formulas in the inverse-Fabius frontier therefore require an independently proved interpolation or frame theorem.
Alias policy. A sample index k, a frequency index, and a Lambert branch index may all be integers; each sum declares which role its index has.

Refinement equations and dilation operators

Dilation. For nonzero real a, $(D_a f)(x) = f(ax)$. A unitary L^2 dilation is instead $(U_a f)(x) = |a|^{1/2} f(ax)$; the two are not conflated.
Translation. $T_b f(x) = f(x - b)$. Thus $D_a T_b = T_{b/a} D_a$ for $a \neq 0$ under this convention.
Refinement equation. A refinable function satisfies

$$\phi(x) = \sum_{k\in\mathbb{Z}} h_k \phi(ax - k)$$

with dilation $|a| > 1$, a declared mask (h_k), and a stated convergence/pointwise interpretation. A factor such as $\sqrt{|a|}$ or $|a|$ is part of the mask normalization and is never implicit.
Mask symbol. $H(z) = \sum_k h_k z^k$ is a Laurent polynomial or series according to the mask support. Its Fourier-domain refinement law depends on the selected Fourier normalization.
Fabius scaling. The equation $F_{\mathrm{gl}}'(x) = 2F_{\mathrm{gl}}(2x)$ is a differential scaling law, not by itself a standard refinability equation for F_{gl}.

Function iterates, orbits, and conjugacy

Iterate. For a self-map $f : X \to X$, $f^{\circ 0} = \mathrm{id}_X$ and $f^{\circ(n+1)} = f \circ f^{\circ n}$. The circle distinguishes iteration from pointwise power $f(x)^n$.
Orbit. The forward orbit of x is $\mathcal{O}_f^+(x) = \{f^{\circ n}(x) : n \in \mathbb{N}_0\}$.
Periodic point. x has least period $p \in \mathbb{N}_{>0}$ when $f^{\circ p}(x) = x$ and no smaller positive exponent does. A fixed point has period one.
Conjugacy. Maps $f : X \to X$ and $g : Y \to Y$ are conjugate through a bijection $h : X \to Y$ when $h \circ f = g \circ h$. If h is only a surjection, the relation is a semiconjugacy.
Fabius inverse. Iterations of `\FabiusQuantile` stay in $[0, 1]$. Iterations of the clamped inverse on all reals are a different dynamical system and use `\FabiusClampedQuantile` explicitly.

Koopman and transfer operators

Koopman. For a measurable map $T : X \to X$, $(\mathcal{K}_T f)(x) = f(Tx)$. Its function space and measure are part of the operator type.
Perron–Frobenius. A transfer operator $\mathcal{P}_T$ is defined by the duality

$$\int (\mathcal{P}_T f) g \, \mathrm{d}\mu = \int f(g \circ T) \, \mathrm{d}\mu$$

when such a density operator exists. It is not automatically pointwise inverse composition.

Invariant measure. μ is T-invariant when $T_{\#}\mu = \mu$, equivalently $\int f \circ T \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu$ for bounded measurable f.
Spectrum. A Koopman eigenfunction satisfies $\mathcal{K}_T f = \lambda f$. Spectral claims state the Banach or Hilbert space because the spectrum changes with it.
Status. Fabius Koopman models in the frontier corpus are **frontier/conjectural** unless an operator, invariant measure, and proved spectral statement are all linked.

Stein operators and zero-bias transforms

Stein operator. An operator $\mathcal{A}$ characterizes a law μ on a test class $\mathcal{T}$ when $\int \mathcal{A}f \, \mathrm{d}\mu = 0$ for every $f \in \mathcal{T}$, together with a uniqueness theorem. The operator alone is not a characterization without the class.
Stein equation. For a test h, $\mathcal{A}f_h = h - \int h \, \mathrm{d}\mu$. A bound on f_h specifies its norm and the class of h.
Zero bias. For mean-zero X with variance $\sigma^2 > 0$, a zero-biased variable X^* satisfies

$$\mathbb{E}[Xf(X)] = \sigma^2 \mathbb{E}[f'(X^*)]$$

for an appropriate class of absolutely continuous f.
Iteration. A zero-bias tower declares how centering and variance renormalization are applied at every stage; the star is a distributional transform, not complex conjugation.
Status. Proposed Fabius Stein characterizations and zero-bias towers retain **frontier/conjectural** status until uniqueness and domain conditions are proved.

Carleman linearization and Carleman matrices

Analytic map. Let $f(z) = \sum_{n\geq 0} a_n z^n$ near a fixed expansion point. The Carleman matrix is

$$C(f)_{mn} \coloneqq [z^n] f(z)^m, \qquad m, n \in \mathbb{N}_0.$$

Zeroth row. Since $f(z)^0 = 1$, $C(f)_{0n} = \delta_{0n}$, including the convention $0^0 = 1$ inside the polynomial identity.
Composition. Formally, $C(f \circ g) = C(f)C(g)$ when both sides are defined with compatible expansion points and the infinite matrix product is coefficientwise finite or convergent.
Truncation. $C_N(f) = (C(f)_{mn})_{0\leq m,n<N}$ is a finite truncation, not the exact Carleman operator. Spectral convergence of truncations requires a theorem.
Collision. This Carleman matrix is distinct from the Carleman condition $\sum_n m_{2n}^{-1/(2n)} = \infty$ for moment determinacy.

Holonomic and D-finite functions

D-finite. A formal power series $f(z)$ over a characteristic-zero field is D-finite when it satisfies

$$p_r(z) f^{(r)}(z) + \cdots + p_0(z) f(z) = 0$$

for polynomials p_j, not all zero.
P-recursive. A sequence (a_n) is P-recursive when $\sum_{j=0}^{r} p_j(n) a_{n+j} = 0$ for all sufficiently large n, with polynomial coefficients $p_j(n)$.
Equivalence. Over characteristic zero, an ordinary generating function is D-finite exactly when its coefficient sequence is P-recursive, subject to the standard formal-series interpretation.

Nonholonomicity. Failure of one guessed differential equation does not prove nonholonomicity. A nonholonomicity claim identifies a rigorous obstruction such as infinitely many singularities or incompatible coefficient asymptotics.
Status. Fabius holonomicity questions in the active frontier reports are **open** unless a theorem is explicitly cited.

Chaos expansions, cumulants, and Edgeworth terms

Orthogonal chaos. A chaos decomposition $L^2(\Omega) = \bigoplus_{n\geq 0} \mathcal{H}_n$ is relative to a specified probability structure. J_n denotes the orthogonal projection onto $\mathcal{H}_n$ only within that scope.
Dyadic-chaos multi-index. In the dyadic-chaos report, $V_1, V_2, \ldots$ are independent uniform coordinates on $[-1, 1]$, and $\ell_n(v) = \sqrt{2n+1}\, P_n(v)$ is the orthonormal Legendre basis in $L^2([-1, 1], \mathrm{d}v/2)$. A tensor index is

$$\alpha = (\alpha_1, \alpha_2, \ldots), \qquad \alpha_j \in \mathbb{N}_0 \quad (j \geq 1).$$

Its support, total degree, and tensor basis element are, without suppressed bounds,

$$\mathrm{supp}(\alpha) := \{j \geq 1 : \alpha_j > 0\}, \qquad |\mathrm{supp}(\alpha)| < \infty, \qquad \|\alpha\|_1 := \sum_{j\geq 1} \alpha_j, \qquad \ell_\alpha(V) := \prod_{j\in\mathrm{supp}(\alpha)} \ell_{\alpha_j}(V_j).$$

The product is finite; equivalently one may include every $j \geq 1$ because $\ell_0 = 1$. The coefficients $c_\alpha(t)$ in $\mathrm{e}^{tX} = \sum_\alpha c_\alpha(t)\ell_\alpha(V)$ inherit exactly this index set. In finite-set monomial expansions the same letter denotes a finite multi-index $(\alpha_j)_{j\in S}$ and $|\alpha| = \sum_{j\in S} \alpha_j$, with the displayed summation bounds determining whether zero components are permitted. The letter α replaces the former ν-family so a tensor degree can never be read as a 2-adic valuation.
Cumulant generating function. For an MGF nonzero near zero,

$$K_X(t) = \log M_X(t) = \sum_{r\geq 1} \kappa_r(X) \frac{t^r}{r!}, \qquad K_X(0) = 0,$$

where the analytic logarithm is the branch normalized by the displayed value at zero.
Standardization. For variance $\sigma^2 > 0$, $Z = (X - \mathbb{E}X)/\sigma$. Standardized cumulants are $\lambda_r = \kappa_r(X)/\sigma^r$ for $r \geq 3$.
Edgeworth expansion. An order-s Edgeworth approximation identifies the normal density or CDF, Hermite-polynomial normalization, scaling parameter, and a uniform remainder regime. The formal polynomial correction alone is not a probability density theorem.
Hermite choice. Probabilists' Hermite polynomials satisfy $\mathrm{He}_n(x) = (-1)^n \mathrm{e}^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} \mathrm{e}^{-x^2/2}$. Physicists' polynomials are separately labelled H_n^{phys}.

Large deviations and rate functions

Log-MGF. $\Lambda(t) = \log \mathbb{E}[\mathrm{e}^{tX}]$ on its effective domain $\mathcal{D}_\Lambda = \{t : M_X(t) < \infty\}$.
Legendre–Fenchel transform. $\Lambda^*(x) = \sup_{t\in\mathbb{R}}\{tx - \Lambda(t)\} \in [0, \infty]$.
LDP. A sequence X_n obeys a large-deviation principle with speed $a_n \to \infty$ and rate I when closed-set upper and open-set lower logarithmic bounds are stated explicitly.
Good rate function. I is good when every sublevel set $\{x : I(x) \leq c\}$ is compact. Convexity is not part of the definition of a general rate function.
Endpoint scaling. An endpoint or small-ball asymptotic for $F_{\mathrm{bd}}(x)$ as $x \downarrow 0$ is not called an LDP until a family, speed, and normalization have been identified.

Entropy, relative entropy, and Fisher information

Differential entropy. For a density f, $h(f) = -\int f(x) \log f(x)\, \mathrm{d}x$ when the positive and negative parts are well defined, with $0 \log 0 = 0$ by continuity. It can be negative.
Relative entropy. For probability measures $\mu \ll \nu$,

$$D_{\mathrm{KL}}(\mu \| \nu) = \int \log\left(\frac{\mathrm{d}\mu}{\mathrm{d}\nu}\right) \mathrm{d}\mu,$$

and it is $+\infty$ when absolute continuity fails or the integral diverges. The order of the arguments matters.
Fisher information. For a differentiable positive density, $J(f) = \int |(\log f)'|^2 f\, \mathrm{d}x = \int |f'|^2/f\, \mathrm{d}x$ when finite. Zeros require a weak or extended-value interpretation.
Mutual information. $I(X;Y) = D_{\mathrm{KL}}(\mathrm{Law}((X,Y)) \,\|\, \mathrm{Law}(X) \otimes \mathrm{Law}(Y))$.
Status. Numerical entropy or Fisher-information tables are **computational evidence** unless accompanied by verified error bounds.

Minkowski sums and zonoids

Minkowski sum. For $A, B \subseteq \mathbb{R}^d$, $A + B = \{a + b : a \in A,\ b \in B\}$ and $\lambda A = \{\lambda a : a \in A\}$.
Zonotope. A finite zonotope generated by vectors $v_1, \ldots, v_N$ is

$$Z_N = \sum_{j=1}^{N} [0, v_j] = \left\{ \sum_{j=1}^{N} t_j v_j : 0 \le t_j \le 1 \right\}.$$

Zonoid. A zonoid is a Hausdorff limit of zonotopes. A random-vector representation such as $\mathbb{E}[0, X]$ requires integrability and a declared selection/Aumann-expectation convention.
Support function. $h_K(u) = \sup_{x \in K} \langle u, x \rangle$ for a nonempty compact convex set K. It satisfies $h_{A+B} = h_A + h_B$.
Common-digit construction. When coordinates share binary or uniform digits, the joint random series and ambient dimension are declared. Marginal Fabius laws alone do not determine the zonoid or dependence structure.

Radon transforms and projections

Hyperplane Radon transform. For suitable $f : \mathbb{R}^d \to \mathbb{C}$,

$$(\mathcal{R}f)(\theta, s) = \int_{\{x : \langle x, \theta \rangle = s\}} f(x)\, \mathrm{d}\mathcal{H}^{d-1}(x), \quad \theta \in \mathbb{S}^{d-1},\ s \in \mathbb{R}.$$

Redundancy. $\mathcal{R}f(-\theta, -s) = \mathcal{R}f(\theta, s)$. A parameter domain may quotient this redundancy or retain it explicitly.
Measure projection. For a measure μ, the one-dimensional projection in direction θ is $(x \mapsto \langle x, \theta \rangle)_{\#}\mu$. Its density, when it exists, is related to the Radon transform but is not assumed.
Fourier slice. With compatible 2π conventions, $\widehat{(\mathcal{R}f)(\theta, \cdot)}_{2\pi}(\xi) = \widehat{f}_{2\pi}(\xi\theta)$, where the transform on the right is d-dimensional and uses exponent $-2\pi\mathrm{i}\langle x, \eta \rangle$.
Status. Dyadic Radon-profile identifications are **frontier/conjectural** unless their measure and slice normalizations are proved.

Heat traces, spectral zeta functions, and determinants

Eigenvalue list. For a nonnegative self-adjoint operator L with discrete spectrum, eigenvalues are $0 \le \lambda_0 \le \lambda_1 \le \cdots$, repeated by multiplicity.
Heat trace. $\Theta_L(t) = \mathrm{tr}(\mathrm{e}^{-tL}) = \sum_j \mathrm{e}^{-t\lambda_j}$ for $t > 0$ when trace class.
Spectral zeta. $\zeta_L(s) = \sum_{\lambda_j > 0} \lambda_j^{-s}$ initially in its convergence half-plane. Zero eigenvalues are excluded and their multiplicity is recorded separately.
Zeta determinant. If ζ_L is regular at zero, $\det{}' L = \exp(-\zeta_L'(0))$. The prime means zero modes are omitted; it is not an ordinary finite determinant.
Heat coefficients. An expansion $\Theta_L(t) \sim \sum_j a_j t^{\alpha_j}$ as $t \downarrow 0$ specifies the exponent order, possible logarithmic terms, and whether the expansion is proved or numerically fitted.
Digital spectrum. A digital or dyadic spectrum defined from a sequence or matrix is not automatically the spectrum of a self-adjoint operator; the construction must supply the operator and Hilbert space.

Noncommutative probability and free cumulants

Noncommutative probability space. A pair $(\mathcal{A}, \tau)$ consists of a unital algebra and a unital linear functional τ; positivity and traciality are additional hypotheses, not automatic.
Moments. For $X \in \mathcal{A}$, $m_n = \tau(X^n)$. Products preserve order; XY need not equal YX.
Free cumulants. Free cumulants (r_n) are defined by

$$m_n = \sum_{\pi \in NC(n)} \prod_{B \in \pi} r_{|B|},$$

where $NC(n)$ is the lattice of noncrossing partitions of $\{1, \ldots, n\}$ and the scalar formula assumes a single variable.
Freeness. Subalgebras are free when alternating centered products from different subalgebras have zero τ. It is not classical independence.
Status. Noncommutative Fabius analogues in the frontier corpus are **proposed definitions** or **conjectural**; they are never crosswalked to the commutative Lean object without an explicit construction.

Matrix dilations and multivariate scaling

Expansive matrix. A real $d \times d$ matrix A is expansive when every eigenvalue has modulus greater than one.
Dilation operator. For $f : \mathbb{R}^d \to \mathbb{C}$, $(D_A f)(x) = f(Ax)$. The unitary L^2 version is $(U_A f)(x) = |\det A|^{1/2} f(Ax)$.
Lattice mask. A matrix refinement equation has the form $\phi(x) = \sum_{k \in \mathbb{Z}^d} h_k \phi(Ax - k)$ with a declared mask and convergence mode.
Fourier scaling. Under the d-dimensional 2π convention, $\widehat{f(A\,\cdot)}_{2\pi}(\xi) = |\det A|^{-1} \widehat{f}_{2\pi}(A^{-\mathsf{T}}\xi)$.
Scalar specialization. For $d = 1$ and $A = (2)$ this reduces to the familiar dyadic scaling. A matrix theorem does not follow from the scalar theorem without controlling anisotropy and lattice geometry.

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16 Namespace governance and collision register

The shared commands above are global semantic names. Every other symbol is local and is governed by the rules in this section. A local declaration is not optional merely because a letter is conventional: the active corpus combines probability, analysis, arithmetic, special functions, approximation theory, spectral theory, and proposed frontier constructions, so nearly every conventional letter has several legitimate meanings.

16.1 Scope hierarchy and declaration schema

The narrowest enclosing declaration wins among local names. Scopes, from narrowest to widest, are: a displayed definition, a theorem/lemma/conjecture environment, an algorithm or table, a subsection explicitly headed "Notation", a section, and finally one standalone document. A fragment included with `\input` does not create an independent preamble or global namespace; it inherits the standalone root's shared input and may introduce only locally scoped names. A local meaning ends at the closing environment or section boundary unless a later sentence explicitly renews it.

Every local notation declaration supplies the following data. "Not applicable" is a permitted answer; silence is not.

Field	Required content
Kind	Scalar, set, proposition, sequence, function, measure, matrix, operator, formal series, equivalence class, or other fully named mathematical kind.
Type	Domain and codomain, including scalar field, measurable/topological structure, and whether equality is pointwise or modulo an equivalence relation.
Indices	Complete index set, origin, inclusivity of endpoints, and empty-index behavior.
Parameters	Which quantities are fixed and varying; admissible domains, inequalities, integrality, primality, nonvanishing, and branch indices.
Formula	Exact definition with every sum, product, integral, supremum, and infimum bound.
Normalization	Fourier frequency/sign, measure mass, polynomial leading coefficient, orthogonality scale, logarithm/power branch, and any factorial or transform prefactor.
Exceptional cases	Values at zero, endpoints, removable singularities, empty sets, empty matrices, divergent parameters, and nonexistence cases.
Status	Definition, Lean-proved, classical, frontier-derived, conjectural, refuted, historical alias, or numerical evidence.
Scope end	The environment, section, or document boundary after which the letter is available for reassignment.

16.2 High-frequency letter collisions

This register does not ban ordinary local letters. It records the qualification needed when two meanings can meet in one document or in cross-document synthesis.

Glyph	Meanings found in the active corpus	Normative resolution
F	Bounded Fabius function, signed global extension, CDF, Fourier transform, polynomial family, matrix, and optimization functional.	Use F_{bd} and F_{gl} for the two Fabius functions, normalization-labelled Fourier commands for transforms, and $\mathrm{Obj}_{\mathrm{decay},n}$ for the decay objective. A local F_a, F_m, or matrix F_N retains its typed subscript and declaration.
Q	Rational numbers, Fabius quantile, quadrature rule, polynomial, matrix, and a q-dependent quantity.	Use $\mathbb{Q}$ for the number field and Q_F or $Q_{F,\mathrm{cl}}$ for inverses. Quadrature is Q_N only inside its declared numerical-analysis scope; the base of a q-series is lower-case q and always explicit.

Glyph	Meanings found in the active corpus	Normative resolution
N	Natural-number type, positive-integer type, truncation level, matrix size, node count, and asymptotic parameter.	Use $\mathbb{N}_0$ or $\mathbb{N}_{>0}$ for sets. A local N carries a declaration such as $N \in \mathbb{N}_{>0}$ and one role; zero-based and one-based counts are never inferred.
A	Event, matrix, Appell family, algebra, feasible set, and dyadic filtration.	Use $\mathcal{A}_n$ for the canonical dyadic filtration. An event is typed $A \in \mathscr{A}$, a matrix has dimensions, and an Appell polynomial has a degree subscript and preferably a semantic superscript.
D	Derivative, differential operator, dilation, denominator, determinant-derived sequence, and domain.	The upright differential is d; derivatives are f' or $D^n f$ after declaring D. Dilations use D_a or D_A, and dyadic denominators retain the superscript D_n^{dyad}.
E	Expectation, event, energy, error, and exponential generating function.	Use $\mathbb{E}$ for expectation. Events are roman-labelled or typed in the event σ-field; energies and errors retain descriptive subscripts; an EGF is labelled “egf”.
G	Cauchy transform, generating function, Gram matrix, graph, group, and inverse alias in historical drafts.	A transform is G_μ, a Gram matrix is G_N or G_X, and a group is typed. The Fabius inverse is never bare G; use the quantile commands.
H	Hilbert space, Hankel matrix, Hermite polynomial, Hessian, entropy, and transfer function.	Use $\mathcal{H}$ for a Hilbert/RKHS space, $H_N(\mu)$ for a finite Hankel matrix, He_n or H_n^{phys} for the two Hermite normalizations, and $\nabla^2 f$ for a Hessian.
I	Identity, indicator, interpolant, interval, information, and inverse.	Use id for an identity map and $\mathbf{1}_A$ for an indicator. Interpolants carry a degree such as I_N, intervals print endpoints, and neither Fabius inverse is denoted by bare I.
K	Kernel, compact set, number field, Koopman operator, and cumulant generating function.	A kernel is $K(x, y)$, a compact set remains $K \subseteq X$, the scalar field is $\mathbb{K}$, the Koopman operator is $\mathcal{K}_T$, and the cumulant generator is $K_X(t)$ only inside its probability scope.
L	Laplace transform, Lagrange/cardinal function, linear operator, likelihood, and slowly varying factor.	Use $\mathcal{L}$ for Laplace. Cardinal functions carry an index, operators declare domain/codomain, and slowly varying functions are labelled L_{sv} when transforms are nearby.
M	Mellin transform, moment-generating function, matrix, moment cutoff, mask, and manifold.	Use $\mathcal{M}$ and M_X for the two transforms. Matrices carry dimensions; cutoffs and masks are locally typed.
P	Probability, polynomial, projection, transition kernel, Padé numerator, and partition.	Use $\mathbb{P}$ for probability. Polynomials carry degree indices, projections are P_V with their range, Markov kernels are $P(x, \mathrm{d}y)$, and Padé numerators are P_L only in a declared $[L/M]$ scope.
R	Real-number type, radius, residue, resolvent, recurrence polynomial, remainder, and Reshetnikov integer.	Use $\mathbb{R}$ and Res for the first two semantic roles. Resolvents are $R_T(z)$, remainders carry an error superscript, and the arithmetic integer is R_n^{Res}.
S	Sum, support, Stieltjes transform, saddle action, spline space, and random series.	Use $\mathrm{supp}(f)$ for support and S_μ for Stieltjes transforms. Saddle actions, spline spaces, and random series carry semantic subscripts and declared domains.
T	Translation, transformation, Toeplitz matrix, random time, Thue–Morse object, and truncation.	Translations are T_b, Toeplitz matrices are $T_N(a)$, and the sign sequence uses ε_n. A dynamical map $T : X \to X$ is typed at first use.
W	Lambert function, Wasserstein distance, Walsh function, and weight.	Use W_k with branch k and $W_p(\mu, \nu)$ with transport exponent p. Walsh functions and weights carry their own indices and domains.

Glyph	Meanings found in the active corpus	Normative resolution
ν, ν_j	Finite p-adic valuation, a signed or probability measure, an integer or complex order in a generalized q-series, a fractional order, and formerly several local indexed families.	Valuations use ν_p or ν_2 and state whether the operand is a nonzero integer or rational and whether zero is undefined, extended to $+\infty$, or software-totalized. A measure is declared as such and appears with measurable-set or differential arguments; q-series and fractional orders state their parameter domains. The former indexed families now print m_r^{Bell}, $\mathsf{c}_k^{\mathrm{F,Newt}}(q)$, $\mathsf{U}_n^{\mathrm{dyad}}$, α, ρ_1, ρ_2, or $\mathsf{R}_M^{\mathrm{law}}$ according to their full local definitions. Outside the canonical implementation of `\TwoAdicValuation`, no raw `\nu_2` remains in active normative prose; the spelling is permitted only in the explicitly marked source-faithful arithmetic paper described in the valuation entry.
ε	Thue–Morse sign, small positive tolerance, perturbation parameter, and error sequence.	Use ε_n for the sign. A tolerance is quantified $\varepsilon > 0$; perturbations and errors receive semantic subscripts when signs occur in the same scope.

Finite-spline, Fourier-product, and endpoint-kernel notation in the inverse-synthesis volume

Scope. The commands `\FiniteSpline`, `\FiniteSplineInverse`, `\Fn`, `\Gn`, `\PhiAng`, `\PhiCyc`, and `\EndpointKernel` are local to the standalone inverse-analyticity, asymptotics, and computability volume and its included chapters. They are not shared-API commands, and their meanings end with that standalone document.

Finite-prefix probability data. For $k \in \mathbb{N}_{>0}$, let $U_1, U_2, \ldots$ be independent random variables uniformly distributed on $[-1, 1]$, set

$$S_k := \sum_{j=1}^{k} 2^{-j} U_j,$$

and let $p_k : \mathbb{R} \to [0, \infty)$ be the Lebesgue density of S_k.

Finite spline family. `\FiniteSpline{k}` has TeX arity one and prints F_k^{spl}. Its mathematical argument is $k \in \mathbb{N}_{>0}$, and

$$F_k^{\mathrm{spl}} : [0,1] \longrightarrow [0, \infty), \qquad F_k^{\mathrm{spl}}(x) := p_k(x-1).$$

The superscript spl marks the finite convolution prefix and is never an iteration exponent.

Exact box-spline form. For $x \in [0,1]$, the same finite spline is the explicit Thue–Morse box spline

$$F_k^{\mathrm{spl}}(x) = \frac{2^{\binom{k}{2}}}{(k-1)!} \sum_{j=0}^{2^k-1} (-1)^{s_2(j)} \left(x - 2^{-k} - j2^{1-k}\right)_+^{k-1}, \qquad s_2(j) := \mathrm{wt}_2(j).$$

Here $\mathrm{wt}_2(j)$ is the number of one-bits in the nonnegative integer j, and $(t)_+ := \max\{t, 0\}$. Thus neither the Thue–Morse sign nor the truncation at zero is implicit.

Local inverse family. `\FiniteSplineInverse{k}` has TeX arity one and prints G_k^{spl}. On a stated interval $J \subset (0,1)$ on which F_k^{spl} is strictly increasing, it denotes the branch

$$G_k^{\mathrm{spl}} : F_k^{\mathrm{spl}}(J) \longrightarrow J, \qquad G_k^{\mathrm{spl}} := \left(F_k^{\mathrm{spl}}\big|_J\right)^{-1}.$$

No inverse value is implied unless the branch interval and monotonicity hypothesis have been supplied. In the asymptotic uses, J is a fixed compact subinterval of $(0,1)$, and the branch exists for all sufficiently large k.

Fixed-index aliases. `\Fn` and `\Gn` each have TeX arity zero. With an ambient $n \in \mathbb{N}_{>0}$, they expand respectively to F_n^{spl} and G_n^{spl}. They do not denote the forward composition iterate $F^{\circ n}$, the inverse composition iterate $(F^{-1})^{\circ n}$, or either globally catalogued Fabius inverse.
Finite-prefix use family. The spline and local-inverse symbols are used for finite-prefix approximation, uniform inverse-transfer asymptotics, and Richardson extrapolation. The prefix depth, evaluation variable, and—for the inverse—the selected monotone branch interval are ambient mathematical data, not additional TeX arguments.
Angular-frequency product. `\PhiAng` has TeX arity zero; its evaluation variable is external. It prints Φ_{ang} and denotes the entire extension of

$$\Phi_{\mathrm{ang}}(\xi) \coloneqq \widehat{\mathrm{up}}_{\mathrm{ang}}(\xi) = \prod_{j=1}^{\infty} \mathrm{sinc}_{\mathrm{rad}}(2^{-j}\xi), \qquad \xi \in \mathbb{C},$$

where $\mathrm{sinc}_{\mathrm{rad}}\, z = \sin z / z$ for $z \neq 0$ and $\mathrm{sinc}_{\mathrm{rad}}\, 0 = 1$.
Cycles-per-unit product. `\PhiCyc` has TeX arity zero and an external variable. It prints $\Phi_{2\pi}$ and denotes the same entire transform in the cycles-per-unit coordinate:

$$\Phi_{2\pi}(z) \coloneqq \widehat{\mathrm{up}}_{2\pi}(z) = \Phi_{\mathrm{ang}}(2\pi z) = \prod_{j=0}^{\infty} \mathrm{sinc}_{\pi}(z/2^j), \qquad z \in \mathbb{C}.$$

Thus neither the factor 2π nor the transform normalization is suppressed.
Fourier-product use families. Φ_{ang} is used for reciprocal-sinc coefficients, moment-generating/Fourier rotation, and polynomial deconvolution. $\Phi_{2\pi}$ is used for integer zero multiplicities, differentiated transform identities, moments, and dyadic self-sampling. Both symbols suppress the fixed normalized Rvachev up-function, but their printed subscripts make the coordinate normalization explicit; they are never silently interchanged.
Endpoint variables. For the endpoint regime $y \downarrow 0$, set $L = \log 2$, $T = \log(1/y)$, $\rho = \sqrt{2T/L}$, $z = \rho^{-1}$, $\ell = \log \rho$, $\beta = L^{-1} + 1/2$, and $\theta = \rho + \beta \pmod 1$. The phase functions $\Psi, A_1, A_2, \ldots$ are the fixed one-periodic coefficient functions in the forward expansion

$$\log F(x) = -\frac{L}{2}\lambda^2 + L\beta\lambda - \frac{1}{2}\log\lambda + C_\sharp + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}),$$

where $\lambda = -W_{-1}(-Lx)/L$ and the remainder is uniform in the real phase.
Endpoint kernel. `\EndpointKernel` has TeX arity zero and prints $\mathcal{K}$; both displayed arguments are mathematical arguments external to the macro. In the formal two-scale coefficient algebra generated by ℓ, the one-periodic functions above, and their derivatives, define

$$\begin{aligned}
\mathcal{R}(z,w) = \; & -\frac{L}{2}\beta^2 + \frac{L}{2}w^2 + \frac{1}{2}\ell + \frac{1}{2}\log(1+\beta z + zw) - C_\sharp - \Psi(\theta + w) \\
& - \sum_{j\geq 1} \frac{A_j(\theta+w) z^j}{(1+\beta z + zw)^j}, \\
\mathcal{K}(z,w) \coloneqq \; & -\frac{1}{L}\mathcal{R}(z,w).
\end{aligned}$$

The inverse endpoint correction $D \in z\mathscr{A}[\ell][[z]]$ is then characterized by $D = z\mathcal{K}(z, D)$. Here w is the second formal argument; it is not a hidden function of z until the fixed-point substitution $w = D$ is made. The same source later writes $\mathcal{K}_W = -\mathcal{R}_W/L$ for a distinct Wright–omega-carrier kernel. The form `\EndpointKernel_W` prints that local symbol; it does not denote $\mathcal{K}(z, W)$, a derivative, or an automatically shared subscripted family.

Endpoint-kernel use family. The kernel is used in the Lagrange diagonal formulas for the fixed point, the all-orders inverse-endpoint recursion, and comparison with the distinct Wright–omega carrier. Its suppressed ambient data are $L, \beta, \ell, \theta, C_\sharp, \Psi$, the full coefficient sequence $(A_j)_{j\geq 1}$, and the chosen two-scale coefficient algebra.
Script composition. `\FiniteSpline`, `\FiniteSplineInverse`, `\Fn`, and `\Gn` already own a subscript and the superscript spl; a derivative or further order must group the whole object, as in $(F_n^{\mathrm{spl}})^{(m)}$. `\PhiAng` and `\PhiCyc` already own their normalization subscripts; powers and derivatives may follow the grouped object, but no second raw subscript may be appended. `\EndpointKernel` owns no script, yet any local suffix such as $_W$ must be declared as a new semantic object rather than inferred as evaluation or differentiation.
Status. The finite-prefix objects and the two Fourier products are exact definitions. The endpoint kernel is an exact formal definition inside the volume's human-proved all-orders coefficient construction; convergence or analytic equality is asserted only where a separate theorem supplies it.

16.3 Accents, scripts, and overloaded punctuation

Hats, tildes, bars, stars, primes, and scripts

Hats. A normalization-labelled hat produced by `\FourierTwoPi` or `\FourierAngular` is a Fourier transform. A unit-period coefficient uses `\FourierSeriesCoefficient`; any other period uses `\FourierSeriesCoefficientAtPeriod` and prints that period. A hat on a polynomial, matrix, estimator, accelerated approximant, or lifted object is local: all use sites call a descriptive document-local macro, and the raw accent occurs only in that macro's one-line definition. The immediately preceding source comment has the form `LOCAL-NON-FOURIER-HAT: \MacroName meaning`, names the same macro, and explains its mathematical role. A generic local alias for `\widehat` or `\hat` is forbidden. Thus no unlabelled hat is silently interpreted as a Fourier transform in a document that also uses monic or normalized polynomials.
Tildes. A tilde may mark centering, normalization, continuation, lifting, or an alternative sign convention. The transformed object's type and the defining map are given at first use. It is never a substitute for the signed global Fabius function.
Bars. For a scalar, bars mean $|x|$; for a finite set they mean $|A|$; a restriction is $f|_S$; divisibility is $a \mid b$; conditional probability is $\mathbb{P}(A \mid B)$ with $\mathbb{P}(B) > 0$. These roles are distinguished by type and spacing, not guessed from visual similarity.
Stars. For a scalar, $\overline{z}$ is conjugation; A^* is an adjoint; X^* in the zero-bias entry is a distributional transform; X^* may also denote a dual space only after a type declaration. None of these implies optimality; an optimizer is labelled x_{opt}.
Primes. f' is a derivative, ζ_L' a derivative of the spectral zeta function, and $\det{}' L$ omits zero modes. A primed variable such as x' is a new local variable only after declaration.
Superscripts. Parenthesized $f^{(n)}$ means the nth derivative; $f^{\circ n}$ means iteration; f^{*n} means convolution power; A^{T} means transpose; A^* means adjoint. Ordinary f^n remains pointwise or algebraic power.
Subscripts. A shared symbol that already owns a semantic subscript is not followed by a second raw subscript. Use the command's arguments or introduce a wrapper whose expansion has one syntactically valid script. This rule covers number systems, quantiles, transform hats, and any future decorated semantic command.
Script-final commands: the rule is a build constraint. Many shared commands expand to a base atom that already carries a script. TeX permits at most one superscript and one subscript per atom, so a directly following script of a type the atom already carries is the fatal error `Double superscript` or `Double subscript`, not a typographic infelicity. A prime counts as a superscript. *Mixed types*

are legal: since \NaturalNumbers expands to $\mathbb{N}_0$, which carries only a subscript, $\mathbb{N}_0^2$ is correct, and so are Q'_F and $\mathrm{sinc}_\pi^{(r)}$. Any audit of this rule that searches for a command followed by a script, without first asking which script types that command's trailing atom carries, reports roughly ten false positives for every real one.

Classification of the shared commands. Twenty-seven commands expand to an atom carrying *both* scripts, so any following ^, _ or prime is fatal: \AssociahedronByDimension, \BellHessenbergMatrixH, \BellHessenbergMatrixK, \BellNumber, \CatalanNumber, \ComplementaryBellNumber, \ExponentialCompleteBellPolynomial, \ExponentialPartialBellPolyn \GenocchiNumber, \GenocchiPolynomial, \GeometricBernoulliPolynomial, \GregoryCoefficient, \InverseFourierAngularOperator, \InverseFourierTwoPiOperator, \LahNumber, \LambertPolynomial, \LinearizedDecayObjective, \NormalizedReversionCoefficient, \OrdinaryGeneratingCoefficient, \OrdinaryPartialBellPolynomial, \ScaledLambertPolynomial, \SecondOrderEulerianPolynomial, \TouchardPolynomial, \TraceBellArgument, \TypeAEulerianPoly \TypeBEulerianNumber, and \TypeBEulerianPolynomial. Seven are superscript-only and break on a following ^ or prime: \CompositionalInverseOf, \MultiplicativeInverseOf, \FormalBigO, \FallingFactorial, \RisingFactorial, \ExponentialGeneratingFunctionOf, and \OrdinaryGeneratingFunctionOf. The remaining thirty-eight script-final commands are subscript-only and break only on a following _. This classification is derived from the expansions, so it must be recomputed whenever a command's definition changes rather than trusted as a fixed list.

Remedy, and document-local aliases. Prefer the command whose meaning is the one intended: writing the power-series ring as \PolynomialLogPowerSeriesRing is better than decorating the Laurent-ring command with a range subscript. When a genuine extra script is needed, brace-wrap the call so the script attaches to the whole group, as in {\LambertPolynomial{n}}', which is valid and renders as intended. A derivative may instead be printed in Leibniz form.

The constraint propagates through *document-local aliases*. A definition such as \newcommand{\PL}{\Polynomial makes \PL script-final too, so \PL_{\ge0} is fatal even though the shared command never appears with a script beside it. An audit that inspects only the shared command names will not see this; alias chains must be resolved first.

16.4 Exhaustive register of document-local non-Fourier hats

This register inventories every distinct command named by an active LOCAL-NON-FOURIER-HAT: declaration. There are 35 marked declarations and 35 distinct command names; each is declared in exactly one active document scope. The phrase *prints a wide hat over X* describes literal output without reintroducing a generic accent command. TeX arguments, mathematical arguments, and scripts appended by a caller are distinguished explicitly. These commands are document-local: their definitions and ambient data do not survive the document or included-fragment scope stated below. Each heading contains one machine-readable \localaccentname token, so the strict audit rejects a missing, extra, or duplicate register entry.

16.4.1 Extensions, certified data, and formal series

`\ClampedExtensionOf`

TeX arity is one; argument 1 is a continuous function f on $I = [0, 1]$. The command prints a wide hat over f. Its mathematical value is the continuous map

$$E_I f : \mathbb{R} \longrightarrow \mathbb{R}, \qquad (E_I f)(x) := f(\operatorname{proj}_I x) = \begin{cases} f(0), & x \le 0, \\ f(x), & 0 \le x \le 1, \\ f(1), & 1 \le x, \end{cases}$$

where $f(x)$ in the middle branch uses the canonical membership $x \in I$. In the Lean construction this is precomposition with $\mathrm{Set.projIcc}(0, 1)$; admissibility of f supplies the bounds and reflection identity used later but is not required to define $E_I f$. The only use family is the fixed-point primitive $A_f(y) = \int_0^y (E_I f)(t)\, \mathrm{d}t$. The accent means endpoint-clamped extension, not Fourier transformation. Status is an exact definition; the later admissibility and fixed-point theorems are separate assertions. The clamping interval, endpoint values, and projection map are suppressed by the printed symbol. The function argument must include any script belonging to f inside the braces; a derivative or other decoration of the resulting map must group the complete expansion before adding the decoration.

`\FullRvachevLegendreCoefficientRat`

TeX arity is one; argument 1 is the full Legendre degree $k \in \mathbb{N}_0$. The command prints a wide hat over c, with subscript k and superscript $\mathbb{Q}$. It denotes the function $c^{\mathrm{full},\mathbb{Q}} : \mathbb{N}_0 \to \mathbb{Q}$ defined by

$$c_k^{\mathrm{full},\mathbb{Q}} := \begin{cases} c_{k/2}^{\mathbb{Q}}, & 2 \mid k, \\ 0, & 2 \nmid k, \end{cases} \qquad c_{2r}^{\mathrm{full},\mathbb{Q}} = c_r^{\mathbb{Q}}, \qquad c_{2r+1}^{\mathrm{full},\mathbb{Q}} = 0.$$

Here $c_r^{\mathbb{Q}}$ is the canonical even Rvachev–Legendre coefficient. Equivalently,

$$c_k^{\mathrm{full},\mathbb{Q}} = \frac{2k+1}{2}\, \mathcal{M}_{\mathrm{Rv}}\left(P_k^{\mathbb{Q}}\right),$$

where $P_k^{\mathbb{Q}}$ is the rational Legendre polynomial and $\mathcal{M}_{\mathrm{Rv}}$ is the moment functional with moment sequence `rvachevRawMomentRat`. The use family consists of the rational Legendre–Gaunt formulas and their real-cast comparison. The accent means parity completion of an even-indexed sequence. The canonical moment sequence, Legendre normalization, and coefficient family are suppressed ambient data. Status is an exact rational sequence definition; comparison with analytic coefficients retains the source's explicit Fabius-function hypotheses. The command already owns both its subscript and its $\mathbb{Q}$-superscript; appending any further raw script would be a script-composition error.

`\CertifiedEnclosureCenter`

TeX arity is one; argument 1 is the exact quantity whose validated numerical center is being named. The command prints a wide hat over that argument. Its sole active use has $Q, Q_{\mathrm{num}} \in \mathbb{C}$ and means only the center component of the certificate

$$\varepsilon_Q \ge 0, \qquad |Q - Q_{\mathrm{num}}| \le \varepsilon_Q.$$

In the validated argument-principle scope, let Ω be a bounded Jordan domain, let f be holomorphic on a neighborhood of $\overline{\Omega}$, assume $0 \notin f(\partial\Omega)$, and decompose the positively oriented boundary into finitely many C^3 panels $\gamma_k : [0, 1] \to \partial\Omega$. Define

$$g_k(s) = \frac{f'(\gamma_k(s))\gamma_k'(s)}{f(\gamma_k(s))}, \quad Q = \sum_k \frac{g_k(0) + g_k(1)}{2},$$
$$M_k \ge \sup_{0 \le s \le 1} \left|g_k''(s)\right|, \qquad E = \frac{1}{12} \sum_k M_k.$$

The panelwise trapezoid estimate gives

$$\left| \oint_{\partial\Omega} \frac{f'(z)}{f(z)}\,\mathrm{d}z - Q \right| \le E.$$

Outward-rounded arithmetic supplies Q_{num} and ε_Q. The resulting certified zero-count disk has center $Q_{\mathrm{num}}/(2\pi\mathrm{i})$ and radius $(E+\varepsilon_Q)/(2\pi)$. Status is therefore a certificate schema: the accented center alone is not a certificate. The ambient normed field, domain Ω, function f, boundary orientation and panelization (γ_k), derivative bounds (M_k), trapezoid rule, rounding method, error radius, and exact target are all suppressed data. The accent means validated numerical center, not Fourier transformation. The complete exact-quantity name belongs inside the single argument; callers must group the expansion before attaching any further script.

`\FormalGevreySeriesSymbol`

TeX arity is one; argument 1 is the name R of a formal series, not its formal variable. The command prints a wide hat over R. In its cyclotomic resurgence scope it denotes

$$R_{\mathrm{for}}(t) \coloneqq \sum_{n\ge 0} c_n t^n \in \mathbb{C}[[t]], \qquad |c_n| \le CA^n n! \quad (n \in \mathbb{N}_0)$$

for some constants $C, A > 0$. The document's exact formal Borel convention is

$$\mathscr{B}R_{\mathrm{for}}(\xi) \coloneqq \sum_{n\ge 0} \frac{c_{n+1}}{n!}\xi^n;$$

the constant c_0 is carried separately by Borel–Laplace summation. Status is a formal-series definition used immediately before a conjectural resurgence statement; the Gevrey estimate itself does not assert Borel summability or analytic continuation. The coefficient field, formal variable, coefficients, Gevrey order, sector, and Borel convention are suppressed by the printed symbol. The accent means formal asymptotic series prior to Borel transformation. The series name, including any semantic script, must be placed inside the one argument; any subsequent decoration must group the expansion.

`\FormalAsymptoticExpansionOf`

TeX arity is one; argument 1 is the name f of an actual function. The command prints a wide hat over f; its evaluation variable remains outside the macro. In the incoming transseries volume, fix $X > 0$, $t = X^{-1}$, $\lambda = \log X$, a characteristic-zero coefficient field K, and polynomial blocks $A_n \in K[\lambda]$. The formal object is

$$\widehat{f}(X) \coloneqq \sum_{n\ge n_0} A_n(\log X) X^{-n} \in K[\lambda]((t)).$$

It is a coefficient array in the t-adic completion. The separate analytic assertion that it is a Poincaré expansion of f means that, for every truncation index N, the difference between $f(X)$ and the sum through the block $N-1$ is little-oh of the last retained scale along a declared ray or sector. The field, limit direction, branch of $\log X$, block degrees, and precise remainder scale are suppressed data. The accent means formal asymptotic expansion, not Fourier transformation and not equality of the formal sum with f.

`\LambertAsymptoticApproximation`

TeX arity is one; argument 1 is the truncation order $N \in \mathbb{N}_{>0}$. It prints $\widehat{W}_N$, already owning the subscript N. On the positive principal-branch ray, let $z = \mathrm{e}^L$, $L > 1$, put $\ell = \log L$, and let P_n be the explicitly defined Lambert polynomial family of that document. Then

$$\widehat{W}_N \coloneqq L - \ell + \sum_{n=1}^{N} \frac{P_n(\ell)}{L^n}.$$

This is a finite real scalar approximation to $W_0(z)$, not the Lambert function itself and not a Fourier transform. Its logarithmic residual is

$$\rho_N \coloneqq \widehat{W}_N + \log \widehat{W}_N - L$$

whenever $\widehat{W}_N > 0$. If $W = W_0(\mathrm{e}^L)$, the mean-value theorem gives $\widehat{W}_N - W = \rho_N/(1 + 1/\theta_N)$ for some θ_N between them, provided that interval avoids zero. The branch, positive ray, variables L, ℓ, polynomial normalization, and residual domain are suppressed. A semantic superscript such as new for one Newton correction may follow the complete command expansion; no second raw subscript may be appended.

`\wideeq`

TeX arity is zero; the command prints a hatted equality sign as a binary relation. For two formal polynomial–logarithmic series

$$A = \sum_{\gamma} a_\gamma \mathfrak{m}_\gamma, \qquad B = \sum_{\gamma} b_\gamma \mathfrak{m}_\gamma$$

written over the same coefficient field, ordered monomial family, and locally finite support convention,

$$A \mathrel{\widehat{=}} B \quad\Longleftrightarrow\quad a_\gamma = b_\gamma \text{ for every index } \gamma.$$

It is coefficientwise formal identity, not equality of convergent numerical sums, asymptotic equivalence, equality modulo a truncation ideal, or Fourier equality. The ambient coefficient field, monomial group, ordering, support condition, and formal variables must already be declared.

`\DulacFormalSeriesOf`

TeX arity is one; argument 1 is the name f of a function or germ. The command prints a wide hat over f, with the small evaluation variable supplied after the expansion. For $x \to 0^+$, put

$$X = x^{-1}, \qquad \ell = \log(1/x) = \log X,$$

choose a strictly increasing exponent sequence $\lambda_0 < \lambda_1 < \cdots \to +\infty$, and polynomial blocks $P_j \in K[\ell]$. Then

$$\widehat{f}(x) \coloneqq \sum_{j\ge 0} x^{\lambda_j} P_j(\ell)$$

is the formal Dulac or power–log series, subject to the declared local-finiteness condition that makes each formal coefficient operation finite. The dominance order compares the exponent λ_j first and logarithmic degree second. The coefficient field, exponent group, branch and sign of the logarithm, support condition, and whether the formal series is an actual asymptotic expansion of a germ are suppressed. The accent means formal Dulac expansion, not Fourier transformation.

16.4.2 Inverse, sampling, and prediction constructions

`\FactorialScaledInverseTransferCoefficient`

TeX arity is one; argument 1 is the coefficient index $m \in \{1, \dots, J-1\}$. The command prints a wide hat over c with subscript m. The evaluation variable y is external to the macro and is suppressed at the bare-symbol use sites.

Fix $0 < \eta < 1/2$, $J \in \mathbb{N}_{>0}$, $q = 1/4$, and $\varepsilon_n = q^n$. In the uniform inverse-transfer expansion

$$G_n^{\mathrm{spl}}(y) = Q_F(y) + \sum_{j=1}^{J-1} c_j(y)\varepsilon_n^j + \mathcal{O}_{\eta,J}(\varepsilon_n^J), \qquad y \in [\eta, 1-\eta],$$

the theorem constructs

$$c_m \in C^\infty([\eta, 1-\eta]; \mathbb{R}) \qquad (1 \le m \le J-1).$$

With $x = Q_F(y)$, the accented coefficient is exactly

$$\widehat{c}_m(y) \coloneqq m!\, c_m(y).$$

The source normally suppresses the displayed argument y and writes only $\widehat{c}_m$ inside partial exponential Bell polynomials.

More explicitly, put

$$a_j(x) \coloneqq (-1)^j b_j F^{(2j)}(x),$$

and let $\mathcal{B}_{r,k}$ be the partial exponential Bell polynomial with $\mathcal{B}_{0,0} = 1$. Then

$$\begin{aligned} c_m(y) = -\frac{1}{F'(x)} \Bigg[& \frac{1}{m!} \sum_{k=2}^{m} F^{(k)}(x) \mathcal{B}_{m,k}(\widehat{c}_1, \dots, \widehat{c}_{m-k+1}) \\ & + \sum_{j=1}^{m} \frac{1}{(m-j)!} \sum_{k=0}^{m-j} a_j^{(k)}(x) \mathcal{B}_{m-j,k}(\widehat{c}_1, \dots, \widehat{c}_{m-j-k+1}) \Bigg], \end{aligned}$$

where the term with $m = j$ is interpreted as $a_m(x)$.

Status is an exact factorial rescaling inside the source's proved uniform inverse-transfer theorem. The source supplies a proof but makes no Lean formalization claim. The printed symbol suppresses y, η, J, q, the spline inverse G_n^{spl}, the base function F, the perturbation coefficients (a_j), and the Bell-polynomial convention. The accent means conversion from ordinary inverse-expansion coefficients to the factorial scaling used by exponential Bell polynomials. It is not a Fourier transform and not the tomographic Fourier-coefficient estimator elsewhere printed as a hatted c.

The macro already owns the subscript m. An explicit evaluation may be written after the complete expansion as $\widehat{c}_m(y)$; callers must not encode y as a second raw subscript. Any derivative, superscript, or other decoration must group the complete expanded symbol.

`\GeometricPrefixReproduction`

TeX arity is one; argument 1 is the prefix depth $m \in \mathbb{N}_{>0}$. The command prints a wide hat over X with subscript m. For

$$X_q = (1-q) \sum_{j \ge 0} q^j U_j, \qquad U_j \overset{\mathrm{iid}}{\sim} \mathrm{Unif}[-1,1], \qquad 0 < q < 1,$$

it denotes the real square-integrable random variable

$$S_{q,m} \coloneqq (1-q) \sum_{j=0}^{m-1} q^j U_j \in L^2(\Omega; \mathbb{R}).$$

This is the deterministic first-m-digit reproduction used in the squared-error test channel, with

$$I(X_q; S_{q,m}) = m\log(1/q), \qquad \mathbb{E}[(X_q - S_{q,m})^2] = q^{2m}\operatorname{Var}(X_q).$$

Status is an exact definition and exact rate/distortion calculation. The parameter q, probability space, digit sequence, digit law, and distortion measure are suppressed. The accent means reproduction or estimation, not Fourier transformation. The argument already becomes the subscript of X; no additional raw subscript or superscript may follow the expanded command.

`\LeastSquaresProjectionEstimate`

TeX arity is one; argument 1 is the complete target random-variable expression. The command prints a wide hat over that target. In its sole scope,

$$X_q = (1-q)\sum_{n\geq 0} q^n U_n, \qquad U_n \overset{\text{iid}}{\sim} \operatorname{Unif}[0,1], \qquad Y_q = \frac{X_q - \frac{1}{2}}{\sqrt{(1-q)/(12(1+q))}}.$$

For $0 < q, q_1, \ldots, q_m < 1$, with the comparison nodes pairwise distinct and $q \notin \{q_1, \ldots, q_m\}$, the accented target is the unique real L^2-orthogonal projection

$$\Pi_{\boldsymbol{q}} Y_q := \operatorname*{argmin}_{Z\in\operatorname{span}(Y_{q_1},\ldots,Y_{q_m})} \mathbb{E}[(Y_q - Z)^2],$$

and

$$\mathbb{E}[(Y_q - \Pi_{\boldsymbol{q}} Y_q)^2] = \prod_{j=1}^{m}\left(\frac{q-q_j}{1-qq_j}\right)^2.$$

Status is an exact Hilbert-space projection and prediction-error identity. The common probability space and digits, standardization, comparison nodes, coefficient field, and projection subspace are suppressed. The accent means least-squares predictor. The full target, including its own subscript, must be enclosed in the one argument; any subsequent decoration must group the complete expansion.

`\NestedLambertLogApproximation`

TeX arity is one; argument 1 is the iterate index $r \in \{0, \ldots, n\}$. The command prints a wide hat over T with subscript r. The endpoint input, recursion depth, truncation order, and evaluation phases are not TeX arguments.

In its sole active scope, $I = Q_F$ on $(0,1)$. Put

$$y_0 = y, \qquad y_{r+1} = I(y_r), \qquad T_r := \log(1/y_r),$$

in the endpoint regime where every attained y_r is positive and belongs to the domain of I. The source does not assert a larger maximal domain for this construction. For a fixed depth $n \geq 1$, put

$$q_n := T_0^{-2^{-n}}, \qquad \ell := \log T_0,$$

and set $L = \log 2$ and $\beta = L^{-1} + 1/2$. For a chosen truncation order $N \in \mathbb{N}_{>0}$ and the one-step coefficient functions h_j supplied by the preceding all-orders theorem, define

$$\mathcal{I}_N(T) := L\rho + L\beta - \log\rho - \sum_{j=1}^{N}\frac{h_j(\log\rho, \rho+\beta)}{\rho^j}, \qquad \rho := \sqrt{2T/L}.$$

The accented sequence is the procedural recursion

$$\widehat{T}_0 := T_0,$$

followed, for each r, by expansion of $\mathcal{I}_N(\widehat{T}_r)$ in powers of q_n and deletion of every term whose q_n-exponent is at least N; the retained expression is called $\widehat{T}_{r+1}$.

The source presents this construction inside the open *Nested Lambert phase closure* problem. It asks for, but does not prove, the uniform estimate

$$T_r - \widehat{T}_r = \mathcal{O}_{n,N}\Big((1+\ell)^{D'_{n,N}} q_n^N\Big) \qquad (0 \le r \le n),$$

the phase-substitution bounds, nonconstant transport of each periodic phase, and closure or minimal enlargement of the exponent lattice $q_n^{\mathbb{Z}}$. Because that closure is itself part of the problem, the source does not yet supply a proved ambient formal-series ring or a proved global function domain and codomain for every $\widehat{T}_r$. It intends real asymptotic expressions along the attained endpoint regime. No Lean formalization is claimed.

The printed symbol suppresses y, I, n, N, T_0, q_n, ℓ, L, β, the coefficient family (h_j), the phase vector, and the expansion/truncation convention. The accent means recursive asymptotic truncation, not Fourier transformation and not the lacunary-product tail extrapolation elsewhere printed with a hatted T. The macro already owns the subscript r; callers must not append a second raw subscript. Any further decoration must group the complete expanded symbol and receive a separate semantic declaration.

16.4.3 Omission marks, endpoint matrices, and lifted operators

`\OmittedSymmetricArgument`

TeX arity is one; argument 1 is the complete variable to delete from a symmetric-polynomial argument list. The command prints a wide hat over that variable. It is omission metanotation, not a map on the scalar q_j. Its exact active meaning is

$$T_k := \sum_{j\ge 1} q_j^2 e_{k-1}(q_1,\ldots,q_{j-1},q_{j+1},\ldots), \qquad q_j = Q^j, \qquad Q = \frac{1}{4},$$

so the j-th entry is absent from the arguments of e_{k-1}. The identity

$$e_1 e_k = T_k + (k+1)e_{k+1}$$

supplies the first correction in the small-field q-binomial chaos asymptotic. Status is exact combinatorial deletion notation. The ambient infinite sequence, deleted position, convergence of the symmetric sums, and value $Q = 1/4$ are suppressed. The accent means omission, not Fourier transformation. A scripted item such as q_j belongs inside the one argument, and the resulting omission marker receives no further script.

`\GeneratedEndpointCoefficientMatrix`

TeX arity is zero. The command prints a wide hat over B; the degree subscript 2 in the generated fragment is appended externally rather than passed as an argument. The fixed evidence object is

$$B_2^{\mathrm{end}} = \begin{bmatrix} \frac{392}{9} & -\frac{496}{9} & \frac{176}{9} \\ -\frac{752}{9} & \frac{1072}{9} & -\frac{392}{9} \\ \frac{680}{9} & -\frac{1072}{9} & \frac{464}{9} \\ -\frac{464}{9} & \frac{496}{9} & -\frac{104}{9} \end{bmatrix} \in \mathbb{Q}^{4\times 3}.$$

It is the endpoint-basis coefficient matrix for degree $d = 2$, scale $M = 2^d = 4$, nodes $x = (-1, 0, 1)$, transported nodes $t = (0, \frac{1}{2}, 1)$, and retained integer centers $k = (1, 2, 3, 4)$. Its columns are the exact endpoint-atom coefficient vectors of the three Lagrange cardinal polynomials. The generator constructs those cardinals, applies the reciprocal-moment differential operator at scale 4, samples the active shifts, removes the annihilator-recurrence null component, and retains the final $d+2$ coefficients. Status is an exact generated companion-evidence fragment, not an independent globally parameterized matrix family. All degree, node, scale, basis, and row/column data are suppressed. The accent means endpoint augmentation/compression. The external degree subscript is presently valid because the zero-argument expansion owns no script; regeneration must preserve that invariant or replace the command with an indexed wrapper.

`\AugmentedEvaluationMatrix`

TeX arity is zero. The command prints a wide hat over U; entry indices such as i, r are appended externally. Fix $d \in \mathbb{N}_0$, distinct nodes $x_0, \ldots, x_d \in [-1, 1]$, and the $d+2$ normalized endpoint atoms

$$c_{d,r} = 2^{d+1} - 2d - 3 + 2r, \qquad \Phi_{d,r}(x) \coloneqq 2^{-d} \operatorname{up}\left(\frac{x - c_{d,r}}{2^{d+1}}\right), \qquad 0 \le r \le d+1.$$

The matrix is

$$U_X^{\text{end}} \coloneqq [\, \Phi_{d,r}(x_i) \,]_{0 \le i \le d,\ 0 \le r \le d+1} \in \mathbb{R}^{(d+1)\times(d+2)}.$$

It maps an endpoint-basis coefficient vector to its $d+1$ nodal values. Status is an exact finite evaluation matrix. The degree, ordered node set, endpoint basis, index ranges, and normalization of the atoms are suppressed. The accent means augmentation of the polynomial evaluation system by one endpoint atom. A single external entry subscript is valid because the zero-argument expansion has no internal script; callers must not create a second subscript, and a future scripted body would require an entry wrapper.

`\AugmentedCoefficientMatrix`

TeX arity is zero. The command prints a wide hat over B; transpose or entry scripts are external. With the degree, nodes, atoms, and evaluation matrix fixed as above, let $\ell_j \in \Pi_d$ be the Lagrange cardinal satisfying $\ell_j(x_i) = \delta_{ij}$. The coefficient matrix

$$B_X^{\text{end}} = [\, b_{rj} \,]_{0 \le r \le d+1,\ 0 \le j \le d} \in \mathbb{R}^{(d+2)\times(d+1)}$$

is defined by the unique endpoint-basis expansions

$$\ell_j = \sum_{r=0}^{d+1} b_{rj} \Phi_{d,r}, \qquad U_X^{\text{end}} B_X^{\text{end}} = I_{d+1}.$$

Define the canonical cofactor vectors by

$$v_r = (-1)^r \det U_X^{(r)}, \qquad \zeta_r = (-1)^r \det B_X^{(r)},$$

where $U_X^{(r)}$ deletes column r from U_X^{end} and $B_X^{(r)}$ deletes row r from B_X^{end}. Then $U_X^{\text{end}} v = 0$, $\zeta^T B_X^{\text{end}} = 0$, and Cauchy–Binet gives the normalization $\zeta^T v = 1$. With

$$\mathcal{M}_X = \begin{pmatrix} U_X^{\text{end}} \\ \zeta^T \end{pmatrix},$$

then the defect-completed inverse formula is

$$B_X^{\text{end}} = \mathcal{M}_X^{-1} \begin{pmatrix} I_{d+1} \\ 0 \end{pmatrix}, \qquad \zeta^T B_X^{\text{end}} = 0.$$

Status is an exact finite right inverse selected by the endpoint-basis polynomiality constraint. The degree, nodes, cardinals, endpoint atoms, polynomiality row, and matrix dimensions are suppressed. The accent means endpoint-atom augmentation. An external transpose is valid only because the zero-argument expansion has no script; do not combine it with an internal or second external superscript.

`\OmittedMatrixIndex`

TeX arity is one; argument 1 is the index $r \in \{0, \dots, d+1\}$ to omit. The command prints a wide hat over r. It is deletion metanotation, not a map on indices. In the evaluation-matrix context,

$$U_X^{(r)} \in \mathbb{R}^{(d+1)\times(d+1)}$$

means U_X^{end} with column r deleted; in the coefficient-matrix context,

$$B_X^{(r)} \in \mathbb{R}^{(d+1)\times(d+1)}$$

means B_X^{end} with row r deleted. The cofactor vectors are

$$v_r = (-1)^r \det U_X^{(r)}, \qquad \zeta_r = (-1)^r \det B_X^{(r)},$$

and Cauchy–Binet yields $\zeta^T v = 1$. Status is exact maximal-minor notation. Which matrix is present, whether a row or column is deleted, the matrix dimensions, and the permitted index set are suppressed. The accent means omission. The argument is used as a nested marker inside the matrix's one subscript; it must not itself acquire any further script.

`\LiftedCoefficientProjector`

TeX arity is two. Argument 1 is the interpolation level q, and argument 2 is the terminal coefficient-space level s, in that order. The command prints a wide hat over $\mathcal{P}$, with subscript q and parenthesized superscript s. Fix nested node sets $X^{(0)} \subset \cdots \subset X^{(s)}$, interpolation projectors $I_q : C[-1,1] \to \Pi_{d_q}$, a terminal coefficient space C_s, a synthesis map $T_s : C_s \to C[-1,1]$, and a canonical coefficient map $E_s : \Pi_{d_s} \to C_s$ satisfying $T_s E_s = I$ on Π_{d_s}. Then

$$\mathcal{P}_q^{(s)} := E_s I_q T_s : C_s \longrightarrow C_s, \qquad 0 \le q \le s.$$

These operators satisfy

$$\mathcal{P}_q^{(s)} \mathcal{P}_r^{(s)} = \mathcal{P}_{\min(q,r)}^{(s)}, \qquad \operatorname{rank} \mathcal{P}_q^{(s)} = d_q + 1.$$

Status is an exact coefficient-space lift of an interpolation projector. The nested grids, degrees, coefficient dictionary, synthesis and analysis maps, and terminal level are suppressed. The accent means coefficient lift. Both available scripts are already built from the two arguments; no raw subscript, superscript, or prime may be appended.

`\LiftedCoefficientDetail`

TeX arity is two. Argument 1 is the detail level q, and argument 2 is the terminal coefficient-space level s, in that order. The command prints a wide hat over Δ, with subscript q and parenthesized superscript s. Using the lifted projectors just defined,

$$\Delta_0^{(s)} := \mathcal{P}_0^{(s)}, \qquad \Delta_q^{(s)} := \mathcal{P}_q^{(s)} - \mathcal{P}_{q-1}^{(s)} \quad (1 \le q \le s).$$

Each $\Delta_q^{(s)} : C_s \to C_s$ is idempotent, distinct details annihilate one another, and

$$\operatorname{rank} \Delta_0^{(s)} = d_0 + 1, \qquad \operatorname{rank} \Delta_q^{(s)} = d_q - d_{q-1} \quad (q \ge 1).$$

Status is an exact finite multiresolution-detail definition. The nested node sets, interpolation projectors, terminal dictionary, degree sequence, and coefficient-space maps are suppressed. The accent means coefficient-space lift, not frequency transformation. Both scripts are owned by the two arguments; appending any further raw script is invalid.

16.4.4 Monic and hypothetical polynomial families

`\MonicDeconvolvedLegendre`

TeX arity is one; argument 1 is the degree $n \in \mathbb{N}_0$. The command prints a wide hat over Q, with subscript n and superscript minus. Let

$$Z := \sum_{j\geq 1} 2^{-j} U_j, \qquad U_j \overset{\text{iid}}{\sim} \operatorname{Unif}[-1,1],$$

so Z has the normalized Rvachev up-density on $[-1,1]$. Let P_n be the classical Legendre polynomial and set

$$M_u(z) := \mathbb{E}[\mathrm{e}^{zZ}] = \prod_{j\geq 1} \frac{\sinh(z/2^j)}{z/2^j}, \qquad \frac{1}{M_u(z)} = \sum_{r\geq 0} \gamma_r \frac{z^r}{r!}, \qquad \mathscr{D}_u := \sum_{r\geq 0} \gamma_r \frac{D^r}{r!}.$$

The deconvolved polynomial is $Q_n^- = \mathscr{D}_u P_n$. Since $[x^n]P_n = 2^{-n}\binom{2n}{n}$ and $\mathscr{D}_u$ preserves leading coefficients, its exact monic normalization is

$$Q_n^{-,\mathrm{mon}} := \frac{2^n}{\binom{2n}{n}} \mathscr{D}_u P_n \in \mathbb{Q}[x].$$

For example,

$$Q_2^{-,\mathrm{mon}} = x^2 - \frac{4}{9}, \qquad Q_3^{-,\mathrm{mon}} = x^3 - \frac{14}{15}x,$$
$$Q_4^{-,\mathrm{mon}} = x^4 - \frac{32}{21}x^2 + \frac{1072}{4725} = xQ_3^{-,\mathrm{mon}} - \frac{62}{105}Q_2^{-,\mathrm{mon}} - \frac{8}{225}Q_0^{-,\mathrm{mon}}.$$

Status is an exact rational polynomial definition; the final equality is the explicit degree-four Favard obstruction. The up-law, moment-generating function, reciprocal moments, differential operator, Legendre normalization, and coefficient field are suppressed. The accent means monic normalization. The command already owns its degree subscript and minus superscript; no additional raw script may follow it.

`\GenericMonicOrthogonalPolynomial`

TeX arity is one; argument 1 is the degree $n \in \mathbb{N}_0$. The command prints a wide hat over Q with subscript n. It is a schema for a hypothetical monic, parity-alternating scalar orthogonal-polynomial family

$$Q_n^{\mathrm{mon}} \in \mathbb{R}[x], \qquad \deg Q_n^{\mathrm{mon}} = n, \qquad [x^n]Q_n^{\mathrm{mon}} = 1.$$

For a scalar quasi-definite moment functional, Favard's theorem would force

$$Q_0^{\mathrm{mon}} = 1, \qquad Q_1^{\mathrm{mon}} = x, \qquad \lambda_n \neq 0 \quad (n \geq 1),$$

and the parity-reduced three-term recurrence

$$Q_{n+1}^{\mathrm{mon}} = xQ_n^{\mathrm{mon}} - \lambda_n Q_{n-1}^{\mathrm{mon}} \qquad (n \geq 1).$$

The symbol does not name a separately constructed global family; it is used only to contrast this three-term schema with the nonzero degree-zero defect in the actual monic deconvolved Legendre family. Status is therefore a generic proof schema, not a definition of new canonical polynomials. The scalar functional, orthogonality normalization, recurrence coefficients, coefficient field, and parity assumptions are suppressed. The accent means monic normalization, not Fourier transformation. The degree is already placed by the one argument, so no second subscript or conflicting superscript may be appended.

16.4.5 Lacunary-product extrapolants

Common lacunary-product data.

For the product extrapolants below, let $b > 1$, $N \in \mathbb{N}_0$, $q = b^{-2} \in (0,1)$, and

$$P_b(t) := \prod_{k=1}^{\infty} \mathrm{sinc}_{\mathrm{rad}}(t/b^k), \qquad P_{b,N}(t) := \prod_{k=1}^{N} \mathrm{sinc}_{\mathrm{rad}}(t/b^k),$$

where $P_{b,0} = 1$. Put

$$T_N(t) := \prod_{k=N+1}^{\infty} \mathrm{sinc}_{\mathrm{rad}}(t/b^k), \qquad T_{N,j}(t) := \prod_{k=N+1}^{N+j} \mathrm{sinc}_{\mathrm{rad}}(t/b^k), \qquad T_{N,0} = 1,$$

and

$$D_N := \{t \in \mathbb{C} : |t| < \pi b^{N+1}\}.$$

Every $T_{N,j}$, including T_N, is zero-free on D_N. Write $L_{N,j}$ for the unique holomorphic logarithm of $T_{N,j}$ on D_N satisfying $L_{N,j}(0) = 0$. For $r \in \mathbb{N}_0$ and $0 \le j \le r$, define

$$(q;q)_m := \prod_{\ell=1}^{m}(1-q^\ell), \qquad (q;q)_0 := 1, \qquad \lambda_{r,j}(q) := \frac{(-1)^{r-j} q^{(r-j)(r-j+1)/2}}{(q;q)_j (q;q)_{r-j}}.$$

`\FactorizedTailExtrapolant`: factorized tail extrapolant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{T}_{N,r}$. The evaluation variable t is written after the macro and is not a TeX argument.
Definition and type. For $N, r \in \mathbb{N}_0$,

$$\widehat{T}_{N,r}(t) := \exp\left(\sum_{j=0}^{r} \lambda_{r,j}(q) L_{N,j}(t)\right), \qquad \widehat{T}_{N,r} : D_N \longrightarrow \mathbb{C}^\times.$$

It extrapolates the exact omitted tail T_N, is holomorphic and zero-free on D_N, and is positive on $D_N \cap \mathbb{R}$.
Scope and status. The printed notation suppresses the fixed base b, hence also $q = b^{-2}$, and leaves t external. The construction, its exact error, and its zero-free properties are proved in the factorized frontier report; that report claims no Lean counterpart at its snapshot.
Accent and scripts. The hat means logarithmic q-Richardson extrapolation, not Fourier transformation. The symbol owns only the subscript N, r, so a later grouped or direct superscript is syntactically available.

`\FactorizedProductExtrapolant`: zero-preserving product extrapolant

TeX interface. Arity 3, with arguments b, N, r in that order; it prints $\widehat{P}^{\mathrm{fac}}_{b,N,r}$. The evaluation variable t is external.
Definition and type.

$$\widehat{P}^{\mathrm{fac}}_{b,N,r}(t) := P_{b,N}(t) \exp\left(\sum_{j=0}^{r} \lambda_{r,j}(b^{-2}) L_{N,j}(t)\right), \qquad \widehat{P}^{\mathrm{fac}}_{b,N,r} : D_N \longrightarrow \mathbb{C}.$$

It has exactly the zero divisor of P_b in D_N, including multiplicities, and agrees with the unfactorized logarithmic product extrapolant wherever $P_{b,N} \neq 0$.

Scope and status. The derived parameter $q = b^{-2}$ and the argument t are not printed as indices. Holomorphic continuation, zero-divisor preservation, exact zero-normalized quotients, and local jet identities are proved in the frontier report but are not claimed there as Lean theorems.
Accent and scripts. The hat means a factorized approximation to P_b, not a Fourier transform. The macro already owns the superscript fac; any derivative, power, or further semantic superscript must be attached to the grouped expression $(\widehat{P}^{\mathrm{fac}}_{b,N,r})$, never appended directly.

`\RelativeTailExtrapolant`: dyadic relative-tail extrapolant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{\mathcal{T}}_{N,r}$. The evaluation variable x is external, and the dyadic base is fixed rather than printed.
Definition and type. Put

$$\mathcal{T}_{N,j}(x) := \prod_{h=N+1}^{N+j} \mathrm{sinc}_{\mathrm{rad}}(\pi x/2^h), \qquad \mathcal{T}_{N,0} = 1,$$

and let $L^{\mathrm{dy}}_{N,j}$ be its holomorphic logarithm on $|x| < 2^{N+1}$, normalized by $L^{\mathrm{dy}}_{N,j}(0) = 0$. Then

$$\widehat{\mathcal{T}}_{N,r}(x) := \exp\left(\sum_{j=0}^{r} \lambda_{r,j}(1/4) L^{\mathrm{dy}}_{N,j}(x)\right), \qquad \widehat{\mathcal{T}}_{N,r} : \{x \in \mathbb{C} : |x| < 2^{N+1}\} \longrightarrow \mathbb{C}^{\times}.$$

Its target is $\mathcal{T}_N(x) = \prod_{h=N+1}^{\infty} \mathrm{sinc}_{\mathrm{rad}}(\pi x/2^h)$.
Scope and status. The fixed data $b = 2$, $q = 1/4$, and the argument x are suppressed. The source introduces this object by cross-reference rather than by a standalone equality; the formula above is the explicit specialization used in its proved dyadic bracket theorem.
Accent and scripts. The hat means extrapolation of a product tail that is zero-free on the complex disk and positive on its real slice, not Fourier transformation. The symbol owns only the subscript N, r.

`\DyadicFactorizedProductExtrapolant`: zero-preserving dyadic Fourier-product approximant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{\Phi}_{N,r}$. The evaluation variable x is external.
Definition and type. Let

$$\Phi(x) := \widehat{\mathrm{up}}_{2\pi}(x) = \prod_{h=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}(\pi x/2^h), \qquad \Phi_N(x) := \prod_{h=0}^{N} \mathrm{sinc}_{\mathrm{rad}}(\pi x/2^h).$$

Then

$$\widehat{\Phi}_{N,r}(x) := \Phi_N(x)\widehat{\mathcal{T}}_{N,r}(x), \qquad \widehat{\Phi}_{N,r} : \{x \in \mathbb{C} : |x| < 2^{N+1}\} \longrightarrow \mathbb{C}.$$

It preserves every included integer zero of Φ, its multiplicity, and the real Thue–Morse lobe sign.
Scope and status. The fixed dyadic data $b = 2$, $q = 1/4$, the relative-tail logarithms, and x are suppressed. The source defines the object by reference to factorization; its zero, jet, bracket, and sign properties are proved in the manuscript but are not claimed as Lean results.
Accent and scripts. Although Φ is itself the 2π-normalized Fourier transform of up, the hat on $\widehat{\Phi}_{N,r}$ means numerical/analytic extrapolation of Φ, not a second Fourier transform. The macro owns only a subscript, so the current derivative notation $\widehat{\Phi}^{(\mu)}_{N,r}$ is syntactically safe.

16.4.6 Radial–angular and unfactorized product estimators

`\RadialJetEstimator`: radial endpoint-coefficient estimator

TeX interface. Arity 1, whose sole argument is the coefficient index j; it prints $\widehat{A}_j$. The phase θ is an external function argument.
Definition and type. Let $a, d \in \mathbb{Z}$ satisfy $d \ge 1$ and $0 \le a < d$, put $\theta = a/d$, and let the integer $\rho > 1$ satisfy $\rho \equiv 1 \pmod d$. Choose $n \in \mathbb{N}_{>0}$ with $n + \theta \ge 1/\log 2$. For fixed $p \in \mathbb{N}_0$, put

$$q = \rho^{-1}, \qquad H = (n+\theta)^{-1}, \qquad y_s = \mathscr{E}(\rho^s(n+\theta)) = \sum_{m=0}^{p} A_m(\theta) H^m q^{sm} + r_s,$$

and

$$\ell_s(z) := \prod_{\substack{0 \le u \le p \\ u \ne s}} \frac{z - q^u}{q^s - q^u}.$$

For $0 \le j \le p$,

$$\widehat{A}_j(\theta) := H^{-j} [z^j] \sum_{s=0}^{p} y_s \ell_s(z).$$

For the real Fabius endpoint expansion this is real-valued; the same finite interpolation formula is valid over $\mathbb{C}$.
Scope and status. The printed symbol suppresses p, n, ρ, q, H, the sample vector $(y_s)_{s=0}^{p}$, and the normalized endpoint residual $\mathscr{E}$. Exact recovery in the polynomial model and the remainder bound are conditional on this externally supplied normalization and sample vector: the source intentionally leaves the precise nonperiodic subtraction inside $\mathscr{E}$ schematic. The resulting conditional theorem is proved in the tomography manuscript, not formalized there in Lean.
Accent and scripts. The hat means recovery of $A_j(\theta)$ by geometric radial interpolation. It is not a Fourier transform. The symbol owns only the subscript j.

`\TomographicCoefficientEstimator`: radial–angular Fourier-coefficient estimator

TeX interface. Arity 1, whose sole macro argument is the Fourier mode k; the base form is $\widehat{c}_k$. The source appends the external configuration superscript (d, p, n), printing $\widehat{c}_k^{(d,p,n)}$.
Definition and type. If

$$A_0(u) = \sum_{m \in \mathbb{Z}} c_m \mathrm{e}^{2\pi i m u}, \qquad \mathcal{R}_{0;p,n,\rho}(\theta) := [z^0] \sum_{s=0}^{p} y_s \ell_s(z),$$

where y_s, ℓ_s are the radial data in the preceding entry, then

$$\widehat{c}_k^{(d,p,n;\rho)} := \frac{1}{d} \sum_{a=0}^{d-1} \mathcal{R}_{0;p,n,\rho}(a/d) \mathrm{e}^{-2\pi i k a/d} \in \mathbb{C}.$$

The semicolon-ρ augmentation is catalogue-only disambiguation: the active source prints $\widehat{c}_k^{(d,p,n)}$ and suppresses ρ. The theorem uses $d \ge 2$, $p \in \mathbb{N}_0$, $n \in \mathbb{N}_{>0}$, $k \in \mathbb{Z}$, $|k| \le (d-1)/2$, and an integer $\rho > 1$ satisfying $\rho \equiv 1 \pmod d$.

Scope and status. The printed superscript records d, p, n but suppresses ρ, the radial samples, and the endpoint normalization. Under the stated uniform endpoint expansion and Wiener-strip hypothesis, the report proves

$$\widehat{c}_k^{(d,p,n)} - c_k = \mathcal{O}(n^{-p-1}) + \mathcal{O}(\exp[-2\pi\sigma(d - |k|)])$$

as $n \to \infty$, uniformly over the grid phases a/d, with d, p, ρ, and the strip width σ fixed. It does not claim a Lean formalization.
Accent and scripts. The hat means an estimate of the already defined Fourier coefficient c_k. The finite Fourier transform occurs inside the construction, but the hat is not the Fourier transform of c_k. Because the macro owns only k as a subscript, the externally appended configuration superscript is currently safe; changing the macro to own a superscript would require a dedicated wrapper.

`\GeometricProductExtrapolant`: multiplicative q-Richardson product extrapolant

TeX interface. Arity 3, with arguments b, N, r in that order; it prints $\widehat{P}_{b;N,r}$. The semicolon separates the base from the two discrete indices. The evaluation variable t is external.
Definition and type. For real t satisfying

$$\frac{|t|}{b^{N+1}} < \pi, \qquad P_{b,N}(t) \neq 0,$$

define

$$\widehat{P}_{b;N,r}(t) := \operatorname{sgn}(P_{b,N}(t)) \exp\left(\sum_{j=0}^{r} \lambda_{r,j}(b^{-2}) \log |P_{b,N+j}(t)|\right) \in \mathbb{R}^{\times}.$$

Equivalently,

$$\left|\widehat{P}_{b;N,r}(t)\right| = \prod_{j=0}^{r} |P_{b,N+j}(t)|^{\lambda_{r,j}(b^{-2})}.$$

Scope and status. The derived value $q = b^{-2}$ and t are not printed as indices. This is the real logarithmic construction away from prefix zeros, unlike the holomorphic factorized continuation. Its exact error series, sign, brackets, and fixed-parameter convergence are proved in the report; the associated analytic claims remain separate Lean obligations.
Accent and scripts. The hat means multiplicative extrapolation of $P_b(t)$, not Fourier transformation. The macro owns only a subscript, so the hybrid-correction superscript $[s]$ used in the source is syntactically safe and denotes correction depth, not differentiation.

`\DyadicProductExtrapolant`: dyadic multiplicative product extrapolant

TeX interface. Arity 2, with arguments N, r in that order; it prints $\widehat{P}_{N,r}$. The base $b = 2$ is fixed, and t is an external function argument.
Definition and type. With

$$P_N(t) := \prod_{k=1}^{N} \operatorname{sinc}_{\mathrm{rad}}(t/2^k),$$

define, whenever $|t|/2^{N+1} < \pi$ and $P_N(t) \neq 0$,

$$\widehat{P}_{N,r}(t) := \operatorname{sgn}(P_N(t)) \exp\left(\sum_{j=0}^{r} \lambda_{r,j}(1/4) \log |P_{N+j}(t)|\right) \in \mathbb{R}^{\times}.$$

Thus it is exactly the $b = 2$, $q = 1/4$ specialization of the preceding geometric estimator.

Scope and status. The base, $q = 1/4$, and t are suppressed. The target is the tail product $P_2(t)$, not the full Fourier product $\Phi(x) = \mathrm{sinc}_{\mathrm{rad}}(\pi x) P_2(\pi x)$. Explicit low-order formulas, error bounds, and the finite Thue–Morse realization are proved in the report. The raw finite complex-exponential identity and the sine/sinc Thue–Morse normalization are Lean-formalized in the complex-product bridge modules; the analytic Richardson estimator, error, bracket, and convergence claims are not.
Accent and scripts. The hat means dyadic q-Richardson extrapolation, not Fourier transformation. The symbol owns only the subscript N, r.

16.4.7 Phase-locked lower-Lambert estimators

Common phase-locked data.
Let $L = \log 2$. For $\lambda > 1/L$, set

$$x(\lambda) := \lambda 2^{-\lambda}, \qquad \lambda(x) := -\frac{1}{L} W_{-1}(-Lx) \quad \left(0 < x < \frac{1}{\mathrm{e}L}\right),$$

so that $\lambda(x(\lambda)) = \lambda$. The phase-locked nodes and weights are

$$x_j(\lambda) := (\lambda + j) 2^{-(\lambda+j)}, \qquad \omega_{r,j}(\lambda) := \frac{(-1)^{r-j}}{r!} \binom{r}{j} (\lambda + j)^r \quad (0 \le j \le r).$$

They satisfy

$$\sum_{j=0}^{r} \omega_{r,j}(\lambda) = 1, \qquad \sum_{j=0}^{r} \omega_{r,j}(\lambda)(\lambda + j)^{-m} = 0 \quad (1 \le m \le r).$$

With the endpoint theorem's fixed constant C_F, define

$$\mathcal{W}_{\mathrm{MSE}}(x) := -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log\lambda(x) + C_F, \qquad Y(x) := \log F_{\mathrm{bd}}(x) - \mathcal{W}_{\mathrm{MSE}}(x).$$

For every fixed $M \in \mathbb{N}_0$, as $x \downarrow 0$ (equivalently $\lambda(x) \to \infty$), uniformly along a fixed compact phase set, the all-orders input has

$$Y(x) = \Psi(\lambda(x)) + \sum_{m=1}^{M} A_m(\lambda(x))\lambda(x)^{-m} + \mathcal{O}(\lambda(x)^{-M-1}),$$

where Ψ and every A_m are one-periodic in λ.

`\PhaseLockedLogEstimator`: phase-locked logarithmic endpoint estimator

TeX interface. Arity 1, whose argument is the extrapolation order r; it prints $\widehat{\Psi}_r$. The lower-Lambert coordinate λ is an external function argument.
Definition and type. For $r \in \mathbb{N}_0$ and $\lambda > 1/\log 2$,

$$\widehat{\Psi}_r(\lambda) := \sum_{j=0}^{r} \omega_{r,j}(\lambda) Y(x_j(\lambda)) \in \mathbb{R}.$$

The weights preserve the periodic constant term and cancel the first r inverse-λ corrections exactly, so the target is $\Psi(\lambda)$.

Scope and status. The printed form suppresses λ, the nodes x_j, the reduced observable Y, the main-saddle normalization C_F, and the weights. At each fixed r and finite residual depth, the logarithmic extraction and its first-omitted-term estimate are Lean-backed under the repository's all-orders endpoint input. No convergence of the infinite residual series and no uniformity in growing r are asserted.
Accent and scripts. The hat means a phase-locked reciprocal-scale estimate of the one-periodic correction $\Psi(\lambda)$, not a Fourier transform. The symbol owns only the subscript r.

`\PhaseLockedExponentialEstimator`: exponentiated phase-locked endpoint estimator

TeX interface. Arity 1, whose argument is the extrapolation order r; it prints $\widehat{\mathcal{A}}_r$. The coordinate λ is external.
Definition and type. For $r \in \mathbb{N}_0$, $\lambda > 1/\log 2$, and the common nodes, weights, and reduced observable,

$$\widehat{\mathcal{A}}_r(\lambda) := \exp\left(\sum_{j=0}^{r} \omega_{r,j}(\lambda) Y(x_j(\lambda))\right) = \prod_{j=0}^{r} \left(\frac{F_{\mathrm{bd}}(x_j(\lambda))}{\exp(\mathcal{W}_{\mathrm{MSE}}(x_j(\lambda)))}\right)^{\omega_{r,j}(\lambda)} \in \mathbb{R}_{>0}.$$

Its exact target is the periodic endpoint amplitude $\mathcal{A}(\lambda) := \exp(\Psi(\lambda))$.
Scope and status. The printed form suppresses λ, x_j, Y, $\mathcal{W}_{\mathrm{MSE}}$, and $\omega_{r,j}$. The displayed equality is exact exponentiation of the logarithmic estimator; the multiplicative estimator's Bell relative-error hierarchy is not part of the closed Lean phase-extraction tranche.
Accent and scripts. The hat means an estimated positive periodic amplitude, not Fourier transformation. The symbol owns only the subscript r.

`\PhaseLockedCoefficientEstimator`: phase-locked endpoint coefficient extractor

TeX interface. Arity 2, with arguments s, r in that exact order: s is the coefficient index and r is the interpolation order. It prints $\widehat{A}_s^{(r)}$. The coordinate λ is external.
Definition and type. For $\lambda > 1/\log 2$, set

$$h_j := (\lambda + j)^{-1}, \qquad \ell_j(h) := \prod_{\substack{0 \le k \le r \\ k \ne j}} \frac{h - h_k}{h_j - h_k}.$$

For $r \in \mathbb{N}_0$ and $0 \le s \le r$,

$$\widehat{A}_s^{(r)}(\lambda) := \frac{1}{s!} \sum_{j=0}^{r} \ell_j^{(s)}(0) Y(x_j(\lambda)) \in \mathbb{R}.$$

The factor $1/s!$ makes this the coefficient of h^s in the interpolating polynomial, rather than its unnormalized sth derivative. Locally, $\widehat{A}_0^{(r)} = \widehat{\Psi}_r$; for $s \ge 1$, the target is the periodic endpoint coefficient $A_s(\lambda)$.
Scope and status. The printed symbol suppresses λ, the nodes, the observable Y, and the Lagrange basis. The extraction formula is exact finite interpolation, but the higher coefficient extractors remain a formalization target; only the logarithmic $s = 0$ tranche has the compiled fixed-order status described above.

Accent and scripts. The hat means coefficient reconstruction, not Fourier transformation. The built-in superscript (r) records interpolation order and is not a derivative. Any additional derivative, power, or semantic superscript must be attached to the grouped expression $(\widehat{A}_s^{(r)})$.

16.4.8 Tail and inverse-root approximations

`\SimpleTailEstimate`: first-order negative-power tail estimate

TeX interface. Arity 1, whose argument is the tail-start index N; it prints $\widehat{R}_N$. The exponent parameter a is an external function argument.
Definition and type. Let X be an $[0,1]$-valued random variable with distribution function F_{bd}, put $d_n := \mathbb{E}X^n$, and, for $a > 0$, define

$$(a)^{\overline{k}} := \begin{cases} 1, & k = 0, \\ a(a+1)\cdots(a+k-1), & k \geq 1, \end{cases} \qquad \binom{a+k-1}{k} = \frac{(a)^{\overline{k}}}{k!},$$

$$t_k(a) := \binom{a+k-1}{k}\frac{d_{k+1}}{k+1}, \qquad R_N(a) := \sum_{k \geq N} t_k(a).$$

With $L = \log 2$, the simple hazard estimate is

$$\widehat{R}_N(a) := \frac{N\, t_N(a)}{\log N/L - (a - \frac{1}{2})}.$$

It is a real number for $N \in \mathbb{N}_{>0}$ whenever the denominator is nonzero, and has the intended positive-tail interpretation for N sufficiently large that the denominator is positive.
Scope and status. The printed form suppresses a, the moment sequence d_n, and the fact that the more precise conjectural hazard also contains $-\Psi'(\log_2 N)/L$. This is a heuristic first-order estimator: as $N \to \infty$ with fixed $a > 0$, the displayed numerical ratios are structural evidence, not certification, and the corresponding uniform tail asymptotic remains conjectural.
Accent and scripts. The hat means approximation of the positive series remainder $R_N(a)$, not Fourier transformation. The symbol owns only the subscript N.

`\LambertRootApproximation`: numerical Lambert-root approximation

TeX interface. Arity 0; it prints the ambient scalar $\widehat{w}$. The input x, branch k, numerical method, and iteration count are not TeX arguments.
Definition and domain. The exact target is

$$r = W_k(x), \qquad r\mathrm{e}^r = x,$$

where

$$W_0 : [-1/\mathrm{e}, \infty) \longrightarrow [-1, \infty), \qquad W_{-1} : [-1/\mathrm{e}, 0) \longrightarrow (-\infty, -1].$$

The value $\widehat{w} \in \mathbb{R}$ is any numerical approximation chosen on the same monotone real inverse interval as r; the symbol itself fixes no iteration. If I is the closed interval joining r and $\widehat{w}$, I avoids -1, and

$$m_I := \min_{t \in I} \left|\mathrm{e}^t(1+t)\right| > 0,$$

then the ordinary residual gives

$$|\widehat{w} - r| \leq \frac{\left|\widehat{w}\mathrm{e}^{\widehat{w}} - x\right|}{m_I}.$$

For $x \neq 0$, if I also avoids 0, define

$$G_x(w) := w + \log|w| - \log|x|, \qquad \mu_I := \min_{t \in I} |1 + 1/t| > 0.$$

The logarithmic residual then gives the alternative certificate

$$|\widehat{w} - r| \leq \frac{|G_x(\widehat{w})|}{\mu_I}.$$

Scope and status. The branch and input are mandatory ambient data. Both residual inequalities are proved a posteriori theorems. Near the branch point their denominators correctly record the conditioning loss. At $x = -1/\mathrm{e}$, where $r = -1$, the interval necessarily meets -1, so neither certificate applies; the guide returns -1 exactly or changes to the signed square-root branch coordinate. No particular Newton, Halley, series, or rational approximation is privileged.
Accent and scripts. The hat means an approximate inverse root, not Fourier transformation. The symbol owns no subscript or superscript, but any later decoration must state whether it records branch, iteration, precision, or another numerical parameter.

16.5 Local alphabet register

The following roles are intentionally local because assigning them global commands would create more collisions than it removes. Each occurrence must nevertheless instantiate the declaration schema above.

Family	Mandatory local data
Polynomials p_n, P_n, A_n	Coefficient field, variable, degree, leading-coefficient normalization, orthogonality measure if any, and index origin.
Matrices A_N, H_N, T_N, G_N	Row and column index types, dimensions, scalar field, entry formula, and zero-size convention.
Kernels $K(x, y)$	Two domain spaces, scalar codomain, symmetry/Hermitian convention, regularity, and positive-definiteness status.
Measures μ, ν	Underlying measurable space, positivity/signedness, total mass, and density only if one exists.
Random variables X, Y, U_j	Probability space or a statement that only their joint law is relevant, codomain, independence/dependence, and distribution.
Generating functions $A(z), G(z), M(z)$	Ordinary, exponential, Dirichlet, or other series type; coefficient indexing; formal versus analytic interpretation; convergence domain.
Recurrences	Initial values, index range on which the recurrence applies, coefficient ring, and treatment of singular leading coefficients.
Optimization variables	Feasible set, topology/norm, objective codomain, existence and uniqueness status, and whether infima are attained.
Asymptotic phases and amplitudes	Limit variable and direction, uniform parameters, branch choices, order of truncation, and quantified remainder.
Frontier constructions	Complete proposed definition plus one of the status labels in section 1.3; a suggestive name does not confer a theorem.

17 Retired-alias index

This index is a reading aid for historical sources and a migration contract, not a second namespace. Names in the first column are printed through `\commandname` so that the retired control sequences are not executed. Migration is semantic: determine the old occurrence's type and normalization first, then choose the target. A textual replacement performed before that classification is invalid whenever the target column contains "or".

Retired source name(s)	Canonical target	Required disambiguation
`\R`	`\RealNumbers`	Number field only; a radius remains a typed local variable.
`\C`	`\ComplexNumbers`	Number field only; constants and contours remain local.
`\Q`	`\RationalNumbers`	Number field only; do not rewrite a quantile, quadrature rule, or polynomial.
`\N`	`\NaturalNumbers` or `\PositiveIntegers`	Decide from whether zero belongs to the set; never append another raw subscript to the chosen command.
`\Z`	`\IntegerNumbers`	Number field only.
`\Up`, `\up`, `\upf`	`\RvachevUp`	The even compactly supported function, not the bounded or global Fabius function.
`\sinc`	`\SincRad` or `\SincPi`	Inspect the defining formula or zero set. Radian zeros are at nonzero multiples of π; normalized zeros are at nonzero integers.
`\sincpi`, `\sincp`, `\sincpiPartc`	`\SincPi`	These source-local names were explicitly normalized by π.
`\sincPartx`, `\sincPartl`, `\sincPartii`, `\sincPartxx`, `\sincPartll`, `\sincPartcc`, `\sincPartvvv`	`\SincRad`	These generated-fragment aliases were classified from their defining formulas as radian sinc.
`\wt`	`\BinaryDigitSum`	Only for binary Hamming weight; a general graph or code weight remains local.
`\tm`, `\TM`	`\ThueMorseSignSymbol` or `\ThueMorseSign`	Use the symbol-only command in declarations and the argument-taking command in formulas.
`\e`, `\ee`	`\EulerE`	Only the exponential base; an error variable remains local.
`\ii`	`\ImaginaryUnit`	The complex unit, not an index.
`\dd`	`\Differential`	Upright integral differential; a derivative operator is declared separately.
`\Li`	`\Polylogarithm`	The special function, with its order printed as a subscript.
`\Var`	`\Variance`	Probability variance only.
`\Cov`	`\Covariance`	Probability covariance only.
`\E`	`\Expectation`	Expectation only; events and energies are local and semantically labelled.
`\Prob`, `\Pp`	`\Probability`	Probability measure, not polynomial or projection.
`\bigO`, `\Oh`, `\OO`, `\bigOPartl`	`\BigO` or `\BigOAt`	Prefer the explicit-regime command; retain the token-only form only when the same sentence quantifies the limit.

Retired source name(s)	Canonical target	Required disambiguation
\smallo, \smalloPartl, \littleo, \littleoh	\LittleO or \LittleOAt	Prefer the explicit-regime command and state uniformity parameters.
\supp	\SupportOperator or \SupportOf	Use the argument-delimited form in prose; distinguish pointwise, topological, measure, and essential support.
\sgn	\Sign	Real sign with value zero at zero.
\TV	\TotalVariationOf or \TotalVariationDistance	Count the mathematical arguments: one typed object means total variation; two probability measures mean the fixed distance.
\dist	\MetricDistance or \EqualInLaw	Inspect the relation: a numeric metric distance is not equality in distribution.
\ord	\OrderOperator	Multiplicative order, vanishing order, or another order operator only after the local arguments define which notion is meant.
\lcm	\LeastCommonMultiple	State the finite family and empty-family convention.
\vtwo, \vTwo, \nuTwo, \nuu, \vtwoPartii, \vtwoPartcc	\TwoAdicValuation	Preserve the operand when replacing any of these source-local aliases. A nonzero integer operand gives a value in $\mathbb{N}_0$; a nonzero rational operand gives a value in $\mathbb{Z}$ and may have negative valuation. Zero is outside both finite mathematical domains. If zero can occur, explicitly select either the extended value $+\infty$ or a named totalized API such as padicValNat, padicValInt, or padicValRat; all three Lean APIs assign zero the finite value 0, but their codomains differ.
\qbinom	\GaussianBinomial	Supply all three arguments: upper index, lower index, and base.
\qpoch, \poch	\QPochhammer	Supply element, base, and finite length or ∞ explicitly.
\Law	\LawOperator or \LawOf	Use the argument-delimited form for a random variable.
\Lop	\LaplaceTransformSymbol or \LaplaceTransformOf	Decide between an operator token and application; mark bilateral transforms separately.
\abs	\AbsoluteValue	Scalar absolute value or complex modulus; do not use for set cardinality.
\norm	\Norm	Declare the normed space and norm species.
\floor, \ceil	\Floor, \Ceiling	Preserve exact endpoint inequalities.
\pospart, \pos	\PositivePart	Ordered-scalar positive part with value zero on nonpositive inputs.
\set	\SetOf	Supply the ambient type and predicate/list grammar.
\angles	\InnerProduct	Declare inner product versus dual pairing and the linear slot.
\diag	\DiagonalOperator	Matrix constructor only; diagonal extraction uses a separately named local operator.
\Span, \spanop	\SpanOperator	State the scalar field.

Retired source name(s)	Canonical target	Required disambiguation
`\Tr`	`\TraceOperator`	Matrix or trace-class operator trace; state the space.
`\rank`	`\RankOperator`	Rank over the stated field.
`\Res`	`\ResidueOperator`	Complex residue; a named arithmetic sequence retains its semantic superscript.
`\Id`	`\IdentityOperator`	Identity map on a stated domain.
`\Log`	`\PrincipalLogarithm`	Only the principal complex branch; a formal logarithm or another branch is separately labelled.
`\defeq`	`\DefinitionEquals`	Introduction of a definition, not a proved equality.
`\Fglobal`, `\Fs`, `\extF`, `\Fext`	`\FabiusGlobal`	Signed global Fabius extension only.
`\InvF`	`\FabiusQuantile` or `\FabiusClampedQuantile`	Decide from the input domain: $[0, 1]$ for the ordinary inverse, all reals for the clamped totalization.
`\fall`, `\falling`	`\FallingFactorial`	Supply the base and nonnegative length as two explicit arguments; the length-zero product is one.
`\rise`, `\rising`	`\RisingFactorial`	Supply the base and nonnegative length as two explicit arguments; the length-zero product is one.
`\Lap`	`\LaplaceTransformSymbol`	Bare transform symbol only; use `\LaplaceTransformOf` when the transformed operand belongs inside the semantic command.
`\Mell`	`\MellinTransformSymbol`	Bare transform symbol only; use `\MellinTransformOf` when the transformed operand belongs inside the semantic command.
`\Mellin`	`\MellinTransformSymbol` or `\MellinTransformOf`	Use the symbol command only for a bare operator token or transform-arrow label. When the operand is present—whether formerly bracketed, postfixed, or encoded as a subscript—put that operand inside `\MellinTransformOf`.
`\stirone`	`\UnsignedStirlingFirstKind`	Preserve the upper and lower triangular indices; this is the unsigned cycle-count triangle.
`\stirtwo`	`\StirlingSecondKind`	Preserve the upper and lower triangular indices; this is the set-partition triangle.
`\eulerian`	`\TypeAEulerianNumber`	The legacy single-angle array is type A, not type B or second order.
`\eulertwo`	`\SecondOrderEulerianNumber`	The legacy double-angle array counts descents of Stirling permutations.
`\BellP`	`\ExponentialPartialBellPolynomial`	Supply the total degree and block/derivative count; retain the exponential factorial normalization.
`\BellC`	`\ExponentialCompleteBellPolynomial`	Supply the total degree; do not replace it by a Bell number.

Retired source name(s)	Canonical target	Required disambiguation
\Bell	\BellNumber, \ExponentialCompleteBellPolynomial, \ExponentialPartialBellPolynomial, or a semantically named non-Bell family	Inspect the defining equation before using it: a scalar indexed Bell sequence is a Bell number, while genuine one- and two-index Bell polynomial families are complete and partial exponential, respectively. If the legacy glyph instead names another object—for example report H's geometric Bernoulli Appell family—rename that object \GeometricBernoulliPolynomial; do not recast it as a Bell object.
\coeff	\CoefficientExtraction	Supply the coefficient variable and exponent as the command's two explicit arguments; the series, polynomial, or other operand follows the command.
\OrdinaryBellPolynomial	\OrdinaryPartialBellPolynomial	Preserve both indices and the ordinary-series, rather than exponential-series, normalization.
\OrdinaryBellSeriesOf, \OrdinaryGeneratingSeriesOf	\OrdinaryGeneratingFunctionOf	Supply the coefficient-family argument and keep the ordinary generating convention explicit.
\NormalizedInverseCoefficient	\NormalizedReversionCoefficient	Supply both the series family and coefficient index; this is compositional reversion, not a reciprocal.
\NormalizedOrdinaryCoefficient	\OrdinaryGeneratingCoefficient	Supply both the coefficient family and index; this is not a reversion coefficient.
\PosetMinimum, \PosetMaximum	\PartitionLatticeMinimum, \PartitionLatticeMaximum	These aliases denoted the discrete and one-block partitions of a partition lattice, not numeric bounds or endpoints of an arbitrary poset.
\TracePowerSum	\TraceBellArgument	Preserve the index and the scaling $-(k-1)!\,\mathrm{tr}(A^k)$, not merely the raw trace.
\cyc	\CycleCountOperator	Permutation cycle count only; a cyclic group or cycle-index polynomial remains locally typed.
\setpart	\SetPartitionOperator	The set of set partitions of the following finite-set operand.
\bitand	\BitwiseAnd	Bitwise AND of declared nonnegative integers, not set intersection or multiplication.

The alias index deliberately does not assign global replacements to bare local letters, generic hats, primes, stars, or calligraphic alphabets. Those collisions require a type- and-scope decision, recorded in section 16, rather than a lexical substitution.

18 Symbol and shared-command indexes

18.1 Rendered-symbol index

The rendered column is intentionally redundant with the semantic command name. The redundancy makes a PDF search by glyph useful while preserving a stable source-level identifier.

Rendered symbol family	Shared command(s)	Distinguishing data
$2026-09-01$	`\FabiusNotationVersion`	Date-valued snapshot stamp for the shared notation contract; it is metadata, not a mathematical variable.
$\mathbb{N}_0, \mathbb{N}_{>0}$	`\NaturalNumbers`, `\PositiveIntegers`	Zero included versus excluded.
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \overline{\mathbb{R}}$	`\IntegerNumbers`, `\RationalNumbers`, `\RealNumbers`, `\ComplexNumbers`, `\ExtendedRealNumbers`	Algebraic number system versus ordered extension by two infinities.
$F_{\mathrm{bd}}, F_{\mathrm{gl}}, \mathrm{up}$	`\FabiusBounded`, `\FabiusGlobal`, `\RvachevUp`	Bounded CDF, signed global extension, and even compactly supported density-shaped function.
$Q_F, Q_{F,\mathrm{cl}}$	`\FabiusQuantile`, `\FabiusClampedQuantile`	Inverse on $[0,1]$ versus clamped totalization on $\mathbb{R}$.
$\mathrm{e}, \mathrm{i}, \mathrm{d}x$	`\EulerE`, `\ImaginaryUnit`, `\Differential`	Exponential base, complex unit, and integration differential.
$\lvert x\rvert, \lVert x\rVert, \lvert A\rvert$	`\AbsoluteValue`, `\Norm`, `\CardinalityOf`	Scalar modulus, norm, and set cardinality; the types disambiguate identical bars.
$\lfloor x\rfloor, \lceil x\rceil, \{x\}, (x)_+$	`\Floor`, `\Ceiling`, `\FractionalPart`, `\PositivePart`	Integer rounding, fractional residue, and ordered positive part.
$\mathbf{1}_A, \{x : P(x)\}, \langle x, y\rangle$	`\IndicatorOf`, `\SetOf`, `\InnerProduct`	Function, set-builder delimiters, and inner-product/declared duality delimiters.
$:=, \mathrm{sgn}, \mathrm{id}$	`\DefinitionEquals`, `\Sign`, `\IdentityOperator`	Definition relation, real sign, and identity map.
$F^{-1}, F^{\langle -1\rangle_\circ}$	`\MultiplicativeInverseOf`, `\CompositionalInverseOf`	Inverse under multiplication versus inverse under composition.
$\mathrm{span}, \mathrm{tr}, \mathrm{rank}, \mathrm{diag}$	`\SpanOperator`, `\TraceOperator`, `\RankOperator`, `\DiagonalOperator`	Linear-algebra operators with an explicit field and space.
$\mathrm{Res}, \mathrm{lcm}, \mathrm{ord}, \mathrm{Log}$	`\ResidueOperator`, `\LeastCommonMultiple`, `\OrderOperator`, `\PrincipalLogarithm`	Complex residue, arithmetic LCM, qualified order species, and principal complex branch.
$(x)^{\underline{n}}, (x)^{\overline{n}}$	`\FallingFactorial`, `\RisingFactorial`	Unit-step falling versus rising products; both equal one at length zero.
$s(n,k), \left[{n \atop k}\right], \left\{{n \atop k}\right\}$	`\SignedStirlingFirstKind`, `\UnsignedStirlingFirstKind`, `\StirlingSecondKind`	Signed first-kind, unsigned cycle, and second-kind partition triangles.
$\mathsf{L}^{\mathrm{Lah}}_{n,k}$	`\LahNumber`	Unsigned Lah number with ordered blocks and an unordered block collection.
$\left\langle{n \atop k}\right\rangle, \mathsf{E}^{\mathrm{Eul},B}_{n,k}, \left\langle\!\left\langle{n \atop k}\right\rangle\!\right\rangle$	`\TypeAEulerianNumber`, `\TypeBEulerianNumber`, `\SecondOrderEulerianNumber`	Type-A, type-B, and second-order Eulerian numbers.
$\mathsf{E}^{\mathrm{Eul},A}_n, \mathsf{E}^{\mathrm{Eul},B}_n, \mathsf{E}^{\mathrm{Eul},2}_n$	`\TypeAEulerianPolynomial`, `\TypeBEulerianPolynomial`, `\SecondOrderEulerianPolynomial`	Corresponding row polynomials, with the evaluation variable following.
$\mathsf{B}^{\mathrm{Bell}}_n, \mathsf{B}^{\mathrm{comp}}_n, \mathsf{T}^{\mathrm{Tou}}_n$	`\BellNumber`, `\ComplementaryBellNumber`, `\TouchardPolynomial`	Set-partition count, signed block-parity sum, and Touchard row polynomial.

Rendered symbol family	Shared command(s)	Distinguishing data
$\mathcal{B}^{\exp}_{n,k}, \mathcal{B}^{\exp}_{n}, \mathcal{B}^{\mathrm{ord}}_{n,k}$	\ExponentialPartialBellPolynomial, \ExponentialCompleteBellPolynomial, \OrdinaryPartialBellPolynomial	Exponential partial/complete and ordinary partial Bell-polynomial normalizations.
$[z^n]$	\CoefficientExtraction	Coefficient of z^n in the following formal or analytic series operand.
$\mathrm{cyc}, \mathrm{Part}, a \,\&\, b$	\CycleCountOperator, \SetPartitionOperator, \BitwiseAnd	Cycle count, set-partition constructor, and nonnegative-integer bitwise AND.
$A^{\mathrm{egf}}, A^{\mathrm{ogf}}$	\ExponentialGeneratingFunctionOf, \OrdinaryGeneratingFunctionOf	Explicit EGF versus OGF labels.
$f_j^{\mathrm{rev,norm}}, x_j^{\mathrm{ogf}}$	\NormalizedReversionCoefficient, \OrdinaryGeneratingCoefficient	Normalized exponential-series reversion input versus ordinary-series coefficient ledger.
$\bot_\Pi, \top_\Pi, s_k^{\mathrm{tr}}$	\PartitionLatticeMinimum, \PartitionLatticeMaximum, \TraceBellArgument	Partition-lattice endpoints and scaled trace input for Bell identities.
$\mathbb{F}_q$	\FiniteField	Finite field with declared prime-power order q.
$\mathsf{C}_n^{\mathrm{Cat}}, \mathsf{G}_n^{\mathrm{Greg}}, \mathsf{G}_n^{\mathrm{Gen}}, \mathsf{G}_n^{\mathrm{Gen}}$	\CatalanNumber, \GregoryCoefficient, \GenocchiNumber, \GenocchiPolynomial	Distinct Catalan, Gregory, and Genocchi families.
$\mathrm{Ber}^{\mathrm{geom}}_{n,q}(x)$	\GeometricBernoulliPolynomial	Geometric-uniform-convolution Appell family; degree and deformation parameter are explicit.
$\mathrm{G}_{\mathrm{Barnes}}$	\BarnesGFunction	Barnes G-function, not a Gregory or Genocchi coefficient.
$\mathsf{H}_n^{\mathrm{Bell}}, \mathsf{K}_n^{\mathrm{Bell}}$	\BellHessenbergMatrixH, \BellHessenbergMatrixK	Two Hessenberg determinant models for complete Bell polynomials.
$\mathsf{K}^{\mathrm{Assoc}}_{\dim=d}, \zeta_{\mathrm{Lag}}$	\AssociahedronByDimension, \LagrangeShiftCoordinate	Dimension-indexed associahedron and scope-local Lagrange shift coordinate.
$\mathrm{supp}, \mathrm{supp}(f)$, $\mathrm{tsupp}(f)$	\SupportOperator, \SupportOf, \TopologicalSupportOf	Operator token, pointwise support, and its topological closure.
$\mathrm{supp}_{\mathrm{coeff}}(F)$	\CoefficientSupportOf	Coefficient-exponent support in an explicitly stated index set.
$*, \mathrm{dist}, \mathrm{TV}(a), d_{\mathrm{TV}}$	\Convolution, \MetricDistance, \TotalVariationOf, \TotalVariationDistance	Algebraic/measure convolution, generic metric, a one-object variation functional, and the fixed two-measure probability TV convention.
$\mathbb{P}, \mathbb{E}, \mathrm{Var}, \mathrm{Cov}$	\Probability, \Expectation, \Variance, \Covariance	Measure, integral operator, and second-order statistics.
$\mathrm{Law}, \mathrm{Law}(X), \overset{\mathrm{law}}{=}, \xrightarrow{\mathrm{law}}$	\LawOperator, \LawOf, \EqualInLaw, \ConvergesInLaw	Law operator/application, law equality, and weak convergence of laws.
$\mathcal{A}_n$	\DyadicSigmaField	Level-n filtration σ-field with explicitly stated generators.
$\mathrm{Obj}_{\mathrm{decay},n}, \mathrm{Obj}^{\mathrm{lin}}_{\mathrm{decay},n}$	\DecayOptimizationObjective, \LinearizedDecayObjective	Exact versus linearized Fourier-decay objectives.
$\mathcal{F}_{2\pi}, \mathcal{F}_{2\pi}^{-1}$	\FourierTwoPiOperator, \InverseFourierTwoPiOperator	Forward and inverse cycles-per-unit operators.
$\mathcal{F}_{\mathrm{ang}}, \mathcal{F}_{\mathrm{ang}}^{-1}$	\FourierAngularOperator, \InverseFourierAngularOperator	Forward and inverse angular-frequency operators.
$\mathcal{F}_{2\pi}[f], \widehat{f}_{2\pi}$	\FourierTwoPiTransformOf, \FourierTwoPi	Bracketed operator application and hat rendering of the same 2π transform.

Rendered symbol family	Shared command(s)	Distinguishing data
$\mathcal{F}_{\mathrm{ang}}[f]$, $\widehat{f}_{\mathrm{ang}}$	`\FourierAngularTransformOf`, `\FourierAngular`	Bracketed operator application and hat rendering of the same angular transform.
$\widehat{h}_{\mathrm{FS}}[k]$, $\widehat{h}_{\mathrm{FS};\,T}[k]$	`\FourierSeriesCoefficient`, `\FourierSeriesCoefficientAtPeriod`	Unit-period and explicitly T-periodic normalized coefficients; neither is a continuous line transform.
$\mathcal{L}, \mathcal{L}[f]$	`\LaplaceTransformSymbol`, `\LaplaceTransformOf`	One-sided operator token and application; bilateral use is subscripted.
$\mathcal{M}, \mathcal{M}[f]$	`\MellinTransformSymbol`, `\MellinTransformOf`	Operator token and application on a stated convergence strip.
M_X, φ_X	`\MomentGeneratingFunction`, `\CharacteristicFunction`	Exponential moment transform and positive-sign probabilistic Fourier transform.
$\mathrm{sinc}_{\mathrm{rad}}, \mathrm{sinc}_{\pi}$	`\SincRad`, `\SincPi`	Radian versus π-normalized entire sinc.
$\mathrm{wt}_2, \varepsilon, \varepsilon_n$	`\BinaryDigitSum`, `\ThueMorseSignSymbol`, `\ThueMorseSign`	Bit count, reserved sequence glyph, and indexed sign.
ν_p, ν_2	`\PadicValuation`, `\TwoAdicValuation`	Prime-parametric and specialized two-adic valuations: $\mathbb{N}_0$-valued on nonzero integers and $\mathbb{Z}$-valued on nonzero rationals. Zero is undefined unless an extended or named totalized convention is stated.
$(a;q)_n$, $\begin{bmatrix} n \\ k \end{bmatrix}_q$, $[n]_q$	`\QPochhammer`, `\GaussianBinomial`, `\QInteger`	Every base and finite/infinite length explicit.
$(a;q)_\nu$, $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}_q$, $[z]_q$	`\GeneralizedQPochhammer`, `\GeneralizedGaussianBinomial`, `\GeneralizedQNumber`	Generalized q-families whose integer or complex order, analytic regime, and branch data are declared locally.
$\begin{bmatrix} n \\ n_1,\dots,n_r \end{bmatrix}_q$	`\QMultinomial`	Gaussian multinomial with the complete lower list and base explicit.
$\mathrm{Li}, W, \Gamma, \zeta, \zeta(s,a)$	`\Polylogarithm`, `\LambertW`, `\GammaFunction`, `\RiemannZeta`, `\HurwitzZeta`	Named special functions; order, branch, shift, and continuation domains printed as applicable.
$\omega, \mathsf{L}_n^{\mathrm{Lam}}, \mathsf{L}_n^{\mathrm{Lam,sc}}$	`\WrightOmega`, `\LambertPolynomial`, `\ScaledLambertPolynomial`	Wright omega, canonical zero-based Lambert polynomial, and its factorial scaling.
$\mathscr{PL}_{\mathrm{pow},K}$, $\mathscr{PL}_{\mathrm{poly},K}$, $\mathscr{PL}_{\mathrm{rat},K}$, $\mathscr{PL}_{\mathrm{itLaur},K}$	`\PolynomialLogPowerSeriesRing`, `\PolynomialLogLaurentRing`, `\RationalLogLaurentField`, `\IteratedLaurentLogField`	Power-series, polynomial-coefficient Laurent, rational-coefficient Laurent, and iterated Laurent-log classes.
$\mathcal{G}_{\mathrm{poly},K}$	`\NearIdentityPolynomialLogGroup`	Near-identity composition group over the named coefficient field.
$\mathcal{O}, o, \mathcal{O}_a(g)$, $o_a(g)$	`\BigO`, `\LittleO`, `\BigOAt`, `\LittleOAt`	Token-only and explicit-limit asymptotic operators.
$\mathcal{O}^{\mathrm{form}}, \mathcal{O}_t^{\mathrm{form}}\left(t^N\right)$	`\FormalBigO`, `\FormalBigOAt`	Formal valuation-tail token and argument-bearing form; neither is an analytic estimate.

18.2 Alphabetical shared-command index

The following is the complete source-level index for version 2026-09-01. Argument counts are part of the API; braces displayed in a command name are mathematical arguments, not optional typography.

Command	Arity	Normative entry
\AbsoluteValue	1	Absolute value, norm, distance.
\AssociahedronByDimension	1	Associahedron indexing.
\BarnesGFunction	0	Barnes G-function.
\BellHessenbergMatrixH	1	Bell Hessenberg matrices.
\BellHessenbergMatrixK	1	Bell Hessenberg matrices.
\BellNumber	1	Bell numbers.
\BigO	0	Asymptotic operators.
\BigOAt	2	Asymptotic operators.
\BinaryDigitSum	0	Binary digit sum.
\BitwiseAnd	0	Bitwise AND.
\CardinalityOf	1	Semantic delimiters.
\CatalanNumber	1	Catalan numbers.
\Ceiling	1	Rounding and positive part.
\CharacteristicFunction	1	Characteristic functions.
\CoefficientExtraction	2	Coefficient extraction.
\CoefficientSupportOf	1	Coefficient support.
\ComplementaryBellNumber	1	Complementary Bell numbers.
\ComplexNumbers	0	Number systems.
\CompositionalInverseOf	1	Compositional inverse.
\ConvergesInLaw	0	Law and weak convergence.
\Convolution	0	Convolution.
\Covariance	0	Expectation and moments.
\CycleCountOperator	0	Permutation cycle count.
\DecayOptimizationObjective	1	Decay objectives.
\DefinitionEquals	0	Semantic delimiters.
\DiagonalOperator	0	Matrix operators.
\Differential	0	Calculus.
\DyadicSigmaField	0	Dyadic filtration.
\EqualInLaw	0	Law and weak convergence.
\EulerE	0	Elementary constants.
\Expectation	0	Expectation and moments.
\ExponentialCompleteBellPolynomial	1	Complete exponential Bell polynomials.
\ExponentialGeneratingFunctionOf	1	Exponential generating functions.
\ExponentialPartialBellPolynomial	2	Partial exponential Bell polynomials.
\ExtendedRealNumbers	0	Number systems.
\FabiusBounded	0	Bounded Fabius function.
\FabiusClampedQuantile	0	Fabius inverses.
\FabiusGlobal	0	Global extension.
\FabiusNotationVersion	0	Contract version.
\FabiusQuantile	0	Fabius inverses.
\FallingFactorial	2	Powers and factorials.
\FiniteField	0	Finite fields.
\Floor	1	Rounding and positive part.
\FormalBigO	0	Formal valuation tails.
\FormalBigOAt	2	Formal valuation tails.
\FourierAngular	1	Angular Fourier transform.
\FourierAngularOperator	0	Angular Fourier transform.
\FourierAngularTransformOf	1	Angular Fourier transform.
\FourierSeriesCoefficient	2	Fourier series.
\FourierSeriesCoefficientAtPeriod	3	Fourier series.
\FourierTwoPi	1	2π Fourier transform.

Command	Arity	Normative entry
`\FourierTwoPiOperator`	0	2π Fourier transform.
`\FourierTwoPiTransformOf`	1	2π Fourier transform.
`\FractionalPart`	1	Rounding and positive part.
`\GammaFunction`	0	Gamma function.
`\GaussianBinomial`	3	Gaussian binomial.
`\GeneralizedGaussianBinomial`	3	Generalized Gaussian binomial.
`\GeneralizedQNumber`	2	Generalized q-numbers.
`\GeneralizedQPochhammer`	3	Generalized q-Pochhammer symbols.
`\GenocchiNumber`	1	Genocchi numbers.
`\GenocchiPolynomial`	1	Genocchi polynomials.
`\GeometricBernoulliPolynomial`	2	Geometric Bernoulli polynomials.
`\GregoryCoefficient`	1	Gregory coefficients.
`\HurwitzZeta`	2	Hurwitz zeta function.
`\IdentityOperator`	0	Maps and the identity map.
`\ImaginaryUnit`	0	Elementary constants.
`\IndicatorOf`	1	Indicators.
`\InnerProduct`	1	Semantic delimiters.
`\IntegerNumbers`	0	Number systems.
`\InverseFourierAngularOperator`	0	Angular Fourier transform.
`\InverseFourierTwoPiOperator`	0	2π Fourier transform.
`\IteratedLaurentLogField`	1	Iterated Laurent-log field.
`\LagrangeShiftCoordinate`	0	Lagrange shift coordinate.
`\LahNumber`	2	Lah numbers.
`\LambertPolynomial`	1	Canonical Lambert polynomials.
`\LambertW`	0	Lambert (W).
`\LaplaceTransformOf`	1	Laplace transforms.
`\LaplaceTransformSymbol`	0	Laplace transforms.
`\LawOf`	1	Law and weak convergence.
`\LawOperator`	0	Law and weak convergence.
`\LeastCommonMultiple`	0	Arithmetic functions.
`\LinearizedDecayObjective`	1	Decay objectives.
`\LittleO`	0	Asymptotic operators.
`\LittleOAt`	2	Asymptotic operators.
`\MellinTransformOf`	1	Mellin transform.
`\MellinTransformSymbol`	0	Mellin transform.
`\MetricDistance`	0	Absolute value, norm, distance.
`\MomentGeneratingFunction`	1	Moment-generating functions.
`\MultiplicativeInverseOf`	1	Multiplicative reciprocal.
`\NaturalNumbers`	0	Number systems.
`\NearIdentityPolynomialLogGroup`	1	Near-identity polynomial-log group.
`\Norm`	1	Absolute value, norm, distance.
`\NormalizedReversionCoefficient`	2	Normalized reversion coefficients.
`\OrderOperator`	0	Logarithm and order.
`\OrdinaryGeneratingCoefficient`	2	Ordinary generating coefficients.
`\OrdinaryGeneratingFunctionOf`	1	Ordinary generating functions.
`\OrdinaryPartialBellPolynomial`	2	Partial ordinary Bell polynomials.
`\PadicValuation`	1	Valuations.
`\PartitionLatticeMaximum`	0	Partition-lattice maximum.
`\PartitionLatticeMinimum`	0	Partition-lattice minimum.
`\Polylogarithm`	0	Polylogarithm.
`\PolynomialLogLaurentRing`	1	Polynomial-log Laurent ring.
`\PolynomialLogPowerSeriesRing`	1	Polynomial-log power-series ring.
`\PositiveIntegers`	0	Number systems.
`\PositivePart`	1	Rounding and positive part.
`\PrincipalLogarithm`	0	Logarithm and order.
`\Probability`	0	Probability spaces.

Command	Arity	Normative entry
`\QInteger`	2	(q)-integers.
`\QMultinomial`	3	q-multinomial coefficients.
`\QPochhammer`	3	(q)-Pochhammer symbols.
`\RankOperator`	0	Matrix operators.
`\RationalLogLaurentField`	1	Rational-log Laurent field.
`\RationalNumbers`	0	Number systems.
`\RealNumbers`	0	Number systems.
`\ResidueOperator`	0	Residues.
`\RiemannZeta`	0	Zeta functions.
`\RisingFactorial`	2	Powers and factorials.
`\RvachevUp`	0	Rvachev up-function.
`\ScaledLambertPolynomial`	1	Scaled Lambert polynomials.
`\SecondOrderEulerianNumber`	2	Second-order Eulerian numbers.
`\SecondOrderEulerianPolynomial`	1	Second-order Eulerian row polynomials.
`\SetOf`	1	Semantic delimiters.
`\SetPartitionOperator`	0	Set partitions.
`\Sign`	0	Rounding and positive part.
`\SignedStirlingFirstKind`	2	Signed first-kind Stirling numbers.
`\SincPi`	0	Sinc normalizations.
`\SincRad`	0	Sinc normalizations.
`\SpanOperator`	0	Matrix operators.
`\StirlingSecondKind`	2	Second-kind Stirling numbers.
`\SupportOf`	1	Support.
`\SupportOperator`	0	Support.
`\ThueMorseSign`	1	Thue–Morse sign.
`\ThueMorseSignSymbol`	0	Thue–Morse sign.
`\TopologicalSupportOf`	1	Support.
`\TotalVariationDistance`	0	Probability metrics.
`\TotalVariationOf`	1	Probability metrics.
`\TouchardPolynomial`	1	Touchard polynomials.
`\TraceBellArgument`	1	Scaled Bell trace inputs.
`\TraceOperator`	0	Matrix operators.
`\TwoAdicValuation`	0	Valuations.
`\TypeAEulerianNumber`	2	Type-A Eulerian numbers.
`\TypeAEulerianPolynomial`	1	Type-A Eulerian row polynomials.
`\TypeBEulerianNumber`	2	Type-B Eulerian numbers.
`\TypeBEulerianPolynomial`	1	Type-B Eulerian row polynomials.
`\UnsignedStirlingFirstKind`	2	Unsigned first-kind Stirling numbers.
`\Variance`	0	Expectation and moments.
`\WrightOmega`	0	Wright omega.

19 Lean crosswalk and formalization boundary

The crosswalk maps semantic mathematical objects to actual current Lean declarations. It does not imply that every identity printed in a frontier document has been formalized. A row marked “notation only” records precisely that no single global Lean object is asserted by the shared command. Lean names are case-sensitive and module paths are relative to `Analysis/FabiusFunction/Lean/FabiusFunction/`.

Catalogue object	Lean declaration and module	Exact boundary
F_{bd}	`fabiusReal` in `Basic.lean`; canonical witness `fabius` and proof `fabius_spec` in `PaperStatements.lean`	`fabiusReal` is the real-valued view of a `BoundedFabius`; the witness and `IsFabius` proof select the canonical object.
Abstract Fabius specification	`IsFabius` in `Basic.lean`	The declaration's actual type is authoritative for hypotheses; the catalogue summary does not manufacture a weaker premise-free theorem.
up	`rvachevUp` in `Basic.lean`	Takes a `BoundedFabius` argument. Evenness, bounds, and other properties are separate theorems, not fields silently attached to an arbitrary function.
F_{gl}	`extendedFabius` in `Basic.lean`	Takes a `BoundedFabius`; translate-cell and calculus results occur in later dedicated modules.
Q_F and $Q_{F,\mathrm{cl}}$	`fabiusOrderIso` and `fabiusInv` in `FabiusInverse.lean`	The formal definitions take both a bounded object and its `IsFabius` proof. Domain behavior must be checked against the exact declaration before claiming ordinary versus clamped inverse laws.
F_n^{spl} and its local inverse G_n^{spl}	`reportFiniteFabiusApproximant` in `QuarterSplineLocalPolynomial.lean`; `fabiusUniformSpline` in `FabiusUniformSpline.lean`	The exact index bridge identifies the report approximant at n with the uniform spline at $n-1$. Strict monotonicity is proved only for $n \geq 3$ on $[1/4, 1/4+2^{-n}]$. No global G_n^{spl} is defined in Lean; the quantile theorem assumes a supplied local left inverse and is not `fabiusInv`.
Φ_{ang} and $\Phi_{2\pi}$	`centeredSincProduct` in `FourierLaplaceRotation.lean`; `rvachevFourierProduct` in `Basic.lean`; bridge `centeredSincProduct_eq_rvachevFourierProduct`	The bridge is $\mathrm{centeredSincProduct}(z) = \mathrm{rvachevFourierProduct}(z/(2\pi))$. Identifying the product with the integral transform uses `rvachevFourier_eq_product` and retains F : `BoundedFabius` and h_F : `IsFabius` F.
Endpoint kernel $\mathcal{K}(z,w)$	Notation only	Formal frontier construction in the two-scale endpoint coefficient algebra. No exact current Lean definition of $\mathcal{K}$, $\mathcal{R}$, or its fixed-point algebra is claimed.
ε_n	`thueMorseSign` in `Arithmetic.lean`	Integer-valued sign sequence on `Nat`; digit, recurrence, and product identities live in their named Thue–Morse modules.
Prime-power Pascal-row valuations	`primePowerChoose_padicValNat_add`, `primePowerChoose_padicValNat`, and `twoPowChoose_padicValNat` in `PrimePowerBinomialValuation.lean`	Arbitrary-prime formulas include $j = p^m$ and $m = 0$; the dyadic theorem uses $0 < j < 2^m$. No theorem for the distinct endpoint-flat weights $\binom{2^m-2}{j-1}$ is inferred.

Catalogue object	Lean declaration and module	Exact boundary
MacMahon's q-Catalan polynomial	`qCatalan`, `qInt_X_dvd_gaussianBinomial_int`, `qInt_X_mul_qCatalan`, `qCatalan_natDegree`, and `qCatalan_eval_one` in `QCatalan.lean`	An exact quotient in $\mathbb{Z}[X]$, its product identity, degree $n(n-1)$, and evaluation at one are proved. Coefficientwise nonnegativity is not a theorem of this module.
Jackson q-beta function	`qBeta`, `qBeta_eq_prod`, `qBeta_eq_qGamma`, `qBeta_comm`, `qBeta_pos`, `qBeta_add_one_left`, and `qBeta_add_one_right` in `QBetaIntegral.lean`	The function is real-valued. Product, gamma-quotient, symmetry, positivity, and recurrence theorems retain $0 < q < 1$ and $x, y > 0$; no meromorphic continuation or $q \uparrow 1$ theorem is inferred.
Cyclotomic carry, primitive-root blocks, and q-Lucas	`cyclotomic_exponent_eq_one_iff` and `cyclotomic_dvd_gaussianBinomial_iff` in `CyclotomicDivisibility.lean`; `finiteQPochhammerIn_isPrimitiveRoot` in `PrimitiveRootBlock.lean`; `gaussianBinomial_q_lucas` in `QLucas.lean`	The carry criterion is over $\mathbb{Q}[X]$; the root-block and q-Lucas theorems assume a primitive root in an integral domain and retain $0 < d$ and the stated residue bounds. They do not define a general `\CyclotomicValuation` function.
Exact full moments	`moment` in `Arithmetic.lean`; `momentIntegral` in `Basic.lean`	Rational recurrence and real integral are distinct objects; `AnalyticMoments.lean` supplies bridge theorems with hypotheses.
Exact half moments	`halfMoment` in `Arithmetic.lean`; `halfMomentIntegral` in `Basic.lean`	Rational recurrence and real integral are not identified by notation alone.
R_n^{Res}	`reshetnikov` in `Arithmetic.lean`	The Lean value is rational at the definition boundary; integrality, positivity, and oddness require their named theorems.
D_n^{dyad}	`dyadicDenominator` in `Arithmetic.lean`	A natural number definition; bounds and divisibility results occur in `DenominatorBound.lean` and related modules.
Number systems and elementary algebra	Mathlib types `Nat`, `Int`, `Rat`, `Real`, `Complex` and their standard APIs	Printed coercions are mathematical exposition. Lean coercions and side conditions remain those required by the elaborated theorem.
Intervals	`Set.Icc`, `Set.Ioo`, `Set.Ico`, `Set.Ioc`	Endpoint order in the Lean constructor matches the four catalogue interval conventions.
Probability laws and pushforwards	Mathlib `Measure.map` and probability-measure APIs	A prose random-series representation is not Lean-proved unless a project theorem states the corresponding measure equality.
Fourier commands	Mathlib Fourier-transform APIs plus project-specific theorems in dedicated Fourier modules	The shared commands choose print normalization. They are not names of one new Lean definition and do not erase Mathlib's hypotheses.

Catalogue object	Lean declaration and module	Exact boundary
Generic Newton interpolation	`newtonCoeff`, `newtonPoly`, `eval_newtonPoly`, `newtonPoly_eq_interpolate`, `newtonCoeff_eq_sum`, `nodal_range_pow`, `prod_erase_pow_sub_pow`, and `newtonCoeff_pow_eq_sum` in `NewtonInterpolation.lean`	Definitions are total over a field; interpolation and divided differences retain their product-nonzero or node-injectivity hypotheses. The geometric formulas do not create a separately named Fabius-specific coefficient.
Finite-node Lagrange–Rvachev decoder	`lagrangeRvachevDecoder` and `lagrangeRvachevAtomCoefficient` in `LagrangeRvachevSynthesis.lean`	Exact synthesis uses $M \neq 0$, $\lvert s\rvert - 1 \le \nu_2(M)$, and $x \in [-1, 1]$. Node injectivity is required for cardinal evaluation, not polynomial reconstruction; row-sum normalization also requires a nonempty node family.
Shifted dyadic polynomial self-sampling	`integral_polynomial_mul_rvachevUp_eq_dyadic_tsum` in `PolynomialCombExactness.lean`	Exact for every polynomial of degree at most N, every real shift, and a bounded Fabius object satisfying `IsFabius`. Compact support makes the displayed bilateral sum finite; exactness failure at degree $N + 1$ is not claimed.
Legendre Gaunt and zero-row Wigner square	`legendreGauntAdmissible` and `legendreWignerThreeJZeroSqRat` in `LegendreGauntClosedForm.lean`; finite Rvachev Gram sums in `FabiusLegendreGauntClosedForm.lean`	Only the total rational square for natural-number indices and zero magnetic row is defined. No signed symbol, phase, half-integer/nonzero-magnetic theory, or general $3j/6j/9j$ API is claimed; the real Gram formula retains `IsFabius`.
$\mathcal{A}_n$	Notation only	Each document declares its generators and probability space; no single project-wide Lean filtration is asserted by the command.
$\mathrm{Obj}_{\mathrm{decay},n}$ and $\mathrm{Obj}^{\mathrm{lin}}_{\mathrm{decay},n}$	Notation only	They name document-local typed functionals in the Fourier-decay frontier. No general formalization or equality between exact and linearized objectives is claimed.
Generic q-series, fractional, matrix, spectral, and frontier entries	Definition or classical notation unless a row or document cites an exact declaration	Their presence in this encyclopedic catalogue is not evidence that the general theory or a proposed Fabius specialization is formalized.

The safe direction of use is Lean-to-prose: inspect a declaration and then state exactly that theorem with all hypotheses. Similar names, compiled object files, an earlier audit, or a catalogue entry are not substitutes for inspecting the current declaration.

20 Active-source manifest and reproducibility

20.1 Snapshot boundary

The inventory below is the recursive active tree observed on 2026-09-01 at the source-validation checkpoint. It excludes every path below `docs/archive/` and counts this catalogue exactly once. It contains 160 mathematical content sources: 62 standalone documents and 98 included or generated fragments. The separate shared contract `docs/fabius-notation.tex` raises the total before this catalogue to 161 active TeX files. Including this standalone catalogue gives 162 active TeX files and 63 standalone documents. "Standalone" means that the uncommented source contains `\documentclass`; every other content source is a fragment inheriting its root preamble.

Manifest group	Standalone	Fragments	Total
`Fabius_Function_and_Rvachev_Up/`	1	0	1
`fabius_lean_walkthrough/`	1	0	1
`FabiusFunction_Mathematical_Glossary/`	1	0	1
`papers/`	7	0	7
Frontier synthesis root	1	0	1
`drafts/combinatorial-coefficient-calculus/`	6	0	6
`drafts/exponents-and-q-series/`	7	20	27
`drafts/frontier-compilations/`	3	1	4
`drafts/integration-and-transforms/`	1	7	8
`drafts/inverse-and-sampling/`	3	30	33
`drafts/lambert-w/`	7	4	11
`drafts/representations/`	12	26	38
`drafts/rvachev_up_fourier_decay/`	1	0	1
`drafts/spectra-and-arithmetic/`	10	10	20
`drafts/thue-morse/`	1	0	1
Mathematical content subtotal	62	98	160
Shared notation contract	0	1	1
This catalogue	1	0	1
Active TeX total including catalogue	63	99	162

The frontier paths in the table are all below `semi-formalized-research-frontiers/`; the root row is its standalone synthesis. Counts describe files, not distinct mathematical claims or publication units. Corrected and source-faithful historical-paper variants are counted separately because each is an active TeX source requiring an interpretable notation namespace.

20.2 Reproducible inventory and strict gate

The repository audit is invoked from the repository root as follows.

```
python Analysis/FabiusFunction/scripts/tex_notation_audit.py --strict --list
python Analysis/FabiusFunction/scripts/tex_notation_audit.py --semantic --strict --list
```

For each active source it records the relative path, byte and line counts, SHA-256 digest, standalone/fragment classification, canonical-input count, input targets, and lexical command counts. It excludes the archive by resolved path, not by a fragile filename pattern. A focused catalogue check supplies this file's path as the positional argument.

The strict gate detects retired commands, local redefinitions of shared semantic commands, missing or duplicated canonical inputs, fragment imports of the shared input, selected literal bypasses of semantic operators, and known script-composition hazards. The opt-in semantic gate additionally requires classification of raw hats, delimiter spellings, number systems, indicator symbols, digit sums, asymptotic operators, sinc normalizations, and probability operators. Its sole raw-hat exemption is a descriptive one-line local macro definition whose immediately preceding `LOCAL-NON-FOURIER-HAT:` comment names that macro and states its meaning; direct raw-hat use remains an error. The audit is deliberately not a TeX parser and cannot infer the meaning of every one-letter variable. The local declaration schema and collision register close that semantic gap. On any scan whose resolved file set is the complete active corpus, the audit also compares valid marker/declaration pairs with the reserved `\localaccentname` metadata in this catalogue and rejects missing, extra, or duplicate rows. The dedicated metadata token is not available for examples or literal listings. Rendered-PDF inspection later closes the composition and layout gap.

21 Completeness and maintenance contract

At shared-contract version 2026-09-01:

- `docs/fabius-notation.tex` defines exactly 91 shared commands.
- Every one of those 91 command names occurs in the canonical quick reference and in the alphabetical command index, with its exact arity.
- Every command belongs to a detailed normative entry defining its type-level role, normalization, exceptional cases, or the local data the command intentionally requires at each use.
- Both line-Fourier conventions, unit-period and explicit-period Fourier-series coefficients, both sinc normalizations, both Fabius inverses, the bounded/global/up-function triad, zero-inclusive/positive naturals, exact/linearized decay objectives, and pointwise/topological support have distinct printed forms.
- The retired-alias index is exhaustive with respect to the audit's retired-command registry. Ambiguous aliases map only after semantic classification.
- The active corpus has 35 validated `LOCAL-NON-FOURIER-HAT:` marker/declaration pairs representing 35 distinct document-local command names. Every distinct name occurs exactly once in the exhaustive accent register, and the whole-corpus audit rejects missing, extra, or duplicate registry metadata.
- Symbols not promoted to the shared API remain governed by the local declaration schema; they have no implicit corpus-wide type or normalization.

A future shared-command change is incomplete unless it updates, in the same logical change, the version stamp, quick reference, detailed entry, rendered-symbol index, alphabetical index, alias mapping if applicable, audit registry, and at least one focused strict-audit result. A future source that introduces a genuinely new mathematical convention updates the relevant encyclopedic entry even when no new shared command is needed. A convention found only in an explicit historical quotation is entered in the retired-alias register, not revived as normative project notation.

This catalogue is a notation contract and an interpretive reference. It does not convert frontier assertions into theorems, does not weaken hypotheses found in Lean, and does not substitute lexical or rendered checks for mathematical verification. Its promise is narrower and exact: every symbol has a declared semantic route, every global command has one normative meaning, and every unavoidable local ambiguity has an explicit scope rule.
